

Jim Ratliff's Graduate-Level Course in Game Theory

Here are lecture notes from a game-theory course I taught to students in their second year of the [economics](#) PhD program at the [University of Arizona](#) during the 1992-1997 period.

The material presented would also be helpful to first-year PhD students learning game theory as part of their microeconomic-theory sequence, as well as to advanced undergraduates learning game theory. I consider the exposition detailed, rigorous, and self-contained.

I no longer teach game theory, so these notes are currently frozen in this state. I'm making them available here because I still get requests for them. I have not updated them to reflect recent advances. I also won't be correcting any errors (though I hope that most of them have already been caught!) or adding any topics.

I would be very interested in and appreciate hearing from anyone who downloads them and finds them useful. Also I *may* eventually post problem sets and their solutions. Let me know if you'd like to be notified of such changes. Please [email me](#).

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Jim Ratliff's Game Theory Course:

1. Strategic-Form Games

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Abstract



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We'll look at noncooperative games which are played only once, which involve only a finite number of players, and which give each player only a finite number of actions to choose from. We'll consider what is called the *strategic* (or *normal*) form of a game. Although our formal treatment will be more general, our exemplary paradigm will be a two-person, simultaneous-move matrix game.

The strategic (or "normal") form of a game is a natural and adequate description of a simultaneous-move game. It is also a useful platform on which to perform at least some of our analysis of games which have a more complicated temporal and information structure than a simultaneous-move game has. (In order to perform the remaining analysis of these games, however, we'll later introduce and use the "extensive form.")

We will define a strategic-form game in terms of its constituent parts: players, actions, and preferences. We will introduce the notion of mixed strategies, which are randomizations over actions. Our first step in the analysis of these games will be to solve the Easy Part of Game Theory, viz. the problem of what choice a rational player would make given her beliefs about the choices of her opponents. Later we will turn to the Hard Part of Game Theory: what beliefs the players can rationally hold concerning the choices of their opponents.

Some specific topics we will discuss: best responses to a (deleted) pure-strategy profile; mixed strategies are chosen from the unit simplex; pure strategies are degenerate mixed strategies; the expected payoff to a mixed-strategy profile; payoff to any player is linear in any one player's mixing probabilities; the best-response correspondence; and best-response mixed strategies.



[Course Outline](#)

[Next section:
§2.1: Strategic Dominance](#)

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Introduction

We'll look at noncooperative games which are played only once, which involve only a finite number of players, and which give each player only a finite number of actions to choose from. We'll consider what is called the *strategic* (or *normal*) form of a game. Although our formal treatment will be more general, our exemplary paradigm will be a two-person, simultaneous-move matrix game.

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preferences. We will introduce the notion of mixed strategies, which are randomizations over actions. Our first step in the analysis of these games will be to solve the Easy Part of Game Theory, viz. the problem of what choice a rational player would make given her beliefs about the choices of her opponents. Later we will turn to the Hard Part of Game Theory: what beliefs the players can rationally hold concerning the choices of their opponents.

Individual strategies

We have a nonempty, finite set I of $n \in \mathbb{N} \equiv \{1, 2, \dots\}$ players

$$I = \{1, \dots, n\}. \quad (1)$$

The i -th player, $i \in I$, has a nonempty set of *strategies*—her *strategy space* S_i —available to her, from which she can choose one strategy $s_i \in S_i$.¹ Note—as indicated by the “ i ” subscript—that each player has her own strategy space S_i . Therefore each player has access to her own possibly unique set of strategies. We will assume that each player’s strategy space is finite.

When necessary we will refer to these as *pure strategies* in order to distinguish them from *mixed strategies*, which are randomizations over pure strategies.

Example:

Consider a two-player game between Robin and Cleever. Suppose Robin has two actions available to her: Up and Down. Then her strategy space S_R would be

$$S_R = \{\text{Up}, \text{Down}\}.$$

When she plays the game she can choose only one of these actions. So her strategy s_R would be either $s_R = \text{Up}$ or $s_R = \text{Down}$. Likewise, suppose that Cleever can move left, middle, or right. Then his strategy space is

$$S_C = \{\text{left}, \text{middle}, \text{right}\}.$$

Strategy profiles

For the time being it will be useful to imagine that all players pick their strategies at the same time: player 1 picks some $s_1 \in S_1$, player 2 picks some $s_2 \in S_2$, etc. We can describe the set of strategies chosen by the n players as the ordered n -tuple:²

¹ A strategy need not refer to a single, simple, elemental action; in a game with temporal structure a strategy can be a very complex sequence of actions which depend on the histories of simple actions taken by all other players. We will see this clearly when we learn to transform an extensive-form description of a game into its strategic form. The name “strategic form” derives precisely because the present formalism ignores all this potential complexity and considers the strategies as primitives of the theory (i.e. as units which cannot be decomposed into simpler constituents).

² In this introduction I’m using boldface notation to represent multicomponent entities in hopes that this will help you keep straight the

$$\mathbf{s} = (s_1, \dots, s_n). \quad (2)$$

This n -dimensional vector of individual strategies is called a *strategy profile* (or sometimes a *strategy combination*). For every different combination of individual choices of strategies we would get a different strategy profile $\mathbf{s}$. The set of all such strategy profiles is called the *space of strategy profiles* S . It is simply the Cartesian product of the strategy spaces S_i for each player.³ We write it as⁴

$$S \equiv S_1 \times \dots \times S_n = \prod_{i=1}^n S_i = \prod_{i \in I} S_i. \quad (3)$$

Example (continued):

Considering Robin as player 1, if she chose $s_R = \text{Down}$ and Cleever chose $s_C = \text{middle}$, then the resulting strategy profile would be:

$$\mathbf{s} = (\text{Down}, \text{middle}).$$

The space of all strategy profiles for this example is

$$S = S_R \times S_C = \{(\text{Up}, \text{left}), (\text{Up}, \text{middle}), (\text{Up}, \text{right}), (\text{Down}, \text{left}), (\text{Down}, \text{middle}), (\text{Down}, \text{right})\}.$$

Player i is often interested in what strategies the other $n - 1$ players choose. We can represent such an $(n - 1)$ -tuple of strategies, known as a *deleted strategy profile*, by⁵

$$\mathbf{s}_{-i} = (s_1, \dots, s_{i-1}, s_{i+1}, \dots, s_n). \quad (4)$$

To each player i there corresponds a *deleted strategy profile space* S_{-i} , which is the space of all possible strategy choices $\mathbf{s}_{-i}$ by her opponents of the form in (4), i.e.⁶

distinction between strategies, for example, and vectors of strategies. However, don't get spoiled: most papers and texts in game theory don't do this. And I'll stop doing it soon.

³ The Cartesian product (or *direct product*) of n sets is the collection of all ordered n -tuples such that the first elements of the n -tuples are chosen from the first set, the second elements from the second set, etc. E.g., the set of Cartesian coordinates $(x, y) \in \mathbb{R}^2$ of the plane is just the Cartesian product of the real numbers $\mathbb{R}$ with itself, i.e. $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$. For another example, let $A = \{1, 2\}$ and $B = \{\alpha, \beta, \delta\}$. Then $A \times B = \{(1, \alpha), (1, \beta), (1, \delta), (2, \alpha), (2, \beta), (2, \delta)\}$. More formally, $A_1 \times \dots \times A_m = \{(a_1, \dots, a_m) : \forall i \in \{1, \dots, m\}, a_i \in A_i\}$. When we form the Cartesian product of m copies of the same set S , we simply write $S^m = S \times \dots \times S$.

⁴ Note that the player set I is ordered. We avoid ambiguity concerning the order in which the Cartesian product is formed when the notation " $\prod_{i \in I}$ " is used by adopting the obvious convention, which is expressed in this case by $\prod_{i=1}^n$.

⁵ In other words this is a strategy profile with one strategy (that of player i) deleted from it. The formal definition obviously does not work quite right if $i = 1$ or $i = n$, but the necessary modifications for these cases should be obvious.

⁶ Let A and B be sets. The *difference* (or *relative complement*) of A and B , denoted $A \setminus B$, is the set of elements which are in A but not in B , i.e. $A \setminus B = \{x \in A : x \notin B\}$. The difference of A and B is also sometimes written as simply $A - B$. The set $I \setminus \{i\} = \{1, \dots, i-1, i+1, \dots, n\}$, when $1 < i < n$. Note that " $\setminus$ " and " $\{I\} \setminus \{i\}$ " are incorrect attempts at expressing this set.

$$S_{-i} \equiv S_1 \times \cdots \times S_{i-1} \times S_{i+1} \times \cdots \times S_n = \prod_{\substack{j=1 \\ j \neq i}}^n S_j = \prod_{j \in \mathcal{I} \setminus \{i\}} S_j. \quad (5)$$

If we want to single out the strategy-decision problem for the particular player i , it is useful to write a strategy profile $s \in S$ as a combination of her strategy $s_i \in S_i$ and the deleted strategy profile $s_{-i} \in S_{-i}$ of the strategies of her opponents. So you will typically see

$$s \equiv (s_i, s_{-i}). \quad (6)$$

As another example of this notation: $(\tilde{s}_i, s_{-i}) = (s_1, \dots, s_{i-1}, \tilde{s}_i, s_{i+1}, \dots, s_n)$. The usage in (6) is notationally abusive.⁷ To avoid this we can define for each player $i \in \mathcal{I}$ a function $\langle \cdot, \cdot \rangle_i: S_i \times S_{-i} \rightarrow S$ defined by $\langle a, (b_1, \dots, b_{i-1}, b_{i+1}, \dots, b_n) \rangle_i = (b_1, \dots, b_{i-1}, a, b_{i+1}, \dots, b_n)$.⁸ Therefore $s = \langle s_i, s_{-i} \rangle_i$.

Example (continued):

Because we're using the letters R and C rather than the numbers 1 and 2 to identify the players in this example and because there are only two players, the correspondence I'm about to draw might seem artificial, but the idea should be clear. Robin's deleted strategy profile is the profile of the strategies chosen by Robin's opponents, viz. by Cleever. Therefore

$$S_{-R} = S_C = \{\text{left}, \text{middle}, \text{right}\},$$

$$S_{-C} = S_R = \{\text{Up}, \text{Down}\}.$$

If $s = (\text{Up}, \text{middle})$, then $s_{-R} = \text{middle}$ and $s_{-C} = \text{Up}$.

Payoffs

The game is played by having all the players simultaneously pick their individual strategies. This set of choices results in some strategy profile $s \in S$, which we call the *outcome* of the game. Each player has a set of preferences over these outcomes $s \in S$. We assume that each player's preferences over lotteries over S can be represented by some von Neumann-Morgenstern utility function $u_i: S \rightarrow \mathbb{R}$.⁹

At the conclusion of the game, then, each player $i \in \mathcal{I}$ receives a *payoff* $u_i(s) = u_i(\langle s_i, s_{-i} \rangle_i)$.¹⁰ Note that

⁷ The strategy-profile notation (s_i, s_{-i}) is abusive because the position into which the first argument is meant to be inserted depends on the “ i ” subscripts. But we can't depend on those subscripts being visible. For example, in a three-player game where $S_1 = S_2 = S_3 = \{U, M, D\}$, what strategy profile does $(U, (M, D))$ represent? It could mean any of (U, M, D) , (M, U, D) , or (M, D, U) for $i = 1, 2, 3$, respectively.

⁸ However, you will not see this $\langle \cdot, \cdot \rangle_i$ notation anywhere else... yet!

⁹ Therefore we can extend the domain of each u_i to be the set of lotteries over outcomes in S . We perform the extension such that u_i has the expected-utility property: For any lottery over S of the form $p \circ s \oplus (1-p) \circ s'$, where $s, s' \in S$ and $p \in [0, 1]$, $u_i(p \circ s \oplus (1-p) \circ s') = pu_i(s) + (1-p)u_i(s')$.

¹⁰ I use the soon-to-be-the-universal-standard notation “ $\langle \dots \rangle$ ” to enclose argument lists of functions, operators, etc., rather than the more

the payoff each player receives depends not only on the strategy she picked but also on the strategies which all the other players picked. In other words the payoff to any one player depends on the entire strategy profile played. The individual payoffs for all the n players for a particular strategy profile s define a payoff vector $\mathbf{u}(s)$ for that strategy profile $\mathbf{u}(s) \equiv (u_1(s), \dots, u_n(s))$; i.e. $\mathbf{u}: S \rightarrow \mathbb{R}^n$.

We can fully describe our game then by the triple $(I, S, \mathbf{u})$, i.e. by a player set I , a space of strategy profiles S , and a vector $\mathbf{u}$ of von Neumann-Morgenstern utility functions defined over S .

Denote by $\#S_i$ the *cardinality* of i 's strategy space S_i (i.e. the number of strategies in S_i). The set of all payoffs to all players can be represented by a $\#S_1 \times \dots \times \#S_n$ matrix.¹¹ Each cell in the payoff matrix corresponds to a particular strategy profile and contains the n -tuple payoff vector which specifies the payoff to each player when the strategy profile corresponding to that cell is played.¹² When there are only two players, this matrix is easily represented in two dimensions. The first player chooses a row and the second player chooses a column. (See the next example.) The payoffs for three-player games can be represented as $\#S_3$ matrices each of which is of dimension $\#S_1 \times \#S_2$. In this case players 1 and 2 choose a row and a column, respectively, and player 3 chooses the particular $\#S_1 \times \#S_2$ matrix.

Example (continued):

We can cook up payoffs to Robin and Cleever for each possible strategy profile (i.e., for each combination of individual strategy choices) to arrive at the following possible payoff matrix:

		C[leever/column]		
		l	m	r
R[obin/ow]	U	2, 8	7, 7	0, 3
	D	1, 5	7, 4	9, 6

Figure 1: A typical payoff matrix.

where the abbreviations for the strategies which label the rows and columns have the obvious meanings. The first payoff of each pair is by convention the one that Row (Robin) receives. The second is that received by Column (Cleever).

To make explicit the connection between the payoff matrix and the payoff function $u_i(s)$ formalism from above, we note two examples:

$$u_R((U, l)) = 2,$$

common standard parentheses "...". I believe my alternative convention helps to quickly distinguish between instances of multiplication and instances of *operation upon*. For example, does " $P(x+a)$ " mean a function P evaluated at $(x+a)$? Or does it mean a number P multiplied by $(x+a)$? This distinction *should* be clear from context alone. However, economists are notably prone to the notational abuses of 1 suppressing arguments and 2 identifying a *function* with its *values*.

¹¹ The multidimensional matrix will have a side for each player, and the number of boxes along each side will be equal to the number of strategies the corresponding player has available to her.

¹² In the two-player case the payoff matrix is often called a *bimatrix*, because each element is an ordered pair.

$$u_C((D, r)) = 6.$$

Best responses to pure strategies

We typically assume in game theory that all players are *rational*. This means that each player will choose an action which *maximizes her expected utility given her beliefs* about what actions the other players will choose. The Easy Part of Game Theory is figuring out what a player will do given her beliefs. This is the problem we focus on now. We ask the question: if player i knows (read “believes with certainty”) which strategy each of her opponents will pick, what strategy should she pick? Obviously, she should pick a *best response* to the plays of her opponents.

Definition

We say that a strategy $s_i^* \in S_i$ for player i is a *best response by i to the deleted strategy profile $s_{-i} \in S_{-i}$* iff¹³

$$(\forall s_i \in S_i) \quad u_i(\langle s_i^*, s_{-i} \rangle_i) \geq u_i(\langle s_i, s_{-i} \rangle_i), \quad (7)$$

or, equivalently,¹⁴

$$s_i^* \in \arg \max_{s_i \in S_i} u_i(\langle s_i, s_{-i} \rangle_i). \quad (8)$$

What is this definition saying? With a few semesters of Varian behind you, this notation should not be too obfuscating, but it’s worthwhile to go through it carefully. We’re fixing our attention on one particular player i and we assume we know the strategy choice $s_j \in S_j$ made by each player $j, j \in I \setminus \{i\}$. If $s_i^* \in S_i$ is a best response by player i to our assumed set of strategies $s_{-i} \in S_{-i}$ played by the other $n - 1$ players, then it must give player i a payoff—this is the left-hand side of the inequality (7)—at least as large as she would get if she played any other strategy $s_i \in S_i$ from her set of allowed strategies S_i . (This is where the universal quantifier “ $\forall s_i \in S_i$ ” comes in.)

Note that the inequality in definition (7) is weak. The best response may not give player i strictly more than any other choice of strategy.¹⁵ But it is at least tied with any other strategies. In other words, you cannot always talk about *the* best response for a player to some set of plays by everyone else, but you can always talk about *a* best response.¹⁶ Therefore we will not necessarily have a best-response *function* which specifies player i ’s unique best response to some deleted strategy profile $s_{-i} \in S_{-i}$, but we

¹³ “iff” $\equiv$ “if and only if.”

¹⁴ Note that $\arg \max$ refers to all values of the argument which maximize the objective function; therefore the $\arg \max$ yields a set.

¹⁵ The weakness of the inequality also makes it possible to write the universal quantifier expression as “ $\forall s_i \in S_i$ ” rather than as “ $\forall s_i \in S_i \setminus \{s_i^*\}$ ”.

¹⁶ For each $s_{-i} \in S_{-i}$, define $g: S_i \rightarrow \mathbb{R}$ by $g(s_i) = u_i(\langle s_i, s_{-i} \rangle_i)$. Player i ’s strategy space S_i is a nonempty, finite set. Therefore its image under g , viz. $g(S_i) \equiv \{g(s_i): s_i \in S_i\}$, is a nonempty, finite, ordered set and thus has a maximum and therefore has a maximizer in S_i . Therefore a best response exists.

will have a *best-response correspondence* for player i , $\text{BR}_i: S_{-i} \rightarrow S_i$, where $\text{BR}_i(s_{-i}) \subset S_i$ specifies the set of best responses for player i to an arbitrary deleted strategy profile $s_{-i} \in S_{-i}$.^{17,18} We can write

$$\text{BR}_i(s_{-i}) = \{s_i^* \in S_i: s_i^* \text{ is a best response by } i \text{ to } s_{-i}\}, \quad (9)$$

or, more formally,

$$\begin{aligned} \text{BR}_i(s_{-i}) &= \{s_i^* \in S_i: \forall s_i \in S_i, u_i(\langle s_i^*, s_{-i} \rangle_i) \geq u_i(\langle s_i, s_{-i} \rangle_i)\}, \\ &= \arg \max_{s_i \in S_i} u_i(\langle s_i, s_{-i} \rangle_i). \end{aligned} \quad (10)$$

(We will soon replace this definition of the best-response correspondence BR_i with a more general one.)

Example (continued):

Let's first find a best response for Robin given that Cleever plays "right." She only has two strategies to choose from: Up and Down. We first compute her payoffs for each alternative given that Cleever plays right. We read these from the first element of the payoff pairs in the rightmost column of the matrix in Figure 1:

$$u_R(U, r) = 0,$$

$$u_R(D, r) = 9.$$

Clearly playing Down gives Robin the strictly higher utility given that Cleever plays right. Therefore her best response to Cleever's $s_C = \text{right}$ is $s_R^* = \text{Down}$; we write $\text{BR}_R(r) = \{D\}$.¹⁹

Note that, against middle by Cleever, Robin is indifferent between Up and Down, because

$$u_R(U, m) = u_R(D, m) = 7.$$

Therefore Robin has two best responses to middle; we write $\text{BR}_R(m) = \{U, D\}$.

Now let's find a best response for Cleever given that Robin will play Down. Cleever has three alternatives, so we compute his utility for each one, given that Robin plays Down, by picking the *second* elements (because Cleever is the second player) from the payoff pairs in the bottom ("Down") row. We get

¹⁷ Recall that a *correspondence* $\varphi: X \rightarrow Y$ is a "set-valued function" or, more properly, a mapping which associates to every element $x \in X$ in the domain a *subset* of the target set Y . In other words, $\forall x \in X, \varphi(x) \subset Y$. The " $\rightarrow$ " symbol is used to distinguish between mapping into subsets of the target set and mapping into the target set itself as the notation " $\rightarrow$ " for a function would imply.

¹⁸ Let A and B be sets. A is a *subset* of B if every element of A is an element of B . If A is a subset of B , we write $A \subset B$. In this case, we also say that B *contains* A , which we write $B \supset A$. Note from the definition that every set is a subset of itself; i.e. $A \subset A$. Warning: Some authors use the symbol " $\sqsubset$ " for this meaning and reserve " $\subset$ " for "proper subset;" viz. " $A \subset B \Rightarrow (A \sqsubset B \text{ and } A \neq B)$."

¹⁹ It might seem somewhat pedantic to include the braces in these cases where the best response correspondence has but a single value; I do so because a correspondence is *set valued*.

$$u_C(D, l) = 5,$$

$$u_C(D, m) = 4,$$

$$u_C(D, r) = 6.$$

We see that Cleever gets his maximum payoff, given that Robin is playing Down, when he plays right. Therefore Cleever's best response to $s_R = \text{Down}$ is $s_C^* = \text{right}$; i.e. $\text{BR}_C(D) = \{r\}$.

Similarly you can find the best response(s) for either player for any choice of strategy by the other player. It is often a useful prelude to the analysis of a game to determine all the best responses to all possible pure-strategy choices of the other players and to indicate them on the game matrix. For example, in Figure 2, I have made boldface the payoffs which are maximal given the opponent's pure strategy. (I.e. Each of Robin's bolded payoffs is maximal in its column; each of Cleever's bolded payoffs is maximal in its row.)

		C[leeve/r/olumn]		
		<i>l</i>	<i>m</i>	<i>r</i>
R[obin/ow]	<i>U</i>	2, 8	7, 7	0, 3
	<i>D</i>	1, 5	7, 4	9, 6

Figure 2: Robin vs. Cleever with best responses to others' pure strategies indicated.

Mixed strategies

So far we have restricted our attention to *pure strategies*; these typically have direct economic interpretations, and these are the actions which define the consequent payoffs. If every player plays a pure strategy, then the payoffs to all players are *deterministic*—there is no uncertainty concerning the payoffs resulting from a specified pure-strategy profile.

However, we will find it useful to expand each player's possible choices to include *mixed strategies*—randomizations over her pure strategies. A player will ultimately execute exactly one of her pure-strategy choices. However, which particular pure strategy she executes is determined by a randomization which is specified by her mixed strategy. Furthermore, we stipulate that each player's randomization is independent of the other players' randomizations.²⁰ When a player $i \in I$ chooses a mixed strategy, every other player $j \in I \setminus \{i\}$ might be uncertain about which pure strategy $s_i \in S_i$ the i -th player will choose.^{21,22}

²⁰ This assumption is sometimes dropped. This leads to the notion of *correlated strategies*. See for example Aumann [1987].

²¹ Mixed strategies will be useful when we study equilibrium concepts for games. Equilibrium may fail to exist if we restrict ourselves to pure strategies; existence is guaranteed if we have access to mixed strategies.

²² You might object to mixed strategies because you don't believe that people would actually pick actions randomly. I advise that you just hold on to your objections for a while and accept for the time being that the concept will prove useful. We will later return to a discussion of the various interpretations which can be given to mixed strategies. This discussion will be more meaningful once you

Formally, we say that a mixed strategy σ_i for player i is a *probability distribution* over player i 's pure-strategy choices S_i ; we write $\sigma_i \in \Delta(S_i)$.²³ If player i is playing the mixed strategy σ_i and if s_i is one of the pure strategies available to her, i.e. $s_i \in S_i$, then the probability that she will choose the pure strategy s_i is denoted by $\sigma_i(s_i)$. (So we could also view the mixed strategy σ_i as a function from player i 's pure-strategy space S_i into the unit interval $[0, 1]$, i.e. $\sigma_i: S_i \rightarrow [0, 1]$.) We denote player i 's mixed-strategy space by Σ_i . (We see that $\Sigma_i = \Delta(S_i)$.)

A mixed strategy specifies a value on $[0, 1]$ for each $s_i \in S_i$. Each player chooses one and only one pure strategy $s_i \in S_i$ in a single play of the game. Therefore any mixed strategy $\sigma_i \in \Delta(S_i)$ must be such that the sum of the probabilities associated with the pure strategies is unity, i.e.

$$\sum_{s_i \in S_i} \sigma_i(s_i) = 1. \quad (11)$$

This property is indeed satisfied as a result of σ_i being a probability distribution over S_i . This is the justification for using a probability distribution to represent a mixed strategy.

Analogously to our treatment of pure strategies: A *mixed-strategy profile* σ is an n -tuple of individual mixed strategies, one for each player; e.g. $\sigma = (\sigma_1, \dots, \sigma_n)$, where $\forall i \in I, \sigma_i \in \Sigma_i$. The *space of mixed-strategy profiles* Σ is just the Cartesian product of the individuals' mixed-strategy spaces, viz. $\Sigma \equiv \prod_{i \in I} \Sigma_i$. Player i 's deleted mixed-strategy space is $\Sigma_{-i} = \prod_{j \in I \setminus \{i\}} \Sigma_j$. We can also extend the domain of the $\langle \cdot, \cdot \rangle_i$ function in the natural way so that $\sigma = \langle \sigma_i, \sigma_{-i} \rangle_i$.²⁴

We say that the *support* of the mixed strategy $\sigma_i \in \Sigma_i$ is the set of pure strategies to which σ_i assigns positive probability,²⁵ i.e.

$$\text{supp}(\sigma_i) \equiv \{s_i \in S_i: \sigma_i(s_i) > 0\}. \quad (12)$$

I.e. the support of the mixed strategy σ_i consists of those pure strategies which player i could conceivably play if she chose the mixed strategy σ_i . For all $\sigma_i \in \Sigma_i$, $\text{supp}(\sigma_i) \subset S_i$.

The unit simplex

At this point we pause to define the $(k-1)$ -dimensional *unit simplex*. The $(k-1)$ -dimensional unit simplex is the set of k -vectors whose components are all nonnegative and sum to one.^{26,27} This

have seen how the theory develops; therefore this is a case where the motivation and interpretation are better left to follow rather than precede the exposition.

²³ When A is a finite set, $\Delta(A)$ is the set of all probability distributions over A .

²⁴ I.e. we extend the domain of $\langle \cdot, \cdot \rangle_i$ to include $\Sigma_i \times \Sigma_{-i}$ so that $\langle \cdot, \cdot \rangle_i: [(S_i \times S_{-i}) \cup (\Sigma_i \times \Sigma_{-i})] \rightarrow (S \cup \Sigma)$, where the restriction of $\langle \cdot, \cdot \rangle_i$ to $S_i \times S_{-i}$ is $\langle \cdot, \cdot \rangle_i: (S_i \times S_{-i}) \rightarrow S$ and the restriction to $\Sigma_i \times \Sigma_{-i}$ is $\langle \cdot, \cdot \rangle_i: (\Sigma_i \times \Sigma_{-i}) \rightarrow \Sigma$. Both restrictions have the same symbolic definition: $\langle a_i, (b_1, \dots, b_{i-1}, b_{i+1}, \dots, b_n) \rangle_i = (b_1, \dots, b_{i-1}, a_i, b_{i+1}, \dots, b_n)$.

²⁵ This is analogous to the support of a random variable, which is the closure of the set of values which are assigned positive probability.

²⁶ Note that these two conditions taken together guarantee that each component is weakly less than one.

²⁷ This simplex is composed of vectors with k components. It gets its name, viz. " $k-1$," because it is a $(k-1)$ -dimensional subspace embedded in a k -dimensional world. $(k-1)$ is also the number of "degrees of freedom" each vector has: Once $k-1$ components have

concept is useful here because any probability distribution over a finite set must belong to a unit simplex.²⁸ Formally, we define the $(k-1)$ -dimensional unit simplex as²⁹

$$\Delta^{k-1} = \left\{ \mathbf{x} \in \mathbb{R}_+^k : \sum_{j=1}^k x_j = 1. \right\} \quad (13)$$

For example the vectors $(\frac{1}{3}, 0, \frac{2}{3})$ and $(0, 0, 1)$ are members of the two-dimensional unit simplex. The vectors $(1, 1, 1)$ and $(\frac{2}{3}, \frac{2}{3}, -\frac{1}{3})$ are not. Figure 3 displays the zero-, one-, and two-dimensional simplices.³⁰

Mixed strategies are chosen from the unit simplex

An alternative to conceiving of a mixed strategy for player i as a function $\sigma_i: S_i \rightarrow [0, 1]$ is to think of it as a $\#S_i$ -vector of probabilities. For example, let $m = \#S_i$ be the number of pure strategies available to player i , and then index player i 's m pure strategies with a superscript: $s_i^1, \dots, s_i^k, \dots, s_i^m$. Then we can write the mixed strategy $\sigma_i \in \Sigma_i$ as the m -tuple

$$\sigma_i = (\sigma_i(s_i^1), \dots, \sigma_i(s_i^m)). \quad (14)$$

We know 1 for every pure strategy $s_i \in S_i$, $\sigma_i(s_i) \in [0, 1]$ and 2 that (11) holds. Referring to definition (13), then, we see that player i 's mixed strategy σ_i belongs to the $(\#S_i - 1)$ -dimensional unit simplex. (In (13), let $k = \#S_i$ and, $\forall j \in \{1, \dots, m\}$, let $x_j = \sigma_i(s_i^j)$.)

been specified, the remaining component is already determined by the requirement that the components sum up to unity.

²⁸ It is useful elsewhere as well. For example in general equilibrium theory prices are often normalized to lie within a unit simplex. Any unit simplex is convex, and this allows the invocation of Brouwer's or Kakutani's fixed-point theorem in order to prove the existence of an equilibrium price vector.

²⁹ Recall that $\mathbb{R}_+^k$ is the nonnegative orthant of $\mathbb{R}^k$, viz. $\{\mathbf{x} \in \mathbb{R}^k : \forall i \in \{1, \dots, k\}, x_i \geq 0\}$.

³⁰ To be perfectly clear... in Figure 3b the one-dimensional simplex is only the line segment connecting $(0, 1) \rightarrow (1, 0)$. In Figure 3c the two-dimensional simplex is the triangular planar region.

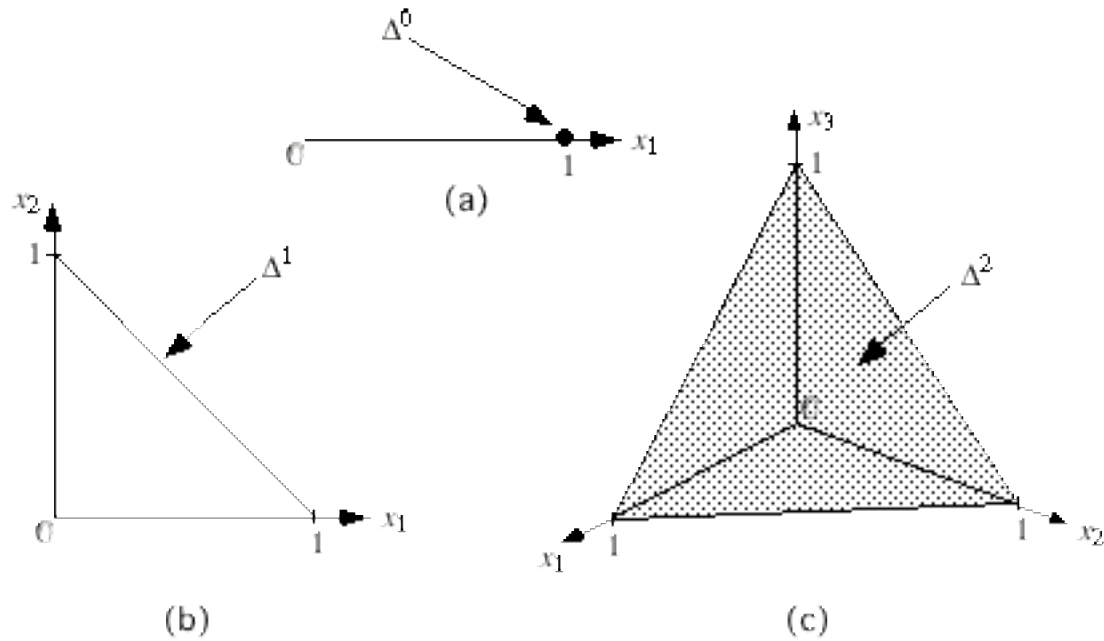


Figure 3: The (a) zero-, (b) one-, and (c) two-dimensional simplices.

Recall that a player's pure-strategy space S_i is the set of possible pure strategies she can choose and that these strategies can be very complex descriptions of contingent actions. A player's *mixed-strategy space* Σ_i is simply the set of $\#S_i$ -vectors belonging to the $(\#S_i - 1)$ -dimensional simplex $\Delta^{\#S_i-1}$.³¹

Pure strategies are degenerate mixed strategies

Choosing a pure strategy s_i is equivalent to choosing the mixed strategy (i.e. probability distribution over pure strategies) which results with probability one in s_i . Therefore we see that every pure strategy "is" a mixed strategy.³² We also say that a pure strategy is a *degenerate* mixed strategy.

To avoid the abuse of notation which results from writing s_i for the degenerate mixed strategy which plays the pure strategy s_i with certainty, we can define $\delta_i(s_i) \in \Delta(S_i)$ to be the player- i mixed strategy which puts unit weight on $s_i \in S_i$ and zero weight on every other player- i pure strategy $s_i' \in S_i \setminus \{s_i\}$. We formally define, for every player- i pure strategy $s_i \in S_i$, the degenerate probability distribution $\delta_i(s_i)$ by specifying, for each player- i pure strategy $s_i' \in S_i$, the probability $\delta_i(s_i)(s_i') \in [0, 1]$ which it attaches to s_i' :

$$\delta_i(s_i)(s_i') = \begin{cases} 1, & s_i' = s_i, \\ 0, & s_i' \neq s_i. \end{cases} \quad (15)$$

(So we see that, for every $s_i \in S_i$, $\delta_i(s_i): S_i \rightarrow \{0, 1\}$. Alternatively, we can write $\delta_i: S_i \rightarrow \Delta(S_i)$ or $\delta_i: S_i^2 \rightarrow [0, 1]$.)

³¹ It might appear at first that a player's mixed-strategy space Σ_i is simpler than her pure-strategy space S_i . Keep in mind however that Σ_i is defined in terms of S_i and so inherits all of S_i 's complexity.

³² I put "is" in quotes because mathematically they are different objects. The pure strategy and the corresponding mixed strategy which puts all probability on that pure strategy are equivalent in the sense that they result in exactly the same action by the player.

A *nondegenerate* mixed strategy is one that is not pure. [Therefore $\exists s_i \in S_i$ such that $\sigma_i(s_i) \in (0, 1)$.] A *completely mixed* or *strictly positive* strategy is one which puts positive weight on every strategy in the player's strategy space; i.e. σ_i is completely mixed if $\forall s_i \in S_i, \sigma_i(s_i) > 0$.³³ (Therefore $\text{supp } \sigma_i = S_i$.)

Example (continued)

Let's return to our game between Robin and Cleever to see how these mixed-strategy concepts are represented in a less notational and less abstract setting. In Figure 4, I have indicated each player's mixed strategies with bracketed probabilities attached to the pure strategies.

		C[leever/column]		
		<i>l</i> : [<i>p</i>]	<i>m</i> : [<i>q</i>]	<i>r</i> : [<i>1 - p - q</i>]
R[obin/ow]	<i>U</i> : [<i>t</i>]	2,8	7,7	0,3
	<i>D</i> : [<i>1 - t</i>]	1,5	7,4	9,6

Figure 4: Robin vs. Cleever with mixed strategies denoted.

Robin has two pure strategies, so the cardinality of her strategy space is 2; i.e. $\#S_R = 2$. Therefore her mixed strategies lie in a one-dimensional unit simplex; they can be described by a single parameter t . We can write any of Robin's mixed strategies as an ordered pair which specifies the probability with which she would choose Up and Down, respectively, i.e. in the form

$$\sigma_R = (t, 1 - t).$$

Alternatively we could write

$$\mathbb{P}(\text{Robin chooses Up}) = \sigma_R(U) = t,$$

$$\mathbb{P}(\text{Robin chooses Down}) = \sigma_R(D) = 1 - t.$$

Robin's mixed-strategy space, i.e. the set of all possible mixed strategies for her, is

$$\Sigma_R = \{(t, 1 - t): t \in [0, 1]\}.$$

Note that we have already seen the graph of Σ_R in Figure 3b.

Cleever has three pure strategies, therefore a mixed strategy for him belongs to the two-dimensional unit simplex and takes the form

$$\sigma_C = (p, q, 1 - p - q),$$

where

³³ The notion of completely mixed strategies is used when discussing some equilibrium refinements, e.g. sequential equilibrium and trembling-hand perfection. When all players choose completely mixed strategies, there is a positive probability of reaching any given node in the game tree. Therefore no node is off the path. (These comments will make more sense after we encounter games in extensive form.)

$$\mathbb{P}(\text{Cleever chooses left}) = \sigma_C(l) = p,$$

$$\mathbb{P}(\text{Cleever chooses middle}) = \sigma_C(m) = q,$$

$$\mathbb{P}(\text{Cleever chooses right}) = \sigma_C(r) = 1 - p - q.$$

His mixed-strategy space is:

$$\Sigma_C = \{(p, q, 1 - p - q) : p, q \geq 0, p + q \leq 1\}.$$

We have already seen the graph of Σ_C in Figure 3c.

Payoffs to mixed-strategy profiles

We have noted that $u_i(s)$ is player i 's payoff when the players choose their parts of the pure-strategy profile $s \in S$. Because the players are not randomizing their actions when they play s , the resultant payoff vector is a certain, deterministic number. Now we ask the question: When the players execute the mixed-strategy profile $\sigma \in \Sigma$, what is the payoff to player i ? Right away we see a problem even with the way this question is phrased. It doesn't make sense to ask *ex ante* what *the* payoff to player i is, because her payoff depends on the precise pure strategies realized as the result of the individuals' randomizations.

We could ask then: What is the *distribution* of payoffs player i would receive if the players executed the mixed-strategy profile σ ? Fortunately, we have no need for such a complicated answer. Because our utility functions $u_i: S \rightarrow \mathbb{R}$ are assumed to be of the von Neumann–Morgenstern variety, we know that each player's preferences over distributions of von Neumann–Morgenstern utilities can be represented by her *expected utility*. Now we need only ask: What is player i 's expected payoff given that the players choose $\sigma \in \Sigma$? We will simplify notation by using the same function name u_i to represent the expected utility to player i from a mixed-strategy profile $\sigma \in \Sigma$ as we used above for pure-strategy profiles. I.e. we write $u_i(\sigma)$, where $u_i: \Sigma \rightarrow \mathbb{R}$.³⁴

The probability of a pure-strategy profile s

How do we calculate this expected utility for player i ? We need to weight player i 's payoff to each arbitrary pure-strategy profile $s = \langle s_i, s_{-i} \rangle_{i \in S}$ by the probability that the profile s will be realized when the players randomize according to the mixed-strategy profile $\sigma \in \Sigma$. Because the players' randomizations are independent of one another's, the probability that $s = (s_1, \dots, s_n)$ will occur is the

³⁴ It is a formal convenience here to use the same function name for two different functions with distinct domains. Although this may appear abusive *prima facie*, the more complete justification is the following: Let A , B , and C be sets such that $A \cap B = \emptyset$. Let $g: A \rightarrow C$ and $h: B \rightarrow C$ be functions. Then we can define $f: (A \cup B) \rightarrow C$ by

$$f(x) = \begin{cases} g(x), & x \in A, \\ h(x), & x \in B. \end{cases}$$

I.e. we can define $u_i: (S \cup \Sigma) \rightarrow \mathbb{R}$. When the argument supplied to u_i is an element of S (respectively, Σ), the function is evaluated using the restriction of u_i to S (respectively, Σ).

product of the probabilities that each player j will play s_j . The probability, according to σ , that j will play $s_j \in S_j$ is $\sigma_j(s_j)$. Therefore the probability that s will occur when the players randomize according to σ is the product of probabilities

$$\begin{aligned} \mathbb{P}(s \text{ is played}) &= \mathbb{P}((s_1, \dots, s_n) \text{ is played}) = \mathbb{P}(1 \text{ plays } s_1) \cdot \dots \cdot \mathbb{P}(n \text{ plays } s_n) \\ &= \sigma_1(s_1) \cdot \dots \cdot \sigma_n(s_n) = \prod_{j=1}^n \sigma_j(s_j) = \prod_{j \in I} \sigma_j(s_j). \end{aligned} \quad (16)$$

Expected payoff to a mixed-strategy profile σ

To complete our calculation of i 's expected utility when the mixed-strategy profile σ is played, we must look at every possible pure-strategy profile $s \in S$, find i 's deterministic payoff for this pure-strategy profile, and weight this payoff according to the profile's probability of occurrence as given by (16). The weighted sum over all these possible pure-strategy profiles is our desired expected payoff. I.e. the expected payoff to player i when the players participate in the mixed-strategy profile σ is³⁵

$$u_i(\sigma) = \sum_{s \in S} \mathbb{P}(s) u_i(s) = \sum_{s \in S} \left(\prod_{j \in I} \sigma_j(s_j) \right) u_i(s). \quad (17)$$

Payoff to i from σ is linear in any one player's mixing probabilities

We can single out for special attention any player $k \in I$ in our calculation of player i 's payoff to a mixed-strategy profile $\sigma \in \Sigma$ and rewrite (17) as

$$\begin{aligned} u_i(\sigma) &= \sum_{s_k \in S_k} \sum_{s_{-k} \in S_{-k}} \left(\sigma_k(s_k) \prod_{j \in I \setminus \{k\}} \sigma_j(s_j) \right) u_i(\langle s_k, s_{-k} \rangle_k) \\ &= \sum_{s_k \in S_k} \sigma_k(s_k) \left[\sum_{s_{-k} \in S_{-k}} \left(\prod_{j \in I \setminus \{k\}} \sigma_j(s_j) \right) u_i(\langle s_k, s_{-k} \rangle_k) \right] \\ &= \sum_{s_k \in S_k} c_k(s_k, \sigma_{-k}) \sigma_k(s_k), \end{aligned} \quad (18)$$

where $\forall k \in I$, $c_k: S_k \times \Sigma_{-k} \rightarrow \mathbb{R}$ is defined by

³⁵ You may be more familiar with writing a summation (and similar remarks hold for products) in the form $\sum_{k=1}^m x_k$, i.e. with an integer index k to indicate a particular element of a finite set of objects $X = \{x_1, \dots, x_m\}$ to be added. We will often find it more convenient—e.g. when there is no natural indexing scheme—to write this summation as $\sum_{x \in X} x$. This simply means to form a sum whose terms consist of every element of X represented once. This is equivalent to the indexed formalism, for both summations and products, because both (finite) addition and multiplication are commutative operations.

$$c_k(s_k, \sigma_{-k}) \equiv \sum_{s_{-k} \in S_{-k}} \left(\prod_{j \in N \setminus \{k\}} \sigma_j(s_j) \right) u_i(\langle s_k, s_{-k} \rangle_k). \quad (19)$$

Equations (18) and (19) say that player i 's expected payoff in the mixed-strategy profile σ is a linear function of player k 's mixing probabilities $\{\sigma_k(s_k)\}_{s_k \in S_k}$. (To see this note that, for each $s_k \in S_k$, the corresponding coefficient $c_k(s_k, \sigma_{-k})$ is independent of $\sigma_k(s_k')$ for all $s_k' \in S_k$.) This observation will be relevant when we determine a player's best-response correspondence. (Note that this analysis includes the case $k=i$.)

Player i 's payoff to a pure strategy s_i against a deleted mixed-strategy profile σ_{-i}

We will see that it will be very useful to determine player i 's payoff against a deleted mixed-strategy profile $\sigma_{-i} \in \Sigma_{-i}$ when player i herself chooses some pure strategy $s_i \in S_i$.

To represent the mixed-strategy profile $\sigma \in \Sigma$ induced by this combination, we again extend the domain of the $\langle \cdot, \cdot \rangle_i$ function to include $S_i \times \Sigma_{-i}$ so that we can make sense of the expression “ $\sigma = \langle s_i, \sigma_{-i} \rangle_i$.” We stipulate the restriction of $\langle \cdot, \cdot \rangle_i$ to $S_i \times \Sigma_{-i}$ to be a function $\langle \cdot, \cdot \rangle_i: S_i \times \Sigma_{-i} \rightarrow \Sigma$ defined by $\langle s_i, \sigma_{-i} \rangle_i = \langle \delta_i(s_i), \sigma_{-i} \rangle_i$.^{36,37} In other words, we replace the pure strategy s_i with the degenerate mixed strategy $\delta_i(s_i)$ which puts all of its weight on s_i .

We calculate the expected payoff to player i when she plays the pure strategy $s_i' \in S_i$ against the deleted mixed-strategy profile $\sigma_{-i} \in \Sigma_{-i}$ using (18), (19), and (15), where we let $k \leftarrow i$ and let $\sigma = \langle s_i', \sigma_{-i} \rangle_i$; i.e. $\sigma_i = \delta_i(s_i')$:

$$\begin{aligned} u_i(\langle s_i', \sigma_{-i} \rangle_i) &= u_i(\langle \delta_i(s_i'), \sigma_{-i} \rangle_i) = \sum_{s_i \in S_i} c_i(s_i, \sigma_{-i}) \sigma_i(s_i) \\ &= \sum_{s_i \in S_i} (\delta_i(s_i')(s_i)) c_i(s_i, \sigma_{-i}) = c_i(s_i', \sigma_{-i}) \\ &= \sum_{s_{-i} \in S_{-i}} \left(\prod_{j \in N \setminus \{i\}} \sigma_j(s_j) \right) u_i(\langle s_i', s_{-i} \rangle_i). \end{aligned} \quad (20)$$

Note then that we have shown that $\forall s_i \in S_i, c_i(s_i, \sigma_{-i}) = u_i(\langle s_i, \sigma_{-i} \rangle_i)$. Therefore from (18), letting $k \leftarrow i$, we can rewrite player i 's expected payoff to a mixed strategy $\sigma_i \in \Sigma_i$ against the deleted mixed-strategy profile $\sigma_{-i} \in \Sigma_{-i}$ as:

³⁶ Let $f: X \rightarrow Z$ be a function and let $Y \subset X$ be a subset of X . Then we can define a function $\tilde{f}$, the *restriction* of f to Y , as a function whose domain is Y and which agrees with f for all points in Y . I.e. $\tilde{f}: Y \rightarrow Z$ and $\forall x \in Y, \tilde{f}(x) = f(x)$.

³⁷ To complete the definition of $\langle \cdot, \cdot \rangle_i$ we also extend its domain to include the set $\Sigma_i \times S_{-i}$, i.e. where player i chooses a mixed strategy and the other players choose pure strategies. (This will be handy in our later analysis of strategic dominance.) We provide the obvious definition for $\langle \cdot, \cdot \rangle_i: \Sigma_i \times S_{-i} \rightarrow \Sigma$: $\langle a_i, (b_1, \dots, b_{i-1}, b_{i+1}, \dots, b_n) \rangle_i = (\delta_1(b_1), \dots, \delta_{i-1}(b_{i-1}), a_i, \delta_{i+1}(b_{i+1}), \dots, \delta_n(b_n))$. Therefore now $\langle \cdot, \cdot \rangle_i: [(S_i \cup \Sigma_i) \times (S_{-i} \times \Sigma_{-i})] \rightarrow (S \cup \Sigma)$.

$$u_i(\langle \sigma_i, \sigma_{-i} \rangle_i) = \sum_{s_i \in S_i} c_i(s_i, \sigma_{-i}) \sigma_i(s_i), \quad (21)$$

or, for more convenient future reference:

$$u_i(\langle \sigma_i, \sigma_{-i} \rangle_i) = \sum_{s_i \in S_i} \sigma_i(s_i) u_i(\langle s_i, \sigma_{-i} \rangle_i) = \sum_{s_i \in \text{supp } \sigma_i} \sigma_i(s_i) u_i(\langle s_i, \sigma_{-i} \rangle_i). \quad (22)$$

In other words a player's payoff to a mixed strategy (against some fixed deleted mixed strategy profile) is a convex combination of the payoffs to the pure strategies in the mixed strategy's support (against that deleted mixed strategy profile). (The set $\{\sigma_i(s_i): s_i \in \text{supp } \sigma_i\}$ of coefficients is a set of convex coefficients because they are nonnegative and sum to unity.)

Example (continued):

Let's employ the very notational expression (17) to compute the payoff to Robin for an arbitrary mixed-strategy profile $\sigma = (\sigma_R, \sigma_C)$. Note that the summation $\sum_{s \in S}$ of (17) generates six terms, viz. one for each member of

$$S = S_R \times S_C = \{(U, l), (U, m), (U, r), (D, l), (D, m), (D, r)\}.$$

For each of these terms the $\Pi_j \sigma_j(s_j)$ product multiplies two factors: $\sigma_R(s_R)$ and $\sigma_C(s_C)$. For example, when $s = (D, l)$,

$$\Pi_j \sigma_j(s_j) = \sigma_R(D) \sigma_C(l) = (1-t)p.$$

This product, then, is the weight attached to $u_R((D, l)) = 1$ when we calculate Robin's expected payoff when the mixed-strategy profile σ is played.

We can use the game matrix from Figure 4 to easily compute the probability coefficients associated with each pure-strategy profile. See Figure 5. This matrix of probability coefficients was formed by multiplying the mixing probability of Robin's associated with a cell's row by the mixing probability of Cleever's associated with that cell's column. Inspection of Figure 5 shows quickly what we had already determined: that the probability coefficient corresponding to (D, l) is $(1-t)p$.

		C[leeve/column]		
		$l: [p]$	$m: [q]$	$r: [1-p-q]$
R[obin/ow]	$U: [t]$	tp	tq	$t(1-p-q)$
	$D: [1-t]$	$(1-t)p$	$(1-t)q$	$(1-t)(1-p-q)$

Figure 5: The probability coefficients which weight the pure-strategy profile payoffs in the calculation of the expected payoff to an arbitrary mixed-strategy profile.

To compute Robin's expected payoff to the mixed-strategy profile σ , then, we multiply her payoff in each cell by the probability coefficient given in that cell of Figure 5, and then sum over all the cells. For example, consider the mixed-strategy profile $\sigma = (\sigma_R, \sigma_C)$ where Robin mixes between Up and Down

according to $\sigma_R = (1/6, 5/6)$ and Cleever mixes according to $\sigma_C = (9/10, 0, 1/10)$. You can easily verify that Robin's expected payoff for this mixed-strategy profile is

$$u_R(\sigma) = \frac{1}{6} \cdot \frac{9}{10} \cdot 2 + \frac{5}{6} \cdot \frac{9}{10} \cdot 1 + \frac{5}{6} \cdot \frac{1}{10} \cdot 9 = \frac{9}{5}.$$

The best-response correspondence

We have earlier considered player i 's problem of deciding on a *best-response pure strategy* $s_i^* \in S_i$ to some deleted pure-strategy profile [i.e. $(n-1)$ -tuple] $s_{-i} \in S_{-i}$ of pure-strategy choices by her opponents. In her calculations for a fixed s_{-i} she was certain that player $j \in I \setminus \{i\}$ would play a particular $s_j \in S_j$.

Now we ask: given that all other players but i are playing the deleted *mixed-strategy* profile $\sigma_{-i} \in \Sigma_{-i}$, what pure strategy is best for i ? The answer to this question is i 's best-response correspondence $BR_i: \Sigma_{-i} \rightarrow S_i$, which maps the space of deleted mixed-strategy profiles Σ_{-i} into subsets of the space of i 's pure strategies S_i . (This definition of the best-response correspondence BR_i is a generalization and replacement of the earlier definition which considered only pure-strategies by the other players.)

Formally we write player i 's problem as finding, for every deleted mixed-strategy profile $\sigma_{-i} \in \Sigma_{-i}$, the set $BR_i(\sigma_{-i}) \subset S_i$ of pure strategies for player i :

$$BR_i(\sigma_{-i}) = \arg \max_{s_i \in S_i} u_i(\langle s_i, \sigma_{-i} \rangle_i). \quad (23)$$

Nonemptiness of $BR_i(\sigma_{-i})$ (i.e. the existence of a best-response pure strategy) is guaranteed for each $\sigma_{-i} \in \Sigma_{-i}$ because S_i is a nonempty and finite set.

Example (continued)

Let's compute Cleever's and Robin's best-response correspondences to the other's arbitrary mixed strategy. Any mixed strategy Robin chooses can be described as a choice of $t \in [0, 1]$. We first seek Cleever's best-response correspondence $BR_C(t)$, which specifies all of Cleever's pure strategies which are best responses to Robin's mixed strategy $\sigma_R = (t, 1-t)$.

To determine this correspondence we compute Cleever's payoffs to each of his three pure strategies against Robin's arbitrary mixed strategy t . Each pure-strategy choice by Cleever corresponds to a column in Figure 4. We then look at the second element of each ordered pair in that column, because that component corresponds to Cleever's payoff, and weight each one by the probability that its row will be chosen by Robin, viz. by t and $(1-t)$ for Up and Down, respectively. This process yields:

$$u_C(l; t) = 8t + 5(1-t) = 5 + 3t,$$

$$u_C(m; t) = 7t + 4(1-t) = 4 + 3t,$$

$$u_C(r; t) = 3t + 6(1-t) = 6 - 3t.$$

We plot Cleever's three pure-strategy payoffs as functions of Robin's mixed strategy $t \in [0, 1]$ in Figure 6.

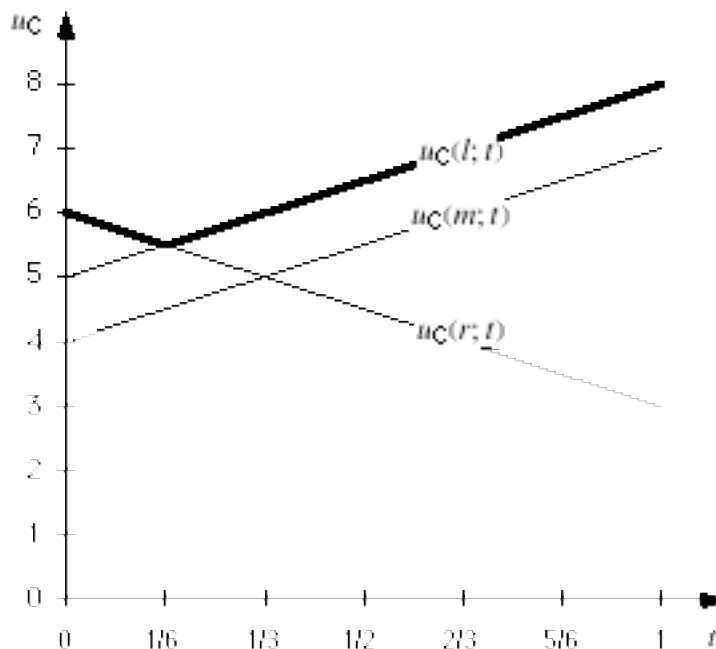


Figure 6: Cleever's pure-strategy payoffs as functions of Robin's mixed strategy.

We first observe that Cleever's payoff to left is strictly above his payoff to middle. (I.e. $\forall t \in [0, 1]$, $u_C(l; t) > u_C(m; t)$.) We will see later that this means that middle is *strictly dominated* by left. We also see that, when $t < 1/6$, right supplies Cleever with a higher payoff than either of the other strategies. When $t > 1/6$, left provides the highest payoff. When $t = 1/6$, both left and right provide Cleever with a payoff of $5\frac{1}{2}$. In the first two cases Cleever has a unique best response to Robin's mixed strategy. In the last case two of Cleever's strategies are best responses.

To summarize we can write Cleever's best-response correspondence as:

$$BR_C(t) = \begin{cases} \{r\}, & t \in [0, 1/6), \\ \{r, l\}, & t = 1/6, \\ \{l\}, & t \in (1/6, 1]. \end{cases}$$

We can represent this best-response correspondence graphically by mapping the relevant intervals describing Robin's mixed strategy into pure-strategy choices by Cleever. (See Figure 7.)



Figure 7: Cleever's best-response correspondence for three subsets of Robin's mixed-strategy space.

The *upper envelope* of these three payoff functions is indicated by heavier line segments in Figure 6.³⁸ This represents the expected payoff Cleever would receive, as a function of Robin's mixed strategy t , if Cleever played a best response to this mixed strategy.

Now we'll determine Robin's best-response correspondence as a function of Cleever's mixed strategy σ_C . Because Cleever has three pure strategies to choose from, we need two parameters to describe his arbitrary mixed strategy, viz. p and q . Analogously as we did above, we compute Robin's expected payoff to each of her two pure strategies as a function of Cleever's mixed-strategy parameters:

$$u_R(U; p, q) = 2p + 7q + 0 \cdot (1 - p - q) = 2p + 7q,$$

$$u_R(D; p, q) = 1 \cdot p + 7q + 9(1 - p - q) = 9 - 8p - 2q.$$

Robin should weakly prefer to play Up whenever

$$u_R(U; p, q) \geq u_R(D; p, q),$$

which occurs when

$$q \geq 1 - \frac{10}{9}p.$$

The isosceles right triangle in Figure 8 represents Cleever's mixed-strategy space in the following sense: Every mixed strategy of Cleever's can be represented by a (p, q) pair satisfying $p, q \geq 0$ and $p + q \leq 1$. Therefore there is a one-to-one correspondence between Σ_C and the points in that triangle.³⁹ Also marked is the line segment of mixed strategies of Cleever's at which Robin is indifferent between playing Up and Down. On that line segment Robin's best-response correspondence contains both pure strategies. Points above that line segment represent Cleever mixed strategies against which Robin strictly prefers to play Up; below that she strictly prefers to play Down. Robin's best-response correspondence can be written as

³⁸ Let $f_1, \dots, f_n$ be functions from some common domain X into the reals, i.e. $f_i: X \rightarrow \mathbb{R}$. Then the *upper envelope* of these functions is itself a function $\tilde{f}: X \rightarrow \mathbb{R}$ defined by: $\tilde{f}(x) \equiv \max\{f_1(x), \dots, f_n(x)\}$.

³⁹ The isosceles triangle is just the projection of $\Sigma_C (= \Delta^2)$ onto the pq -axis. [In Figure 3(c) just drop each point on the shaded triangle perpendicularly down onto the x_1x_2 -axis to see where the isosceles triangle in Figure 8 comes from.]

$$\text{BR}_R(p, q) = \begin{cases} \{U\}, & q > 1 - \frac{10}{9}p, \\ \{U, D\}, & q = 1 - \frac{10}{9}p, \\ \{D\}, & q < 1 - \frac{10}{9}p, \end{cases}$$

with the understanding that $(p, q) \in \mathbb{R}_+^2$ and $p + q \leq 1$.

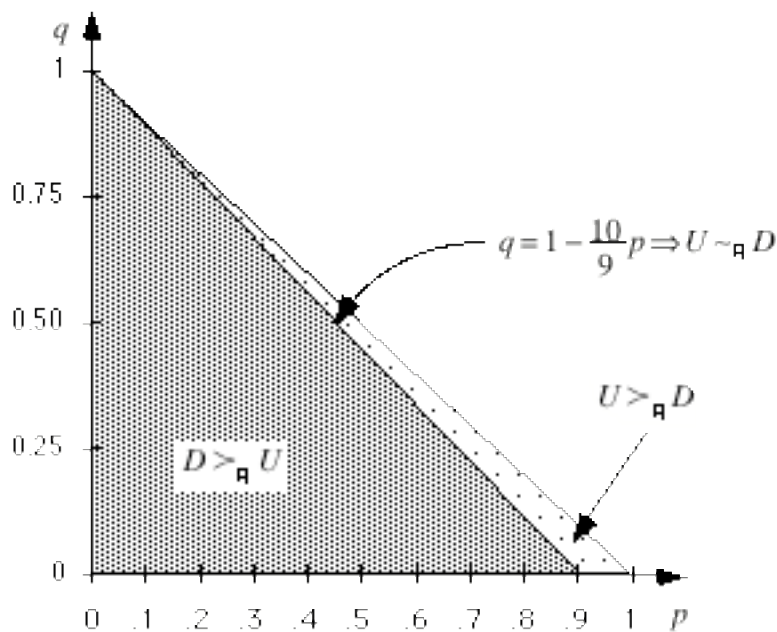


Figure 8: Robin's best-response correspondence for three subsets of Cleever's mixed-strategy space.

Best-response mixed strategies

Above we defined a player's best *pure-strategy* response(s) to a given deleted profile of other players' mixed strategies. (To be even more precise... we determined which of a player's pure strategies were best responses *within her set of pure strategies*.) But how do we know that a player's best response is a pure strategy? Could she do better by playing a mixed strategy? We will see that, *given her opponents' strategies*, a player would never strictly prefer to play a mixed strategy over one of her pure-strategy best responses.⁴⁰ In fact the only time when—again, against a particular $\sigma_{-i} \in \Sigma_{-i}$ —a player would even be willing to mix is when her best-response correspondence for that deleted strategy profile contains more than one pure strategy; i.e. when $\#\text{BR}_i(\sigma_{-i}) > 1$. When that is true, she is willing to put positive

⁴⁰ That doesn't mean that mixed strategies aren't useful. Even if a player is indifferent between playing a mixed strategy and a pure strategy against some particular set of opponents' strategies, playing a mixture has the effect of making her opponents uncertain about what she will do. This can cause them to choose their strategies more beneficially for her.

weight on any pure-strategy best response.

Formally... a player- i mixed strategy $\sigma_i^* \in \Sigma_i$ is a best response for player i against the deleted mixed-strategy profile $\sigma_{-i} \in \Sigma_{-i}$ if

$$\sigma_i^* \in \arg \max_{\sigma_i \in \Sigma_i} u_i(\langle \sigma_i, \sigma_{-i} \rangle_i). \quad (24)$$

It might help slice through the notational fog if we index with k the possible pure strategies for i and let $m = \#S_i$ be the number of pure strategies for i . For every $k \in \{1, \dots, m\}$, let 1 s_i^k be i 's k -th pure strategy, 2 p_i^k be the probability of s_i^k according to the mixed strategy $\sigma_i \in \Sigma_i$ (i.e. $p_i^k \equiv \sigma_i(s_i^k)$), and 3 u_i^k be the payoff to i against σ_{-i} when she plays her k -th pure strategy s_i^k ; i.e. $u_i^k \equiv u_i(\langle s_i^k, \sigma_{-i} \rangle_i)$. Now we can write the maximization problem for a player seeking an optimal mixed-strategy response as that of choosing an m -vector $\mathbf{p}_i = (p_i^1, \dots, p_i^m) \in \Delta^{m-1}$ of probabilities which solves

$$\max_{\mathbf{p}_i \in \Delta^{m-1}} p_i^1 u_i^1 + p_i^2 u_i^2 + \dots + p_i^m u_i^m. \quad (25)$$

Note, as we showed before in (18), that player i 's payoff is linear in her mixing probabilities $p_i^1, \dots, p_i^m$. And as we showed in (22) player i 's payoff to her mixed strategy $\mathbf{p}_i$ is a convex combination of her pure strategy payoffs $\{u_i^k\}_{k=1}^m$.

In the above optimization problem player i must assign probabilities to her different pure-strategy choices in such a way as to maximize the probability-weighted sum of her payoffs from those pure strategies. In this type of problem probability is a scarce resource: the more of it one pure strategy gets, the less another receives. Therefore you want to put your probability where it counts the most.

Consider the case in which one pure strategy s_i^k is strictly better than any of the other pure strategies; i.e. u_i^k is strictly larger than all of the other pure-strategy payoffs. Then the pure strategy s_i^k should receive all of the probability; i.e. p_i^k should be unity and all of the other probabilities should be zero. (Otherwise the objective function could be increased by shifting probability away from a pure strategy whose payoff is lower.) This would correspond to playing the pure strategy s_i^k .

Now consider the case in which several pure strategies are best. E.g. s_i^k and s_i^r , $k \neq r$, both result in the payoff $\bar{u}_i \equiv u_i^k = u_i^r$ which is strictly larger than all of the other pure-strategy payoffs. We should definitely not waste probability on any of these other, low-performance pure strategies, because we could increase our expected payoff by shifting that probability to either of these best pure strategies. However, we are indifferent to how much of our probability we assign to the various best pure strategies. E.g., because s_i^k and s_i^r both result in the payoff $\bar{u}_i$, it does not matter whether we put all of the probability on s_i^k , put all of the probability on s_i^r , or split the probability by putting $\alpha \in [0, 1]$ on s_i^k and $(1 - \alpha)$ on s_i^r . [In this third case, our expected payoff is still $\alpha u_i^k + (1 - \alpha) u_i^r = \alpha \bar{u}_i + (1 - \alpha) \bar{u}_i = \bar{u}_i$.] In this last case we see that the best-response mixed-strategy correspondence for this deleted mixed-strategy profile contains a continuum of mixed strategies corresponding to all the possible mixtures over the best pure strategies.

So we see that any mixed strategy which allocates probability only to best-response pure strategies is a best-response mixed strategy and vice versa.⁴¹ We can express this conclusion in the following theorem:

Theorem

The player- i mixed strategy $\sigma_i^* \in \Sigma_i$ is a best-response for player i to the deleted mixed-strategy profile $\sigma_{-i} \in \Sigma_{-i}$ if and only if

$$\text{supp } \sigma_i^* \subset \text{BR}_i(\sigma_{-i}). \quad (26)$$

Sketch of Proof

To prove $\mathcal{P} \Leftrightarrow \mathcal{Q}$ (i.e. “ $\mathcal{P}$ if and only if $\mathcal{Q}$ ”), where $\mathcal{P}$ and $\mathcal{Q}$ are propositions, we must prove both A $\mathcal{P} \Rightarrow \mathcal{Q}$ (i.e. “ $\mathcal{P}$ only if $\mathcal{Q}$ ”) and B $\mathcal{Q} \Rightarrow \mathcal{P}$ (i.e. “ $\mathcal{P}$ if $\mathcal{Q}$ ”).⁴²

A: $(\sigma_i^* \text{ is a best response to } \sigma_{-i}) \Rightarrow \text{supp } \sigma_i^* \subset \text{BR}_i(\sigma_{-i})$.

The conditional proposition $\mathcal{P} \Rightarrow \mathcal{Q}$ is equivalent to $\sim \mathcal{Q} \Rightarrow \sim \mathcal{P}$.⁴³ Therefore to prove A we assume that $\text{supp } \sigma_i^* \not\subset \text{BR}_i(\sigma_{-i})$ and try to deduce that σ_i^* is not a best response to σ_{-i} . The fact that $\text{supp } \sigma_i^* \not\subset \text{BR}_i(\sigma_{-i})$ implies that $\exists s_i' \in S_i \setminus \text{BR}_i(\sigma_{-i})$ such that $\sigma_i^*(s_i') > 0$. We show that σ_i^* is not a best response to σ_{-i} by exhibiting a mixed strategy $\sigma_i^\dagger \in \Sigma_i$ such that $u_i(\langle \sigma_i^\dagger, \sigma_{-i} \rangle_i) > u_i(\langle \sigma_i^*, \sigma_{-i} \rangle_i)$. To construct the better mixed strategy $\sigma_i^\dagger$, we arbitrarily pick some best-response pure strategy $\tilde{s}_i \in \text{BR}_i(\sigma_{-i})$ on which to shift all the probability which the original mixed strategy σ_i^* bestowed upon the non-best-response strategy s_i' . Formally... define $\sigma_i^\dagger: S_i \rightarrow [0, 1]$ for all $s_i \in S_i$ by

$$\sigma_i^\dagger(s_i) = \begin{cases} 0, & s_i = s_i', \\ \sigma_i^*(\tilde{s}_i) + \sigma_i^*(s_i'), & s_i = \tilde{s}_i, \\ \sigma_i^*(s_i), & s_i \notin \{s_i', \tilde{s}_i\}. \end{cases}$$

I leave it as an exercise for you to show that indeed $u_i(\langle \sigma_i^\dagger, \sigma_{-i} \rangle_i) > u_i(\langle \sigma_i^*, \sigma_{-i} \rangle_i)$.

B: $\text{supp } \sigma_i^* \subset \text{BR}_i(\sigma_{-i}) \Rightarrow (\sigma_i^* \text{ is a best response to } \sigma_{-i})$.

We first observe that every player- i pure-strategy best response yields player i the same expected utility; i.e. $\forall s_i', s_i'' \in \text{BR}_i(\sigma_{-i}), u_i(\langle s_i', \sigma_{-i} \rangle_i) = u_i(\langle s_i'', \sigma_{-i} \rangle_i)$.⁴⁴ Denote by $\bar{u}_i \in \mathbb{R}$, this common expected utility; i.e. $\forall s_i \in \text{BR}_i(\sigma_{-i}), u_i(\langle s_i, \sigma_{-i} \rangle_i) = \bar{u}_i$.

Now we show that $\bar{u}_i$ is an upper bound on the utility which any mixed strategy can achieve. To see this we refer to (22) which shows that the payoff to player i from any mixed strategy $\sigma_i \in \Sigma_i$ against σ_{-i} is a convex combination of the payoffs to the pure strategies in the support of σ_i . A convex combination

⁴¹ We have already argued that a best-response pure-strategy exists. The degenerate mixed strategy which puts unit weight on any such pure-strategy best response exists and is a mixed-strategy best response. Therefore a mixed-strategy best response exists.

⁴² A *proposition* is a statement which is either true or false.

⁴³ For any proposition $\mathcal{P}$, we denote by $\sim \mathcal{P}$ the *negation* of $\mathcal{P}$. $\sim \mathcal{P}$ is also a proposition. Its truth value is the opposite of the truth value of $\mathcal{P}$.

⁴⁴ If one yielded a strictly higher expected utility, the other would not be a best response.

of a set of real numbers must be weakly less than the maximum of that set.⁴⁵ Therefore any mixed strategy σ_i must yield an expected utility such that $u_i(\langle \sigma_i, \sigma_{-i} \rangle_i) \leq \bar{u}_i$.

Any mixed strategy which yields an expected utility of $\bar{u}_i$ must be a best response. (If it were not, there would exist another mixed strategy which yielded a higher utility, but this would contradict that $\bar{u}_i$ is an upper bound.)

Now use $\text{supp } \sigma_i^* \subset \text{BR}_i(\sigma_{-i})$ and (22), to show that $u_i(\langle \sigma_i^*, \sigma_{-i} \rangle_i) = \bar{u}_i$. Therefore σ_i^* is a best-response mixed strategy. ☺⁴⁶

The expected payoff to i from playing such a best-response mixed strategy is exactly the expected payoff she would receive from playing any one of her best pure strategies. Therefore a player never strictly prefers to mix rather than to play one of her best pure strategies against a particular profile of opponents' strategies.

For a given deleted profile of opponents' mixed strategies $\sigma_{-i} \in \Sigma_{-i}$, we now know which mixed strategies are best responses given the pure-strategy best-responses $\text{BR}_i(\sigma_{-i})$. This gives us an alternative and often useful way to graphically represent the players' best-responses, viz. in terms of the mixing probabilities which are optimal given the opponents' mixed strategies.

We can now define player i 's mixed-strategy best-response correspondence $\text{MBR}_i: \Sigma_{-i} \rightarrow \Sigma_i$, which specifies, for any deleted mixed-strategy profile $\sigma_{-i} \in \Sigma_{-i}$ by i 's opponents, a set $\text{MBR}_i(\sigma_{-i}) \subset \Sigma_i$ of player- i mixed strategies which are best responses to σ_{-i} . This definition follows directly from (26):

$$\text{MBR}_i(\sigma_{-i}) = \{\sigma_i \in \Sigma_i: \text{supp } \sigma_i \subset \text{BR}_i(\sigma_{-i})\}. \quad (27)$$

Example:

We previously determined Robin's and Cleever's (pure-strategy) best-response correspondences BR_R and BR_C . We can use each player's pure-strategy best-response correspondence to express the corresponding mixed-strategy best response correspondence, viz. MBR_R and MBR_C , respectively.

Every mixed-strategy for Robin can be written as an element of the one-dimensional simplex $\{(t, 1-t): t \in [0, 1]\}$, where we adopt the convention that the mixed strategy $(t, 1-t)$ corresponds to playing Up with probability t . Robin's mixed-strategy best-response correspondence is:

⁴⁵ More formally, you can show the following: For some integer n , let $\{x_1, \dots, x_n\}$ and $\{\alpha_1, \dots, \alpha_n\}$ be sets of real numbers such that $(\alpha_1, \dots, \alpha_n) \in \Delta^{n-1}$; i.e. the $\{\alpha_j\}$ are convex coefficients. Then $\alpha_1 x_1 + \dots + \alpha_n x_n \leq \max \{x_1, \dots, x_n\}$.

⁴⁶ This smiley-face symbol indicates the end of the proof, as in Aumann and Sorin [1989] Aumann and Sorin [1989: 14].

$$\text{MBR}_R(p, q) = \begin{cases} \{(1, 0)\}, & q > 1 - \frac{10}{9}p, \\ \{(t, 1-t): t \in [0, 1]\}, & q = 1 - \frac{10}{9}p, \\ \{(0, 1)\}, & q < 1 - \frac{10}{9}p, \end{cases}$$

with the understanding that $(p, q) \in \mathbb{R}_+^2$ and $p + q \leq 1$.

Every mixed strategy for Cleever can be written as an ordered triple belonging to the two-dimensional simplex $\{(p, q, 1-p-q): p, q \geq 0, p+q \leq 1\}$, where we adopt the convention that the mixed strategy $(p, q, 1-p-q)$ corresponds to playing left and middle with probabilities p and q , respectively. Cleever's mixed-strategy best-response correspondence is:

$$\text{MBR}_C(t) = \begin{cases} \{(0, 0, 1)\}, & t \in [0, 1/6), \\ \{(p, 0, 1-p): p \in [0, 1]\}, & t = 1/6, \\ \{(1, 0, 0)\}, & t \in (1/6, 1]. \end{cases}$$

Example:

Consider the two-player game of Figure 9. Each player has two pure strategies, so each player's mixed strategy can be described by a single number on the unit interval. I'll assign p and q to Row for Up and to Column for left, respectively.

	$l: [q]$	$r: [1-q]$
$U: [p]$	2, -1	-1, 2
$D: [1-p]$	-3, 2	0, -1

Figure 9: A simple two-player game.

We first compute Row's mixed-strategy best-response correspondence $p^*: [0, 1] \Rightarrow [0, 1]$, where $p^*(q)$ returns the set of all optimal mixing probabilities of playing Up for a given probability q with which Column plays left. To do this we compute Row's expected payoff to each of her pure strategies as a function of Column's mixed strategy q :

$$u_R(U; q) = 2q - (1 - q) = 3q - 1,$$

$$u_R(D; q) = -3q.$$

Comparing these two pure-strategy payoffs, we see that Row strictly prefers Up when $q > 1/6$, is indifferent between Up and Down when $q = 1/6$, and strictly prefers Down when $q < 1/6$. Row's pure-strategy best-response correspondence is

$$BR_R(q) = \begin{cases} \{U\}, & q \in [0, 1/6), \\ \{U, D\}, & q = 1/6, \\ \{D\}, & q \in (1/6, 1]. \end{cases}$$

We can therefore write Row's mixed-strategy best-response correspondence as

$$MBR_R(q) = p^*(q) = \begin{cases} \{0\}, & q \in [0, 1/6), \\ [0, 1], & q = 1/6, \\ \{1\}, & q \in (1/6, 1]. \end{cases}$$

The graph of this correspondence appears as the solid heavy line in Figure 10.

We similarly compute Column's mixed-strategy best-response correspondence $q^*(p)$, which returns the probabilities of choosing left which are optimal given the probability p with which Row chooses Up. Computing Column's pure-strategy expected payoffs against Row's mixed-strategy p we obtain

$$u_C(l; p) = -p + 2(1 - p) = 2 - 3p,$$

$$u_C(r; p) = 2p - (1 - p) = -1 + 3p.$$

Column strictly prefers left if $p < \frac{1}{2}$, is indifferent between left and right if $p = \frac{1}{2}$, and strictly prefers right if $p > \frac{1}{2}$. Column's pure-strategy best-response correspondence is

$$MBR_C(q) = q^*(p) = \begin{cases} \{U\}, & p \in [0, 1/2), \\ \{U, D\}, & p = 1/2, \\ \{D\}, & p \in (1/2, 1]. \end{cases}$$

We can write Column's mixed-strategy best-response correspondence as

$$q^*(p) = \begin{cases} \{1\}, & p \in [0, 1/2), \\ [0, 1], & p = 1/2, \\ \{0\}, & p \in (1/2, 1]. \end{cases}$$

The graph of this correspondence appears as the heavy shaded line in Figure 10.

A foreshadowing: We will see later that the intersection of the graphs of these correspondences is the *Nash equilibrium* of the game.

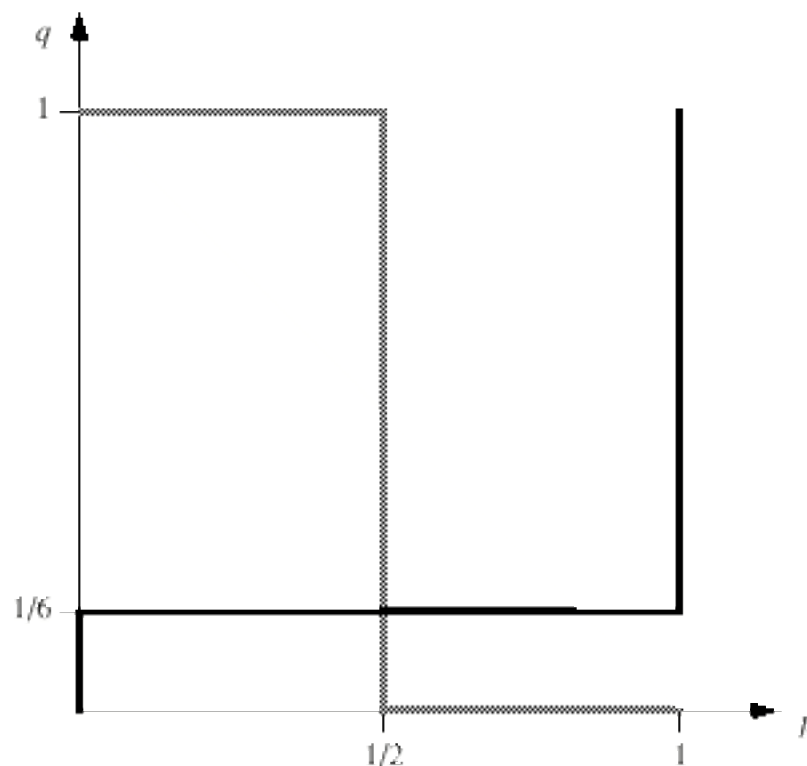


Figure 10: The mixed-strategy best-response correspondences for the game in Figure 9.

Summary

A strategic-form game is determined by a set of players, a pure-strategy space for each player, and a von Neumann-Morgenstern utility function for each player, the arguments of which are the pure strategies chosen by all the players. Such games can be conveniently represented by a matrix which includes a side for each player and whose cells contain payoff vectors.

Rational players choose actions which maximize their expected utility given their beliefs about the actions of their opponents. We set out to solve the Easy Part of Game Theory, viz. the problem of what choice a rational player would make given her beliefs about the choices of her opponents. We saw how to determine each player's pure-strategy best response(s) to the pure-strategy choices of her opponents.

We then expanded the set of choices available to the players by introducing mixed strategies, which are probability distributions over pure strategies. We saw that mixed strategies are members of a unit simplex and that pure strategies are degenerate mixed strategies. We computed players' expected payoffs to arbitrary mixed-strategy profiles by weighting the payoffs to pure-strategy profiles by the probability that each pure-strategy profile would be realized by the players' independent randomizations.

We determined each player's pure-strategy best response(s) against arbitrary mixed strategies by the opponent. We saw that, for a given deleted profile of other players' strategies, a nondegenerate mixed strategy can be a best response only when there are at least two pure-strategy best responses and that a

mixed strategy is a best response if and only if it puts positive probability only upon best-response pure strategies. We then saw how the pure-strategy best-response correspondences yield the mixed-strategy best-response correspondences.

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Jim Ratliff's Game Theory Course:

2.1 Strategic Dominance

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Abstract



What predictions can we make about a game's outcome if we only assume that the players are rational? We typically won't be able to make a unique prediction; instead we'll usually find a multiplicity of outcomes which are consistent with the rationality of all the players.

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To begin our analysis we discuss the concept of strategic *dominance*. Then we will turn to the more precisely relevant concept of "never a best response." These notions will be the foundation for our study of *nonequilibrium* solution concepts. These concepts are nonequilibrium in the sense that they typically admit outcomes which are not Nash equilibria. If we can make a useful

prediction using only nonequilibrium analysis, our conclusion can be much more compelling than if we had achieved the same result using the much stronger (and frequently more dubious) assumptions required by equilibrium analysis. Furthermore, by applying nonequilibrium techniques in our initial analysis of a game we will frequently greatly simplify our subsequent equilibrium analysis.

A strategy for a player is dominated if there exists another strategy for her which is better for her no matter what choices the opponents make. A rational player would never play a dominated strategy. We define what it means for a pure strategy to be dominated by another pure strategy, for a pure strategy to be dominated by a mixed strategy, and then for a mixed strategy to be dominated by a pure or mixed strategy.

Because a rational player would never play a dominated strategy, we can sometimes use a dominance analysis to rule out some outcomes as possibilities when the game is played by rational players. In some games a dominance analysis leads to a unique prediction of the outcome when players are rational; we say that these games are dominance solvable. More generally, dominance analysis may reject some outcomes as impossible when the game is played by rational players but still leaves a multiplicity of outcomes.

Closely related to the concept of a strategy being dominated for a player is the idea that this strategy is "never a best response" for that player: No matter what beliefs she has about the actions of her opponents, she could not rationally choose to play that strategy. If a strategy is dominated, it can never be a best response. However, it is not obvious that the implication holds in the reverse direction. Therefore the set of strategies which are never best responses is weakly larger than the set of dominated strategies. An analysis based on whether strategies are possibly best responses does exhaust the implications of all players being rational. We will see that in two-player games a strategy is never a best response if and only if it is dominated. However, for games with three or more players, it is possible that an undominated strategy will yet never be a best response. Therefore we can sometimes rule out as a plausible choice a strategy even when it is undominated. For more-than-two-player games, then, a dominance argument need not fully exploit the assumption that all players are rational.



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Rationalizability](#)

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Strategic Dominance

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Introduction

We return to the terrain of strategic-form (*aka* normal-form) games and ask: what predictions can we make about a game's outcome if we only assume that the players are rational? (To be definite... at this point we are *not* making assumptions about what the players believe about the other players' rationality or believe about the strategic choices the other players will make.) We typically won't be able to make a unique prediction; instead we'll usually find a multiplicity of outcomes which are consistent with the rationality of all the players. In fact, with this relatively weak assumption we often will be unable to *refine* the set of possible outcomes at all.¹ (In a later handout we'll ask the question: what predictions can we make about a game's outcome if we also assume that this rationality is "common knowledge?"² This will often yield more precise predictions than merely assuming that all players are rational.)

To begin our analysis we discuss the concept of strategic *dominance*. Then we will turn to the more precisely relevant concept of "never a best response." These notions will be the foundation for our study of *nonequilibrium* solution concepts. These concepts are nonequilibrium in the sense that they typically admit outcomes which are not Nash equilibria.³ (Each player *will* be playing a best response to her

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- 1 To refine a set of outcomes is to remove some outcomes from further consideration through the application of some acceptability criterion.
- 2 A loose definition of common knowledge will suffice for now: A fact is *common knowledge* if each player knows it, knows that all other players know it, knows that all other players know that all other players know it, *ad infinitum*.
- 3 Of course, we haven't encountered Nash equilibrium in this course yet. Many of you are nevertheless familiar with that concept and it is for those folk I include this comment.

beliefs about what strategies the other players choose. However, her beliefs may be false.) These nonequilibrium techniques are of interest for the following reason: If we can make a useful prediction using only nonequilibrium analysis, our conclusion can be much more compelling than if we had achieved the same result using the much stronger (and frequently more dubious) assumptions required by equilibrium analysis. Furthermore, by applying nonequilibrium techniques in our initial analysis of a game we will frequently greatly simplify our subsequent equilibrium analysis.

A strategy for a player is *dominated* if there exists another strategy for her which is better for her no matter what choices the opponents make. A rational player would never play a dominated strategy. We define what it means for a pure strategy to be dominated by another pure strategy, for a pure strategy to be dominated by a mixed strategy, and then for a mixed strategy to be dominated by a pure or mixed strategy.

Because a rational player would never play a dominated strategy, we can sometimes use a dominance analysis to rule out some outcomes as possibilities when the game is played by rational players. In some games a dominance analysis leads to a unique prediction of the outcome when players are rational; we say that these games are dominance solvable. In other games a dominance analysis results in no refinement of the set of possible outcomes. Other games lie between these two extremes: dominance analysis rejects some outcomes as impossible when the game is played by rational players but still leaves a multiplicity of outcomes.

Although we have established that a rational player would never play a dominated strategy, we have *not* established that any undominated strategy could be plausibly chosen by a rational player. I.e. we have not shown that a dominance analysis fully exhausts the implications of all players being rational.

Closely related to the concept of a strategy being dominated for a player is the idea that this strategy is “never a best response” for that player: No matter what beliefs she has about the actions of her opponents, she could not rationally choose to play that strategy. If a strategy is dominated, it can never be a best response. However, it is not obvious that the implication holds in the reverse direction. I.e. it’s not obvious that a strategy which is never a best response is also a dominated strategy. Therefore the set of strategies which are never best responses is weakly larger than the set of dominated strategies. An analysis based on whether strategies are possibly best responses *does* exhaust the implications of all players being rational: A strategy cannot be plausibly chosen by a rational player if and only if it is never a best response.

We will see—perhaps surprisingly—that in two-player games a strategy is never a best response if and only if it is dominated. For two-player games, then, a dominance analysis fully exploits the assumption that all players are rational. However, for games with three or more players, it is possible that an undominated strategy will yet never be a best response. Therefore we can sometimes rule out as a plausible choice a strategy even when it is undominated. For more-than-two-player games, then, a dominance argument need not fully exploit the assumption that all players are rational.

Recapitulation

Let's review the standard paradigm and notation. We have a finite set I of n players, $I = \{1, \dots, n\}$. The i -th player has a nonempty finite set of pure strategies—her pure-strategy space S_i —available to her, from which she can choose one strategy $s_i \in S_i$. In general her choice of pure strategy will be probabilistically determined by a mixed strategy $\sigma_i \in \Delta(S_i) \equiv \Sigma_i$, which is a probability distribution over her pure-strategy space S_i ; the probability that she chooses a particular pure strategy $s_i \in S_i$ is specified by $\sigma_i(s_i)$.

The players simultaneously choose their strategies. If they choose pure strategies, the resulting pure-strategy profile is some ordered n -tuple $s = (s_1, \dots, s_n)$, where $s \in S = \prod_{i \in I} S_i$. S is called the space of pure-strategy profiles. If they choose mixed strategies, the resulting mixed-strategy profile is some ordered n -tuple $\sigma = (\sigma_1, \dots, \sigma_n)$, where $\sigma \in \Sigma \equiv \prod_{i \in I} \Sigma_i$. Σ is called the space of mixed-strategy profiles.

When we want to focus on the strategy choices of all players except i , we write player i 's deleted strategy profile $s_{-i} = (s_1, \dots, s_{i-1}, s_{i+1}, \dots, s_n)$ for the pure-strategy case or $\sigma_{-i} = (\sigma_1, \dots, \sigma_{i-1}, \sigma_{i+1}, \dots, \sigma_n)$ for the mixed-strategy case. We denote the space of player i 's deleted pure- and mixed-strategy profiles by S_{-i} and Σ_{-i} , respectively. We write $\langle a_i, b_{-i} \rangle_i$ to indicate the strategy profile resulting when player i chooses strategy $a_i \in (S_i \cup \Sigma_i)$ and her opponents choose the deleted strategy profile $b_{-i} \in (S_{-i} \cup \Sigma_{-i})$.

After all the players simultaneously pick their individual strategies, a pure-strategy profile $s \in S$ is realized.⁴ Each player i receives an *ex post* payoff which depends on this s , viz. $u_i(s)$.

The expected (*ex ante*) payoff to player i when the players all participate in the mixed-strategy profile $\sigma \in \Sigma$ is

$$u_i(\sigma) = \sum_{s \in S} \left(\prod_{j \in I} \sigma_j(s_j) \right) u_i(s). \quad (1)$$

As a special case of (1) you can show that player i 's expected payoff when she plays the pure strategy $s_i \in S_i$ against the deleted mixed-strategy profile $\sigma_{-i} \in \Sigma_{-i}$ is

$$u_i(\langle s_i, \sigma_{-i} \rangle_i) = \sum_{s_{-i} \in S_{-i}} \left(\prod_{j \in I \setminus \{i\}} \sigma_j(s_j) \right) u_i(\langle s_i, s_{-i} \rangle_i). \quad (2)$$

When player i believes that her $n - 1$ opponents are playing the deleted mixed-strategy profile $\sigma_{-i} \in \Sigma_{-i}$, the pure strategies which are best responses for i are given by i 's best-response correspondence $BR_i: \Sigma_{-i} \rightarrow S_i$,⁵

⁴ If all the players choose pure strategies, the realized pure-strategy profile s is simply the strategy profile the players choose. More generally, the players choose mixed strategies. Each player's mixed strategy results in a pure-strategy realization for her. The collection of these pure-strategy realizations is a realization of a pure-strategy profile.

$$BR_i(\sigma_{-i}) = \arg \max_{s_i \in S_i} u_i(\langle s_i, \sigma_{-i} \rangle_i), \quad (3)$$

which maps the space of deleted mixed-strategy profiles Σ_{-i} into subsets of the space of i 's pure strategies S_i ; i.e. $BR_i: \Sigma_{-i} \rightarrow S_i$.

Fix some deleted mixed-strategy profile $\sigma_{-i} \in \Sigma_{-i}$ by i 's opponents. When player i has a unique pure-strategy best response—i.e. when $BR_i(\sigma_{-i})$ is a singleton (alternatively, when $\#BR_i(\sigma_{-i}) = 1$)—she has no choice but to play that pure strategy. However, if there are two or more pure-strategy best responses, she can randomize over those pure-strategy best responses in any fashion, and she cannot put positive weight on any pure strategy which is not a best response. In other words, a mixed strategy $\sigma_i \in \Sigma_i$ is a best response for i against the deleted mixed-strategy profile $\sigma_{-i} \in \Sigma_{-i}$ if and only if it puts positive weight only on pure-strategy best responses; i.e. if and only if its support is contained in the best-response set:⁶

$$\text{supp } \sigma_i \subset BR_i(\sigma_{-i}). \quad (4)$$

Strong dominance

Often a player's optimal strategy depends crucially on what she believes her opponents are doing. However, this is not always the case. We say that a strategy is *strongly* (or *strictly*) *dominant* for a player if it is strictly better than all of her other choices *regardless of the actions of her opponents*. When a rational player—someone who chooses her action so as to maximize her expected utility given her beliefs—has a dominant strategy, we can be quite confident in predicting that she would choose that dominant strategy. If every player had a dominant strategy, predicting the outcome of the game would be a trivial exercise: The predicted outcome would be the strategy profile in which each player chose her strongly dominant strategy. Fortunately or unfortunately, it is rarely the case that a player has a dominant strategy; however, notions of strategic dominance often remain useful. (In what follows, if I leave out the qualifier “strongly” or “strictly,” interpret the statement as if the qualifier were explicit.)

First we will focus on when a pure strategy might be dominated. We will discuss what it means for one pure strategy for player i to dominate another of her pure strategies. We will see that we can assess dominance even if we restrict attention only to the opponents' pure strategies. After that we will see that sometimes a mixed strategy can dominate a pure strategy even though no pure strategy did. Lastly, we will how a mixed strategy can be dominated by another pure or mixed strategy.⁷

⁵ Recall that a correspondence yields a *set* for each value of its argument. “arg max” refers to the set of values of the argument which maximizes the maximand.

⁶ Recall that the support of a mixed strategy is the set of pure strategies to which it allocates positive probability; i.e. $\text{supp } \sigma_i = \{s_i \in S_i: \sigma_i(s_i) > 0\}$.

⁷ A good reference concerning strong domination is Fudenberg and Tirole [1991].

Pure-strategy strong dominance

Consider two pure strategies for player i : $s_i, s_i' \in S_i$. We say that s_i' *strictly dominates* s_i if s_i' gives player i a strictly higher expected utility than does s_i for every possible deleted pure-strategy profile $s_{-i} \in S_{-i}$ which her opponents could play, i.e. if

$$\forall s_{-i} \in S_{-i}, \square u_i(\langle s_i', s_{-i} \rangle_i) > u_i(\langle s_i, s_{-i} \rangle_i). \quad (5)$$

If s_i' strictly dominates s_i , a rational player would never choose s_i (if she believes her opponents will all choose pure strategies) because, regardless of her beliefs concerning her opponents' strategies s_{-i} , she could increase her expected utility by choosing s_i' instead.

If there exists a strategy $s_i' \in S_i$ which strictly dominates $s_i \in S_i$ for player i , then we say that s_i is *strictly dominated* for player i . If there does not exist a pure strategy which dominates s_i , then we say that s_i is *undominated* (by pure strategies) for player i . If there exists an $s_i' \in S_i$ which dominates every other strategy $s_i \in S_i \setminus \{s_i'\}$,⁸ then we say that s_i' is a *strictly dominant* strategy for player i . (We will see that a strategy can be dominated even if there does not exist a dominant strategy.) If a rational player has a dominant strategy, she must choose that strategy (if she believes that her opponents are all choosing pure strategies). (We will soon remove this “opponents are all choosing pure strategies” qualification.)

Note from (5) that in assessing whether one of her strategies is dominated, dominant, or undominated a player only needs to know her own payoff function $u_i: S \rightarrow \mathbb{R}$ —i.e. how her payoff depends on the choices of all the players; she does not need to know anything about her opponents' payoffs or even that her opponents are rational.

Example: Analyzing pure-strategy dominance.

Pure-strategy domination is easy to analyze just by inspection of the payoff matrix. Consider Row's payoffs in the game in Figure 1.⁹ We observe that c dominates b for Row, because $3 > 1$, $4 > 3$, and $0 > -1$. Similarly, a dominates b . So, for either reason, b is a dominated strategy. However, neither a nor c dominates each other. (Against d , c is better than a , but against e , a is better than c .) Therefore a and c are both undominated by pure strategies. Further, Row doesn't have a dominant strategy, because she doesn't have a single strategy which dominates both of the remaining strategies. (A necessary, but not sufficient, condition for Row to have a dominant strategy is that every Row payoff in exactly one row appear in boldface. A sufficient condition is that no Row payoffs in any other row appears in boldface.)

⁸ Let A and B be sets. The *difference* (or *relative complement*) of A and B , denoted $A \setminus B$, is the set of elements which are in A but not in B , i.e. $A \setminus B = \{x \in A: x \notin B\}$. The difference of A and B is also sometimes written as $A - B$.

⁹ Row payoffs which are maximal within their columns and Column payoffs which are maximal within their row appear in boldface.

	<i>d</i>	<i>e</i>	<i>f</i>
<i>a</i>	2, 1	5, 2	0, -1
<i>b</i>	1, 0	3, 3	-1, 2
<i>c</i>	3, 7	4, 8	0, 6

Figure 1: Column has a dominant strategy; Row does not.

Now consider Column's payoffs. Strategy *e* dominates *d*, because $2 > 1$, $3 > 0$, and $8 > 7$. Similarly *e* dominates *f*. As a consequence both *d* and *f* are dominated strategies, and *e* is a dominant strategy for Column because it dominates all of his other strategies. (Note that every Column payoff in column *e* appears in boldface and that no other Column payoff appears in boldface.)

Restricting attention to opponents' pure strategies

Look again at the definition of (5). If $s_i' \in S_i$ strictly dominates $s_i \in S_i$ in this sense, we are sure that s_i' will outperform s_i for player *i* as long as her opponents are choosing pure strategies (because the universal quantifier says for all $s_{-i} \in S_{-i}$). But what if the opponents are playing mixed strategies? Can she still be sure that s_i' is a better choice than s_i ? In general we want to allow for players choosing mixed strategies. It seems we would actually want the definition of (5) to be that s_i' strictly dominates s_i if the inequality holds for all possible mixed strategies by her opponents, i.e. if

$$\forall \sigma_{-i} \in \Sigma_{-i}, \square u_i(\langle s_i', \sigma_{-i} \rangle_i) > u_i(\langle s_i, \sigma_{-i} \rangle_i). \quad (6)$$

Prima facie, the definition in (6) looks more difficult to satisfy than (5) because the inequality must hold in a larger set of cases. However, the two definitions are actually equivalent and the version in (5) is easier to verify because we only need to check a finite number of deleted pure-strategy profiles rather than all possible deleted mixed-strategy profiles. The two definitions are equivalent in the sense that, if s_i' strictly dominates s_i according to (5), then s_i' also strictly dominates s_i according to (6), and vice versa. Let's see why this is so. Clearly, satisfaction of the inequality in (6) implies satisfaction in (5) because the set of deleted pure-strategy profiles S_{-i} is included in the set of deleted mixed-strategy profiles Σ_{-i} .¹⁰

The other direction—that satisfaction of the inequality against all deleted pure-strategy profiles by the opponents implies satisfaction against all possible deleted mixed-strategy profiles—requires a little work to see. The idea is that player *i*'s payoff to any pure strategy against a particular combination of mixed strategies by her opponents is a convex combination of her payoffs against deleted pure-strategy profiles by her opponents, where the coefficients of that convex combination depend only on her opponents' strategies. By playing the better (against pure strategies) pure strategy she will receive a higher convex combination than she would have by playing the inferior (against pure strategies) pure strategy.

¹⁰ This wording is so abusive as to be untrue. However it is so convenient an abuse that I (and the rest of the game theory community) permit it to stand. It is not true that $S_{-i} \subset \Sigma_{-i}$. The following, however, is true: For every $s_{-i} \in S_{-i}$, form the deleted mixed-strategy profile $\sigma_{-i} = (\delta_1(s_1), \dots, \delta_{i-1}(s_{i-1}), \delta_{i+1}(s_{i+1}), \dots, \delta_n(s_n)) \in \Sigma_{-i}$. Then s_{-i} and σ_{-i} are equivalent in the sense that, for all $\sigma_i \in \Sigma_i$, $u_i(\langle \sigma_i, s_{-i} \rangle_i) = u_i(\langle \sigma_i, \sigma_{-i} \rangle_i)$.

In order to make this argument clearer consider an arbitrary deleted mixed-strategy profile $\sigma_{-i} \in \Sigma_{-i}$. We can use (2) to express $u_i(\langle s_i', \sigma_{-i} \rangle_i)$ as a convex combination of $u_i(\langle s_i', s_{-i} \rangle_i)$ terms, one for each $s_{-i} \in S_{-i}$. Now assume that s_i' strictly dominates s_i in the sense of (5). Then we replace each $u_i(\langle s_i', s_{-i} \rangle_i)$ term by something we know from (5) to be smaller, viz. $u_i(\langle s_i, s_{-i} \rangle_i)$. The result is equal to $u_i(\langle s_i, \sigma_{-i} \rangle_i)$, and we thereby establish the inequality (6) for an arbitrary σ_{-i} and therefore for all $\sigma_{-i} \in \Sigma_{-i}$. In other words symbols,

$$u_i(\langle s_i', \sigma_{-i} \rangle_i) = \sum_{s_{-i} \in S_{-i}} \left(\prod_{j \in I \setminus \{i\}} \sigma_j(s_j) \right) u_i(\langle s_i', s_{-i} \rangle_i) > \sum_{s_{-i} \in S_{-i}} \left(\prod_{j \in I \setminus \{i\}} \sigma_j(s_j) \right) u_i(\langle s_i, s_{-i} \rangle_i) = u_i(\langle s_i, \sigma_{-i} \rangle_i). \quad (7)$$

Therefore if s_i' strictly dominates s_i in the sense of (5), then it does in the sense of (6) as well.

Earlier I said that, if a rational player believed that all of her opponents were choosing pure strategies, she would never play a dominated strategy, as defined by (5), and she would choose her dominant strategy if she had one. Now that we have seen that we can without loss of generality restrict our attention to opponents' pure strategies when assessing questions of dominance, we can dispense with this qualification that the player believes that all of her opponents are choosing pure strategies: We can categorically say that a rational player would never play a dominated strategy [as defined in (5)] and that she would choose her dominant strategy if she had one.

When all players have dominant strategies

Consider the case in which each player $i \in I$ is rational, 2 knows her own payoffs $u_i: S \rightarrow \mathbb{R}$, and 3 has a dominant strategy s_i^* . Then each player must choose her dominant strategy s_i^* , and the result of the game must be the strategy profile $(s_1^*, \dots, s_n^*)$. We say that such a game is *dominance solvable*.¹¹

Example: The Prisoners' Dilemma is dominance solvable.

Consider the Prisoners' Dilemma game of Figure 2.¹² Each prisoner can either Fink or stay Mum. If they both stay Mum they are released for lack of evidence and get to keep their loot. Any prisoner who Finks is rewarded for squealing; he is also set free unless his partner also squeals. Any prisoner who is squealed upon is given a prison sentence. The reward to squealing does not compensate for the suffering of prison life. So if they both stay Mum, each is better off than if they both Finked. However, if one player stays Mum, the other gains by being the sole Fink.

	M	F
M	1, 1	-1, 2
F	2, -1	0, 0

Figure 2: A Prisoners' Dilemma

¹¹ We will also apply this term later with regard to the iterated elimination of strictly dominated strategies.

¹² The earliest written description of the Prisoners' Dilemma appears in Tucker [1950], which is reprinted in Straffin [1980].

Consider the Row suspect. Finking is better if Column stays Mum, because Row not only goes free (as she would if she stayed Mum too) but also pockets the reward for testifying. Finking is better if Column Finks, because jail is a foregone conclusion so Row might as well benefit from the fruits of testifying. So we see that, regardless of Column's action, Finking is a strictly better choice for Row. Therefore Finking is a dominant strategy for Row. Exactly the same analysis applies to Column's choice. Because each player has the dominant strategy Fink, the only outcome for the game with rational prisoners is (Fink, Fink).

Note that this analysis only requires that each player is rational and knows the deal *he* is offered by the authorities (i.e. his own payoffs). A prisoner does not need to know what deal the other prisoner is promised and does not even need to know that the other prisoner is rational.

However, game theory is not typically this easy. (Games where all players have dominant strategies are uninteresting game theoretically.) Consider the matching pennies game of Figure 3. Each player decides which side of a coin to show. Row prefers that the coins match; Column prefers that they be opposite. Here a player's best choice depends crucially on the choice of her opponent. (E.g. Column gains by playing *T* if Row plays *H*, but Column gains by playing *H* if Row plays *T*.) No strategy is dominant for either player.

	<i>H</i>	<i>T</i>
<i>H</i>	1, -1	-1, 1
<i>T</i>	-1, 1	1, -1

Figure 3: Neither player has a dominant strategy in matching pennies.

Mixed-strategy dominance

We can use the above definition of pure-strategy dominance to rule out of consideration any pure strategy s_i which is dominated by some other pure strategy s_i' . Can we do more? In other words are there cases in which a pure strategy is dominated by some mixed strategy $\sigma_i' \in \Sigma_i$ of player i 's but is not dominated by any pure strategy? The answer is yes.

So we define a more general notion of domination. Consider a pure strategy $s_i \in S_i$ and a mixed strategy $\sigma_i' \in \Sigma_i$ for player i . We say that σ_i' strictly dominates s_i if σ_i' gives player i a strictly higher expected utility than does s_i for every possible pure-strategy profile s_{-i} which her opponents could play, i.e. if

$$\forall s_{-i} \in S_{-i}, \square u_i(\langle \sigma_i', s_{-i} \rangle_i) > u_i(\langle s_i, s_{-i} \rangle_i). \quad (8)$$

We use the obvious terminology, parallel to that we defined above in connection with pure-strategy dominance, concerning when a strategy is dominated, undominated, and dominant. (Exactly the same argument we used with respect to pure-strategy domination above works here to justify our restriction of attention to deleted pure-strategy profiles by the opponents.)

It's probably not obvious to you that allowing mixed strategies could increase the set of pure strategies which are dominated and therefore irrational. I.e. how do we know that it's not the case that any pure strategy s_i which is dominated by a mixed strategy σ_i' wouldn't be dominated more simply by some pure strategy s_i' ? Let's look at a simple game to provide an example.

Example: A mixed strategy can dominate where no pure strategy can.

Consider the game in Figure 4. It's easy to verify that none of Row's pure strategies dominate each other. In particular neither Up nor Middle dominates Down. Even though Up is better than Down when Column plays left, Down is better than Up when Column plays right. Similar remarks can be made about the relationship between Middle and Down.

	$l: [q]$	$r: [1-q]$
$U: [p]$	6,0	0,6
$M: [1-p]$	0,6	6,0
D	2,0	2,0

Figure 4: A mixed strategy can dominate where no pure strategy can.

However, consider the mixed strategy for Row in which she plays Up one-half the time and Middle the remainder, i.e. $\sigma_R' = \frac{1}{2} \circ U \oplus \frac{1}{2} \circ M$. We will see that, even though neither Up nor Middle dominates Down, this mixture of Up and Middle *does* dominate Down. Let's compute Row's expected utility from this mixed strategy against both of Column's pure-strategy choices, left and right:

$$u_R(\sigma_R'; l) = \frac{1}{2}(6) + \frac{1}{2}(0) = 3 > 2 = u_R(D; l), \quad (9)$$

$$u_R(\sigma_R'; r) = \frac{1}{2}(0) + \frac{1}{2}(6) = 3 > 2 = u_R(D; r). \quad (10)$$

So we see that this mixture strictly outperforms Down regardless of which strategy Column chooses; therefore σ_i' dominates Down even though Down was undominated by pure strategies alone.

How did I know to choose the $(\frac{1}{2}, \frac{1}{2})$ mixing probabilities as I did? Let's write a mixture between Up and Middle more generally as

$$\sigma_R' = p \circ U \oplus (1-p) \circ M, \quad (11)$$

where $p \in [0, 1]$. We want to choose p such that Row's payoff to the mixed strategy exceeds her payoff to the pure strategy Down both when Column chooses left and when he chooses right. I.e. we want p to satisfy the two inequalities¹³

$$u_R(\sigma_R'; l) = 6p + 0 \cdot (1-p) > u_R(D; l) = 2, \quad (12)$$

$$u_R(\sigma_R'; r) = 0 \cdot p + 6(1-p) > u_R(D; r) = 2. \quad (13)$$

¹³ You can see how using the definition of dominance which only requires comparisons against a finite number of opponents' pure-strategy choices simplifies life tremendously!

Inequality (12) is satisfied whenever $p > \frac{1}{3}$; inequality (13) is satisfied whenever $p < \frac{2}{3}$. Therefore the mixture between Up and Middle $\sigma_{R'}$ will dominate Down as long as we choose

$$p \in (\frac{1}{3}, \frac{2}{3}). \quad (14)$$

My choice of $p = \frac{1}{2}$ was just one of many I could have made.

The intuition for the successful domination of Down by a mixture of Up and Middle can be more clearly explained when we consider Column's choice between left and right as a mixed strategy, $\sigma_C = q \circ l \oplus (1 - q) \circ r$. In Figure 5 we plot Row's expected payoff to Up, Middle, Down, and to the $(\frac{1}{2}, \frac{1}{2})$ mixture of Up and Middle, all as a function of Column's mixing parameter q . It is easy to calculate that

$$u_R(U; q) = 6q + 0 \cdot (1 - q) = 6q, \quad (15)$$

$$u_R(M; q) = 0 \cdot q + 6(1 - q) = 6 - 6q. \quad (16)$$

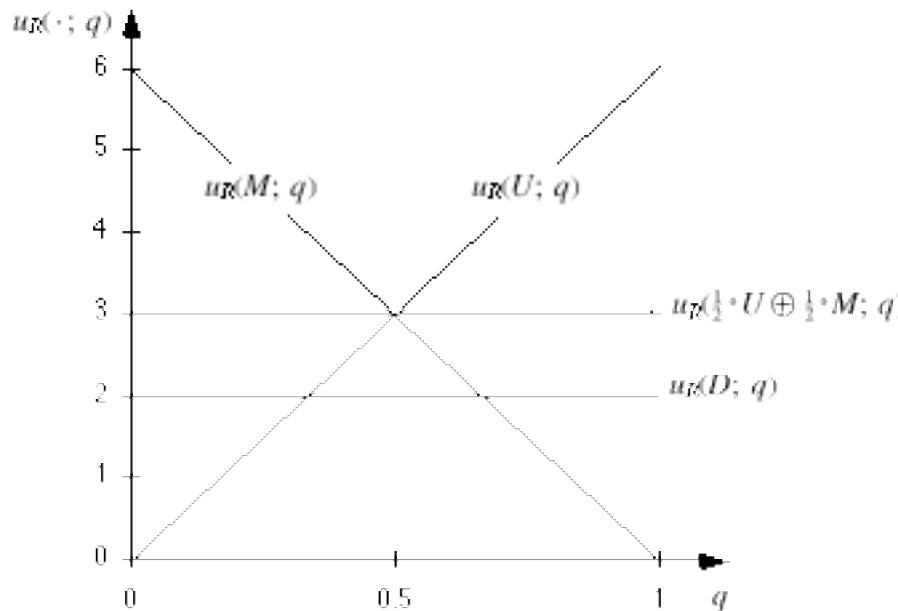


Figure 5: Row's pure- and mixed-strategy payoffs as a function of Column's mixed strategy.

Now we can ask the question: why was the mixture successful at dominating Down when neither Up nor Middle was successful on its own? We can see that Up is very good for Row against high q , i.e. when Column puts a lot of weight on left; but Up is very bad against low q . Conversely, Middle is bad against high q but good against low q . So we see that Up and Middle are complementary: one's regime of strength is the other's regime of weakness. By mixing between these two highly variable performers, we obtain a mixed strategy which is uniformly mediocre. In this example uniformity was a virtue when finding dominating strategies—it doesn't matter how good a strategy is at its best if it is too bad at its worst. Mediocrity is tolerable as long as it is good enough. Even if Row's payoffs to Down were somewhat higher, we would still have been able to find a dominating mixture. You should be able to show that this would be possible as long as Row's common payoff to Down were less than 3.

Dominated mixed strategies

Let's summarize our discussion of strict dominance so far. First we defined what it meant for one pure strategy to dominate another pure strategy. We justified restricting attention to pure strategies of the opponents when assessing the performance of a conjectured dominating strategy. We then defined strict dominance of a pure strategy by a mixed strategy. We saw that a pure strategy could be dominated by a mixed strategy even if it is not dominated by any pure strategy.

Note that thus far we have only discussed what it meant for a pure strategy to be dominated. But that raises a further question: If we know what pure strategies are dominated and therefore not rationally chosen, what can we say about the mixed strategies which are dominated and therefore not rationally chosen by a rational player?

Part of the answer is simple: Any mixed strategy which puts positive probability on a dominated strategy is itself dominated. I leave it as an exercise to generalize (6) in the natural way in order to define what it would mean for a mixed strategy to be strictly dominated (by another mixed strategy). Then show that, if some mixed strategy σ_i has a dominated pure strategy in its support, you could construct another mixed strategy σ_i' which strictly dominates σ_i .

However, this does not mean that any mixed strategy which puts positive probability only upon undominated pure strategies is necessarily undominated itself. We saw above that a pure strategy could be dominated by a mixed strategy even though it was undominated by any pure strategy. Very similarly, a nondegenerate mixed strategy σ_i can be dominated by another mixed strategy σ_i' (even by a pure strategy) even though σ_i puts no weight on dominated pure strategies.

Example: A mixed strategy over undominated pure strategies can be dominated.

To see that a mixed strategy can be dominated even when it puts no weight on dominated pure strategies we can modify the game in Figure 4 slightly, by changing Row's payoff to Down from $2 \rightarrow 4$. See Figure 6.

	$l: [q]$	$r: [1-q]$
$U: [p]$	6,0	0,6
$M: [1-p]$	0,6	6,0
D	4,0	4,0

Figure 6: Down is now better than it was.

Now we again consider the mixture $\sigma_R' = \frac{1}{2} \circ U \oplus \frac{1}{2} \circ M$, and plot Row's payoff to this mixed strategy and to her pure strategies U , M , and D as functions of Column's mixed strategy q . As we can see in Figure 7, D now dominates the mixed strategy even though neither pure strategy U or M was dominated.

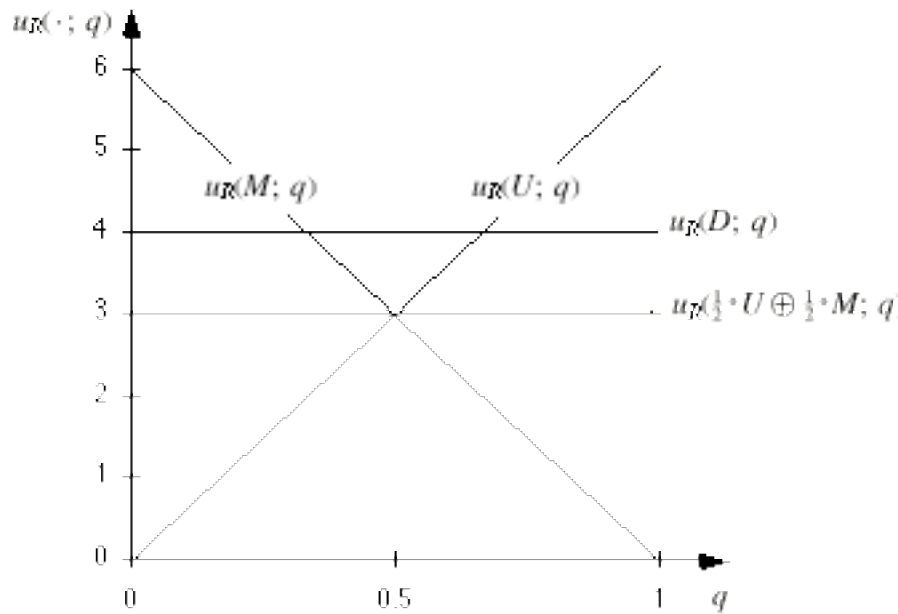


Figure 7: An equal mixture of U and M is dominated by D even though neither U nor M is dominated.

In fact we can show that *every* nondegenerate mixture of Up and Middle is dominated. It will be useful to plot the expected payoff to Row from a mixture of Up and Middle as a function of Column's mixed strategy q . Let's compute Row's expected payoff to the mixed strategy $p \circ U \oplus (1-p) \circ M$:

$$u_R(p \circ U \oplus (1-p) \circ M; q) = 6[pq + (1-p)(1-q)] = 3 + 6(2p-1)(q - \tfrac{1}{2}). \quad (17)$$

I wrote the payoff in this form to show that the graph of Row's expected payoff, as a function of q , of every such mixed strategy is simply a rotation of her pure-strategy payoff to U (or M) about the intersection $(q = \frac{1}{2}, u = 3)$ of the two pure-strategy payoff graphs.¹⁴ To see this note, that when $q = \frac{1}{2}$, $u_R(p \circ U \oplus (1-p) \circ M) = 3$, for all p . Note also that the slope of graph of $u_R(p \circ U \oplus (1-p) \circ M)$ varies smoothly with p , coinciding with the slopes of the graphs of $u_R(M; q)$ and $u_R(U; q)$ for $p = 0$ and $p = 1$, respectively. See Figure 8.

It is obvious from Figure 8 that Row's pure strategy Down dominates every mixture of Up and Middle such that $\frac{1}{3} < p < \frac{2}{3}$. However, for $p \leq \frac{1}{3}$, the mixture is at least as good as Down for small q ; for $p \geq \frac{2}{3}$, the mixture is least as good as Down for q near one. So we still need to show that these small and large p mixtures of Up and Middle which are undominated by a pure strategy *are* dominated by mixed strategies.

Let's consider the case where $p = \frac{5}{6}$. (See Figure 9.) I will show that the payoff to this mixture of Up and Middle is dominated by a mixture of Up and Down. Exactly as we saw before concerning the graph

¹⁴ The payoff to the mixed strategy is a convex combination of the pure-strategy payoffs; therefore the graph of the mixed-strategy payoff is a vertical convex combination of the pure-strategy graphs. Therefore the mixed-strategy graph must pass through the intersection of the pure-strategy graphs.

of the payoff to a mixture of Up and Down, you can show that the graph of the payoff to a mixture of Up and Down is a rotation of the payoff to Up about the point of intersection of the graphs of the pure-strategy payoffs to Up and Down. The mixture of Up and Down I propose as a dominating mixed strategy chooses Up with probability $\frac{2}{3}$. You can show that the graph of Row's expected payoff to this mixture has the same slope as her payoff to the mixture of Up and Middle, but is strictly higher, as shown in Figure 9. Therefore this mixture of Up and Down dominates the mixture of Up and Middle.

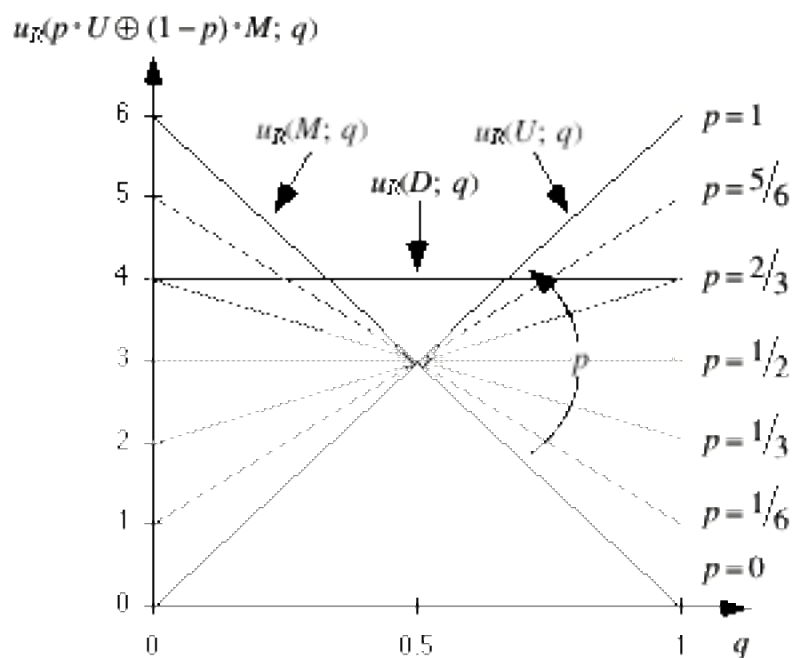


Figure 8: The payoff to a mixture of Up and Middle, as a function of q , is a rotation of a pure-strategy payoff function about the intersection of the two pure-strategy payoffs.

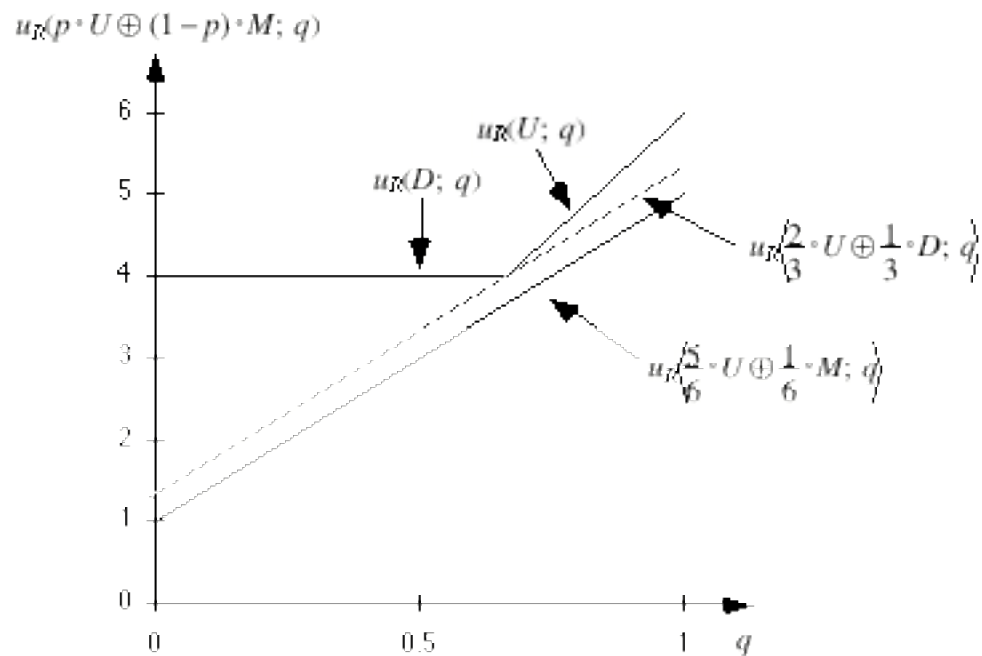


Figure 9: A mixed strategy, undominated by pure strategies, can be dominated by a mixed strategy.

Of course this example doesn't show that *every* nondegenerate mixture of Up and Middle is dominated as I claimed earlier. How do I know that I can *always* find a mixture which dominates any given mixture of Up and Middle? Now I'll give a more general argument.

In Figure 10 I've indicated the upper envelope of Row's pure-strategy payoff functions.¹⁵ Consider any line tangent to this upper envelope. Such a tangent might coincide with one of the three pure-strategy segments; if not, it must be tangent at the intersection between either a the Middle and Down pure-strategy payoff functions or b the Down and Up payoff functions. Therefore any line tangent to the upper envelope corresponds to either 1 a pure strategy, 2 a mixture between Middle and Down, or 3 a mixture of Down and Up.

¹⁵ Let $f_1, \dots, f_n$ be functions from some common domain X into the reals, i.e. $f_i: X \rightarrow \mathbb{R}$. Then the *upper envelope* of these functions is itself a function $\tilde{f}: X \rightarrow \mathbb{R}$ defined by: $\tilde{f}(x) \equiv \max\{f_1(x), \dots, f_n(x)\}$.

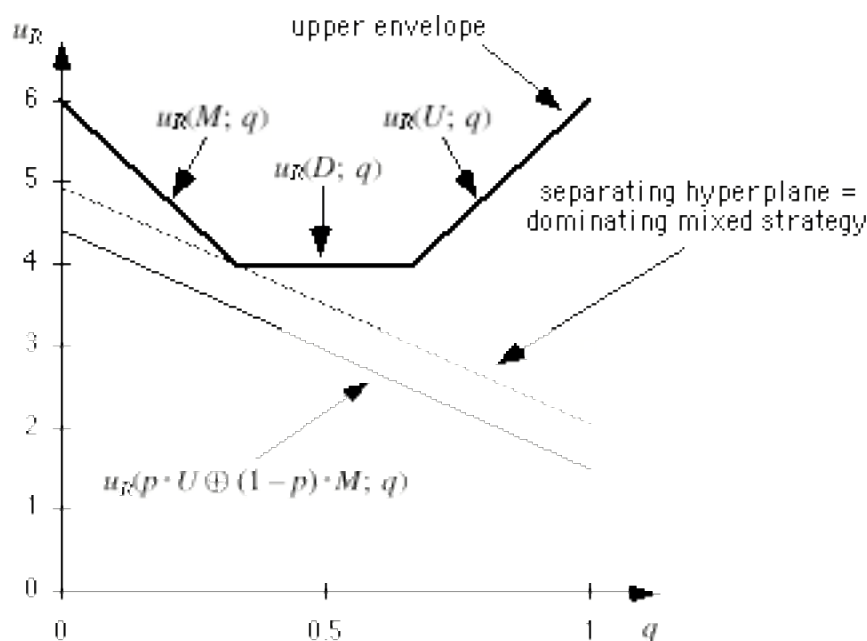


Figure 10: The separating hyperplane theorem guarantees the existence of a mixed strategy which dominates any nondegenerate mixture of Up and Middle.

Note from Figure 8 that any nondegenerate mixture of Up and Middle lies strictly below the upper envelope displayed in Figure 10. What I need to show then is that, for any such nondegenerate mixture, there exists a line tangent to the upper envelope which lies strictly above the mixture of Up and Middle. Then, because every such line corresponds to some mixed strategy, I will have shown that there exists a mixed strategy which dominates the mixture of Up and Middle.

To show that there indeed exists a line tangent to the upper envelope and above the mixture of Up and Middle, I pull out the separating hyperplane theorem.¹⁶ First note that the graph above the upper envelope is a convex set.¹⁷ The set of points below the mixture of Up and Middle is also a convex set.¹⁸ These two convex sets are disjoint, therefore there exists a hyperplane which separates the two sets and is tangent to the first.¹⁹

¹⁶ There are several versions of the separation theorem; see Debreu [1959] or Debreu [1959]. The theorem I would want would say: “Let A and B be disjoint closed convex sets. Then there exists a hyperplane which supports A (i.e. includes a point on the boundary of A) and strictly separates B (i.e. so that B is contained within the interior of one of the closed half planes determined by the hyperplane).” I haven’t found a reference for this exact theorem; typically you find a supporting hyperplane theorem which addresses the issue of a point on the boundary of A and you can find separating theorems which don’t guarantee that the hyperplane intersects the boundary of one of the sets. Nevertheless I assert the theorem in quotes with confidence.

¹⁷ The set of points above the graph of a convex function is convex. Each of the pure-strategy bounding lines is a convex function. The points above the upper envelope is the intersection of the points above each bounding line; therefore the points above the upper envelope are the intersection of convex sets and, hence, are a convex set.

¹⁸ The set of points below the graph of a concave function is convex. A hyperplane is both convex and concave.

¹⁹ We will see later that this type of construction relies crucially on there only being two players.

Domination and never-a-best-response

Consider a strategy $\sigma_i \in \Sigma_i$ for player $i \in I$ and beliefs $\sigma_{-i} \in \Sigma_{-i}$ which player i holds about the actions of the other players. If σ_i is not a best response for player i to these beliefs σ_{-i} , then she must have available a better strategy $\sigma_i' \in \Sigma_i$; i.e. $\exists \sigma_i' \in \Sigma_i$ such that $u_i(\langle \sigma_i', \sigma_{-i} \rangle_i) > u_i(\langle \sigma_i, \sigma_{-i} \rangle_i)$. It's conceivable that there are no beliefs to which σ_i is a best response; we say that σ_i is *never a best response* for i if

$$\forall \sigma_{-i} \in \Sigma_{-i}, \exists \sigma_i' \in \Sigma_i, u_i(\langle \sigma_i', \sigma_{-i} \rangle_i) > u_i(\langle \sigma_i, \sigma_{-i} \rangle_i). \quad (18)$$

If σ_i is a dominated strategy for player i , then there exists a strategy $\sigma_i' \in \Sigma_i$ which is better-for- i than σ_i regardless of the actions σ_{-i} of the other players; i.e.²⁰

$$\exists \sigma_i' \in \Sigma_i, \forall \sigma_{-i} \in \Sigma_{-i}, u_i(\langle \sigma_i', \sigma_{-i} \rangle_i) > u_i(\langle \sigma_i, \sigma_{-i} \rangle_i). \quad (19)$$

Note that these two conditions for being never a best response and being dominated, respectively, though similar, are not identical. The order of appearance of the universal and existential quantifiers is reversed. From (19) you can easily deduce (18); i.e. a dominated strategy is never a best response. However, (18) does not simply imply (19); we leave open the possibility that a strategy may never be a best response but is yet undominated by any other strategy.

Despite the nonexistence of a simple, logical, syntactical deduction of (19) from (18), we will see that in two-player games it is indeed the case that any strategy which is never a best response is also dominated; i.e. when $n = 2$, $(19) \Leftrightarrow (18)$.²¹ Then we will see an example which shows that this relationship need not hold in general when there are more than two players.

In two-player games: never-a-best-response $\Leftrightarrow$ dominated

Without loss of generality consider player 1. Here's a preview sketch of our demonstration that, for two-player games, a strategy is never a best response if and only if it is dominated: We have already observed that if a strategy is dominated it is never a best response. Now we show that if a strategy is never a best response then it is dominated. The mathematical space we will consider is player 1's expected utility plotted against player 2's mixed strategy. The milestones of the demonstration are: 1 The graph of player 1's payoff to any player-1 mixed strategy as a function of player 2's mixed strategy will be a hyperplane in this space.²² 2 The hyperplane corresponding to a dominated strategy lies everywhere strictly below the hyperplane corresponding to a dominating strategy. 3 The upper envelope

²⁰ Of course, we know that this set of inequalities could be equivalently written with the substitutions $s_{-j} \rightarrow \sigma_{-j}$ and $S_{-j} \rightarrow \Sigma_{-j}$. I write these this way to emphasize the relationship to the previous set of inequalities.

²¹ Pearce [1984: Appendix B, Lemma 3] provides an elegant and short proof of this result by constructing a new, zero-sum game from the original, not-necessarily-zero-sum game and exploiting the existence of a Nash equilibrium. This proof however does not deliver the same graphical intuition as the approach taken here. Myerson [1991] proves a theorem, which implies this result in two-player games, using a linear-programming formulation. Fudenberg and Tirole [1991] sketch the proof of a related theorem which has the more explicitly separating-hyperplane flavor of our approach.

²² Let $f: X \rightarrow Y$. The *graph* of f is $\{(x, f(x)): x \in X\} \subset X \times Y$. Therefore the graph of a function lies in the Cartesian product of its domain and target set.

of the hyperplanes corresponding to player 1's pure strategies describes player 1's expected utility when she chooses her strategy optimally given player 2's strategy. The hyperplane corresponding to any strategy which is never a best response must lie everywhere strictly below this upper envelope. 4 Any hyperplane which is tangent to this upper envelope corresponds to some mixed strategy for player 1. 5 If a strategy is never a best response, there exists a hyperplane which is tangent to the upper envelope and which lies everywhere strictly above the hyperplane for the never-a-best-response strategy. Therefore there exists a strategy which dominates the never-a-best-response strategy, and therefore the never-a-best-response strategy is dominated. (Figure 11 illustrates this theorem in a case where player 2 has only two pure strategies. Therefore his mixed-strategy space is the one-dimensional unit simplex, which we parameterize by $q \in [0, 1]$.)

1□ The graph of player 1's payoff to any mixed strategy will be a hyperplane in this space.

We have previously seen that the payoff to a player from a mixed-strategy profile is a linear function of the mixing probabilities of any one player:²³ For any $i, k \in I$,

$$u_i(\sigma) = \sum_{s_k \in S_k} \sigma_k(s_k) \left[\sum_{s_{-k} \in S_{-k}} \left(\prod_{j \in I \setminus \{k\}} \sigma_j(s_j) \right) u_i(s) \right]. \quad (20)$$

In particular, player 1's payoff to the mixed-strategy profile (σ_1, σ_2) is the linear (and therefore convex and concave) function of the $\{\sigma_2(s_2)\}_{s_2 \in S_2}$

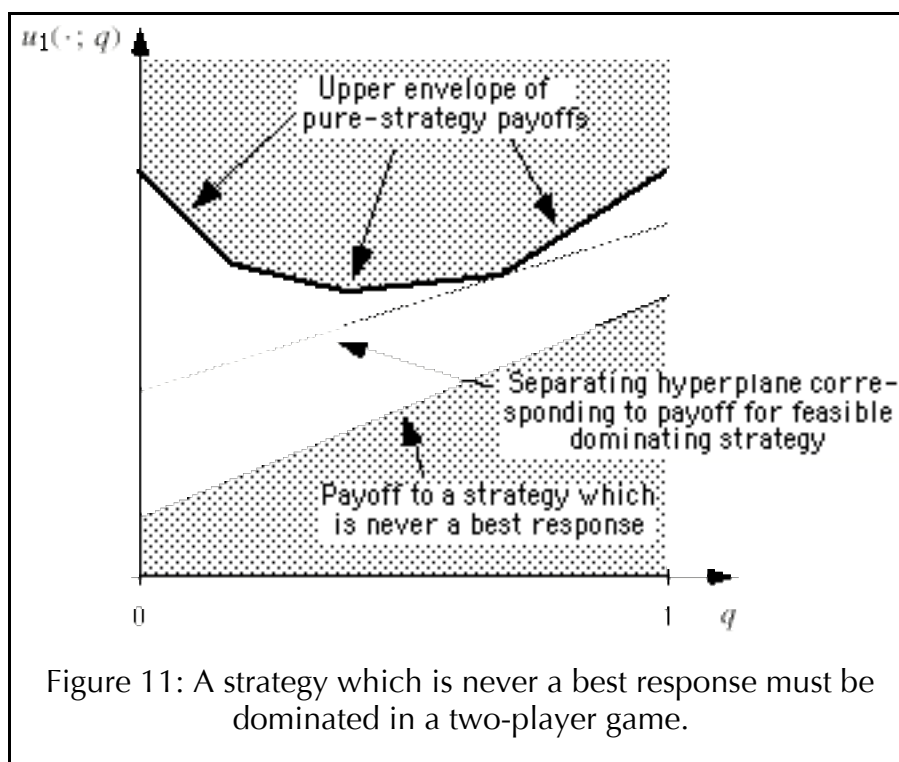
$$u_1((\sigma_1, \sigma_2)) = \sum_{s_2 \in S_2} c(s_2; \sigma_1) \sigma_2(s_2), \quad (21)$$

where the coefficients $c(s_2; \sigma_1)$ are given by $c: S_2 \times \Sigma_1$ defined by

$$c(s_2; \sigma_1) = \prod_{s_1 \in S_1} \sigma_1(s_1) u_1((s_1, s_2)). \quad (22)$$

Therefore the graph of $u_1(\sigma_1, \cdot): \Sigma_2 \rightarrow \mathbb{R}$ (i.e. for fixed $\sigma_1 \in \Sigma_1$) is a hyperplane in the space $\Sigma_2 \times \mathbb{R}$, which we denote by $\mathcal{H}(\sigma_1) = \{(\sigma_2, u((\sigma_1, \sigma_2))): \sigma_2 \in \Sigma_2\}$.

²³ See the "Strategic-Form Games" handout.



2□ The hyperplane corresponding to a dominated strategy lies everywhere strictly below the hyperplane corresponding to a dominating strategy.

If σ_1' dominates σ_1 , then, for all $\sigma_2 \in \Sigma_2$, $u_1((\sigma_1', \sigma_2)) > u_1((\sigma_1, \sigma_2))$ and therefore the hyperplane corresponding to σ_1' lies everywhere above σ_1 ; i.e. $\mathcal{H}(\sigma_1')$ lies everywhere above $\mathcal{H}(\sigma_1)$.

3□ The upper envelope of the hyperplanes corresponding to player 1's pure strategies describes player 1's expected utility when she chooses her strategy optimally given player 2's strategy. The hyperplane corresponding to any strategy which is never a best response must lie everywhere strictly below this upper envelope.

Let $f_1, \dots, f_n$ be functions from some common domain X into the reals, i.e. $f_i: X \rightarrow \mathbb{R}$. Then the *upper envelope* of these functions is itself a function $\bar{f}: X \rightarrow \mathbb{R}$ defined by: $\bar{f}(x) \equiv \max\{f_1(x), \dots, f_n(x)\}$. Consider the finite set of functions $\{u_1(s_1, \cdot)\}_{s_1 \in S_1}$, where each $u_1(s_1, \cdot): \Sigma_2 \rightarrow \mathbb{R}$. (These are player 1's expected-payoff functions to her pure strategies.) Let $\bar{u}_1$ be the upper envelope of the $\{u_1(s_1, \cdot)\}_{s_1 \in S_1}$, i.e. $\bar{u}_1: \Sigma_2 \rightarrow \mathbb{R}$ and $\forall \sigma_2 \in \Sigma_2$, $\bar{u}_1(\sigma_2) = \max\{u_1(s_1, \sigma_2): s_1 \in S_1\}$. For a given $\sigma_2 \in \Sigma_2$, if player 1 chooses her strategy optimally she receives the expected payoff $\max_{s_1 \in S_1} u_1(s_1, \sigma_2) = \bar{u}_1(\sigma_2)$. Therefore the upper envelope function $\bar{u}$ describes player 1's expected payoff as a function of player 2's strategy when player 1 chooses a best response to player 2's strategy.

Denote the graph of this upper envelope function by $\mathcal{U}$; i.e. $\mathcal{U} = \{(\sigma_2, \bar{u}_1(\sigma_2)): \sigma_2 \in \Sigma_2\}$. This upper-envelope surface is composed of a finite number of linear segments (i.e. pieces of hyperplanes).²⁴ Each

²⁴ Consider any pure strategy $s_1 \in S_1$ for player 1. Let $F(s_1)$ be the set of player-2 mixed strategies for which s_1 is a best response for player

of these faces corresponds to some player-1 pure strategy s_1 in the sense that for all σ_2 directly below this face (i.e. in the projection of this face upon Σ_2) s_1 is a best response by player 1 to player 2's choice of σ_2 . The intersections of two or more faces are linear manifolds (i.e. intersections of hyperplanes). Such a linear manifold might be a single point, a line, or a set of higher dimension. Consider the projection of such a linear manifold onto Σ_2 . For each σ_2 belonging to this projection, any of the pure strategies corresponding to the intersecting faces is a best response by player 1 to player 2's choice of σ_2 .

Note that, for every player-1 strategy $\sigma_1 \in \Sigma_1$, the hyperplane corresponding to σ_1 , viz. $\mathcal{H}(\sigma_1)$, must lie everywhere weakly below the upper envelope $\mathcal{U}$. I.e. $\forall \sigma_1 \in \Sigma_1, \forall \sigma_2 \in \Sigma_2, u_1((\sigma_1, \sigma_2)) \leq \bar{u}_1(\sigma_2)$. If, to the contrary, for some $\sigma_2 \in \Sigma_2$, it were true that $\mathcal{H}(\sigma_1)$ lay strictly above $\mathcal{U}$, then we would have the contradiction that $u_1((\sigma_1, \sigma_2)) > \bar{u}_1(\sigma_2) = \max \{u_1((s_1, \sigma_2)): s_1 \in S_1\} = \max \{u_1((\sigma_1, \sigma_2)): \sigma_1 \in \Sigma_1\}$.²⁵

If $\sigma_1 \in \Sigma_1$ were a best response by player 1 to some beliefs $\sigma_2 \in \Sigma_2$, then $u_1((\sigma_1, \sigma_2)) = \bar{u}_1(\sigma_2)$.²⁶ Therefore, if σ_1 were never a best response, then $\forall \sigma_2 \in \Sigma_2, u_1((\sigma_1, \sigma_2)) < \bar{u}_1(\sigma_2)$, and therefore the hyperplane $\mathcal{H}(\sigma_1)$ corresponding to this never-a-best-response strategy must lie everywhere strictly below the upper envelope $\mathcal{U}$.

4□ Any hyperplane which is tangent to this upper envelope corresponds to some mixed strategy for player 1.

Consider a hyperplane which is tangent to the upper envelope $\mathcal{U}$. If this hyperplane is coincident with the upper envelope along an entire face of the upper envelope then it corresponds to (i.e. is the graph of) the expected payoff to player 1, as a function of player 2's mixed strategy, from the pure strategy which corresponds to that face. If this hyperplane is tangent at a linear manifold which is the intersection of two faces, the tangent hyperplane corresponds to the expected payoff to player 1 when she plays some mixture over the two pure strategies corresponding to the intersecting faces. More generally, if the hyperplane is tangent to the upper envelope at some point $(\sigma_2, \bar{u}_1(\sigma_2))$, then this tangent hyperplane corresponds to the payoff to player 1 from some mixed strategy σ_1 which puts weight only on those pure strategies which correspond to the faces which intersect at the point of tangency. I.e. $\exists \sigma_1 \in \Sigma_1$ such that $\text{supp } \sigma_1 \subset \text{BR}_1(\sigma_2)$ and such that the tangent hyperplane is the graph of $u_1(\sigma_1, \cdot)$ over Σ_2 .²⁷

5□ If a strategy is never a best response, there exists a hyperplane which is tangent to the upper envelope and which lies everywhere strictly above the hyperplane for the never-a-best-response strategy.

1; i.e. $F(s_1) = \{\sigma_2 \in \Sigma_2: s_1 \in \text{BR}_1(\sigma_2)\}$. You can show that for all $s_1 \in S_1$, $F(s_1)$ is a convex set. Therefore $\{F(s_1)\}_{s_1 \in S_1}$ is a finite partition of Σ_2 , because the cells of the collection are pairwise disjoint and exhaust Σ_2 . Therefore, for any $s_1 \in S_1$, the graph of the upper envelope $\mathcal{U}$ over $F(s_1)$ is a hyperplane.

²⁵ Make sure you understand why $\max \{u_1(s_1, \sigma_2): s_1 \in S_1\} = \max \{u_1(\sigma_1, \sigma_2): \sigma_1 \in \Sigma_1\}$!

²⁶ Recall that a mixed strategy $\sigma_1 \in \Sigma_1$ is a best response by player 1 to the mixed strategy $\sigma_2 \in \Sigma_2$ if and only if $\text{supp } \sigma_1 \subset \text{BR}_1(\sigma_2)$; i.e. it puts positive weight only upon pure-strategy best responses.

²⁷ I'm not offering an accessible proof for this claim at the present time. See me this time next year to see whether I've worked out an elegant argument!

Therefore there exists a strategy which dominates the never-a-best-response strategy, and therefore the never-a-best-response strategy is dominated.

Now I pull out the separating hyperplane theorem.²⁸ First note that the graph above the upper envelope is a convex set.²⁹ Consider the graph of the payoff to player 1 from a strategy which is never a best response. It is a hyperplane and therefore concave. The set of points below this hyperplane is also a convex set.³⁰ These two convex sets are disjoint; therefore there exists a hyperplane which is tangent to the higher set and strictly separates the lower set. Because this separating hyperplane is tangent to the upper envelope $\mathcal{U}$, it corresponds to a mixed strategy for player 1. Because the separating hyperplane strictly separates the lower convex set, it lies everywhere strictly above the hyperplane corresponding to the never-a-best-response strategy. Therefore there exists a mixed strategy for player 1 which dominates the never-a-best-response strategy. 😊

Three or more players: never-a-best-response $\nRightarrow$ dominated

We have seen that in two-player games any strategy which is never a best response must also be dominated. Now we show that this equivalence between never being a best response and being dominated does not extend generally to games with more than two players. We show this by exhibiting a three-player game in which player 3 will have a strategy which is never a best response to any pair of mixed strategies by the two opponents yet this strategy will not be dominated by any other strategy of player 3's.

Consider the game in Figure 12.³¹ There are three players; player 1 chooses a row (U or D); player 2 chooses a column (l or r); and player 3 chooses a matrix (A , B , C , or D). We will only be concerned with the payoffs to player 3, so we only include her payoffs in the matrices of Figure 12. Each of players 1 and 2 have two pure strategies, therefore a mixed strategy for each of them can be written in terms of a single parameter. We indicate the probability that player 1 chooses U by p and the probability that player 2 chooses l by q .

²⁸ There are several versions of the separation theorem; see Takayama [1985: 39–45] or Debreu [1959]. The theorem I would want would say: “Let A be a closed convex set and B be a convex set, where A and B are disjoint. Then there exists a hyperplane which supports A (i.e. includes a point on the boundary of A) and strictly separates B (i.e. so that B is contained within the interior of one of the closed half planes determined by the hyperplane).” I haven’t found a reference for this exact theorem; typically you find a supporting hyperplane theorem which addresses the issue of a point on the boundary of A and you can find separating theorems which don’t guarantee that the hyperplane intersects the boundary of one of the sets. Nevertheless I assert the theorem in quotes with confidence.

²⁹ The set of points above a convex function is convex. Each of the pure-strategy bounding lines is a convex function. The points above the upper envelope is the intersection of the points above each bounding line; therefore the points above the upper envelope are the intersection of convex sets and, hence, are a convex set.

³⁰ The set of points below a concave function is convex. (A hyperplane is both convex and concave.)

³¹ This game is given in problem 2.7 of Fudenberg and Tirole [1991].

		$[q] \quad [1-q]$		$[q] \quad [1-q]$		$[q] \quad [1-q]$		$[q] \quad [1-q]$	
		l	r	l	r	l	r	l	r
$[p]$	U	9	0	0	9	0	0	6	0
$[1-p]$	D	0	0	9	0	0	9	0	6
		A		B		C		D	

Figure 12: Player 3's payoffs in a three-player game as a function of Row's and Column's mixed strategies p and q , respectively. Player 3 chooses the matrix.

We first compute player 3's expected payoff to each of her four pure strategies as functions of the mixed strategies of her two opponents.

$$u_3(A; p, q) = 9pq, \quad (23a)$$

$$u_3(B; p, q) = 9[p(1-q) + (1-p)q] = 9(p+q-2pq), \quad (23b)$$

$$u_3(C; p, q) = 9(1-p)(1-q), \quad (23c)$$

$$u_3(D; p, q) = 6[pq + (1-p)(1-q)] = 6(1+2pq-p-q). \quad (23d)$$

Player 3's expected payoff to each of her four pure strategies is plotted as a function of p and q in Figure 13.

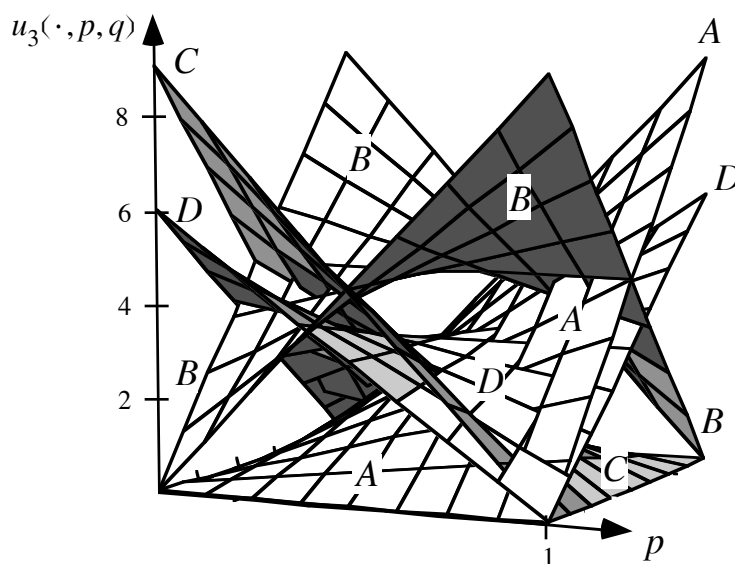


Figure 13: Player 3's pure-strategy payoffs as functions of player 1 and 2's deleted mixed-strategy profile (p, q) in the game from Figure 12.

We notice immediately that, unlike a two-player game, the graphs of the selected player's payoffs when plotted against the deleted strategy profile of her opponents are not hyperplanes.³² Although we previously observed that a player's payoff is linear in a specified other player's mixing probabilities, her payoff is not linear in the mixing probabilities of more than one players.

I want to show that D is not dominated by any of the pure strategies A , B , or C or in fact by any mixture of A , B , and C . Yet we will see that D is not a best response to any deleted mixed-strategy profile $\sigma_{-3} = (p, q)$ of the opponents.

It is clear from Figure 13 or from equations (23) that D is not dominated by any of the other three pure strategies. When $p = q = 1$, D gives a higher payoff than B or C . When $p = q = 0$, D gives a higher payoff than A (or B , but we've already shown that B doesn't dominate D).

Now we want to show that there is no mixture of A , B , and C which dominates D .³³ Call our candidate mixed strategy $\sigma_3 = r \circ A \oplus (1 - r - s) \circ B \oplus s \circ C$, where $r, s \geq 0$ and $r + s \leq 1$. In order that σ_3 dominate D , it must yield a higher payoff against all deleted strategy profiles $(p, q) \in [0, 1]^2$. In particular this must be the case when $p = q = 1$. Against this deleted mixed-strategy profile player 3's payoff to the mixed strategy σ_3 is $9r$, because B and C yield a zero payoff there, and D pays off 6. In order that $9r > 6$, we must choose $r > 2/3$. Similarly, when $p = q = 0$, C provides the only nonzero payoff to σ_3 . Therefore, in order that the payoff from σ_3 exceed the payoff of 6 from D , the weight on C must be such that $s > 2/3$. Therefore the two requirements, in order that σ_3 dominate D , viz. $r, s > 2/3$ are incompatible with the requirement that $r + s \leq 1$. Therefore there does not exist any strategy which dominates D .

You can show algebraically that D is never a best response. However, for our purposes it will suffice to clearly show graphically that the payoff from D lies everywhere below the upper envelope of the pure-strategy payoffs. (See Figure 14.) This example shows that the equivalence between never being a best response and being dominated does not extend to more than two-player games.

Summary and preview of coming attractions

We began our preparation for the study of nonequilibrium solution concepts by introducing the notion of strategic dominance. We defined what it means for one pure strategy for a player to dominate another of her pure strategies. We showed that to evaluate questions of strategic dominance for a player we can with total generality assume that her opponents are playing pure strategies. We generalized the notion of strategic dominance so as to define what it means for a mixed strategy to dominate another mixed strategy. This generalization is useful because it is possible for a strategy to be dominated by a mixed strategy even when it is undominated by pure strategies.

A rational player would never play a dominated strategy. Therefore we can rule out as observed outcomes any which involves any player playing a dominated strategy. (When we later make a stronger

³² In a two-player game, a deleted strategy profile is simply a strategy, because there is only one opponent.

³³ Thanks to Sam Dinkin for supplying this part of the demonstration!

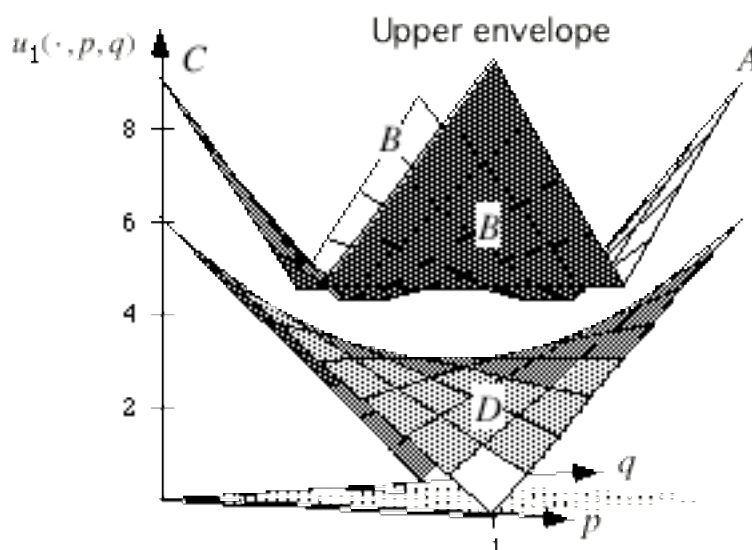


Figure 14: The payoff to D lies everywhere below the upper envelope of player 3's pure-strategy payoffs.

assumption that the rationality of all players is *common knowledge*, we will apply a stronger technique called the *iterated elimination of dominated strategies* to further refine the set of possible observed outcomes.)

Although we established that a rational player would never play a dominated strategy, it need not be the case that any undominated strategy could be plausibly chosen by a rational player. I.e. a dominance analysis need not fully exhaust the implications of all players being rational. We then introduced what it means for a strategy to be *never a best response*. We argued that a strategy cannot be plausibly chosen by a rational player if and only if it is never a best response.

We saw that in two-player games a strategy is never a best response if and only if it is dominated. For two-player games, then, a dominance analysis does fully exploit the assumption that all players are rational. (And the iterated elimination of dominated strategies will fully exhaust the implications of common knowledge of rationality.)

However, for games with three or more players, it is possible that an undominated strategy will yet never be a best response. Therefore we can sometimes rule out as a plausible choice a strategy even when it is undominated. For more-than-two-player games, then, a dominance argument will not fully exploit the assumption that all players are rational. (And therefore the iterated elimination of dominated strategies will not fully exhaust the implications of common knowledge of rationality. We will instead iteratively eliminate strategies which are never best responses. This process will yield the nonequilibrium solution concept of *rationalizability*.)

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Jim Ratliff's Game Theory Course:

2.2 Iterated Dominance & Rationalizability

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Abstract



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We can often make sharper predictions about the possible outcomes of a game if we are willing to make stronger assumptions. Up until now we have assumed that the players are rational but we haven't even assumed that each knows that the others are rational. Beyond that we could further assume that each player knows that the other players know that the others are all rational. We could continue adding additional layers of such assumptions *ad nauseam*. We summarize the entire infinite hierarchy of such assumptions by saying that the rationality of the players is common knowledge.

Rationality constrains players to choose best responses to their beliefs but does not restrict those beliefs. Common knowledge of rationality imposes a consistency requirement upon players' beliefs about others' actions.

By assuming that the players' rationality is common knowledge, we can justify an iterative process of outcome rejection--the iterated elimination of strictly dominated strategies--which can often sharpen our predictions. Outcomes which do not survive this process of elimination cannot plausibly be played when the rationality of the players is common knowledge.

A similar, and weakly stronger, process--the iterated elimination of strategies which are never best responses--leads to the solution concept of rationalizability. The surviving outcomes of this process constitute the set of rationalizable outcomes. Each such outcome is a plausible result and these are the only plausible results when the players' rationality is common knowledge. In two-player games the set of rationalizable outcomes is exactly the set of outcomes which survive the iterated elimination of strictly dominated strategies. In three-or-more-player games, the set of rationalizable outcomes can be strictly smaller than the set of outcomes which survives the iterated elimination of strictly dominated strategies.

In a rationalizable outcome players' beliefs about the same question can differ--and hence some are incorrect; and a player can find--after the others' choices are revealed--that she would have preferred to have made a different choice.



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Nonequilibrium Solution Concepts: Iterated Dominance and Rationalizability

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Introduction

We began our preparation for the study of nonequilibrium solution concepts by introducing the notion of strategic dominance. We defined what it means for a strategy of one player to be dominated by another of her strategies. Because a rational player would never play a dominated strategy, we can sometimes use a dominance analysis to rule out some outcomes as possibilities when the game is played by rational players. In some games, e.g. the prisoners' dilemma, a dominance analysis leads to a unique prediction of the outcome when players are rational; we say that these games are dominance solvable. In other games, e.g. matching pennies, a dominance analysis results in no refinement of the set of possible outcomes. Other games lie between these two extremes: dominance analysis rejects some outcomes as impossible when the game is played by rational players but still leaves a multiplicity of outcomes.

Closely related to the concept of a strategy being dominated for a player is the idea that this strategy is "never a best response" for that player: No matter what beliefs she has about the actions of her opponents, she could not rationally choose to play that strategy. If a strategy is dominated, it can never be a best response. However, it is not obvious that the implication holds in the reverse direction. I.e. it's not obvious that a strategy which is never a best response is also a dominated strategy. Therefore the set of strategies which are never best responses is weakly larger than the set of dominated strategies. An analysis based on whether strategies are possibly best responses *does* exhaust the implications of all players being rational: A strategy cannot be plausibly chosen by a rational player if and only if it is never a best response.

We have seen that in two-player games a strategy is never a best response if and only if it is dominated. For two-player games, then, a dominance analysis fully exploits the assumption that all players are rational. However, for games with three or more players, it is possible that an undominated strategy will yet never be a best response. Therefore we can sometimes rule out as a plausible choice a strategy even when it is undominated. For more-than-two-player games, then, a dominance argument need not fully exploit the assumption that all players are rational.

We can often make sharper predictions about the possible outcomes of a game if we are willing to make stronger assumptions. Up until now we have assumed that the players are rational but we haven't even assumed that each knows that the others are rational. Beyond that we could further assume that each player knows that the other players know that the others are all rational. We could continue adding additional layers of such assumptions *ad nauseam*. Fortunately we can summarize the entire infinite hierarchy of such assumptions by simply saying that the rationality of the players is *common knowledge*. Rationality constrains players to choose best responses to their beliefs but does not restrict those beliefs. Common knowledge of rationality imposes a consistency requirement upon players' beliefs about others' actions.

By assuming that the players' rationality is common knowledge, we can justify an iterative process of outcome rejection—the *iterated elimination of strictly dominated strategies*—which can often sharpen our predictions. Outcomes which do not survive this process of elimination cannot plausibly be played when the rationality of the players is common knowledge. A similar, and weakly stronger, process—the *iterated elimination of strategies which are never best responses*—leads to the solution concept of rationalizability.¹ The surviving outcomes of this process constitute the set of *rationalizable* outcomes. Each such outcome is a plausible result—and these are the only plausible results—when the players' rationality is common knowledge. In two-player games the set of rationalizable outcomes is exactly the set of outcomes which survive the iterated elimination of strictly dominated strategies. In three-or-more-player games, the set of rationalizable outcomes can be strictly smaller than the set of outcomes which survives the iterated elimination of strictly dominated strategies. In a rationalizable outcome players' beliefs about the same question can differ—and hence some are incorrect; and a player can find—after the others' choices are revealed—that she would have preferred to have made a different choice.

Recapitulation

Let's briefly review the standard paradigm and notation. We have a finite set I of n players, $I = \{1, \dots, n\}$. The finite pure-strategy space for player i is S_i ; her mixed-strategy space is $\Sigma_i \equiv \Delta^{\#S_i-1}$; typical elements are $s_i \in S_i$ and $\sigma_i \in \Sigma_i$. When player i chooses the mixed strategy σ_i , the probability with which she plays the pure strategy $s_i \in S_i$ is $\sigma_i(s_i)$. When we omit the subscript on a set defined for each player, we mean the Cartesian product of all the player sets: e.g., $S \equiv \prod_{i \in I} S_i$. A subscript “ $-i$ ” means “ $\prod_{j \neq i} S_j$ ”. An element $s_{-i} \in S_{-i}$ is called a deleted pure-strategy profile. Player i 's von Neumann-Morgenstern utility function is $u_i: S \rightarrow \mathbb{R}$. (We often abuse notation and consider this utility function to

¹ “Weakly stronger” seems a little oxymoronic!

take a mixed-strategy profile as its argument instead of a pure-strategy profile. In such a case it represents the player's expected utility when the players randomize independently according to their component mixed strategies in the mixed-strategy profile.)

We began our preparation for the study of nonequilibrium solution concepts by introducing the notion of strategic dominance. Let $\sigma_i, \sigma_i' \in \Sigma_i$ be two mixed strategies for player i . We say that σ_i' strictly dominates σ_i if σ_i' gives player i a strictly higher expected utility than does σ_i for every possible deleted pure-strategy profile s_{-i} which her opponents could play, i.e. if²

$$\forall s_{-i} \in S_{-i}, \square u_i(\sigma_i', s_{-i}) > u_i(\sigma_i, s_{-i}). \quad (1)$$

Iterated strict dominance

We saw above that in some games, e.g. the Prisoners' Dilemma, each player has a dominant strategy and we could therefore make a very precise prediction about the outcome of the game. To achieve this conclusion we only needed to assume that each player was rational and knew her own payoffs. We also saw an example, viz. matching pennies, where dominance arguments got us nowhere—no player had any dominated strategies. There are games which lie between these two extremes in the degree to which and ease with which dominance arguments can refine the set of possible outcomes.

The technique we'll discuss now is called the *iterated elimination of strictly dominated strategies*.³ In order to employ it we will need to make stronger informational assumptions than we have up until now. For example, we won't merely assume that each player is rational. We might need to assume as well, in a two-player game for example, that player 1 *knows* that player 2 is rational; and player 2 knows that player 1 knows that player 2 is rational, etc. In some games application of the iterated elimination of strictly dominated strategies can require that these hierarchies of beliefs about beliefs be quite deep.

Consider a two-player game between Row and Column, whose pure-strategy spaces are S_R and S_C , respectively. Prior to a dominance analysis of a game, we know only that the outcome will be one of the strategy profiles from the space of strategy profiles $S = S_R \times S_C$. We reasoned above that a rational player would never play a dominated strategy. If Row has a dominated strategy, say $\tilde{s}_R$, but Column does not, then Row, being rational, would never play this strategy. We could therefore confidently predict that the outcome of the game must be drawn from the smaller space of strategy profiles

$$S' = (S_R \setminus \{\tilde{s}_R\}) \times S_C. \quad (8)$$

Here is the interesting point and the key to the utility of the iterative process we're developing: Although Column had no dominated strategy in the original game, he may well have a dominated

² We showed that satisfaction of this condition was equivalent to satisfaction of the same condition but with the substitution of $\sigma_{-i} \rightarrow s_{-i}$ and $\Sigma_{-i} \rightarrow S_{-i}$. I.e. without loss of generality, in order to assess questions of dominance for player i , we can restrict attention to deleted pure-strategy profiles by her opponents.

³ See Fudenberg and Tirole [1991].

strategy $\tilde{s}_C$ in the new, smaller game S' .⁴ If so, and if we make sufficient assumptions, we can rule out as possible outcomes all which involve such newly dominated strategies from Column's strategy space; this again results in a smaller space of strategy profiles. And in this smaller game additional row strategies may now be revealed to be dominated. In some cases this process can continue until a unique strategy for each player survives this elimination process. In this case we say—as we did when each player had a dominant strategy in the original game—that the game is *dominance solvable*.

Common knowledge of rationality

I just said that we had to make assumptions to justify the deletion of Column's dominated strategy $\tilde{s}_C$. What assumptions are necessary for this step? First, Column must be rational. Additionally, in order for Column to see that $\tilde{s}_C$ is dominated for him, he must see that Row will never play $\tilde{s}_R$. Row will never play $\tilde{s}_R$ if she is rational; therefore we must assume that Column *knows* that Row is rational. With these additional assumptions we can confidently predict that any outcome of the game must be drawn from

$$S'' = (S_R \setminus \{\tilde{s}_R\}) \times (S_C \setminus \{\tilde{s}_C\}). \quad (9)$$

Let's not get too tedious, but let's carry this out one more level. It may be the case that in the game defined by the strategy-profile space S'' there is now a strategy of Row's which is newly dominated, call it $\hat{s}_R$. However, we can't rule out that Row will play $\hat{s}_R$ unless we can assure that Row knows that the possible outcomes are indeed limited to S'' , i.e. that Column will not choose $\tilde{s}_C$. Column won't choose $\tilde{s}_C$ if he is rational and knows that Row is rational. Therefore we must assume that Row knows that Column is rational and knows that Column knows that Row is rational.

In any finite game this chain of assumptions can only be usefully carried out to a finite depth. To ensure that we can make such assumptions to an arbitrary depth we often make a convenient assumption: that it is *common knowledge* that all players are rational.⁵

What does it mean for something to be common knowledge?⁶ Let $\mathcal{P}$ be a proposition, e.g. that “player 1 is rational.” If $\mathcal{P}$ is common knowledge, then

Everyone knows $\mathcal{P}$;
 Everyone knows that (Everyone knows $\mathcal{P}$);
 Everyone knows that [Everyone knows that (Everyone knows $\mathcal{P}$)];
 Etc.

⁴ The new game is defined by the strategy spaces $S'_R = S_R \setminus \{\tilde{s}_R\}$ and $S'_C = S_C$ and by the utility functions $u'_i: S' \rightarrow \mathbb{R}, i \in \{R, C\}$, where each u'_i is the restriction of $u_i: S \rightarrow \mathbb{R}$ to the smaller domain S' , viz. $\forall s \in S', u'_i(s) = u_i(s)$.

⁵ When we predicted that Row would not play $\tilde{s}_R$ we implicitly assumed that Row knew her own payoffs but not that Row knew Column's payoffs. When we predicted that Column would never play $\tilde{s}_C$, we implicitly assumed that Column knew that Row knew Row's payoffs *and* that Column knew Row's payoffs. So if we wanted to be perfectly explicit about our assumptions about the players' knowledge of their payoffs, there's another whole hierarchy of beliefs to merge into the hierarchy of beliefs concerning rationality. We can summarize this hierarchy by assuming that “the players' payoffs are common knowledge.”

⁶ Aumann [1976] gives the definition for two players; Myerson [1991] and Pearce [1984] provide it for many players.

In other words, if $\mathcal{P}$ is common knowledge, then every statement of the form

$$(\text{Everyone knows that})^k \text{ everyone knows } \mathcal{P}, \quad (10)$$

is true for all $k \in \{0, 1, 2, \dots\}$.

Prima facie, it might seem unreasonable to expect real-life players to be able to keep track of such unbounded hierarchies of beliefs about beliefs. Actually, common knowledge can be achieved rather simply. Suppose you and I are both present when an event E (assumed so salient that it cannot be ignored) occurs. Suppose further that then we both immediately make eye contact with one another. It is then common knowledge (between the two of us) that E occurred.⁷

Example: Solving a game by the iterated elimination of strictly dominated strategies.

Consider the game in Figure 5(a). Note that Column has no strategies which are dominated by pure strategies. Note further that none of Column's pure strategies can be dominated by a mixture of the other two. However, for Row, Up dominates Middle, and Down is undominated. So a rational Row would never play Middle. Therefore if

$$\text{Row is rational}, \quad (11)$$

then we can strike Middle from Row's strategy space and collapse the bimatrix, resulting in the game in Figure 5(b).

	<i>l</i>	<i>c</i>	<i>r</i>		<i>l</i>	<i>c</i>	<i>r</i>
<i>U</i>	4,11	3,6	5,12	<i>U</i>	4,11	3,6	5,12
<i>M</i>	3,4	2,8	4,6	<i>D</i>	3,10	4,6	3,8
<i>D</i>	3,10	4,6	3,8				
(a)				(b)			
	<i>l</i>	<i>r</i>			<i>l</i>	<i>r</i>	
<i>U</i>	4,11	5,12	<i>U</i>	4,11	5,12	<i>U</i>	5,12
<i>D</i>	3,10	3,8					
(c)			(d)		(e)		

Figure 5: Solving a game by the iterated elimination of strictly dominated strategies.

If Column knows that the possible outcomes must belong to the game in Figure 5(b)—which requires that he know (11), then he would see that center is dominated by both left and right. (Left and right are both undominated.) Therefore if in addition to (11) both

$$\text{Column is rational}, \quad (12)$$

⁷ For example, let E be the event that two particular people are in the same room at a particular time.

Column knows Row is rational, (13)

we can strike center, which results in the game of Figure 5(c).

If Row sees that the possible outcomes must be drawn from the game in Figure 5(c), which requires that she know (12) and (13), then she would see that Up dominates Down. Therefore if also

Row knows Column is rational, (14)

Row knows Column knows Row is rational, (15)

we could strike Down, which results in the game of Figure 5(d).

If Column knows that Figure 5(d) is the relevant game, which requires that he also knows (14) and (15), then Column would recognize that right dominates left. Hence, if also

Column knows Row knows Column is rational, (16)

Column knows Row knows Column knows Row is rational, (17)

the only strategies which survive the iterated elimination of strictly dominated strategies are Up for Row and right for Column. This strategy profile is shown in Figure 5(e). So we see that the *necessary* assumptions to solve this game were (13) $\rightarrow$ (17). (We always assume that the players are rational. What is new here are the assumptions about the players' higher-order beliefs about rationality.) All of these assumptions about beliefs are implied by the sufficient assumption that it is common knowledge that both players are rational.

Example: Iterated strict dominance can require mixed-strategy domination.

I wanted to make the logical reasoning in the previous example as transparent as possible, so I chose the game such that elimination required only domination by pure strategies. However, a more general mixed-strategy analysis can be necessary. Consider the game in Figure 6.

	<i>l</i>	<i>r</i>
<i>U</i>	6,4	0,2
<i>M</i>	0,3	6,1
<i>D</i>	2,1	2,4

Figure 6: A mixture of Up and Middle dominates Down; then left dominates right.

There are no pure-strategy dominance relationships in the original game. However, the mixed strategy $\frac{1}{2} \circ U \oplus \frac{1}{2} \circ M$ dominates Down. After deleting Down, left dominates right for Column. After deleting right, Up dominates Middle. Therefore the only possible outcome under common knowledge of rationality is (U, l) .

Iterated strict dominance: formal definition

In order to support rigorous proofs of later claims regarding the surviving outcomes of the iterated elimination of strictly dominated strategies we'll now define this solution concept more formally. The process is iterative; we can think of it as an algorithm (which is depicted in flowchart form in Figure 7).⁸

Very loosely to begin with.... We start with the original game S . We delete all the dominated strategies for each player, which results in a smaller game S^1 . More generally, we consider the game defined by some set of not-yet-rejected outcomes S^{t-1} . By rejecting any player's strategy which is dominated, we reach the weakly smaller game S^t .⁹ (Therefore S_i^t is the set of player i 's pure strategies which are undominated in the game S^{t-1} .) When we reach a point where the resulting game cannot be further shrunk by the elimination of strictly dominated strategies, then our process has concluded. We say that this set of outcomes, denoted S^∞ , has survived the iterated elimination of strictly dominated strategies.

More formally now.... We use $t \in \mathbb{Z}_+ \equiv \{0, 1, 2, \dots\}$ as a counter. We denote by $S_i^t \subset S_i$ the set of player- i pure strategies which are unrejected after t rounds of this iterative procedure. Therefore the period-0 game is just the original game: $\forall i \in I, S_i^0 = S_i$. We denote by $\Sigma_i^t \subset \Sigma_i$ the set of player- i mixed strategies which are mixtures only over the pure strategies which are unrejected after t rounds; i.e. $\Sigma_i^t = \{\sigma_i \in \Sigma_i: \text{supp } \sigma_i \subset S_i^t\}$.^{10,11} In particular, $\Sigma_i^0 = \Sigma_i$.

Now consider the game resulting after $t-1$ rounds of elimination, viz. $S^{t-1} = \prod_{i \in I} S_i^{t-1}$. Consider some player $i \in I$ and consider each of her not-yet-rejected pure strategies $s_i \in S_i^{t-1}$. The strategy s_i is dominated in the game S^{t-1} if there exists a mixed strategy σ_i over these not-yet-rejected pure strategies S_i^{t-1} , i.e. $\sigma_i \in \Sigma_i^{t-1}$, such that σ_i dominates s_i in the game S^{t-1} (i.e. for all deleted strategy profiles $s_{-i} \in S_{-i}^{t-1}$ by her opponents). Therefore the set S_i^t of strategies which are undominated in S^{t-1} is

$$S_i^t = \{s_i \in S_i^{t-1}: \nexists \sigma_i \in \Sigma_i^{t-1}, \forall \hat{s}_{-i} \in S_{-i}^{t-1}, u_i(\sigma_i, \hat{s}_{-i}) > u_i(s_i, \hat{s}_{-i})\}. \quad (18)$$

⁸ Just a note of explanation about the “ $i \in I$ loop” $\rightarrow$ “end i loop” constructions in the flowchart for those who haven't programmed computers: When an “ $i \in I$ loop” box is initially encountered from above, i is set to the first element of I and control passes downward. When the “end i loop” box is encountered, one of two things can happen. 1 If there are still elements $i \in I$ which haven't been processed in this loop, control returns to the preceding “ $i \in I$ loop” and the i counter is incremented to the next element of I . 2 If this was the last element i in the set I , then control drops out of the loop to the next lower box.

⁹ The previous round of eliminations may have revealed newly dominated strategies for one or more players.

¹⁰ Note that we are not at this point requiring that the mixed strategies in Σ_i^t be undominated.

¹¹ Note that Σ_i^t is not a mixed strategy for player i in her smaller strategy space S_i^t . Rather it is a mixed strategy over her original strategy space S_i but whose support includes only pure strategies in the smaller strategy space S_i^t .

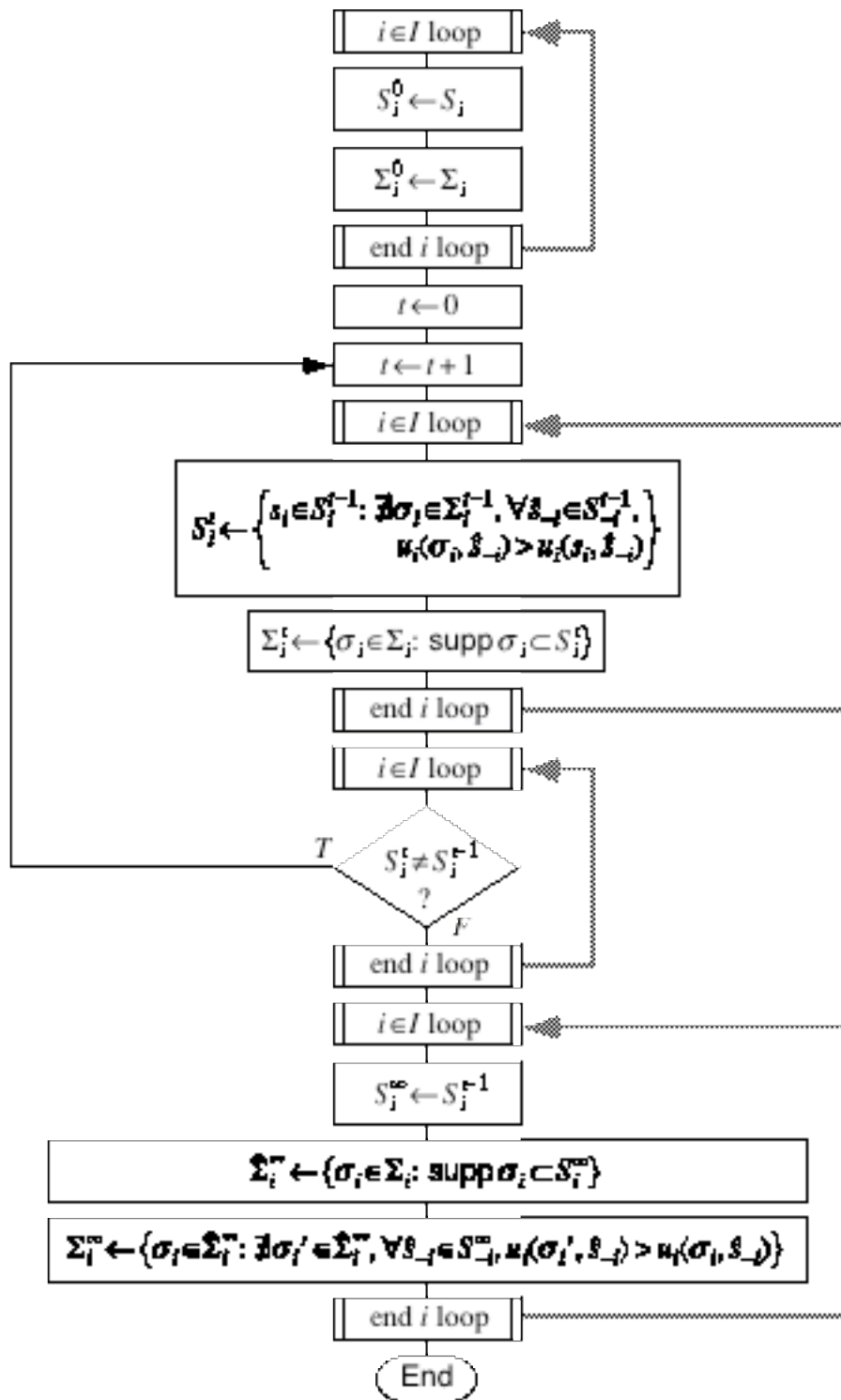


Figure 7: A flowchart definition of the iterated elimination of strictly dominated strategies.

We then form the set of mixed strategies

$$\Sigma_i^t = \{\sigma_i \in \Sigma_i: \text{supp } \sigma_i \subset S_i^t\} \quad (19)$$

which put weight only upon the still-admissible pure strategies of S_i^t ; these constitute our arsenal with which to try to dominate strategies in the new game S^t . We note that the sequence of player- i strategy spaces $\{S_i^t\}_{t \in \mathbb{Z}_+}$ is nested:

$$\forall t \in \mathbb{Z}_+, S_i^{t+1} \subset S_i^t. \quad (20)$$

We must eventually get to a stage in the iterations such that the surviving strategy set for each player is unchanged from the previous round, i.e. $\exists \tau \in \mathbb{N} \equiv \{1, 2, \dots\}, \forall i \in I, S_i^\tau = S_i^{\tau-1}$. (Otherwise at least one pure strategy from at least one player must be eliminated in each round. The number of pure strategies for each player is finite. Therefore it's impossible to remove strategies forever.) If we were to continue the algorithm beyond this stage, we'd find that the strategy sets remained unchanged, i.e. for all $i \in I, S_i^\tau = S_i^{\tau+1} = S_i^{\tau+2} = \dots$. [This is clear from examination of (18). Consider any $i \in I$ and $s_i \in S_i^\tau$. This s_i was undominated in $S^{\tau-1}$ (which is why it survived to belong to S_i^τ). But $S^\tau = S^{\tau-1}$, so s_i must be undominated in S^τ as well, and therefore deserves membership in $S_i^{\tau+1}$. And so on for $S_i^{\tau+k}, k \in \mathbb{N}$.]

Once we reach a stage in this iterative process at which the strategy sets are no longer shrinking, say at period τ as in the above paragraph, we have exhausted the implications of an iterative dominance analysis for behavior in the game. We set, for each $i \in I$,

$$S_i^\infty = S_i^\tau, \quad (21)$$

where S_i^∞ is the set of player- i pure strategies which survive the iterated elimination of strictly dominated strategies.¹² When the game is played under the conditions of common knowledge of rationality, every player i would choose some strategy $s_i \in S_i^\infty$. The Cartesian product of these player-strategy sets, viz. $S^\infty = \prod_{i \in I} S_i^\infty$ is the set of strategy profiles which survive the iterated elimination of strictly dominated strategies. When the game is played under the conditions of common knowledge of rationality, any pure-strategy profile must be within the set S^∞ , i.e. $s \in S^\infty$.

To determine the set of mixed strategies Σ_i^∞ for player i which are compatible with an iterated dominance analysis based on the common knowledge of rationality, we first find all the mixed strategies $\hat{\Sigma}_i^\infty$ which put positive weight only upon the unrejected pure strategies S_i^∞ ,

$$\hat{\Sigma}_i^\infty = \{\sigma_i \in \Sigma_i: \text{supp } \sigma_i \subset S_i^\infty\}, \quad (22)$$

However, we have seen before that a mixed strategy which spreads all its weight only among undominated pure strategies can still be itself dominated.¹³ Therefore we must filter this set of mixed

¹² We could alternatively write that S_i^∞ is the intersection of the infinite sequence of player- i strategy spaces, viz. $S_i^\infty = \bigcap_{t \in \mathbb{Z}_+} S_i^t$.

¹³ See the example on pages 10–13 in the “Dominance” handout of September 7, 1993.

strategies to remove any which are dominated by other mixed strategies in that set. This results in the set Σ_i^∞ of mixed strategies for player i which are not rejected by an iterated dominance argument,

$$\Sigma_i^\infty = \{\sigma_i \in \hat{\Sigma}_i^\infty : \nexists \sigma_i' \in \hat{\Sigma}_i^\infty, \forall s_{-i} \in S_{-i}^\infty, u_i(\sigma_i', s_{-i}) > u_i(\sigma_i, s_{-i})\}. \quad (23)$$

Example: A dominance-solvable two-player game

Let's perform iterated elimination of strictly dominant strategies on the game in Figure 8. U dominates M . With M removed, l dominates r . With r removed, D dominates U . Therefore the iterated-dominance outcome is (D, l) .

	l	r
U	10,5	10,4
M	8,4	8,5
D	12,5	6,4

Figure 8: A dominance-solvable two-player game.

In terms of our formalism, the nested pure-strategy sets for each player at each stage of the iterative process are

$$S_R^0 = \{U, M, D\}, \quad S_C^0 = \{l, r\},$$

$$S_R^1 = \{U, D\}, \quad S_C^1 = \{l, r\}$$

$$S_R^2 = \{U, D\}, \quad S_C^2 = \{l\},$$

$$S_R^3 = \{D\}, \quad S_C^3 = \{l\},$$

$$S_R^\infty = \{D\}, \quad S_C^\infty = \{l\}.$$

The sets of mixed strategies which survive the iterated elimination of strictly dominated strategies, viz. Σ_R^∞ and Σ_C^∞ , are trivial in this example. Each player has only one surviving pure strategy and therefore the only mixture over that strategy is the corresponding degenerate mixed strategy, viz.

$$\Sigma_R^\infty = \{(0 \circ U \oplus 0 \circ M \oplus 1 \circ D)\}, \quad \Sigma_C^\infty = \{(1 \circ l \oplus 0 \circ r)\}.$$

Iterated strict dominance: looking more closely

Let's revisit the formal specification of the iterated elimination of strictly dominated strategies to hunt for and hopefully resolve possibly problematic issues. Consider again the example from Figure 8. Why did we reject r for Column? Because we had previously rejected M for Row on account of being dominated by U . But we later rejected U itself for Row. Since we used U to reject M , but later decided that Row would never actually play U , perhaps we should call into question our rejection of M and therefore of r for Column. How can we think about issue more clearly?

Perhaps one justification for reconsidering our rejection of M would be: if, starting with our final game $S_R^\infty \times S_C^\infty = \{D\} \times \{I\}$, we reintroduced M into Row's strategy set and found that M was no longer dominated, then we might conclude that our rejection of M was mistaken—that we were misled by the later-to-be-rejected-itself strategy U . However, if we perform that experiment we have the game of Figure 9, and we see that M is still dominated, albeit by D now rather than U .

	I
M	8,4
D	12,5

Figure 9: The dominance-solved game of Figure 8 with M restored to Row's strategy space.

So we see that, in this example at least, a strategy which was rejected as dominated in an early stage of the iterative process was still dominated when reinjected into its player's strategy space at the end of the iterative process, even though the originally dominating strategy which justified its rejection had been later itself rejected as dominated. We will now see that this is a general result: any strategy which is dominated at some stage of the iterative process would still be dominated at any later stage if reintroduced into its player's strategy space.

Let's first establish a relevant fact. For some player $i \in I$ consider the game $S_i \times S_{-i}^\infty$. We'll now see that S_i^∞ is the set of player i 's pure strategies which are undominated in the game $S_i \times S_{-i}^\infty$. To prove this equality we need to show that 1 S_i^∞ contains all the strategies which are undominated in $S_i \times S_{-i}^\infty$ and 2 any strategy which is dominated in $S_i \times S_{-i}^\infty$ does not belong to S_i^∞ .

The set S_i^∞ contains all of player i 's strategies which are never rejected during the iterative elimination process. Therefore to show that S_i^∞ contains all the undominated strategies in $S_i \times S_{-i}^\infty$ we need to show that all of the strategies which are rejected during the iterative elimination process are in fact dominated in $S_i \times S_{-i}^\infty$. So we consider a rejected strategy $s_i \in S_i \setminus S_i^\infty$ and let $t \in \mathbb{Z}_+$ be the stage of the process in which it is rejected. I.e.

$$\exists \sigma_i \in \Sigma_i^t, \forall \hat{s}_{-i} \in S_{-i}^t, u_i(\sigma_i, \hat{s}_{-i}) > u_i(s_i, \hat{s}_{-i}). \quad (24)$$

If s_i is dominated in $S_i \times S_{-i}^\infty$, then

$$\exists \sigma_i' \in \Sigma_i, \forall \hat{s}_{-i} \in S_{-i}^\infty, u_i(\sigma_i', \hat{s}_{-i}) > u_i(s_i, \hat{s}_{-i}). \quad (25)$$

We need to show that satisfaction of (24) implies satisfaction of (25). But we note from (19) and (20) that $\Sigma_i^t \subset \Sigma_i$ and $S_{-i}^\infty \subset S_{-i}^t$. Therefore in attempting to satisfy (25), compared to satisfying (24), we can choose a mixed strategy from a larger set Σ_i and dominance need hold only in fewer cases (viz. for all $\hat{s}_{-i} \in S_{-i}^\infty$ rather than for all $\hat{s}_{-i} \in S_{-i}^t$). Therefore if (24) is satisfied, we can set $\sigma_i' \leftarrow \sigma_i$ in order to satisfy (25). This shows 1.

Before we show 2, I need to state (and you can establish!) a useful result:

Theorem

Let $S_i' \subset S_i$ be the set of player i 's undominated pure strategies. Let $\underline{S}_i$ be the set of player i 's dominated pure strategies. Let $\sigma_i \in \Sigma_i$ be a mixed strategy which puts positive weight on at least one dominated pure strategy; i.e. $\text{supp } \sigma_i \cap \underline{S}_i \neq \emptyset$.

Then there exists a mixed strategy $\sigma_i' \in \Sigma_i$ such that 1 σ_i' dominates σ_i and 2 σ_i' puts no weight on dominated pure strategies; i.e. $\text{supp } \sigma_i' \subset S_i'$.

Proof

It's up to you. This is extra-credit challenge #1.

Now we want to show 2, i.e. that any strategy which is dominated in $S_i \times S_{-i}^\infty$ does not belong to S_i^∞ . Therefore we need to show that any strategy which is dominated in $S_i \times S_{-i}^\infty$ is rejected at some stage $t \in \mathbb{Z}_+$ in the iterative elimination process. Let $\tau \in \mathbb{Z}_+$ be the “final” period of the process (i.e. $S_i^\infty = S_i^\tau$). I'll show that if s_i survives until the last stage, viz. until stage τ , then it will be rejected there. So assume $s_i \in S_i^\tau$. In order to be rejected in period τ we must have

$$\exists \sigma_i \in \Sigma_i^\tau, \forall \hat{s}_{-i} \in S_{-i}^\tau, u_i(\sigma_i, \hat{s}_{-i}) > u_i(s_i, \hat{s}_{-i}). \quad (26)$$

Because s_i is dominated in $S_i \times S_{-i}^\infty$; i.e. there exists a $\sigma_i' \in \Sigma_i$ such that (25) holds. If it's already the case that $\sigma_i' \in \Sigma_i^\tau$, then we're done. [Just take $\sigma_i \leftarrow \sigma_i'$ and note (21) to see that (26) is satisfied.] So consider the remaining case where $\sigma_i' \notin \Sigma_i^\tau$. Therefore σ_i' must put positive weight on some pure strategy which is dominated in $S_i \times S_{-i}^\infty$ (because S_i^∞ contains all the undominated strategies in this game and therefore $\Sigma_i^\infty = \Sigma_i^\tau$ contains all the mixtures over the undominated strategies). Therefore we can use the above theorem to assert the existence of another mixed strategy $\sigma_i \in \Sigma_i^\tau$ which dominates σ_i' and therefore satisfies (26).

So we have shown that, for each player $i \in I$, the set of her pure strategies which survive the iterated elimination of strictly dominated strategies, viz. S_i^∞ , is exactly the set of pure strategies which are undominated when her opponents can only choose deleted strategy profiles in S_{-i}^∞ . Now that this is established you can construct an argument to show that, if any combination of rejected strategies $s_i \in S_i \setminus S_i^\infty$ is reinjected into player i 's strategy space at the last stage of the iterated elimination process, it would still be rejected at that stage.

Rationalizability

We concluded earlier that a rational player would never play a dominated strategy (because she could do strictly better in all cases by choosing instead a strategy which dominated that strategy). By assuming that the rationality of all players is common knowledge, we developed the technique of the iterated elimination of strictly dominated strategies in order to determine a set of strategies for each player which survive this elimination procedure. We concluded that no player would ever choose a strategy outside of her surviving set. The implication here was one way: common knowledge of rationality implies that the game's outcome must survive the iterated elimination of strictly dominated strategies. We did *not* show that every surviving strategy could be reasonably chosen by a rational player.

An alternative expression of the common knowledge of rationality is closely related to our domination discussion: A rational player would never choose a strategy which is not a best response to some beliefs about opponents' choices. Further, a player's beliefs about the choices of others are constrained in that the other players must be believed to also be playing strategies which are best responses to their own beliefs. The *rationalizable* outcomes are those which survive the iterated elimination of strategies which are never best responses.¹⁴ However, this argument *does* go both ways. Bernheim [1984] argues not only that a rational player must choose her strategies from their rationalizable set, but also that every strategy in this set can be consistently justified as a rational choice.

A rational player must choose a best response to her beliefs about the actions of the other players. For example, in Figure 10, *E* is dominated by *D*; therefore there is no possible belief which Row could hold about Column's strategy to which *E* would be a best response.¹⁵ Therefore a rational Row would never play *E*.

Although a player's rationality constrains her action to be a best response to her beliefs, it does not restrict what her beliefs about others' actions can reasonably be. (After all, an irrational opponent might do anything.) A rational Row player could play *B* because it is a best response to *z*. But a rational Column would never play *z*, because it is dominated by *y*. If Row knew that Column is rational, Row would realize that Column would never choose *z*, and Row would further deduce that *B* is not a best response to anything Column might rationally do. (*B* is dominated by *C* when Column's strategy space is reduced to $\{w, x, y\}$.) Row's knowledge of Column's rationality restricts Row's beliefs to put zero weight on Column choosing *z*.

	<i>w</i>	<i>x</i>	<i>y</i>	<i>z</i>
<i>A</i>	7,5	-8,4	0,4	99,3
<i>B</i>	5,0	4,1	15,9	100,8
<i>C</i>	6,0	5,8	20,4	10,2
<i>D</i>	2,6	7,-10	3,9	10,8
<i>E</i>	1,6	2,-10	1,7	8,6

Figure 10

In summary, if Row and Column are both rational and if Row knows that Column is rational, then we can restrict our attention to the smaller game $\{A, C, D\} \times \{w, x, y\}$ shown in Figure 11. You can convince yourself that no further elimination of strictly dominated strategies is possible; hence (because this is a two-player game) all of the outcomes in this smaller game are rationalizable.¹⁶

¹⁴ See Fudenberg and Tirole [1991].

¹⁵ I have indicated in boldface type the row payoffs which are maximal in each column and the column payoffs which are maximal in each row.

¹⁶ Recall that in two-player games the rationalizable outcomes are exactly those which survive the iterated elimination of strictly dominated strategies. In three-or-more-player games the set of rationalizable outcomes is a weakly smaller set than those survivors of iterated elimination of strictly dominated strategies.

	w	x	y
A	7,5	-8,4	0,4
C	6,0	5,8	20,4
D	2,6	7,-10	3,9

Figure 11: The rationalizable subset of the game from Figure 10.

Rationalizability as a consistent system of beliefs

We defined the rationalizable outcomes as those which survived the iterated elimination of strategies which were never best responses. In order to focus explicitly on the constraints which common knowledge of rationality imposes upon players' beliefs I will now discuss rationalizability from a different perspective. Consider the strategy profile (C, x) in the game of Figure 11. I will show that this profile is rationalizable by showing that C and x are rationalizable strategies for Row and Column, respectively. To do this I will show that there exists a *consistent system of beliefs* for the players which justifies their choices—i.e. which shows that these choices do not conflict with the common knowledge of rationality assumption. (See Bernheim [1984].)

Let's establish some notation so that we can tractably talk about beliefs about beliefs about beliefs about.... Let $\mathcal{R}$ and $\mathcal{C}$ stand for the Row and Column players, respectively. If Row chooses A , we write $\mathcal{R}(A)$, and similarly for other choices by either player. If Column believes that Row will choose A , we write $\mathcal{C}\mathcal{R}(A)$. If Column believes that Row believes that Column will choose y , we write $\mathcal{C}\mathcal{R}\mathcal{C}(y)$, etc.

A rational Row player would play C , i.e. $\mathcal{R}(C)$, if she believes that Column will play y ; i.e. if $\mathcal{R}\mathcal{C}(y)$. Is this belief by Row reasonable? Column would play y if he thought that Row would play D ; therefore we assume $\mathcal{R}\mathcal{C}\mathcal{R}(D)$. Would Row do this? Row would play D if she thought that Column would play x ; therefore we assume $\mathcal{R}\mathcal{C}\mathcal{R}\mathcal{C}(x)$. Finally, Column would choose x if he believes that Row would choose C , viz. $\mathcal{R}\mathcal{C}\mathcal{R}\mathcal{C}\mathcal{R}(C)$. But C is justified by the sequence of beliefs we have just described. We summarize the beliefs of Row's which justify her playing C :¹⁷

$$\mathcal{R}(C) \quad \mathcal{R} \text{ plays } C, \quad (27a)$$

$$\mathcal{R}\mathcal{C}(y) \quad \mathcal{R} \text{ believes } \mathcal{C} \text{ will play } y, \quad (27b)$$

$$\mathcal{R}\mathcal{C}\mathcal{R}(D) \quad \mathcal{R} \text{ believes } \mathcal{C} \text{ believes } \mathcal{R} \text{ will play } D, \quad (27c)$$

$$\mathcal{R}\mathcal{C}\mathcal{R}\mathcal{C}(x) \quad \mathcal{R} \text{ believes } \mathcal{C} \text{ believes } \mathcal{R} \text{ believes } \mathcal{C} \text{ will play } x, \quad (27d)$$

$$\mathcal{R}\mathcal{C}\mathcal{R}\mathcal{C}\mathcal{R}(C) \quad \mathcal{R} \text{ believes } \mathcal{C} \text{ believes } \mathcal{R} \text{ believes } \mathcal{C} \text{ believes } \mathcal{R} \text{ will play } C. \quad (27e)$$

This hierarchy of beliefs establishes a cycle of strategies (C, y, D, x, C) , all of which are thus shown to be rationalizable. To see how the above argument is sufficient to show that x is rationalizable, let's look explicitly at Column's beliefs which would make his choice of x rationalizable. Column would play x , $\mathcal{C}(x)$, if he believed that Row would play C , i.e. if $\mathcal{C}\mathcal{R}(C)$. Row would play C if she believed that Column would play y ; so we assume the belief for Column that $\mathcal{C}\mathcal{R}\mathcal{C}(y)$. Column would play y if Row

¹⁷ $\mathcal{R}(C)$ is not a belief; but the $\mathcal{R}\mathcal{C}\dots(\cdot)$ expressions below it are beliefs.

were playing D ; therefore we assume $\mathcal{CR}\mathcal{CR}(D)$. Row would play D if Column were choosing x . So we have the same cycle (but shifted) of rationalizable strategies $\{x, C, y, D, x\}$. We summarize Column's beliefs:

$$\mathcal{C}(x) \tag{28a}$$

$$\mathcal{C}\mathcal{R}(C) \tag{28b}$$

$$\mathcal{C}\mathcal{R}\mathcal{C}(y) \tag{28c}$$

$$\mathcal{C}\mathcal{R}\mathcal{C}\mathcal{R}(D) \tag{28d}$$

$$\mathcal{C}\mathcal{R}\mathcal{C}\mathcal{R}\mathcal{C}(x) \tag{28e}$$

Comparing notes

We have shown that the strategy profile (C, x) is rationalizable and have explicitly determined the beliefs each player must hold in order to justify her strategy choice. Let's examine the properties of this outcome from two different perspectives. First, we ask what the players would find if they got together to compare notes about their belief systems. Second, we'll ask what their assessments of the wisdom of their strategy choices would be after the game was played; we'll ask them: "would you do it over again the same way?"

If Row and Column, planning to play C and x , respectively, met to discuss truthfully and candidly their perspectives on the game being played, Row would find that her beliefs were all wrong but Column would have correctly anticipated everything.

We see from (28b) that Column correctly conjectured (27a)—that Row would play C . Further, we see from (28c) $\rightarrow$ (28e) that Column also correctly intuited Row's higher-order beliefs (27b) $\rightarrow$ (27d). For example, Column correctly believed that Row believed that Column believed that Row would choose D .

Row was not so clairvoyant (or lucky). From (27b) we see that Row believed that Column would play y . In fact, Column played x instead. Not surprisingly, Row also got Column's higher-order beliefs all wrong. For example, from (27c), we see that Row thought that Column thought that Row would play D ; from (28b) we see that Column actually thought that Row would play C .

After the game is played, and the actual strategy choices revealed, would the players be happy with the choices they had made? Would they do it over again the same way?¹⁸ Column correctly forecast that Row would choose C , and Column played his best response to C , viz., x . So Column would be satisfied with the choice he made.

¹⁸ There is a subtlety here. In general the players choose mixed strategies. I am *not* asking whether, after the game is played, a player is happy with the pure-strategy realization of her mixed strategy given the pure-strategy realizations of others' mixed strategies. I am asking whether she would be happy with her choice of mixed strategy given her opponents' mixed-strategy choices. To "do it over again the same way" means to once again choose the same lottery but have the roulette wheel spun again. The way I frame this scenario, it is implicit that the players would observe their opponents' mixed strategies, not just the pure-strategy realizations. This issue will arise again.

Row on the other hand incorrectly forecast Column's choice, thinking that Column would choose y instead of x . Row's best response to y , viz. C , was not a best response to the actually played x . Therefore Row could have received a higher payoff, 7 instead of 5, by playing the best response to x , viz. D , instead.

When beliefs are held in common

Let's contrast the properties we identified above for the outcome (C, x) with those of another rationalizable outcome: (A, w) . The demonstration that A and w are rationalizable requires a much less circuitous hierarchy of beliefs. Row would choose A if she thought that Column would choose w , i.e. $\mathcal{RC}(w)$. Column would choose w if he thought that Row would choose A ; i.e. we assume that $\mathcal{RC}(\mathcal{R}(A))$. This yields our desired cycle of strategy profiles (A, w, A) :

$$\mathcal{R}(A) \quad (29a)$$

$$\mathcal{RC}(w) \quad (29b)$$

$$\mathcal{RC}(\mathcal{R}(A)) \quad (29c)$$

Similarly, Column's choice of w can be justified by the beliefs:

$$\mathcal{C}(w) \quad (30a)$$

$$\mathcal{C}(\mathcal{R}(A)) \quad (30b)$$

$$\mathcal{C}(\mathcal{R}(\mathcal{C}(w))) \quad (30c)$$

We can easily summarize the belief system generated by $(29a) \rightarrow (30c)$ by saying: It is common knowledge that Row will play A and Column will play w .

Let's ask the same questions for this rationalizable strategy profile that we asked for (C, x) . First, what misconceptions would Row and Column discover if they met at Gentle Ben's to trade their deepest secrets? Absolutely none. Both players not only correctly anticipated the other's action but also correctly divined the other's beliefs. For example, from (30c) and (29b) we see that Column correctly believed that Row believed that Column would indeed choose w .

After Row and Column played the game and observed each other's strategy choice, would either want to change her strategy? No. Because each player correctly anticipated her opponent's action, each played a best response to the opponent's actual choice. Neither player could improve upon her payoff given the choice of her opponent.

So we see a striking qualitative difference between the profile (C, x) and (A, w) . The key lies in the following observation: Consider (C, x) . Although x is Column's best response to C , C is not Row's best response to x . However, consider (A, w) . A is Row's best response to w , and w is Column's best response to A . This is evident immediately from observing the boldface type in Figure 11. The payoff vector $(5, 8)$ corresponding to the strategy profile (C, x) had only one element in boldface, indicating that only one player was picking a best response to the other's choice. On the other hand, both of the elements in the

payoff vector (**7, 5**) corresponding to (A, w) appear in boldface, indicating that both players were picking a best response to the other's choice.

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Jim Ratliff's Game Theory Course:

3.1 Nash Equilibrium

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Abstract



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When all players are rational and their rationality is common knowledge, the resulting outcome will be rationalizable. If a player does not correctly forecast the play of an opponent, her beliefs to which she played a best response were incorrect. Her actual action may not be a best response to the actual play of her opponents. Therefore in a rationalizable outcome one or more players can have *ex post* regret about their choices.

When rational players correctly forecast the strategies of their opponents they are not merely playing best responses to their *beliefs* about their opponents' play; they are playing best responses to the *actual* play of their opponents. When all players correctly forecast their opponents' strategies, and play best responses to these forecasts, the resulting strategy profile is a *Nash equilibrium*.

Nash equilibrium has been justified as representing a *self-enforcing agreement*. I present an example of Aumann's intended to show that a Nash equilibrium need not be self-enforcing. Nash equilibria can also be vulnerable to multiplayer deviations. We discuss a difficulty of justifying Nash equilibrium as the outcome of a dynamic process: if it is known that the game is repeated, then the repeated game is itself a new, more-complicated game.

A game need not have a pure-strategy Nash equilibrium. However, every strategic-form game does have a (possibly degenerate) mixed-strategy Nash equilibrium. We learn (thanks to John Nash) that a Nash equilibrium exists if and only if there exists a fixed point of a particular best-response correspondence. Kakutani's fixed-point theorem guarantees the existence of this fixed point when the correspondence satisfies certain conditions. We verify that the best-response correspondence satisfies the Kakutani conditions and hence prove the existence of a Nash equilibrium.



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Nash Equilibrium

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When players beliefs are correct

Consider the two-player game of Figure 1. There are no dominance relationships for either player and therefore all pure outcomes survive the iterated elimination of strictly dominated strategies. Because there are only two players, this is also the set of rationalizable pure strategy profiles.¹ Consider the strategy profile (U, r) . We can alternatively establish that this outcome is rationalizable by performing an explicit analysis of the players' beliefs: U is a best response by Row if she believes that Column is choosing l . Column's choice of l would be rationalized by his belief that Row were playing U . Therefore the consistent set of beliefs which rationalizes U is^{2,3}

$$\mathcal{R}(U) \quad \mathcal{R} \text{ plays } U, \quad (1a)$$

$$\mathcal{R} \mathcal{C}(l) \quad \mathcal{R} \text{ believes } \mathcal{C} \text{ will play } l, \quad (1b)$$

$$\mathcal{R} \mathcal{C} \mathcal{R}(U) \quad \mathcal{R} \text{ believes } \mathcal{C} \text{ believes } \mathcal{R} \text{ will play } U. \quad (1c)$$

Similarly, r is rationalizable for Column because it is a best response if he believes that Row will play D , and D is a best response by Row if she believes that Column will choose right. Therefore the consistent set of beliefs which rationalizes r is

$$\mathcal{C}(r) \quad \mathcal{C} \text{ plays } r, \quad (2a)$$

$$\mathcal{C} \mathcal{R}(D) \quad \mathcal{C} \text{ believes } \mathcal{R} \text{ will play } D, \quad (2b)$$

$$\mathcal{C} \mathcal{R} \mathcal{C}(r) \quad \mathcal{C} \text{ believes } \mathcal{R} \text{ believes } \mathcal{C} \text{ will play } r. \quad (2c)$$

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¹ See the handout "Nonequilibrium Solution Concepts: Iterated Dominance and Rationalizability."

² See the handout "Nonequilibrium Solution Concepts: Iterated Dominance and Rationalizability."

³ $\mathcal{R}(U)$ is not a belief; but the $\mathcal{R}\mathcal{C}\dots(\cdot)$ expressions below it are beliefs.

	<i>l</i>	<i>r</i>
<i>U</i>	5,5	4,4
<i>D</i>	4,4	5,5

Figure 1.

After this game is played in this way—viz. Row plays Up and Column plays right—each player will realize *ex post* that her beliefs about her opponent’s play were incorrect and, further, each will regret her own choice in the light of what she learned about her opponent’s strategy. Specifically, from (1b) we see that Row believed that Column would play *l*, but Column instead chose *r*. Had Row known that Column would choose *r*, she would have chosen Down instead. Similarly, from (2b) we see that Column believed that Row would play Down, but Row played Up instead. Had Column known that Row would play Up, he would have preferred to have chosen left. In this (U, r) outcome, then, each player was choosing a best response to her beliefs about the strategy of her opponent, but each player’s beliefs were wrong.

Now consider the strategy profile (U, l) . We have already seen [from (1a) $\rightarrow$ (1c) above] that Up is rationalizable for Row. To see that left is rationalizable for Column we need only exhibit the following consistent set of beliefs:

$$\mathcal{C} \text{ } (l) \quad \mathcal{C} \text{ plays } l, \quad (3a)$$

$$\mathcal{C} \text{ } \mathcal{R} \text{ } (U) \quad \mathcal{C} \text{ believes } \mathcal{R} \text{ will play } U, \quad (3b)$$

$$\mathcal{C} \text{ } \mathcal{R} \text{ } \mathcal{C} \text{ } (U) \quad \mathcal{C} \text{ believes } \mathcal{R} \text{ believes } \mathcal{C} \text{ will play } l. \quad (3c)$$

When the game is played this way—viz. Row plays Up and Column plays left—each player’s prediction of her opponent’s strategy was indeed correct. And since each player was playing a best response to her correct beliefs, neither player regrets her own choice of strategy.

In other words, when rational players correctly forecast the strategies of their opponents they are not merely playing best responses to their *beliefs* about their opponents’ play; they are playing best responses to the *actual* play of their opponents. When all players correctly forecast their opponents’ strategies, and play best responses to these forecasts, the resulting strategy profile is a *Nash equilibrium*.⁴ (See Nash [1951].)

Before defining Nash equilibrium, let’s quickly recap our notation. The player set is $I = \{1, \dots, n\}$. Each player i ’s pure-strategy space is S_i and her mixed-strategy space is Σ_i (the set of probability distributions over S_i). When these symbols lack subscripts, they refer to Cartesian products over the player set. A subscript of “ $-i$ ” indicates the set $I \setminus \{i\}$. Her expected utility from a mixed-strategy profile σ is $u_i(\sigma)$.

⁴ Note in Figure 1 that both players payoffs in the (U, l) box are bolded. This indicates that Row’s payoff is maximal giving Column’s choice and that Column’s choice is maximal given Row’s choice. A pure-strategy profile is a Nash equilibrium if and only if its payoff vector has every element in boldface. Similarly, (D, r) is also a Nash equilibrium of this game.

Definition A *pure-strategy Nash equilibrium* of a strategic-form game is a pure-strategy profile $s^* \in S$ such that “every player is playing a best response to the strategy choices of her opponents.” More formally, we say that s^* is a Nash equilibrium if

$$(\forall i \in I) \square s_i^* \text{ is a best response to } s_{-i}^*, \quad (4a)$$

or, equivalently,

$$(\forall i \in I) \square s_i^* \in \mathbf{BR}_i(s_{-i}^*), \quad (4b)$$

or, more notationally,

$$(\forall i \in I) (\forall s_i \in S_i) \square u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*). \quad (4c)$$

Note well that when a player i judges the optimality of her part of the equilibrium prescription—i.e. decides whether *she* will play her part of the prescription—she does assume that her *opponents* will play their part s_{-i}^* of the prescription. Therefore in (4c) she is asking herself the question: Does there exist a *unilateral* deviation s_i for me such that I would *strictly* gain from such defection given that the opponents held truly to their prescriptions.

A game need not have a pure-strategy Nash equilibrium. Consider the matching pennies game of Figure 2. Each player decides which side of a coin to show. Row prefers that the coins match; Column prefers that they be opposite. We can see from the figure that this game has no pure-strategy equilibrium.⁵ No matter how the players think the game will be played (i.e. what pure-strategy profile will be played), one player will always be distinctly unhappy with her choice and would prefer to change her strategy.

	<i>H</i>	<i>T</i>
<i>H</i>	1, -1	-1, 1
<i>T</i>	-1, 1	1, -1

Figure 2: Matching pennies does not admit a pure-strategy Nash equilibrium.

This nonexistence problem when we restrict ourselves to pure strategies was historically a major motivation for the introduction of mixed strategies into game theory: We will see that the existence of a (possibly degenerate) mixed-strategy Nash equilibrium *is* guaranteed. Here’s the natural generalization to mixed strategies of the previous definition:

Definition

A *Nash equilibrium* of a strategic-form game is a mixed-strategy profile $\sigma^* \in \Sigma$ such that “every player is playing a best response to the strategy choices of her opponents.” More formally, we say that σ^* is a Nash equilibrium if

⁵ In no cell of the matrix are both payoffs bolded.

$$(\forall i \in I) \square \sigma_i^* \text{ is a best response to } \sigma_{-i}^*, \quad (5a)$$

or, equivalently,

$$(\forall i \in I) \square \text{supp } \sigma_i^* \subset \text{BR}_i(\sigma_{-i}^*), \quad (5b)$$

or, more notationally,⁶

$$(\forall i \in I) (\forall s_i \in S_i) \square u_i(\sigma_i^*, \sigma_{-i}^*) \geq u_i(s_i, \sigma_{-i}^*). \quad (5c)$$

Game Theory for \$60: If Nash equilibrium is the answer, what is the question?

Nash equilibria as self-enforcing agreements

Even though rationalizability would seem to be logically prior to the Nash equilibrium concept (owing to fewer assumptions), Nash equilibrium chronologically predates rationalizability in the development of game theory. Nash equilibrium has been and still is widely employed in applications. So strong is the bias toward Nash equilibrium as virtually a definition of rationality that a common theme in the literature holds that a player who does not play her part of the relevant equilibrium must be making a mistake. Only relatively recently have game theorists regularly expressed serious concern over its justification. (See Kreps [1989] and Kreps [1990].)

The most serious attempted justification of Nash equilibrium is its interpretation as a necessary condition for a *self-enforcing agreement*. Consider a scenario where you and I have an opportunity prior to playing the game to communicate and to reach a nonbinding agreement about how we will play. If we do indeed reach such an agreement, there is a strong argument that the agreement we reach should constitute a Nash equilibrium: Because the agreement is nonbinding, we each have the opportunity, regardless of what we agreed to, to take the selfishly best action given our expectations of what the other will do. If the agreement were not a Nash equilibrium, at least one of us would have an incentive to deviate from the agreement (assuming that that person still believed that the other would carry through with her part of the agreement's specification).

⁶ To see why (5b) is a translation of (5a), recall that a mixed strategy σ_i is a best response to a deleted mixed-strategy profile σ_{-i} if and only if it puts positive weight only upon pure strategies which are themselves best responses to σ_{-i} . It might not be obvious why (5c) is a sufficient characterization of what we mean when we say that σ_i^* is a best response to σ_{-i} . It requires that σ_i^* is at least as good as any other pure strategy which i has; however, it doesn't address the possibility that player i might have some even better *mixed* strategy. The key is that player i 's payoff to any mixed strategy is a convex combination of her payoffs to her pure strategies. If her payoff to the mixed strategy σ_i^* is weakly greater than each of her pure strategy payoffs, it weakly exceeds any convex combination of these pure-strategy payoffs.

	<i>l</i>	<i>r</i>
<i>U</i>	9,9	0,8
<i>D</i>	8,0	7,7

Figure 3: A Nash equilibrium need not be self-enforcing.

Robert Aumann [1990] has offered an example to show that a Nash equilibrium need not be self-enforcing. I.e. although being a Nash equilibrium may be necessary for an outcome to be self-enforcing, it is not sufficient. Consider the game of Figure 3. There are two pure-strategy Nash equilibria: (U, l) and (D, r) . Both pure-strategy equilibrium profiles have merit as a prediction of play. (U, l) is Pareto dominant, but (D, r) is safer because each player would guarantee herself at least 7 by conforming with this equilibrium profile, while conforming with (U, l) risks a zero payoff if her opponent does not conform. I.e. unless Row is quite certain that her opponent will choose his part, viz. l , of the Pareto-dominant equilibrium, Down yields Row a higher expected payoff than Up. Specifically, playing Up requires that Row attaches a probability of at least $\frac{1}{8}$ to the event that Column chooses left. A symmetric argument shows that Column must be very certain, in order that he play left, that Row will choose Up.

In fact, Row realizes that Column is reasoning in the same way that she is: that Column will play r unless he is highly certain that Row will play U . This increases Row's scepticism that Column will play l and therefore makes D even more tempting for Row. And of course Column realizes this, so r is even more tempting for him. And Column realizes this, etc. Aumann's point is not that (D, r) must be played by rational players, but rather that it can be plausibly played.

Can preplay communication help the players transcend their lack of confidence and achieve the Pareto-dominant equilibrium? Aumann says no. Consider the case where each player is skeptical enough of the other's intention that in the absence of a preplay agreement they would each choose their part of the (D, r) equilibrium. Regardless of what Column chooses to play, he always prefers that Row choose Up (because $9 > 0$ and $8 > 7$); and Row always prefers that Column choose left. Even if the two players agree prior to the game to play the (U, l) equilibrium, Row might reason this way: If we hadn't had our chat, I would have chosen Down. Now that we have had our chat, do I have reason to be more confident that Column will play left so that I can play Up? No. Regardless of Column's true intentions he would want me to *believe* that he would choose left so that I would play Up. Therefore I have learned nothing from his professed agreement to play left. Column would reason similarly about Row's professed agreement to playing the (U, l) equilibrium. Neither player learns anything from the preplay communication and they each know that the other has learned nothing, etc. Therefore they each remain as skeptical as they were initially and therefore the Pareto-dominated equilibrium (D, r) is played.

This argument does not apply to all games. Consider the game of Figure 4. There are two Nash equilibria which are Pareto unranked; i.e. Row prefers one to the other and Column prefers the other to the one. Each Nash equilibrium Pareto dominates every nonequilibrium outcome. The players would prefer to coordinate on any Nash equilibrium—even the player for whom that equilibrium is not the better—than fail to coordinate on any equilibrium. (This is why this is called a coordination game. If one

player believes that one equilibrium is being played and the other player believes the other equilibrium is being played, then no equilibrium will be played.⁷⁾

	<i>l</i>	<i>r</i>
<i>U</i>	2,1	0,0
<i>D</i>	0,0	1,2

Figure 4: A coordination game, where Nash equilibria are self-enforcing.

Assume that the two players agree prior to the game to play the best-for-Row equilibrium (U, l) . Should Row believe Column's claim that he will play left? In this game it is not the case that Column wants Row to play Up regardless of what Column himself plans to do. If Column were to play right, he would prefer that Row play Down rather than Up. By agreeing to (U, l) Column is signaling that he wants Row to play Up. Unlike the game of Figure 3, however, he is also signaling that he plans to keep his part of the agreement as well. Therefore both of the pure-strategy Nash-equilibrium profiles are self-enforcing in this game.

Nash equilibrium as the result of a dynamic process

Nash equilibrium is also widely applied in games where there is no explicit communication and negotiation phase. Justifications for the application of the Nash concept in these contexts is less well developed but is a subject of current research. Recall our study of the rationalizable outcome (U, r) , which was not a Nash equilibrium in the game of Figure 1. The occurrence of such an outcome can only be explained if at least one of the players harbors misconceptions about how the game will be played. Furthermore, at least one of the players will express regret about her strategy choice after the end of the game. This suggests that Nash equilibrium would be the relevant solution concept if there were some pregame dynamic which assured that players' beliefs were in agreement prior to their strategy selection. It is problematic to construct such examples, however. As Bernheim [1984] argues⁸

Specifically, it is fruitless to argue that repetitions of a game generate convergence to equilibrium. Unlike the stylized dynamics of competitive equilibrium, where the movement of prices forms a structural link between repetitions of an economy, there is nothing structural tying together successive plays of a game. Thus, if players are unaware that the game is being repeated, there is no meaningful dynamic. On the other hand, if they are aware of the repetitions, then the repeated game is itself a new game, entirely distinct from its components. Convergence of component choices may then have nothing whatsoever to do with attainment of equilibrium in the game actually played. Attempts to introduce equilibrating forces simply generate larger composite games, and the nature of strategic choices in these larger games remains inherently one-shot.

⁷ Note that this occurred in the play generated by the system of beliefs (1) and (2) in the game of Figure 1: Row believed the (U, l) was being played and Column believed that (D, r) was being played.

⁸ We will study in detail games played in a dynamic context later in the semester.

There are three reasons I can think of why you should be very serious about learning about Nash equilibrium: 1 Even though the opportunity for pregame communication and negotiation is not universally available, the class of games in which it is a possibility is an important one; 2 Current attempts to satisfactorily justify the application of Nash equilibrium to a wider class of games may ultimately prove successful (in which case the equilibria of these games will be relevant); and 3 Nash equilibrium is widely applied in economics; any serious economist needs to understand the concept and related techniques very well.

Nash equilibria can be vulnerable to multiplayer deviations

As noted above, the definition of Nash equilibrium only requires the absence of any profitable unilateral deviations by any player. A Nash equilibrium is not guaranteed to be invulnerable to deviations by coalitions of players however. Consider the three-player game of Figure 5. Player 1 chooses a row, player 2 chooses a column, and player 3 chooses a matrix. There are two pure-strategy Pareto-ranked Nash equilibria: (U, l, A) and (D, r, B) , where (U, l, A) Pareto dominates (D, r, B) .

	<i>l</i>	<i>r</i>		<i>l</i>	<i>r</i>
<i>U</i>	0, 0, 10	-5, -5, 10	<i>U</i>	-2, -2, 0	-5, -5, 0
<i>D</i>	-5, -5, 0	1, 1, -5	<i>D</i>	-5, -5, 0	-1, -1, 5
Matrix <i>A</i>			Matrix <i>B</i>		

Figure 5: A three-player game (Row, Column, Matrix)

Consider the (U, l, A) equilibrium. No player wants to deviate unilaterally: Given that Column is choosing l and Matrix is choosing A , Row would be worse off to switch to D . Given that Row and Matrix are choosing U and A , respectively, Column would be worse off choosing r . And given that Row and Column are choosing U and l , respectively, Matrix would be worst off choosing B .

However, fix Matrix's choice at A and consider the joint deviation by Row and Column from (U, l) to (D, r) . Both would profit from such a shift in their strategies, yet (U, l, A) is still a Nash equilibrium, because Nash equilibrium is only concerned with the existence of profitable unilateral deviations. A strategy profile is a *strong equilibrium* if no coalition (including the grand coalition, i.e. all the players collectively) can profitably deviate from the prescribed profile. (See Aumann [1959], Aumann [1960].) The definition immediately implies that any strong equilibrium is both Pareto efficient and a Nash equilibrium. A strong equilibrium need not exist.⁹ However, note that (D, r, A) to which the coalition of Row and Column might defect is itself not even a Nash equilibrium. Therefore one could question whether it should be used as the basis for rejecting (U, l, A) . See Bernheim, et al. [1987] for more on *coalition-proof* Nash equilibrium.

⁹ In this game it is clear that a pure-strategy strong equilibrium does not exist. We already showed how (U, l, a) is ruled out. The only other pure-strategy Nash equilibrium, viz. (D, r, B) , is ruled out because it is not Pareto efficient.

Existence of Nash equilibrium

Nash equilibrium $\Leftrightarrow$ fixed point of the best-response correspondence

We will now prove that every game has a Nash equilibrium when we allow mixed strategies.¹⁰ We first show that a strategy profile is a Nash equilibrium if and only if it is a *fixed point* of a best-response correspondence. Then we show that this correspondence must have a fixed point.

Let's briefly recall what a fixed point of a function is. Consider a function $f: X \rightarrow X$ whose domain is identical with its target set. We say that the element $x \in X$ of the domain is a *fixed point* of the function f if $f(x) = x$. In other words, the function f leaves the point x untransformed.

The concept of a fixed point can be generalized to correspondences.¹¹ Let the correspondence $\varphi: X \rightrightarrows X$ have a domain identical to its target set. We cannot usefully stipulate that $x \in X$ is fixed point of φ if $\varphi(x) = x$ because $\varphi(x)$ is a subset of X and x is an element of X and therefore this equality cannot possibly hold. Instead we say that $x \in X$ is a fixed point of the correspondence $\varphi: X \rightrightarrows X$ if $x \in \varphi(x)$. In other words the correspondence's values at a fixed point include the fixed point itself. See Figure 6.

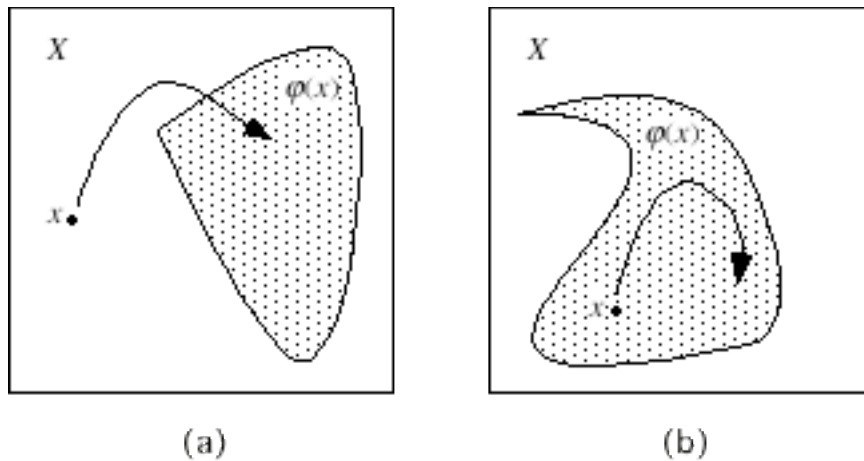


Figure 6: Representations of a correspondence $\varphi: X \rightrightarrows X$ when (a) x is not a fixed point of φ and (b) x is a fixed point of φ .¹²

We have previously defined for each player $i \in I$ a pure-strategy best-response correspondence $BR_i: \Sigma_{-i} \rightrightarrows S_i$, which specifies for every deleted mixed-strategy profile $\sigma_{-i} \in \Sigma_{-i}$ by player i 's opponents a set $BR_i(\sigma_{-i}) \subset S_i$ of player- i pure strategies which are best responses. Then we saw that any player- i mixed strategy which put positive weight upon only these pure-strategy best responses was itself a best-response mixed strategy for player i . To prove the existence of a Nash equilibrium we will find it useful

¹⁰ Fudenberg and Tirole [1991] is a good reference for the proof of the existence of a Nash equilibrium.

¹¹ Recall that a *correspondence* $\varphi: X \rightrightarrows Y$ is a “set-valued function” or, more properly, a mapping which associates to every element $x \in X$ in the domain a *subset* of the target set Y . In other words, $\forall x \in X, \varphi(x) \subset Y$.

¹² The arrow is not meant to indicate that the point $x \in X$ gets mapped to a single point in the shaded region. Rather x is mapped by φ into the entire shaded region. If the arrows are obfuscating, just ignore them.

to work directly with the mixed-strategy best-response correspondences implied by the pure-strategy best-response correspondences BR_i , $i \in I$.

For each player $i \in I$, we define her mixed-strategy best-response correspondence $\psi_i: \Sigma \rightarrow \Sigma_i$ by

$$\psi_i(\sigma) = \{\sigma'_i \in \Sigma_i: \text{supp } \sigma'_i \subset BR_i(\sigma_{-i})\}. \quad (6)$$

In other words given any mixed-strategy profile $\sigma \in \Sigma$ we can extract the deleted mixed-strategy profile $\sigma_{-i} \in \Sigma_{-i}$ from it.¹³ Then we determine player i 's pure-strategy best responses to σ_{-i} and form the set of player- i mixed strategies which put positive weight only upon these pure-strategy best responses. This set of player- i mixed strategies is the value of the player- i mixed-strategy best-response correspondence ψ_i evaluated at σ .

Now we form a new correspondence ψ by forming the Cartesian product of the n personal mixed-strategy best-response correspondences ψ_i . We define for every $\sigma \in \Sigma$,

$$\psi(\sigma) = \prod_{i \in I} \psi_i(\sigma). \quad (7)$$

For each $i \in I$, $\psi_i(\sigma) \subset \Sigma_i$, so for each $\sigma \in \Sigma$, $\psi(\sigma)$ is a subset of the Cartesian product of the individual-player mixed-strategy spaces Σ_i ; i.e. $\psi(\sigma) \subset \Sigma$. Therefore we see that ψ is a correspondence itself from the space of mixed-strategy profiles into the space of mixed-strategy profiles; i.e. $\psi: \Sigma \rightarrow \Sigma$.¹⁴

Consider any Nash equilibrium $\sigma \in \Sigma$. Each player i 's mixed strategy σ_i is a best response to the other players' deleted mixed-strategy profile σ_{-i} . Therefore σ_i satisfies the requirements for inclusion in the set $\psi_i(\sigma)$ as defined in (6); i.e. σ_i belongs to player i 's mixed-strategy best-response correspondence evaluated at the Nash-equilibrium profile σ . Because this inclusion must hold for all players, we have

$$\sigma = (\sigma_1, \dots, \sigma_n) \in \prod_{i \in I} \psi_i(\sigma) = \psi(\sigma), \quad (8)$$

or in other words if σ is a Nash equilibrium we must have

$$\sigma \in \psi(\sigma). \quad (9)$$

I.e. a Nash equilibrium profile is a fixed point of the best-response correspondence ψ . This logic is reversible: any fixed point of the best-response correspondence is a Nash equilibrium profile. Therefore a mixed-strategy profile is a Nash equilibrium if and only if it is a fixed point of the best-response correspondence ψ .

¹³ We *could* define the domain for the correspondence ψ_i to be Σ_{-i} rather than Σ . However, we are free to define it the way we do; the definition ignores the extraneous information $\sigma_i \in \Sigma_i$. You will see the formal advantage to this definition as we proceed.

¹⁴ Creating a correspondence which maps one set into itself was the motivation behind defining the domains of the personal best-response correspondences ψ_i to be Σ rather than Σ_{-i} .

The best-response correspondence has a fixed point

To prove the existence in general of a Nash equilibrium, we will prove the existence of a fixed point of the best-response correspondence using Kakutani's Kakutani [1941] fixed-point theorem:¹⁵

Theorem

Let $K \subset \mathbb{R}^m$ be compact and convex.¹⁶ Let the correspondence $\varphi: K \rightrightarrows K$ be upper hemicontinuous with nonempty convex values.¹⁷ Then φ has a fixed point.

In our application of Kakutani's theorem the space Σ of mixed-strategy profiles will play the role of K and the best-response correspondence $\psi: \Sigma \rightrightarrows \Sigma$ will play the role of φ . We need to verify that Σ and ψ do indeed satisfy the hypotheses of Kakutani's theorem. Then we can claim that the best-response correspondence has a fixed point and therefore every game has a Nash equilibrium.

We need to show that Σ is compact and convex. The space Σ of mixed-strategy profiles is the Cartesian product of the players' mixed-strategy spaces Σ_i , each of which is compact and convex. This will imply that Σ is itself compact and convex. We will prove here that the convexity of the Σ_i implies the convexity of Σ . You can prove as an exercise that the compactness of the Σ_i implies that Σ is compact.

Lemma

Let $A_1, A_2, \dots, A_m$ be convex sets. Let A be their Cartesian product $A \equiv \prod_{k=1}^m A_k$. Then A is a convex set.

Proof

To show that A is convex, we take an arbitrary pair of its members, viz. $a', a'' \in A$, and show that an arbitrary convex combination of this pair, viz. $\hat{a} \equiv \alpha a' + (1 - \alpha)a''$, for $\alpha \in [0, 1]$, also belongs to A . Both a' and a'' are m -tuples of elements which belong to the constituent sets A_i ; i.e. $a' = (a'_1, \dots, a'_m)$ and $a'' = (a''_1, \dots, a''_m)$ where, for each $i \in \{1, \dots, m\}$, $a'_i, a''_i \in A_i$. The convex combination $\hat{a}$ is the m -tuple each of whose i -th elements is defined by $\hat{a}_i \equiv \alpha a'_i + (1 - \alpha)a''_i$. Each of these $\hat{a}_i$ is a convex combination of a'_i and a''_i and hence belongs to A_i because A_i is assumed to be convex. Therefore the original convex combination $\hat{a}$ is an m -tuple each of whose i -th elements belongs to A_i and therefore $\hat{a} \in A$. ☺

Now we need to show that the best-response correspondence ψ is nonempty valued, convex valued, and upper hemicontinuous. We have earlier seen that, for all players $i \in I$ and all deleted mixed-strategy profiles σ_{-i} , the pure-strategy best-response correspondence $\text{BR}_i(\sigma_{-i})$ is nonempty.¹⁸ Therefore there

¹⁵ Good references for fixed-point theorems are Border [1985], Debreu [1959], and Green and Heller [1981].

¹⁶ A subset of a Euclidean space is *compact* if and only if it is closed and bounded.

¹⁷ To say that the correspondence $\varphi: K \rightrightarrows K$ is nonempty valued means that $\forall x \in K, \varphi(x) \neq \emptyset$. To say that φ is convex valued is to say that $\forall x \in K, \varphi(x)$ is a convex set. We will define upper hemicontinuity soon!

¹⁸ See the handout "Strategic-Form Games."

exists a (possibly degenerate) best-response mixed strategy σ_i such that $\text{supp } \sigma_i \subset \text{BR}_i(\sigma_{-i})$; hence, for all $i \in I$ and all $\sigma \in \Sigma$, $\psi_i(\sigma) \neq \emptyset$ and therefore $\psi(\sigma) \neq \emptyset$.

Now we show that the best-response correspondence ψ is convex valued. We do this by first showing that, for each $i \in I$, ψ_i is convex valued, i.e. for all $\sigma \in \Sigma$, $\psi_i(\sigma)$ is a convex set. Then $\psi(\sigma)$ is convex by the above lemma because it is the Cartesian product of the $\psi_i(\sigma)$. Consider two player- i mixed strategies $\sigma_i', \sigma_i'' \in \Sigma_i$ both of which are best responses to the deleted mixed-strategy profile σ_{-i} extracted from some mixed-strategy profile $\sigma \in \Sigma$, i.e. $\sigma_i', \sigma_i'' \in \psi_i(\sigma)$. We need to show that any convex combination of these two mixed strategies is also a best response to σ_{-i} , i.e. for all $\alpha \in [0, 1]$, $\hat{\sigma}_i \equiv [\alpha \sigma_i' + (1 - \alpha) \sigma_i''] \in \psi_i(\sigma)$. Clearly this holds for $\alpha \in \{0, 1\}$, so we focus on $\alpha \in (0, 1)$. Because $\forall \sigma_i', \sigma_i'' \in \psi_i(\sigma_i)$, $\text{supp } \sigma_i' \subset \text{BR}_i(\sigma_{-i})$ and $\text{supp } \sigma_i'' \subset \text{BR}_i(\sigma_{-i})$. For $\alpha \in (0, 1)$,

$$\text{supp } \hat{\sigma}_i = \text{supp } \sigma_i' \cup \text{supp } \sigma_i'' \subset \text{BR}_i(\sigma_{-i}).^{19,20}$$

Therefore $\hat{\sigma}_i \in \psi_i(\sigma)$; therefore $\psi_i(\sigma)$ is a convex set; therefore $\psi(\sigma)$ is a convex set; and therefore ψ is convex valued.

Now we show that the best-response correspondence ψ is upper hemicontinuous.²¹

Definition Let $\varphi: K \rightrightarrows K$ be a correspondence where K is compact. Then φ is *upper hemicontinuous* if φ has a closed graph. I.e. φ is upper hemicontinuous if all convergent sequences in the graph of the correspondence converge to a point in the graph of the correspondence. I.e. if [for every sequence $\{(x^k, y^k)\}$ in $K \times K$ and point $(x^0, y^0) \in K \times K$ such that $(x^k, y^k) \rightarrow (x^0, y^0)$ and such that, for all k , $y^k \in \varphi(x^k)$], then it is the case that $y^0 \in \varphi(x^0)$.

To show that ψ is upper hemicontinuous we assume to the contrary that there exists a convergent sequence of pairs of mixed-strategy profiles $(\sigma^k, \hat{\sigma}^k) \rightarrow (\sigma, \hat{\sigma})$ such that, for all $k \in \mathbb{N} \equiv \{1, 2, \dots\}$, $\hat{\sigma}^k \in \psi(\sigma^k)$ but $\hat{\sigma} \notin \psi(\sigma)$. Along the sequence, because $\hat{\sigma}^k \in \psi(\sigma^k)$, for each player $i \in I$,

$$\hat{\sigma}_i^k \in \psi_i(\sigma^k). \quad (10)$$

At the limit, because $\hat{\sigma}$ is not a best response to σ , some player i must have a better strategy than $\hat{\sigma}_i$ against σ_{-i} , i.e. because $\hat{\sigma} \notin \psi(\sigma)$, $\exists i \in I$, $\exists \sigma_i' \in \Sigma_i$, such that

$$u_i(\sigma_i', \sigma_{-i}) > u_i(\hat{\sigma}_i, \sigma_{-i}). \quad (11)$$

We now exploit the continuity of player i 's utility function.²² Because $\sigma^k \rightarrow \sigma$, it is true that $\sigma_{-i}^k \rightarrow \sigma_{-i}$.

¹⁹ Consider a pure strategy $s_i \in S_i$. Then $s_i \in \text{supp } [\alpha \sigma_i' + (1 - \alpha) \sigma_i'']$ if $\alpha \sigma_i'(s_i) + (1 - \alpha) \sigma_i''(s_i) > 0$. Because $\alpha \in (0, 1)$, this occurs if and only if either $\sigma_i'(s_i) > 0$ or $\sigma_i''(s_i) > 0$.

²⁰ For any sets A , B , and C , $(A \subset C \text{ and } B \subset C)$ implies $(A \cup B) \subset C$.

²¹ Hemicontinuity of correspondences has in the past also been called semicontinuity. Hemicontinuity is the more modern usage and intended to prevent any confusion with semicontinuity of functions. Debreu [1959], Border [1985], and Green and Heller [1981] are good references concerning the continuity of correspondences.

²² The function u_i is a continuous function of continuous functions.

and therefore we can take k sufficiently large to make $u_i(\sigma_i', \sigma_{-i}^k)$ arbitrarily close to left-hand side of (11), viz. $u_i(\sigma_i', \sigma_{-i})$. Because $(\sigma^k, \hat{\sigma}^k) \rightarrow (\sigma, \hat{\sigma})$, we can take k sufficiently large to make $u_i(\hat{\sigma}_i^k, \sigma_{-i}^k)$ arbitrarily close to the right-hand side of (11), viz. $u_i(\hat{\sigma}_i, \sigma_{-i})$. Therefore for all k sufficiently large we have

$$u_i(\sigma_i', \sigma_{-i}^k) > u_i(\hat{\sigma}_i^k, \sigma_{-i}^k).^{23} \quad (12)$$

But this is tantamount to saying that $\hat{\sigma}_i^k$ is not a best response to σ_{-i}^k , and this contradicts (10). Therefore ψ must be upper hemicontinuous.

So we have verified that the space Σ of mixed-strategy profiles and the best-response correspondence ψ satisfy the hypotheses of Kakutani's fixed-point theorem. Therefore the best-response correspondence has a fixed point and therefore every game has a Nash equilibrium. ☺

²³ The general argument here is the following: Let $x > y$ and let $\{x^k\}$ and $\{y^k\}$ be sequences such that $x^k \rightarrow x$ and $y^k \rightarrow y$. Then there exists a $\bar{k}$ such that, for all $k > \bar{k}$, $x^k > y^k$.

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Jim Ratliff's Game Theory Course:

3.2 Computing Mixed-Strategy Nash Equilibria of 2 x 2 Strategic-Form Games

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Abstract



We'll now see explicitly how to find the set of (mixed-strategy) Nash equilibria for general two-player games where each player has a strategy space containing two actions (i.e. a "2x2 matrix game").

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We first compute the best-response correspondence for a player. We partition the possibilities into three cases: The player is completely indifferent; she has a dominant strategy; or, most interestingly, she plays strategically (i.e., based upon her beliefs about her opponent's play).

The Nash equilibria of the game are the strategy profiles in the intersection of the two players' best-response correspondences. We use this fact and the possible forms of players' best-response correspondences to explore the possible sets of Nash equilibria in these games

We then apply this technique to two particular games. The first game is a typical and straightforwardly solved example; the second is nongeneric in the sense that it has an infinite number of equilibria. For each game we will compute the graph of each player's best-response correspondence and identify the set of Nash equilibria by finding the intersection of these two graphs.



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[§3.1: Nash Equilibrium](#)

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[§4.1: Introduction to Extensive-Form Games](#)

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Introduction

We'll now see explicitly how to find the set of (mixed-strategy) Nash equilibria for two-player games where each player has a strategy space containing two actions (i.e. a " 2×2 matrix game"). After setting up the analytical framework and deriving some general results for such games, we will apply this technique to two particular games. The first game is a typical and straightforwardly solved example; the second is *nongeneric* in the sense that it has an infinite number of equilibria.¹ For each game we will compute the graph of each player's best-response correspondence and identify the set of Nash equilibria by finding the intersection of these two graphs.

The canonical game

We consider the two-player strategic-form game in Figure 1. We assign rows to player *A* and columns to player *B*. *A*'s strategy space is $S_A = \{U, D\}$ and *B*'s is $S_B = \{l, r\}$. Because each player has only two actions, each of her mixed strategies can be described by a single number (p for *A* and q for *B*)

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¹ Here *nongeneric* means that the phenomenon depends very sensitively on the exact payoffs. If the payoffs were perturbed the slightest bit, then the phenomenon would disappear.

belonging to the unit interval $[0, 1]$. A mixed-strategy profile for this game, then, is an ordered pair $(p, q) \in [0, 1] \times [0, 1]$. We denote the players' payoffs resulting from pure-strategy profiles by subscripted a 's and b 's, respectively. E.g. the payoff for A when A plays D and B plays r is a_{Dr} .

		B	
		$l: q$	$r: 1-q$
A	$U: p$	a_{Ul}, b_{Ul}	a_{Ur}, b_{Ur}
	$D: 1-p$	a_{Dl}, b_{Dl}	a_{Dr}, b_{Dr}

Figure 1: The canonical two-player, two-action-per-player strategic-form game.

The pure-strategy equilibria, if any, of such a game are easily found by inspection of the payoffs in each cell, each cell corresponding to a pure-strategy profile. A particular pure-strategy profile is a Nash equilibrium if and only if 1 that cell's payoff to the row player (viz. A) is a (weak) maximum over all of A 's payoffs in that column (otherwise the row player could profitably deviate by picking a different row given B 's choice of column) and 2 that cell's payoff to the column player (viz. B) is a (weak) maximum over all of B 's payoffs in that row. For example, the pure-strategy profile (U, r) would be a Nash equilibrium if and only if the payoffs were such that $a_{Ur} \geq a_{Dr}$ and $b_{Ur} \geq b_{Ul}$.

Best-response correspondences

Finding the pure-strategy equilibria was immediate. Finding the mixed-strategy equilibria takes a little more work, however. To do this we need first to find each player's best-response correspondence. We will show in detail how to compute player A 's correspondence. Player B 's is found in exactly the same way.

Player A 's best-response correspondence specifies, for each mixed strategy q played by B , the *set* of mixed strategies p which are best responses for A . I.e. it is a correspondence p^* which associates with every $q \in [0, 1]$ a set $p^*(q) \subset [0, 1]$ such that every element of $p^*(q)$ is a best response by A to B 's choice q . The *graph* of p^* is the set of points

$$\{(p, q): q \in [0, 1], p \in p^*(q)\}. \quad (1)$$

A's payoff as a function of the mixed-strategy profile

To find A 's best-response correspondence we first compute her expected payoff for an arbitrary mixed-strategy profile (p, q) by weighting each of A 's pure-strategy profile payoffs by the probability of that profile's occurrence as determined by the mixed-strategy profile (p, q) :¹

$$u_A(p; q) = pq a_{Ul} + p(1-q) a_{Ur} + (1-p)q a_{Dl} + (1-p)(1-q) a_{Dr}. \quad (2)$$

A 's utility maximization problem is

¹ The semicolon in " $u(p; q)$ " is used to denote that, while p is a choice variable for A , q is a parameter outside of A 's control.

$$\max_{p \in [0,1]} u_A(p; q) \quad (3)$$

Because p is A 's choice variable, it will be convenient to rewrite equation (2) as an affine function of p :¹

$$u_A(p; q) = p[(a_{UL} - a_{UR} - a_{DL} + a_{DR})q + (a_{UR} - a_{DR})] + [(a_{DL} - a_{DR})q + a_{DR}], \quad (4a)$$

$$= \delta(q)p + [(a_{DL} - a_{DR})q + a_{DR}]. \quad (4b)$$

Of interest here is the sign of the coefficient of p ,

$$\delta(q) \equiv (a_{UL} - a_{UR} - a_{DL} + a_{DR})q + (a_{UR} - a_{DR}), \quad (5)$$

which is itself an affine function of q .

A 's best-response correspondence

For a given q , the function $u_A(p; q)$ will be maximized with respect to p either 1 at the unit interval's right endpoint (viz. $p = 1$) if $\delta(q)$ is positive, 0 at the interval's left endpoint (viz. $p = 0$) if $\delta(q)$ is negative, or 3 for every $p \in [0, 1]$ if $\delta(q)$ is zero, because $u_A(p; q)$ is then constant with respect to p .

Now we consider the behavior of A 's best response as a function of q . There are three major cases to consider.

Case 1: complete indifference

A 's payoffs could be such that $\delta(q) = 0$ for all q .² In this case A 's best-response correspondence would be independent of q and would simply be the unit interval itself: $\forall q \in [0, 1], p^*(q) = [0, 1]$. In other words A would be willing to play any mixed strategy regardless of B 's choice of strategy. The graph of p^* in this case is the entire unit square $[0, 1] \times [0, 1]$. (See Figure 2.)

¹ I would be happy to say *linear* here instead of *affine*. The strict definition of linear seems to be made consistently in linear algebra, but the less restrictive definition seems to be tolerated in other contexts.

² This would require that in each column A receives the same payoff in each of the two rows.

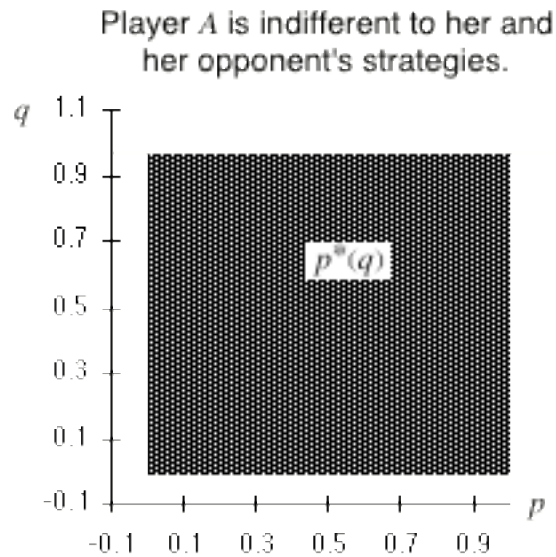


Figure 2: Best-response correspondence when A is completely indifferent.

Case 2: A has a dominant pure strategy

If this is not the case, i.e. if $\delta(q)$ is not identically zero, then—because $\delta(q)$ is affine—there will be exactly one value $q^\dagger$ at which $\delta(q^\dagger) = 0$. For all q to one side of $q^\dagger$, $\delta(q)$ will be positive; for all q on the other side of $q^\dagger$, $\delta(q)$ will be negative. However, this $q^\dagger$ need not be an element of $[0, 1]$. If $q^\dagger \notin [0, 1]$, then all $q \in [0, 1]$ will lie on a common side of $q^\dagger$ and therefore $\delta(q)$ will have a single sign throughout the interval $[0, 1]$. Therefore A will have the same best response for every q (viz. $p = 1$ if $\delta(q) > 0$ on $[0, 1]$; $p = 0$, if $\delta(q) < 0$ on $[0, 1]$); i.e. A has a strongly dominant pure strategy. (See Figure 3.)

Case 3: A plays strategically

Now we consider the case where A plays strategically: her optimal strategy depends upon her opponent's strategy. If $q^\dagger \in (0, 1)$, then the unit interval is divided into two subintervals—those points to the right of $q^\dagger$ and those points to the left of $q^\dagger$ —in each of which A plays a different pure strategy. At exactly $q^\dagger$, A can mix with any $p \in [0, 1]$. (See Figure 4.)

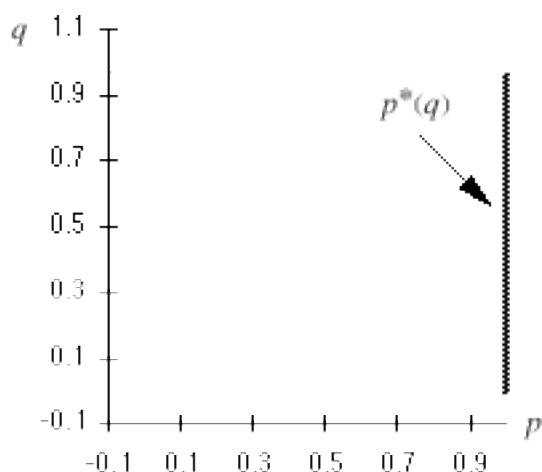


Figure 3: Best-response correspondence when A has a dominant pure strategy ($p = 1$).

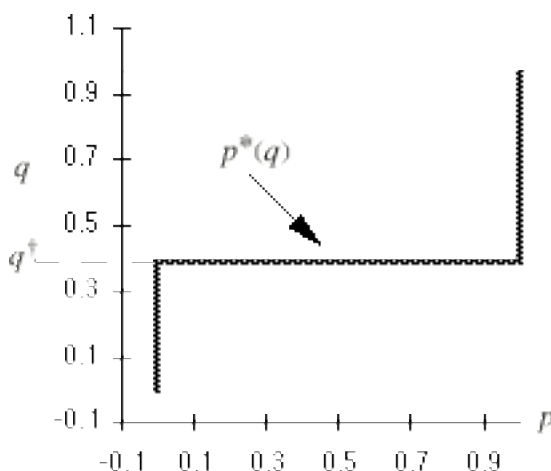


Figure 4: Best-response correspondence when A plays strategically.

If $q^\dagger = 0$ or $q^\dagger = 1$, then A can mix at $q^\dagger$ and will play a common pure strategy for every other $q \in [0, 1]$. In this case the pure strategy which A plays against $q \in (0, 1)$ is a weakly dominant strategy. (It is strictly best against $q \in [0, 1] - \{q^\dagger\}$ and as good as anything else against $q^\dagger$.)

Summary of the best-response correspondence

We see that—in a 2×2 matrix game—each player will either 1 be free to mix regardless of her opponent's strategy (a case which 'almost never' occurs), 2 play a dominant pure strategy, or 3 be free to mix in response to exactly one of her opponent's possible choices of strategy but for every other choice will play a pure strategy.

B's best-response correspondence

Player B's best-response correspondence specifies, for each mixed strategy p played by A, the set of mixed strategies q which are best responses for B. I.e. it is a correspondence q^* which associates with

every $p \in [0, 1]$ a set $q^*(p) \subset [0, 1]$ such that every element of $q^*(p)$ is a best response by B to A 's choice p . The graph of q^* is the set of points

$$\{(p, q) : p \in [0, 1], \square q \in q^*(p)\}. \quad (6)$$

This correspondence is found using the same method of analysis we used for A 's. You will easily show that

$$u_B(q; p) = \gamma(p)q + [(b_{Ur} - b_{Dr})p + b_{Dr}], \quad (7)$$

where

$$\gamma(p) \equiv (b_{Ul} - b_{Ur} - b_{Dl} + b_{Dr})p + (b_{Dl} - b_{Dr}). \quad (8)$$

The Nash equilibria are the points in the intersection of the graphs of A 's and B 's best-response correspondences

We know that a mixed-strategy profile (p, q) is a Nash equilibrium if and only if 1 p is a best response by A to B 's choice q and 2 q is a best response by B to A 's choice p . We see from (1) that the first requirement is equivalent to (p, q) belonging to the graph of p^* , and from (6) we see that the second requirement is equivalent to (p, q) belonging to the graph of q^* . Therefore we see that (p, q) is a Nash equilibrium if and only if it belongs to the intersection of the graphs of the best-response correspondences p^* and q^* . We can write the set of Nash equilibria, then, as

$$\{(p, q) \in [0, 1] \times [0, 1] : p \in p^*(q), q \in q^*(p)\}. \quad (9)$$

A typical example

Consider the 2×2 game in Figure 5. First we immediately observe that there are two pure-strategy Nash equilibria: (U, r) and (D, l) .

		$l: q$	$r: 1-q$
$U: p$		1, -1	3, 0
$D: 1-p$		4, 2	0, -1

Figure 5: A typical 2×2 game.

A 's best-response correspondence

Now we find A 's best-response correspondence. From (5) we see that

$$\delta(q) = -6q + 3, \quad (10)$$

which vanishes at $q^\dagger = 1/2$. Because $\delta(q)$ is decreasing in q we see that A will choose the pure strategy $p = 1$ against q 's on the interval $[0, 1/2)$ and the pure strategy $p = 0$ against q 's on the interval $(1/2, 1]$. Against $q = q^\dagger = 1/2$, A is free to choose any mixing probability. Player A 's best-response correspondence p^* is plotted in Figure 6.

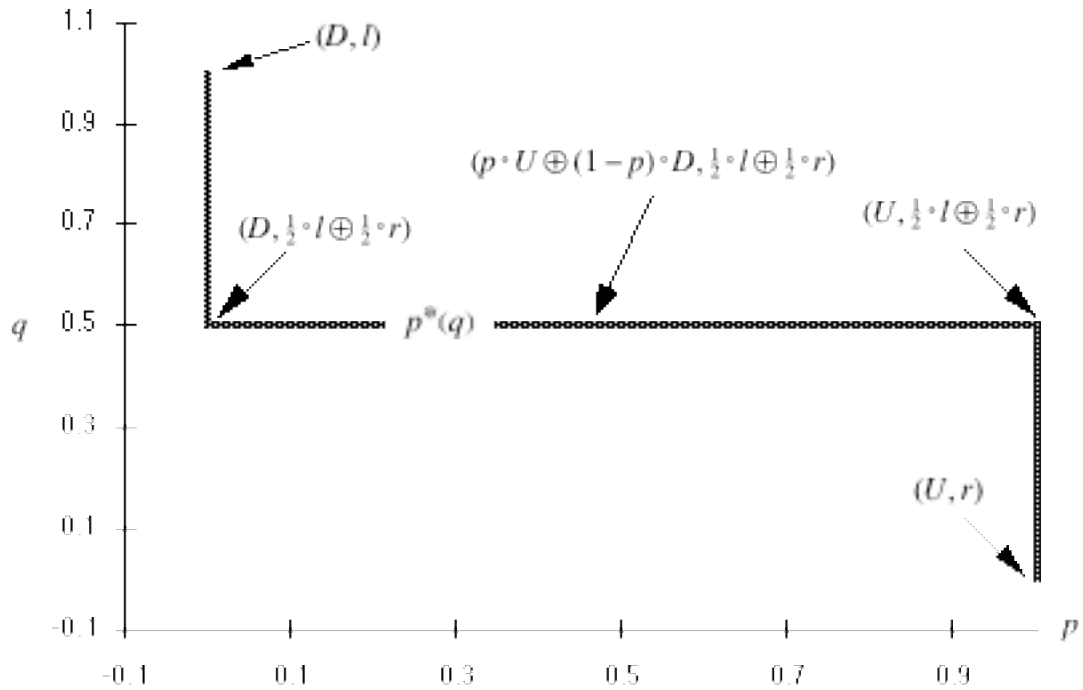


Figure 6: A 's best-response correspondence for the game of Figure 5.

We can also derive A 's best-response correspondence graphically by plotting her payoff to her different pure strategies as a function of B 's mixed-strategy choice q . Using (4b) and (10) we have

$$u_A(p; q) = (-6q + 3)p + 4q. \quad (11)$$

Evaluating this payoff function at A 's pure-strategy choices $p = 1$ and $p = 0$, respectively, we have

$$u_A(U; q) = 3 - 2q, \quad (12)$$

$$u_A(D; q) = 4q. \quad (13)$$

Both of these functions are plotted for $q \in [0, 1]$ in Figure 7. These two lines intersect when $q = q^\dagger = 1/2$; i.e. they intersect at the mixed strategy for B at which we earlier determined A would be willing to mix. To the left of this point, A 's payoff to U is higher than her payoff to D ; the reverse is true on the other side of $q^\dagger$. Therefore A 's best response is to play U against $q \in [0, 1/2)$ and D against $q \in (1/2, 1]$. At the intersection point $q^\dagger$, A is indifferent to playing U or D , so she is free to mix between them. This is exactly the best-response correspondence we derived analytically above.

B 's best-response correspondence

We similarly find B 's best-response correspondence. From (8) we find that

$$\gamma(p) = -4p + 3, \quad (14)$$

which decreases in p and vanishes at $p^\dagger = 3/4$. Player B , then, chooses the pure strategy $q = 1$ against p 's on the interval $[0, 3/4)$ and the pure strategy $q = 0$ against p 's on the interval $(3/4, 1]$. Against $p = p^\dagger = 3/4$,

B is free to choose any mixing probability. Player B 's best-response correspondence q^* is plotted in Figure 8.

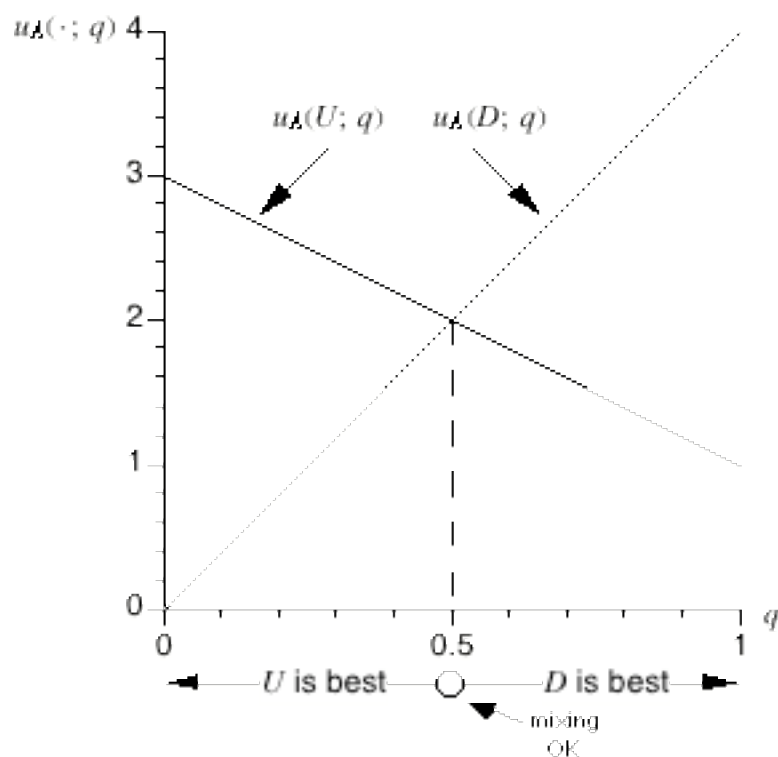


Figure 7: A's pure-strategy payoffs as a function of B's mixed strategy q .

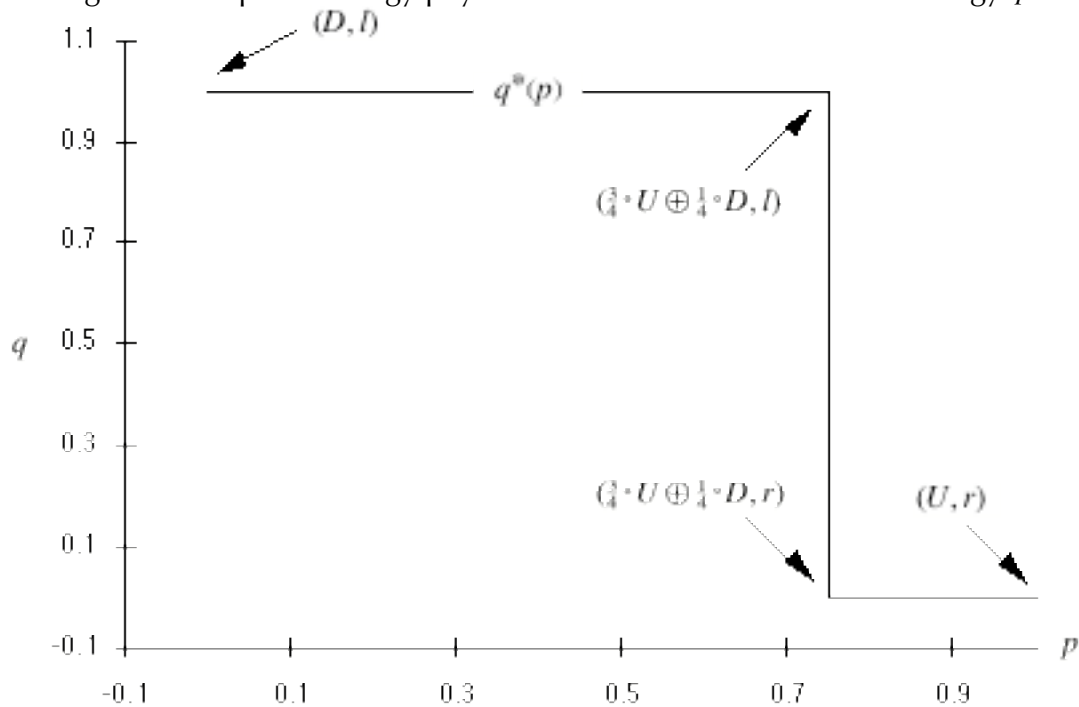


Figure 8: B's best-response correspondence for the game of Figure 5.

Using (7) and (14) we compute B 's payoff to the mixed-strategy profile (p, q) to be

$$u_B(q; p) = (-4p + 3)q + (p - 1). \quad (15)$$

Evaluating this at B 's pure-strategy choices we get

$$u_B(l; p) = 2 - 3p, \quad (16)$$

$$u_B(r; p) = p - 1. \quad (17)$$

These functions are plotted for $p \in [0, 1]$ in Figure 9. We again note that their intersection—in this case at $p^\dagger = 3/4$ —occurs at the opponent's mixed strategy which allows mixing by the player choosing a best response. The same reasoning we used above to graphically derive A 's best-response correspondence works here, and we arrive at the same behavior rules we found analytically.

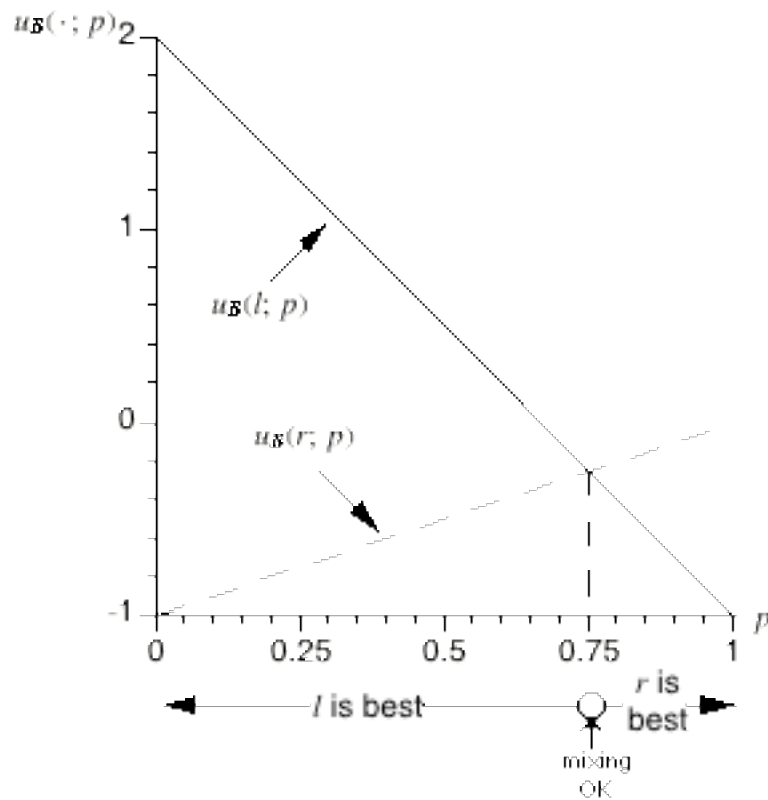


Figure 9: B 's pure-strategy payoffs as a function of A 's mixed strategy p .

The Nash set

Both A 's and B 's best-response correspondences are plotted together in Figure 10. We see that the intersection of the graphs of the two best-response correspondences contains exactly three points, each corresponding to a mixed-strategy profile (p, q) : $(0, 1)$, $(3/4, 1/2)$, and $(1, 0)$. The first and last of these correspond to the two pure-strategy Nash equilibria we identified earlier. Note that the additional equilibrium we found is the strategy profile $(p^\dagger, q^\dagger)$. This strategy profile will in general be the only

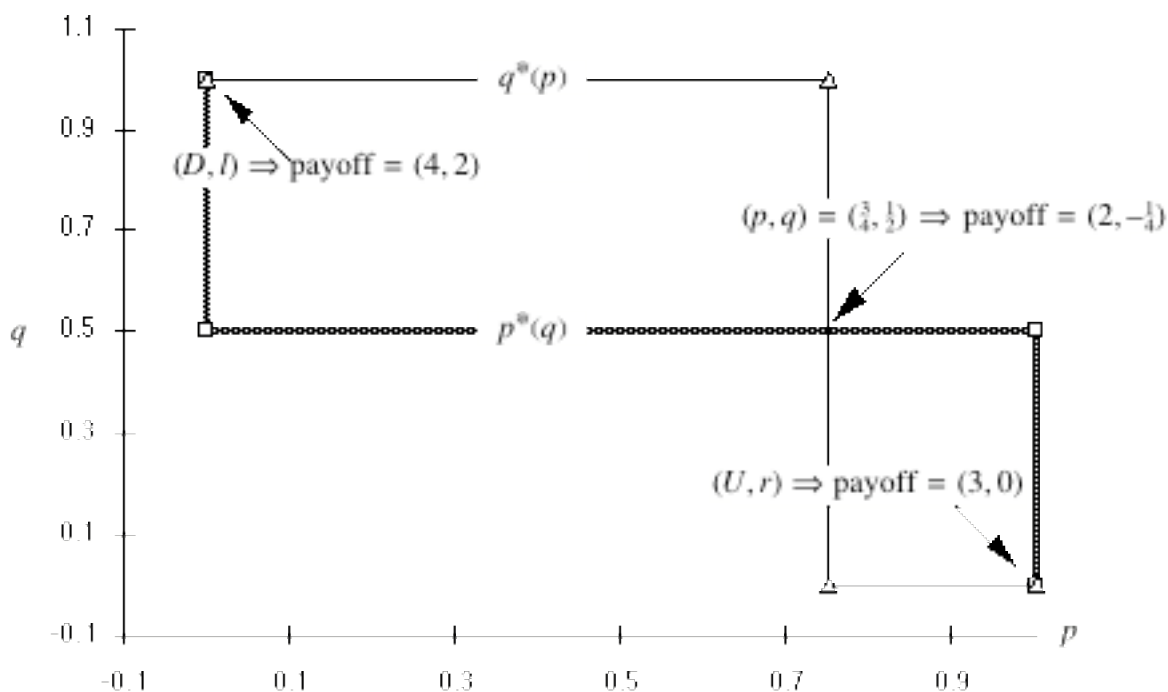


Figure 10: The players' best-response correspondences, the Nash set, and equilibrium payoffs.

non-pure equilibrium strategy profile when $p^\dagger$ and $q^\dagger$ both lie in the interior of the unit interval.¹ Note as well that there is an odd number of Nash equilibria for this game, as is “almost always” the case. The payoff vectors for these equilibria are $(4, 2)$, $(2, -\frac{1}{4})$, and $(3, 0)$, respectively. [The mixed-strategy profile payoffs are computed using (11) and (15).] Note that the equilibrium payoffs are completely Pareto ranked.

A nongeneric example

We now consider the two-player normal form game in Figure 11. We immediately determine that the unique pure-strategy equilibrium is (D, r) .

		$l: q$	$r: 1-q$
$U: p$		1, $\frac{3}{2}$	3, 1
$D: 1-p$		4, 2	3, 3

Figure 11: A nongeneric two-player game.

To find A 's best-response correspondence we use (5) to compute that

$$\delta(q) = -3q, \quad (18)$$

¹ It is not possible when $p^\dagger, q^\dagger \in (0, 1)$ that in equilibrium only one player mixes. Assume, say, that A mixes. We know that A will mix only when $q = q^\dagger \in (0, 1)$. Therefore B is mixing as well.

which decreases in q and vanishes at $q^\dagger = 0$. Therefore A plays the pure strategy $p = 0$ against any $q \in (0, 1]$ and she is willing to mix against $q = q^\dagger = 0$. Therefore D is a weakly dominant strategy for A . Her best-response correspondence $p^*(q)$ is plotted in Figure 12.

To find B 's best-response correspondence we use (8) and find that

$$\gamma(p) = \frac{3}{2}p - 1. \quad (19)$$

This function increases in p and vanishes at $p^\dagger = \frac{2}{3}$. Therefore B will play the pure strategy $q = 0$ against any $p \in [0, \frac{2}{3})$, will play the pure strategy $q = 1$ against any $p \in (\frac{2}{3}, 1]$, and will be free to mix for $p = p^\dagger = \frac{2}{3}$. B 's best-response correspondence $q^*(p)$ is also plotted in Figure 12.

Inspection of Figure 12 shows that the intersection of the graphs of A 's and B 's best-response correspondences is a line segment along which B plays $q = 0$ and A mixes with any probability p on $[0, \frac{2}{3}]$. We note that the unique pure-strategy Nash equilibrium we identified earlier is the left endpoint of this set. This example is nongeneric in that we have an infinity (in fact, a continuum) of equilibria: a situation which generically never happens.

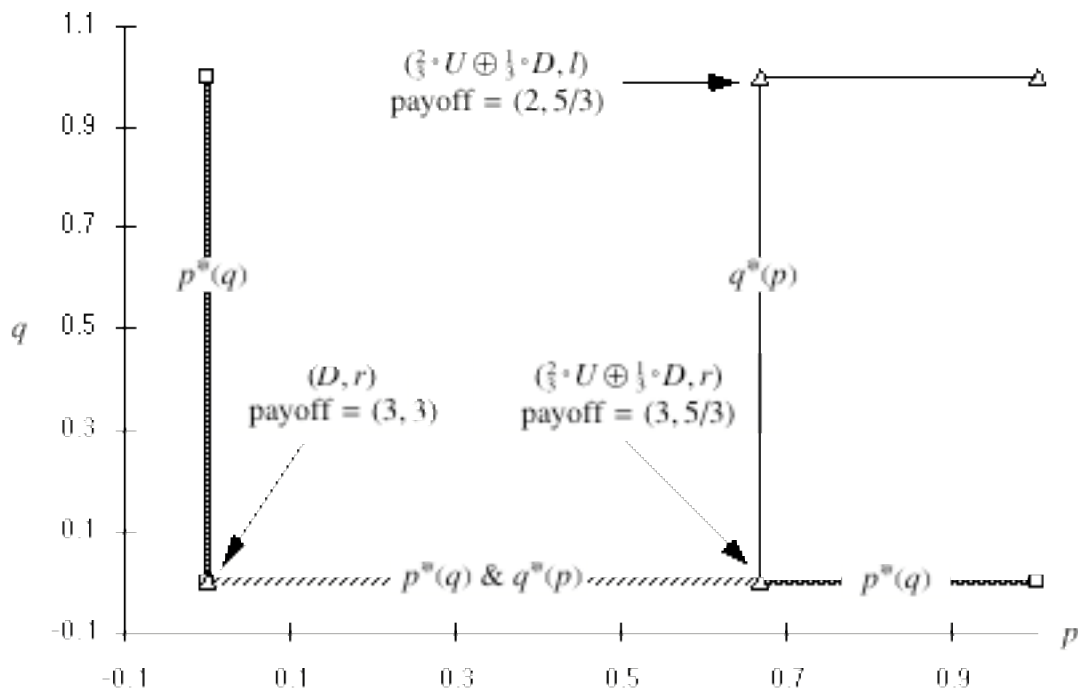


Figure 12: A game with a continuum of equilibria.

Before leaving this example we should also take note of the equilibrium payoffs. At the pure-strategy equilibrium each player gets 3. (See Figure 12.) Player A is indifferent to mixing between U and D , given that B is playing r . However, this mixing hurts B . At the right-hand endpoint of the Nash set, A still receives 3 but B 's payoff, which has been decreasing linearly with A 's mixing probability p , has declined to $\frac{5}{3}$. Note that, at the alternative strategy profile in which B plays l and A mixes with $p = \frac{2}{3}$, B would get the same payoff as playing r , but A would get only 2, rather than 3. If A mixed any more

strongly toward U than $p = 2/3$, B would defect to the alternative strategy l , giving A less than in this equilibrium. This is what determines the location of the right-hand endpoint of the Nash set. In this game the set of equilibria are Pareto ranked. Obviously, player B would prefer to coordinate on the pure-strategy equilibrium, and there is no reason A should disagree.

Jim Ratliff's Game Theory Course:

4.1 Introduction to Extensive-Form Games

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Abstract



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When we model a strategic economic situation we want to capture as much of the relevant detail as tractably possible. A game can have a complex temporal and information structure; and this structure could well be very significant to understanding the way the game will be played. These structures are not acknowledged explicitly in the game's strategic form, so we seek a more inclusive formulation. It would be desirable to include at least the following: 1) the set of players, 2) who moves when and under what circumstances, 3) what actions are available to a player when she is called upon to move, 4) what she knows when she is called upon to move, and 5) what payoff each player receives when the game is played in a particular way.

Components 2 and 4 are additions to the strategic-form description; component 3 typically involves more specification in the extensive form than in the strategic form.

We can incorporate all of these features within an *extensive-form* description of the game. The foundation of the extensive form is a *game tree*. First I'll discuss what a tree is and then describe the additional specifications and interpretations we need to make in order to transform it into a full-fledged extensive-form description of the game. We carefully discuss the concept of an *information set*. We'll discuss additional restrictions upon the information structure to reflect a common assumption, viz. *perfect recall*, which asserts that players don't forget. We'll discuss the relation between this traditional, graph-theoretic exposition and a more recent one based on the *arborescence*. After that we'll learn how to ease our analysis of complicated games by breaking them up into simpler *subgames*.



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Extensive-Form Games: Introduction

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Introduction

When we model a strategic economic situation we want to capture as much of the relevant detail as tractably possible. A game can have a complex temporal and information structure; and this structure could well be very significant to understanding the way the game will be played. These structures are not acknowledged explicitly in the game's strategic form, so we seek a more inclusive formulation. It would be desirable to include at least the following: 1) the set of players, 2) who moves when and under what circumstances, 3) what actions are available to a player when she is called upon to move, 4) what she knows when she is called upon to move, and 5) what payoff each player receives when the game is played in a particular way. Components 2 and 4 are additions to the strategic-form description; component 3 typically involves more specification in the extensive form than in the strategic form.

We can incorporate all of these features within an *extensive-form* description of the game. The foundation of the extensive form is a *game tree*. First I'll discuss what a *tree* is and then describe the additional specifications and interpretations we need to make in order to transform it into a full-fledged extensive-form description of the game. We'll discuss additional restrictions upon the information structure to reflect a common assumption, viz. *perfect recall*, which asserts that players don't forget. We'll discuss the relation between this traditional, graph-theoretic exposition and a more recent one based on the *arborescence*. After that we'll learn how to ease our analysis of complicated games by breaking them up into simpler *subgames*.

Game Trees

We'll construct our definition of a tree by first defining more primitive graph-theoretic entities. (Iyanaga and Kawada [1980: 234–235]). A *graph* $G = (V, E)$ consists of a finite set V of *vertices* and a finite set E of *edges* of V . An edge $e \in E$ is an unordered pair of distinct elements of V ; e.g. the edge $e = (v, v')$ *joins* the two vertices $v, v' \in V$.

Consider an alternating sequence of vertices and edges of a graph G , $P = \{v_0, e_1, v_1, e_2, \dots, e_n, v_n\}$. P is a *path* if 1 each edge e_i joins its neighbors v_{i-1} and v_i and 2 no edge is encountered more than once (i.e. $e_i \neq e_j$ when $i \neq j$).¹ In this case we say that v_i and v_j , for $i, j \in \{1, \dots, n\}$, are *connected* by P and that P *runs from* v_0 *to* v_n .² (Note that a path may encounter the same vertex more than once.) A path is *closed* if it begins and ends with the same vertex, i.e. $v_0 = v_n$. A closed path which never reencounters a vertex, except for the first, is called a *circuit*. I.e. a closed path P is a circuit if $v_i \neq v_j$ whenever $i \neq j$ and $\{i, j\} \neq \{0, n\}$. In other words a circuit is a single “loop.” A graph G is *connected* if for every pair of distinct vertices $v, v' \in V$, $v \neq v'$, there exists a path in G which connects v and v' .

Now we’re ready to define a tree. A *tree* is a connected graph which contains no circuit.³ In game theory we often refer to the vertices as *nodes* and, consistent with our dendro-metaphor, to edges as *branches*.⁴ Because a tree is a connected graph, any node can be reached from any other node by traversing a sequence of branches. Because it has no circuits, a tree is *unicursal*: for any pair of nodes, there is exactly one path which runs from the first to the second.

Example: graphs and trees

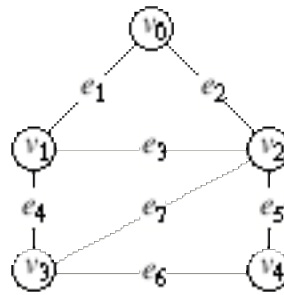


Figure 1: A graph.

The sets of vertices and edges in Figure 1 form a graph. The edges depicted graphically are defined by

$$\begin{aligned} e_1 &= (v_0, v_1), & e_2 &= (v_0, v_2), & e_3 &= (v_1, v_2), & e_4 &= (v_1, v_3), \\ e_5 &= (v_2, v_4), & e_6 &= (v_3, v_4), & e_7 &= (v_2, v_3). \end{aligned}$$

The following two alternating sequences of vertices and edges are paths:

$$\{v_0, e_1, v_1\}, \square \{v_0, e_2, v_2, e_5, v_4, e_6, v_3, e_7, v_2, e_3, v_1\}.$$

Note in the second case that the vertex v_2 is encountered twice. The sequence $\{v_0, e_1, v_2\}$ is not a path. Neither is

¹ Let $e = (v, v')$ and $e' = (v', v)$ be edges, where v and v' are vertices. Because edges are *unordered* pairs, we have $e = e'$.

² This “runs from... to” terminology is not standard, but I find it useful for expository purposes here.

³ I.e., no circuit can be constructed from vertices and edges belonging to the graph.

⁴ My treatment roughly follows that of Kuhn [1953], which is the hugely influential, seminal paper concerning extensive-form games.

$\{v_0, e_1, v_1, e_3, v_2, e_7, v_3, e_4, v_1, e_3, v_2\},$

because the edge e_3 is encountered twice.

The two sequences

$\{v_0, e_2, v_2, e_5, v_4, e_6, v_3, e_4, v_1, e_1, v_0\}, \square \{v_0, e_1, v_1, e_3, v_2, e_7, v_3, e_6, v_4, e_5, v_2, e_2, v_0\}$

are closed paths. The first is a circuit; the second is not, because the vertex v_2 is encountered twice.

Figure 2 shows two connected graphs. The first is a tree; the second is not.

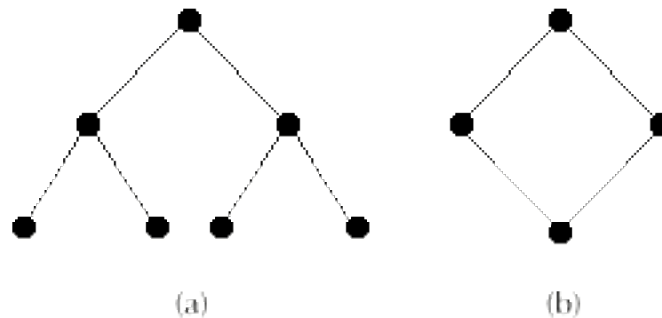


Figure 2: Two connected graphs: (a) a tree and (b) a nontree.

Now we want to enhance our tree to become a *game tree*. We henceforth assume that the tree has at least two vertices. We bestow one vertex of the tree with the honor of being the *initial node*, $\mathcal{O}$. (See Figure 3.) This is where the game begins. It is a *decision node*.

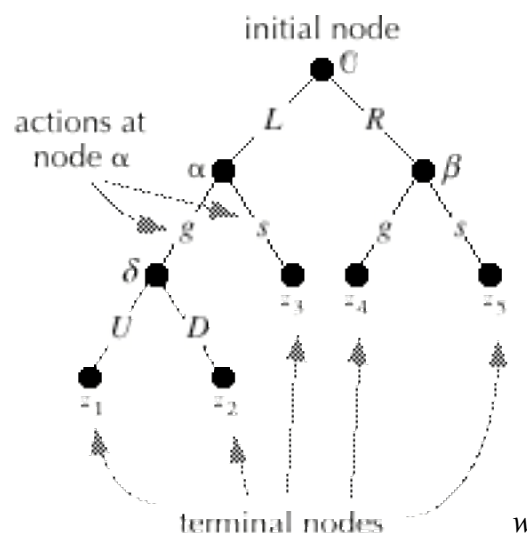


Figure 3: A game tree.

If a noninitial node x is incident with two or more branches, it is also a decision node. Let $X \subset V$ be the set of decision nodes. (In Figure 3, $X = \{\mathcal{O}, \alpha, \beta, \delta\}$.) One of these branches leads *to* the node; the remaining branches lead *away* from the node. Since branches are unordered pairs of nodes, how do you

know which branch leads to the node and which lead away? It's simple: For some noninitial decision node $x \in X \setminus \{\mathbb{O}\}$, find the unique path which runs from the initial node $\mathbb{O}$ to node x . That path will contain exactly one of the branches incident with x ; this branch is the one which leads *to* x . The other branches incident at x are the branches which lead away from x . For example, in Figure 3, there are three edges incident at node δ : g , U , and D . The unique path running from the initial node to node δ is $\{\mathbb{O}, L, \alpha, g, \delta\}$. This path contains only the g edge incident at δ ; therefore g is the branch leading to δ and U and D are the edges leading away.

We interpret each edge $e \in E$ as an *action*. Let $\bar{A}$ be the set of all actions and let $f: E \rightarrow \bar{A}$ be the function which assigns an action to each edge. The branches which lead away from a decision node represent the actions available at that node. We require that, for a given decision node $x \in X$, the function f assign a unique action to each branch leading away from x . (Otherwise it would not be well specified what path would result from a particular action taken at x .) The actions available at any decision node $x \in X$ are given by the set $A(x) = f(\{e \in E: e \text{ leads away from } x\})$.⁵ (So A is a correspondence $A: X \rightarrow \bar{A}$.) In Figure 3, $A(\mathbb{O}) = \{L, R\}$, $A(\alpha) = A(\beta) = \{g, s\}$, and $A(\delta) = \{U, D\}$. More than one edge may correspond to the same action (as long as any such pair of edges does not lead away from the same node). For example, in Figure 3, there are two edges which correspond to the action g . In what follows I will often refer to an edge $e \in E$ by its associated action $f(e)$.⁶

Any noninitial node with only one incident branch necessarily has no actions available. Such a node is a *terminal node*. Let $Z \subset V$ be the set of terminal nodes. The tree in Figure 3 has five terminal nodes: $Z = \{z_1, z_2, z_3, z_4, z_5\}$. The decision and terminal nodes partition the game tree's nodes: i.e. $V = X \cup Z$ and $X \cap Z = \emptyset$. The game ends whenever a terminal node is reached. We identify the terminal nodes with *outcomes* of the game. (Sometimes *outcome* is used to refer not just to a terminal node but to the unique path from the initial node to that terminal node.)

So we have established that the game begins at the initial node and play ends at a terminal node. In what order are the intermediate decisions made? Consider two distinct nodes $v, v' \in V$, $v \neq v'$. We say that v *precedes* v' , or that v is a *predecessor* of v' , if v lies on the unique path which runs from the initial node $\mathbb{O}$ to v' . We write $v < v'$. Equivalently, we say that v' is a *successor* of v , and write $v' > v$. For example, in Figure 3, the unique path running from the initial node $\mathbb{O}$ to δ is $\{\mathbb{O}, L, \alpha, g, \delta\}$. This path contains α but not contain β ; therefore α is a predecessor of δ , but β is not.

Predecession, then, defines a binary relation $<$ on the set of nodes V , viz.

$$< = \{(v, v') \in V^2: v < v'\}.^{7,8} \quad (1)$$

⁵ Recall that if $f: X \rightarrow Y$ is a function and $S \subset X$, then $f(S)$ is the *image* of S under f , i.e. $f(S) = \{f(x): x \in S\}$.

⁶ For example, in the previous paragraph I referred to the edge connecting α and δ by its action g .

⁷ A *binary relation* R on a set X can be thought of as a subset of the Cartesian product of X with itself, i.e. $R \subset X \times X$, which includes all those pairs $(x, y) \in X^2$ such that xRy . I.e. $(x, y) \in R \Leftrightarrow xRy$.

⁸ The *succession* binary relation $>$ is the *inverse relation* of $<$; i.e. $\forall v, v' \in V$, $v < v' \Leftrightarrow v' > v$; it could also be called the *dual order* (except for a minor problem that $<$ is irreflexive and therefore not actually providing an *ordering*). (See later footnote.)

It is intuitively obvious—and you can show rigorously—that the precedence relation is *irreflexive* ($\forall v \in V$, not $v < v$), *asymmetric* [$v < v' \Rightarrow \text{not } (v' < v)$], and *transitive* [$(v < v' \text{ and } v' < v'') \Rightarrow v < v''$]. This binary relation need not be complete; there may exist a pair of nodes $v, v' \in V$ such that neither $v < v'$ nor $v' < v$. For example in Figure 3, α and β are unordered by precedence. A partial listing of the precedence relations in Figure 3 is: $\mathbb{O} < z_3$, $\alpha < z_2$, and $\alpha < \delta$.

The initial node is a predecessor of every noninitial node and has no predecessor itself. For every noninitial node $x \in V \setminus \{\mathbb{O}\}$, let $P(x)$ be the set of predecessors of x ; i.e. $P(x) = \{v \in V: v < x\}$. For example in Figure 3, $P(\delta) = \{\mathbb{O}, \alpha\}$ and $P(z_4) = \{\mathbb{O}, \beta\}$. Although precedence provides only a partial ordering in general on the set V of all nodes, you can show that it totally orders the set $P(x)$ of predecessors of any noninitial node $x \in V \setminus \{\mathbb{O}\}$. (I.e. if $v, v' \in P(x)$ and $v \neq v'$, then either $v < v'$ or $v' < v$.)⁹

We say that v is the *immediate predecessor* of v' if there is an action available at v (i.e. a branch incident at v which leads away from the initial node) which leads directly to v' . (This requires that $v < v'$ and that a single edge joins v and v' . You can prove that each noninitial node has a unique immediate predecessor.) Alternatively we could say that v is the immediate predecessor of v' if $v = \max_{<} P(v')$, which means that all other predecessors of v' must also precede v .¹¹ In Figure 3, for example, $\mathbb{O}$ is the immediate predecessor of α but not of δ , and α is the immediate predecessor of δ .

We have thus far identified the locations in the tree at which decisions are made—the decision nodes—and created a structure which relates the various decisions by specifying what actions lead to which other decisions and in what temporal order. But we still haven't said who makes which decisions. We retain our standard player set $I = \{1, \dots, n\}$, but we also allow Nature to be player 0 when required. (We will see later that Nature can be given the job of introducing new information into the game.)

So now we assign to each decision node $x \in X$ exactly one player $i \in I \cup \{0\}$. We specify this assignment by a node-ownership function $\iota: X \rightarrow (I \cup \{0\})$ where each decision node $x \in X$ is assigned to the player specified by $\iota(x)$. The set of decision nodes assigned to player $i \in I \cup \{0\}$ is $X_i = \{x \in X: \iota(x) = i\}$. The node-ownership function ι thus induces a *player partition* $\mathcal{X} = \{X_0, X_1, \dots, X_n\}$ of the set X of decision nodes.¹² We say that a node *belongs to* a particular player or that a player *owns* a particular node.

Every terminal node $z \in Z$ represents some outcome which affects the players. Perhaps it's success for

⁹ Note that if a node $x \in V$ has only one predecessor, so that $P(x)$ is a singleton set, then $P(x)$ is trivially totally ordered by the binary relation $<$.

¹⁰ In my experience it is conventional to require that a binary relation R on a set X be reflexive (i.e. $\forall x \in X, xRx$) in order that it impose a preordering upon X . (See Debreu [1959: 7] and Iyanaga and Kawada [1980: 958, 960].) Kreps [1990: 364] explicitly weakens this notion by defining that “a set is totally ordered by a binary relation if any two *distinct* elements of the set are ordered in one direction or the other.” [emphasis added] This allows the concept of totally ordered to be usefully applied to irreflexive binary relations. Note that for reflexive binary relations the two definitions are equivalent. Kreps [1988: 8] provides alternative terminology for this concept by saying that a binary relation R on a set X is *weakly connected* if for all $x, y \in X$, either $x = y$, xRy , or yRx .

¹¹ Because $P(v')$ is totally ordered by the binary relation $<$, we can arrange its elements such that $v^1 < v^2 < \dots < v^k$. Then $v^k = \max_{<} P(v')$.

¹² A *partition* of a set A is a collection of cells $A_1, \dots, A_n$, such that 1 each $A_i \subset A$, 2 the collection is exhaustive, i.e. $A_1 \cup A_2 \cup \dots \cup A_n = A$, and 3 the cells are mutually exclusive, i.e. when $i \neq j$, $A_i \cap A_j = \emptyset$.

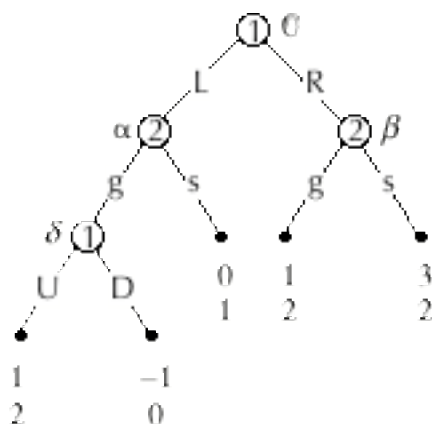


Figure 4: An extensive-form game indicating the player partition and payoffs as a function of outcome.

one and failure for another; the outcomes may represent profit levels for each firm or lengths of prison sentences for each criminal suspect. Whatever the outcomes are, they are events over which each player has preferences. We assume that each player has von Neumann-Morgenstern preferences, which can be represented by a utility function $\mu_i: Z \rightarrow \mathbb{R}$ with the expected-utility property, and seeks to maximize her expected utility.¹³

Figure 4 shows an extensive-form game elaborated from the tree in Figure 3. I have indicated the player partition, which assigns players to nodes, by numerical labels and assigned payoff vectors to each terminal node. By convention the upper entry in each payoff vector is the first player's utility derived from that outcome; the second entry is the second player's utility. The player partition is given by $X_1 = \{\emptyset, \delta\}$ and $X_2 = \{\alpha, \beta\}$. (Nature does not appear in this game.)

Information sets

In the first paragraph I identified five aspects of an economic scenario we'd capture using the extensive form. So far we've captured the set of players, who moves when and under what conditions, what actions are available to a player when it's her turn to move, and what payoffs are received at the end of the game. The remaining challenge is to represent the information which a player knows when called upon to move.

What kind of information do I have in mind? We need to be concerned about things which have changed during the course of the game as a result of the actions of the players.¹⁴ The key to representing information in the game tree is realizing the connection between nodes and *history*. At any point in the play of the game the history is a record of who did what when. Remember that game trees are

¹³ I'm using μ_i to represent the utility to player i of an outcome $z \in Z$ because I want to reserve the symbol u_i for later use as her utility to a strategy profile. (We will see that an outcome $z \in Z$ could be the result of any of several strategy profiles.)

¹⁴ If information is entering the game in some other fashion, this can be modeled by the addition of Nature as a nonstrategic player who chooses her strategy according to a probability distribution. The revelation of Nature's moves then communicates the exogenously generated information to the players.

unicursal—there’s precisely one path which runs from the initial node to any given node. Therefore, if you know the node you have reached, you also know precisely the history of play that got you there.^{15,16}

We can divide extensive-form games into two classes: In games of *perfect information*, whenever it is your turn to move you know precisely where in the tree you are located. Therefore you know the history of play completely. In games of *imperfect information*, you might be uncertain about the history of play when you are called on to play. This uncertainty about history corresponds to uncertainty about which of several nodes you might be at.

To express this uncertainty we employ the *information set*. An information set $h \subset X$ is a set of decision nodes. In the course of playing the game the players’ actions may cause one of your nodes to be reached. When this occurs, you might not know precisely which of your nodes has been reached. This uncertainty is captured by saying that you only know that a particular information set has been reached; all you know is that you’ve reached one of the nodes $x \in h$ in that information set. It is as if you are sent a message, whenever it is your move, specifying the information set you have reached.

There are some obvious restrictions we need to impose on the composition of information sets. First, each decision node must belong to exactly one information set. (If a node belonged to more than one information set, it wouldn’t be well defined what message its owner should receive when that node is reached.) Therefore the set H of all information sets forms a partition of the decision nodes: $X = \bigcup_{h \in H} h$ and $\forall h, h' \in H, h \cap h' = \emptyset$. We can also define an information-set membership function $\hat{h}: X \rightarrow H$, which specifies an information set $h \in H$ for each decision node $x \in X$ according to $\hat{h}(x) = h \Leftrightarrow x \in h$.

Secondly, all of the nodes in an information set must belong to the same player. (I.e. $\forall h \in H, \forall x, x' \in h, \iota(x) = \iota(x')$.) Otherwise, two or more players may think it is their turn to move.¹⁷ (Therefore the information partition H is a *refinement* of the player partition $\mathcal{X} = \{X_0, X_1, \dots, X_n\}$, because for each information set $h \in H$ there exists a player $i \in I \cup \{0\}$ such that this information set is entirely contained within player i ’s set of nodes, i.e. such that $h \subset X_i$.¹⁸)

We can partition the set H of information sets into sets of information sets which belong to each player: For every player $i \in I \cup \{0\}$, the set $H_i = \{h \in H: \forall x \in h, \iota(x) = i\}$ contains the information sets

¹⁵ Note that this wouldn’t be the case if we were playing the game represented by the nontree of Figure 1b. If we had reached the bottom node, we wouldn’t know whether we had arrived there via the clockwise or counterclockwise path from the top initial node.

¹⁶ You might object that in the real world there often are many different ways to reach the same result. Such a situation can be modeled within this framework. You would still have a different terminal node for each of these histories; but you would assign the same payoff vector to each of these terminal nodes.

¹⁷ “But what’s wrong with that? Perhaps it’s a simultaneous move game. Then it would be perfectly kosher to have two players believing it’s their turn,” you say. In an extensive-form representation a simultaneous-move game is modeled as a sequential game where the second mover doesn’t get to observe the first mover’s choice at least until after the second mover has played. We’ll discuss simultaneity further later.

¹⁸ One partition is a *refinement* of a second partition if each cell of the first is a subset of some cell of the second. More formally.... Let $\mathcal{M} = \{M_i\}_{i \in \{1, \dots, m\}}$ and $\mathcal{N} = \{N_j\}_{j \in \{1, \dots, n\}}$ be two partitions. $\mathcal{M}$ is a refinement of $\mathcal{N}$, denoted $\mathcal{M} < \mathcal{N}$, if $\forall M_i \in \mathcal{M}, \exists N_j \in \mathcal{N}$ such that $M_i \subset N_j$. In such a case we say that $\mathcal{N}$ is *coarser* than $\mathcal{M}$.

owned by player i . Therefore, for all $i \in I \cup \{0\}$, $\bigcup_{h \in H_i} h = X_i$.

In general, for any information set $h \in H$, $\iota(h) \equiv \{\iota(x) : x \in h\}$ would refer to the image of the set of nodes h under the node-ownership function ι ; i.e. $\iota(h)$ would be the set of players who owned one or more nodes in h . However, since all nodes in h are owned by the same player, we see that $\iota(h)$ is single valued; i.e. $\forall h \in H$, $\#\iota(h) = 1$. Therefore we can also consider, with some abuse of notation, ι to be an information-set ownership function $\iota(h) = \{i \in I \cup \{0\} : \forall x \in h, \iota(x) = i\}$. As with decision nodes, we say that an information set belongs to a particular player or that a player owns a particular information set.

Thirdly, every node in an information set must have the same set of available actions. Otherwise the player wouldn't be sure which actions were truly available to her. I.e. $\forall h \in H, \forall x, x' \in h, A(x) = A(x')$. It is useful then to refer to the set of actions available at an information set $h \in H$, viz. the image of h under the correspondence A : $A(h) = \bigcup_{x \in h} A(x)$, because for all nodes in the information set $A(h)$ is the set of actions available there: $\forall x \in h, A(x) = A(h)$.

We can summarize these last two restrictions by

$$\forall h \in H, \forall x, x' \in h, \iota(x) = \iota(x') \text{ and } A(x) = A(x'). \quad (2)$$

An information set may be a *singleton*—it may consist of only a single node. If every information set of every player is a singleton, we have a game of perfect information; i.e. $\forall h \in H, \#h = 1$, or equivalently: $\forall i \in I \cup \{0\}, \forall h \in H_i, \#h = 1$. If any information set contains more than one node—i.e. $\exists i \in I \cup \{0\}, \exists h \in H_i$, such that $\#h > 1$ —then the game is one of imperfect information. When an information set contains more than one node, we indicate this graphically on the extensive form by connecting all of the nodes in that information set with a dashed line. Singleton information sets require no special ornamentation. For example, the game in Figure 4 has perfect information; therefore

$$H_1 = \{\{\odot\}, \{\delta\}\}, \square H_2 = \{\{\alpha\}, \{\beta\}\}. \quad (3)$$

Return to the perfect information game of Figure 4. If we change the information structure so that player 1's first move is unobserved by player 2 by the time player 2 makes his first move, then he doesn't know at which of his two nodes he is located. We indicate this information imperfection by connecting these two nodes with a dashed line. (See Figure 5a.) Each of player 1's two nodes belongs to its own information set. In this game then there are three information sets: two belonging to the first player and one belonging to the second. Note that, as required, player 2 has the same two actions, viz. g and s , available at both nodes of his information set; $A(\alpha) = A(\beta) = \{g, s\}$. Contrast each player's collection of information sets in this game, viz.

$$H_1 = \{\{\odot\}, \{\delta\}\}, \square H_2 = \{\{\alpha, \beta\}\}, \quad (3)$$

with those given in (3) for the game in Figure 4.

The pseudo-extensive form of Figure 5b has two violations of the restrictions upon information sets. One information set consists of nodes belonging to different players. (Consider the information set $h = \{\alpha, \beta\}$, where $\iota(\alpha) = 2$ but $\iota(\beta) = 3$.) A second information set has different actions available to its

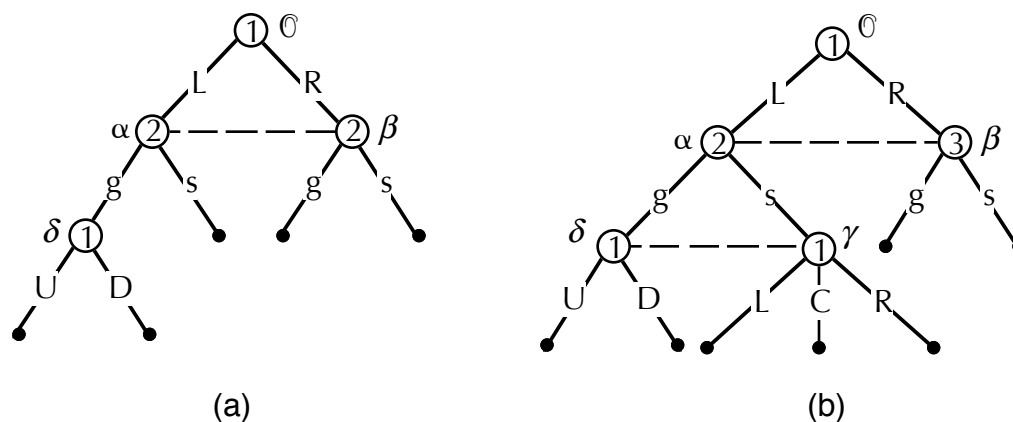


Figure 5: A game of (a) imperfect information and (b) implausible information.

player at different nodes. (Consider the information set $h' = \{\delta, \gamma\}$, where $A(\delta) = \{U, D\}$ but $A(\gamma) = \{L, C, R\}$.)

In summary, then, an extensive-form game Γ is defined by a tree (V, E) , a distinguished vertex $\mathbb{O}$, a set of actions $\bar{A}$, an action-for-edge assignment function $f: E \rightarrow \bar{A}$, an information partition H of the decision nodes X , a player set I , a node-ownership function $\iota: X \rightarrow I \cup \{0\}$, and player payoff functions $\mu_i: Z \rightarrow \mathbb{R}$ for $i \in I$. From this definition the following entities can be derived: the set X of decision nodes, the set Z of terminal nodes, the set of actions $A(x)$ available at each decision node $x \in X$, the player partition $\mathfrak{X}$, and the precedence relation $<$ on V . It is further required that, for each information set $h \in H$, every node in h shares the same player-owner and shares the same set of available actions: $\forall h \in H, \forall x, x' \in h, \iota(x) = \iota(x')$ and $A(x) = A(x')$.

Games of perfect recall

We've established some restrictions on what nodes we can include in the same information set: We can only draw dotted lines which connect nodes which belong to the same player and at which the same set of actions is available. Now I'll present two extensive-form fragments which might suggest additional restrictions.

In Figure 6a, at node α player 1 can choose Left and turn the move over to player 2 or she can choose Right and immediately move again. If she chooses Right, she reaches node β , which is in the same information set as the node α from which she had just come. Note what this particular information set construction implies: When player 1 chose Right at node α , she realized what she was doing. When she then reaches β , she doesn't know whether she's at β or back at α . Therefore she no longer knows that she chose Right when at α ; she has forgotten the action she just took.

It is typical in game theory to restrict ourselves to games of *perfect recall*. In such games a player always know what actions she has taken and never forgets anything she ever knew. How do we

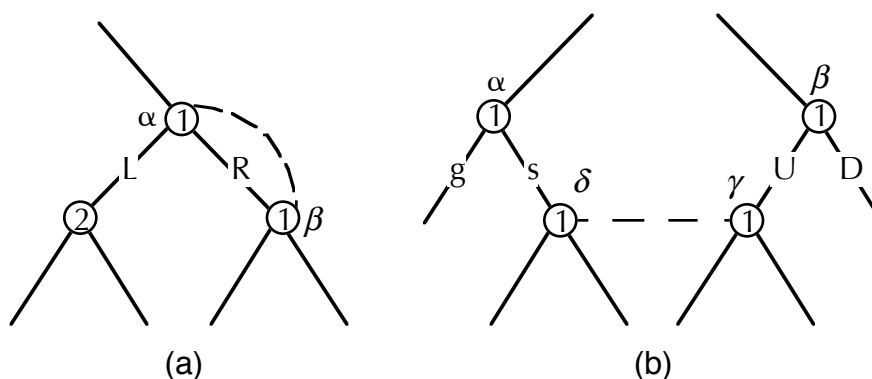


Figure 6: Two game fragments with forgetful players.

formalize this restriction?¹⁹ We can rule out the game fragment in Figure 6a with the following requirement: If two nodes are in the same information set, then neither can precede the other. ($\forall h \in H$, $\forall x, x' \in h$, neither $x < x'$ nor $x' < x$.)

However, that requirement isn't sufficient. Consider Figure 6b. Nodes δ and γ are both in the same information set. There are two ways player 1 can reach that information set: She can be at node α and choose stop, or she can start at node β and play Up. Note that α and β are not in the same information set; therefore they represent different information—different histories. If player 1 reaches α she knows that she reached α but never passed through β ; if she reaches β she knows she reached β but never passed through α . But if she reaches the $\{\gamma, \delta\}$ information set via node α , say, she cannot possibly know that she didn't pass through β . (If she did know this, then she would know for certain that she is at node δ . But that would violate the information set's knowledge restriction that she doesn't know which of the two nodes δ and γ she is at.) Therefore she has forgotten something she knew earlier; so this would be a violation of perfect recall.

The extensive-form fragment of Figure 6b doesn't violate the prohibition that members of the same information set should not be ordered by precedence; so we need a stronger restriction. The formal characterization of this restriction is extremely arcane and abstruse.²⁰ I'll try to motivate it the best I can.

Consider two nodes δ and γ which belong to the same information set for player 1. Now consider a player-1 node α which precedes δ when player 1 chooses the action a at α . (See Figure 7a.²¹) When

¹⁹ There are two reasons to be interested in perfect recall. The first is that most if not all games we would be interested in analyzing would satisfy this property. (But for precisely this reason there would be little point in formalizing the restriction.) More important theoretically, however, is that the restriction to games of perfect recall allows a simplification in the analysis of these games: We will see later that in games of perfect recall there is an equivalence between mixed strategies and *behavior* strategies such that we can choose whichever analytical perspective is more convenient.

²⁰ See Fudenberg and Tirole [1991: 81], Kreps and Wilson [1982: 867], Kreps [1990: 374–375], and Myerson [1991: 43–44]. The property so defined will prove sufficient for a later demonstration of the equivalence between mixed and behavior strategies. Kuhn [1953: 213] defines a weaker property, which he also calls perfect recall. Although this is a misnomer, his property is necessary and sufficient for the equivalence.

²¹ The “” indicates that α is a predecessor of δ but not necessarily an immediate predecessor.

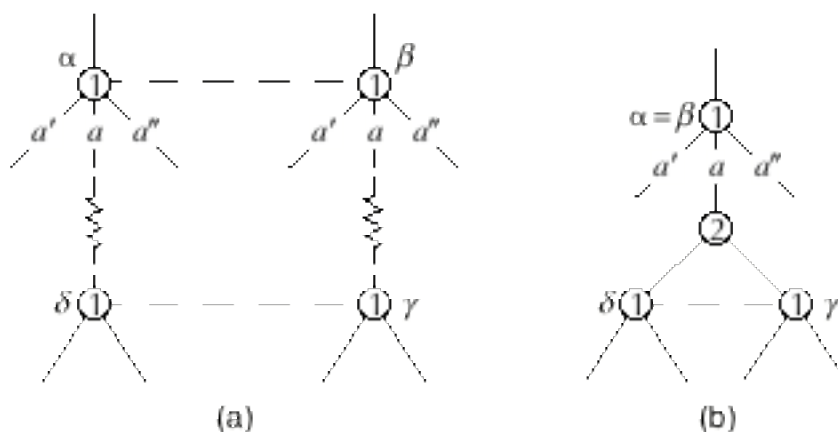


Figure 7: Establishing anti-amnesia requirements.

player 1 chooses a at α , she knows 1 that she has passed through α 's information set and 2 that she chose the action a at α 's information set. In order that the requirements of perfect recall be satisfied, she must still know these two facts when she gets to δ .

Indeed, even if she instead arrives at a different node, say γ , in δ 's information set, she must still know exactly the same information she would know at δ . (If she knew different facts at δ than at γ , she would be able to distinguish between the two nodes, contradicting that they are in the same information set.) Therefore at γ she must also know that she passed through α 's information set and chose a there. So we require that every node γ in δ 's information set have a predecessor β in α 's information set such that choosing a at β leads eventually to δ . (We don't rule out that α and β may be the same node. This case is shown in Figure 7b.)

Hopefully the following requirement which expresses that an extensive form satisfies perfect recall will now make sense: For any player- i nodes α , δ , and γ such that 1 δ and γ belong to the same information set and 2 α precedes δ via action a at α , it must be the case that there exists a player- i node β which belongs to α 's information set and which precedes γ via action a at β .²²

Before we state this more formally, let's develop one additional piece of notation. Consider a node α which precedes a node β . Let a be the action at node α which is required in order for play to arrive at β . Then we say that $\alpha \oplus a \Rightarrow \beta$. In other words the unique path which runs from α to β is of the form: $\{\alpha, a, \dots, \beta\}$.

Now we can say that a game satisfies perfect recall if: for all triples of decision nodes, $\alpha, \delta, \gamma \in X$, and for all actions $a \in A(\alpha)$ available at node α such that 1 $\iota(\alpha) = \iota(\delta) = \iota(\gamma)$, 2 $\hat{h}(\delta) = \hat{h}(\gamma)$, and 3 $\alpha \oplus a \Rightarrow \delta$, there exists a node $\beta \in \hat{h}(\alpha)$ such that $\beta \oplus a \Rightarrow \gamma$.

²² We earlier required that, if two nodes belong to the same information set, neither can precede the other. You can show that this original requirement is implied by the newer condition.

Arborescences

The graph-theoretic formulation of the extensive form we have presented here originated with Harold Kuhn [1953]. Kreps and Wilson [1982] offered an alternative formulation, where they replaced the standard definition of a tree with an entity called an *arborescence*.²³ They claim that their formulation is equivalent to Kuhn's. I will briefly present the relevant part of the Kreps-Wilson formulation in order to demonstrate that the arborescence requirement is unintentionally too weak to guarantee that Krepsian extensive forms are legitimate game trees. (Ratliff [1991])

We earlier defined a tree as a set of nodes and a set of branches (each of which joined a pair of nodes). Kreps and Wilson define an arborescence as a set of nodes and a binary relation $<$, called *precedence*, which is asymmetric, transitive, and such that the set of predecessors of a given node is totally ordered by $<$. This “totally ordered” condition means that: if two nodes α and β both precede a third node γ , then either α precedes β or β precedes α .²⁴

These properties of the precedence relation $<$ are well suited to rule out many node configurations which would not be legitimate trees. For example, transitivity combined with asymmetry rules out the “cycle in play” configuration in Figure 8a: From the diagram we see that $\alpha < \beta$, $\beta < \gamma$, $\gamma < \delta$ and $\delta < \alpha$. Transitivity then implies that $\alpha < \alpha$, which violates asymmetry.

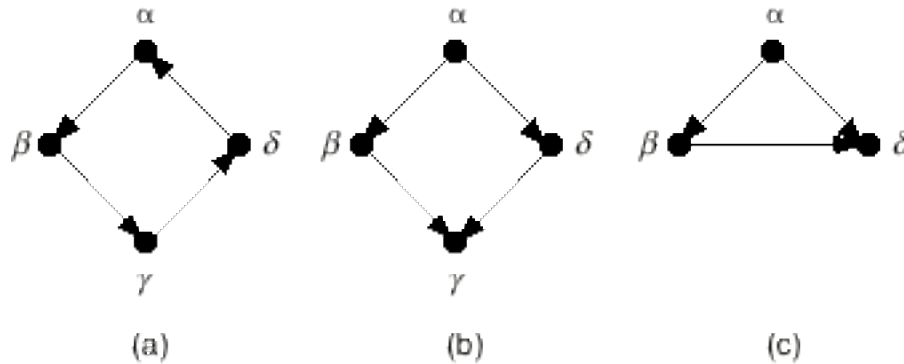


Figure 8: Three nontrees; two are captured by the Kreps-Wilson arborescence filter; one escapes.

The connected graph in Figure 8b is not unicursal; there are two paths joining α and γ , for example. This configuration is ruled out by the requirement that the precedence relation totally order the predecessors of any given node. To see this we observe that α , β , and δ are all predecessors of γ . However, the set $\{\alpha, \beta, \delta\}$ is not totally ordered by $<$, because β and δ are not related by precedence; i.e. we have neither $\beta < \delta$ nor $\delta < \beta$.

An even simpler nontree does slip through their filter, however. Consider the three-node connected

²³ See also Kreps [1990: section 11.2, especially pages 363–365].

²⁴ Note that these are exactly the properties we have already attributed to the precedence relation we defined in the graph-theoretic tree framework. What Kreps and Wilson are trying to do here is to go the other way: derive a tree from the binary relation itself.

graph in Figure 8c. We observe that the predecessors of each node are totally ordered by the precedence relation $<$: The node α has no predecessors, so satisfaction is trivial. The node β has α as its single predecessor; a singleton set is trivially ordered. The node δ has two predecessors, viz. α and β . These are ordered because $\alpha < \beta$. Therefore the connected graph of Figure 8c is an arborescence but is definitely not a tree, because it contains a circuit. Therefore there exist arborescences which are not trees; therefore an arborescence need not be a legitimate model for an extensive-form game.

Simultaneity

Consider the case in which two players will each take an action at the same time. Clearly an important implication of this temporal structure is that neither player can observe the other's choice before making her own choice. For example in Figure 9a we depict a game in which player 1 chooses from Up and Down and player 2 chooses from left and right, and these choices are made simultaneously.

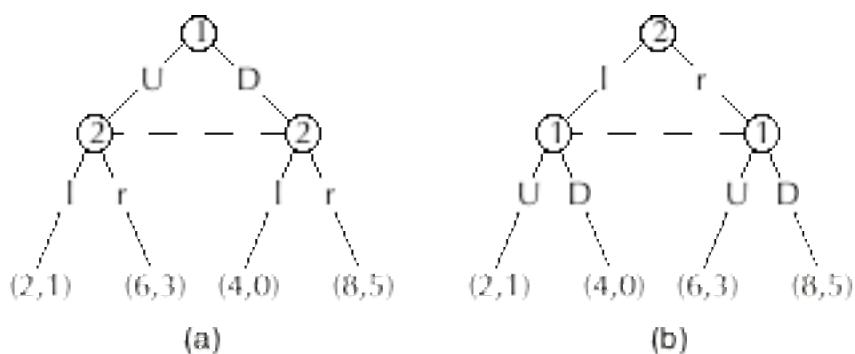


Figure 9: Two extensive forms of the same simultaneous-move game.

Our tree formalism requires that we represent the choices as being sequential. In order to provide equivalence between a sequential representation and a nonsequential game, we extract the relevant information structure from the simultaneous-move game and replicate it in the extensive form. In other words we connect player 2's nodes into a single information set. In this way we ensure that, even though his move is depicted as coming after player 1's, he does not observe player 1's move before he makes his own choice. By making his move in ignorance of his opponent's, player 2 is making his choice with exactly the same information he would have if the two players moved simultaneously. Note that it was arbitrary that we depicted player 1's choice as coming before player 2's. We could just as easily have reversed the roles. This would result in the extensive form of Figure 9b.

Divide and conquer: subgames

Extensive-form games can be very complicated. Tackling a huge game tree in a single bite can be an impossibly formidable undertaking. Fortunately, we can often identify pieces of large games which are themselves simpler games. If we first take the time to identify and understand these simpler pieces we can greatly ease our analytical burden. We will also be able to strengthen our to-be-introduced

extensive-form solution concepts by imposing them not only on the game as a whole but also upon these pieces of a game. The pieces of a game we seek are called *subgames*.²⁵

The concept of a subgame is intended to capture the notion of “today is the first day of the rest of your life.” A subgame is a subset of the original game. In order to define a subgame we impose conditions which guarantee that 1 the subset makes sense as a game in its own right and 2 it must be played under the same informational conditions under which that subset would be played if encountered in the original game. (I.e. if this subgame were reached in the larger game, it would be common knowledge that this is the game remaining to be played.)

Consider an extensive-form game Γ . In order to define the subgames of Γ we first define how we would *decompose* the game Γ at some decision node $v^* \in X$ to create the extensive-form object Γ^* .²⁶ (Γ^* may not be a legitimate game for our purposes, so I describe it agnostically as merely an object.) If v^* happens to be the initial node $\mathcal{O}$, then let $\Gamma^* = \Gamma$. Otherwise, find the unique edge which immediately precedes v^* and delete it. Now we’re left with two trees, only one of which contains v^* . (See Figure 10.) Let this tree be the graph $(\tilde{V}, \tilde{E})$, where $\tilde{V} \subset V$ and $\tilde{E} \subset E$. This is the subtree we want. Designate $v^* \in \tilde{V}$ as the initial node of Γ^* . We note that $\tilde{V}$ consists only of v^* and its successors in the original game. This implies that Γ^* is *closed under succession*: i.e. if $v \in \Gamma^*$ and $v' \succ v$ in Γ , then $v' \in \Gamma^*$.

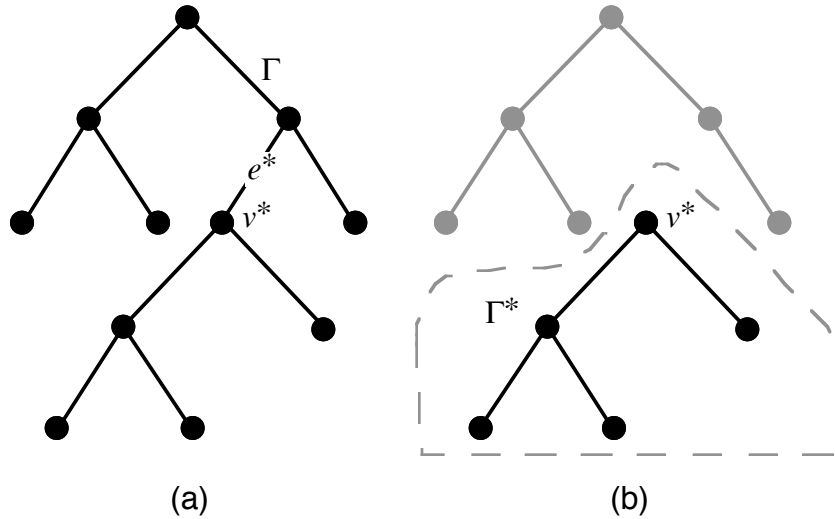


Figure 10: (a) A tree Γ . (b) Γ decomposed at v^* into two components by deleting the edge e^* .

To complete the specification of Γ^* we stipulate that it inherits much of the structure of the original extensive-form game Γ by *restriction* when necessary of the original entities to the new game tree.²⁷ Specifically we restrict the original action-for-edge assignment function f to the residual set of edges $\tilde{E}$,

²⁵ See Kreps and Wilson [1982: 865–869] and Fudenberg and Tirole [1991: 94–95].

²⁶ See Kuhn [1953: 203–204].

²⁷ Let $f: X \rightarrow Z$ be a function and let Y be a set with a nonempty intersection with the domain X , i.e. $X \cap Y \neq \emptyset$. Then we can define a function $\tilde{f}$, the *restriction* of f to Y , as a function whose domain is $X \cap Y$ and which agrees with f for all points in $X \cap Y$. I.e. $\tilde{f}: (X \cap Y) \rightarrow Z$ and $\forall x \in (X \cap Y), \tilde{f}(x) = f(x)$. In the special case where $Y \subset X$, $\tilde{f}: Y \rightarrow Z$ and $\forall x \in Y, \tilde{f}(x) = f(x)$.

and we restrict the node-ownership function ι and the payoff functions μ_i to the residual set of nodes $\tilde{V}$. We create a new information partition $\tilde{H}$ by replacing each information set $h \in H$ with $\tilde{h} = h \cap \tilde{V}$. The remaining entities—the residual set of decision nodes $\tilde{X} = X \cap \tilde{V}$ (note that any decision node $x \in X$ in the original game which also belongs to the new game is also a decision node in the new game), the residual set of terminal nodes $\tilde{Z} = Z \cap \tilde{V}$ (ditto for terminal nodes), the correspondence $A: \tilde{X} \rightarrow \tilde{A}$ describing the actions available at each decision node, and the binary relation $< \subset \tilde{V} \times \tilde{V}$ —can be derived from these restricted primitive objects.

Although Γ^* as we have defined it here *is* a well-formed game, it might not be a legitimate subgame for our purposes. We want to ensure that, if this new game were reached in the course of play of the original game Γ , it would be common knowledge to the players that it was indeed this game Γ^* remaining to be played. In particular we want a subgame to “preserve” or “respect” information sets: If $h \in H$ is an information set in the original game, then we want this information set to either appear completely intact as an information set in the new game or be totally excluded from the new game. We can state this requirement as $h \in H \Rightarrow (h \in \tilde{H} \text{ or } h \cap \tilde{X} = \emptyset)$.²⁸

Note that this implies—as long as Γ is a game of perfect recall—that the new initial node v^* must have been a singleton information set in the original game.²⁹ Note that the entire game is always a subgame of itself.³⁰

In summary, a subset of an extensive-form game (as formally defined by restriction above) forms a subgame if you can find a node v^* in the subset such that all of its successors and only its successors in the original game belong to this subset and “information sets are preserved.”³¹ This last condition requires that any information set of the original game is either totally absent from the subset or appears completely intact in the subset. The payoffs to the terminal nodes of the subgame are obtained by restriction of the payoffs of the original game to the terminal nodes of the subgame.

Figure 11 shows a game in extensive form with four conjectured subgames— A , B , C , and D —outlined.³² A and B are true subgames. However, C is not a subgame because it cannot be formed from the original tree through decomposition by the deletion of a single edge. (Alternatively: because for

²⁸ The extensive-form object Γ^* derived by decomposition from Γ is often called a *subgame* and, after imposing the restriction that information sets are preserved, becomes a *proper subgame*. However, proper subgames are the only kind of subgames we will be interested in. Therefore it would be quite inconvenient to have to attach the “proper” modifier every time we wanted to refer to one. Therefore I will use subgame only to refer to proper subgames in this sense.

²⁹ Γ^* contains only v^* and its successors in the original game. If v^* shared a common information set in Γ with a distinct node v , then v would be required to belong to Γ^* (by preservation of information sets) and hence v must be a successor of v^* in Γ . However, if Γ is a game of perfect recall, then all nodes within an information set in Γ must be unordered by precedence, contradicting that $v^* < v$.

³⁰ There is an occasional confusion about the meaning of “proper subgame”: the definitions in Kreps and Wilson [1982: 868] and Kreps [1990: 423] agree with the explicit statement in Fudenberg and Tirole [1991: 94–95] that “proper” does *not* refer to strict inclusion as it does in the term “proper subset.” Friedman [1990: 44] says indeed that a proper subgame *is* a subgame which is strictly smaller than the original game. Reader beware!

³¹ An alternative statement of the requirement that the subset contain exactly v^* and its successors would be: the subset is closed under succession and v^* precedes every other node in the subset.

³² The outlining boxes are not meant to indicate that the edge going into the first node is part of the conjectured subset. The conjectured subset begins with the first node itself.

any choice of decision node in C there would be nonsuccessors in C .³³) D is not a true subgame because it does not preserve information sets: the information set to which α belongs in the original game is neither wholly contained nor wholly excluded from the conjectured subgame.

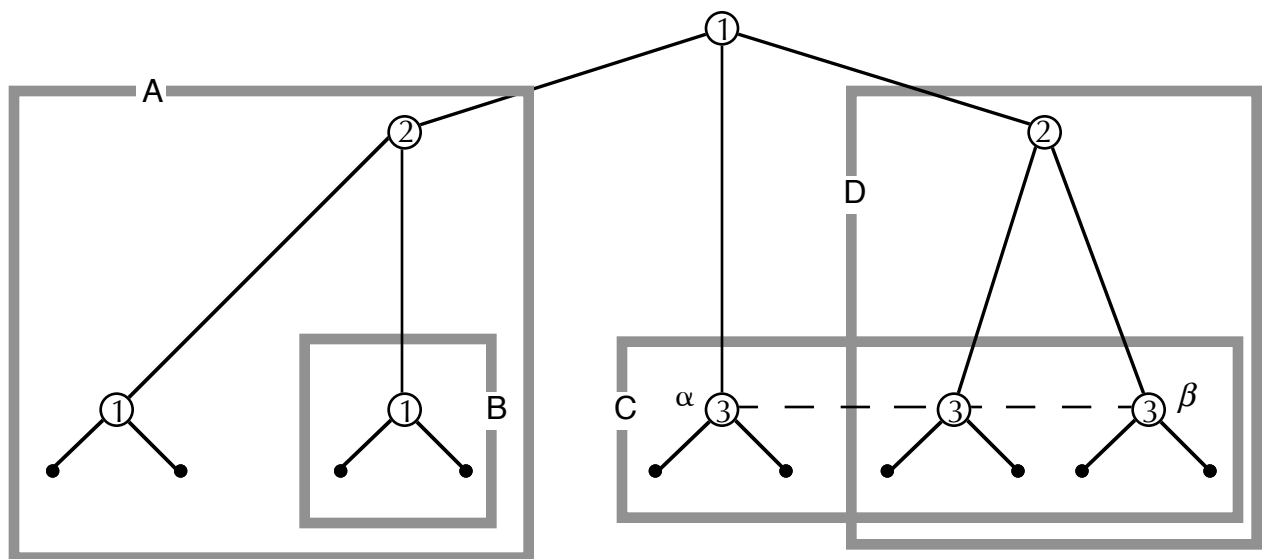


Figure 11: An extensive form and four conjectured subgames.

³³ Another alternative argument: although C is closed under succession, there does not exist a node in C which precedes all other nodes in C .

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Jim Ratliff's Game Theory Course:

4.2: Strategies in Extensive-Form Games

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Abstract



We define a strategy for a player in an extensive-form game as a specification for each of her information sets of the (pure or mixed) action she would take at that information set. One such strategy for each player constitutes a strategy profile for the extensive-form game.

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Every extensive-form game can be expressed as a strategic-form game.

We incorporate uncertain exogenous events into the extensive form by introducing Nature as a nonstrategic player who acts randomly. We learn how to construct the strategic-form of an extensive-form game when Nature takes a turn at bat.

We study two different types of randomized strategies in extensive-form games. The behavioral strategy specifies randomizations at each information set independently. The mixed strategy specifies randomizations over pure extensive-form strategies. Behavior strategies do not permit correlations across information sets that mixed strategies allow. However, we show that, in extensive-form games satisfying *perfect recall*, mixed and behavior strategies can be used interchangeably in a precise sense. We learn how to convert back and forth between the two types of randomized strategies.

We define how to restrict an extensive-game strategy to a particular subgame.



[Previous section:
§4.1: Introduction to Extensive-
Form Games](#)

[Course Outline](#)

[Next section:
§4.3: Solution Concepts in
Extensive-Form Games](#)

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Recapitulation

We are now studying *dynamic* games—in which some decisions are made after others. We use the extensive form because it allows us to make explicit this temporal structure as well as to define the game's information structure. (The results of previous decisions by one player may not be observable immediately, if ever, to her opponents.)

We introduced the *game tree* as the supporting framework for the extensive form. Choices are made at *decision nodes*, which belong to the set X . One of these, viz. $\odot$, is designated the *initial node*, at which the game begins. It is assigned either to one of the players in the player set $I = \{1, \dots, n\}$ or to Nature; Nature would make a random choice representing any exogenous uncertainty by some or all of the players. Play progresses from node to node based on the players' decisions. The game ends when any one of the *terminal nodes*, which belong to the set Z , is reached, at which point the players are awarded the payoffs corresponding to that node. Player i 's preferences over terminal nodes are represented by her von Neumann-Morgenstern utility function $\mu_i: Z \rightarrow \mathbb{R}$; namely, if terminal node $z \in Z$ is reached, her utility is $\mu_i(z)$.

When called upon to move, a player might not know precisely at which of her nodes she is located. Her uncertainty is modeled by *information sets*. An information set $h \in H$ is a set of nodes, all belonging to the same player and at all of which the same set of actions is available. When called upon to act, a

player is informed only of the information set which has been reached, i.e. that she is located at some node in that information set. We typically impose additional constraints on the structure of the information partition to assure that the game is one of *perfect recall*—one in which a player always knows what move she takes and she never forgets anything she ever knew.

We saw that an extensive game could often be decomposed into smaller parts called *subgames*, which are subtrees which respect information sets. A subgame makes sense as a game by itself, and the subgame embodies the informational conditions under which it would be played if reached in the original game. (If the subgame were reached in the larger game, it would be common knowledge that this was the game remaining to be played.)

Strategy profiles of extensive-form games

A player's *strategy* specifies what action she would take whenever called upon to move. To be more precise: A player's strategy is a specification of what action she would implement *at each of her information sets*. We require the strategy to be in general a function of the information set because knowledge of the information set which has been reached is all the information a player has when called upon to move, and we certainly couldn't let a player's decision depend on information she doesn't have.

For example, in the game depicted in Figure 1, player 1 has four information sets: α , δ_1 , δ_2 , and δ_3 , so $H_1 = \{\alpha, \delta_1, \delta_2, \delta_3\}$.¹ (Note that information set δ_3 contains two nodes, while the other three information sets are singletons.) Therefore a strategy for player 1 would be a 4-tuple of strategies, containing one strategy for each of her information sets, which we could write in the following form:

$$s_1 = (s_1(\alpha), s_1(\delta_1), s_1(\delta_2), s_1(\delta_3)), \quad (1)$$

where each $s_1(h)$ is a specification of her action at the information set $h \in H_1$. For example consider the strategy

$$s_1 = (Y, D, D, U), \quad (2)$$

which is graphically depicted by the thick gray line segments in Figure 2. (Note that the “U” for the doubleton information set δ_3 generates two thick line segments, because there are two nodes in that information set and therefore two nodes at which that information set's action would be taken.)

¹ Note that in the previous handout: “Extensive-Form Games,” we commonly labeled nodes in a tree. Here I'm labeling information sets. E.g. “ α ” here is the singleton information set which contains the initial node: $\alpha = \{\odot\}$.

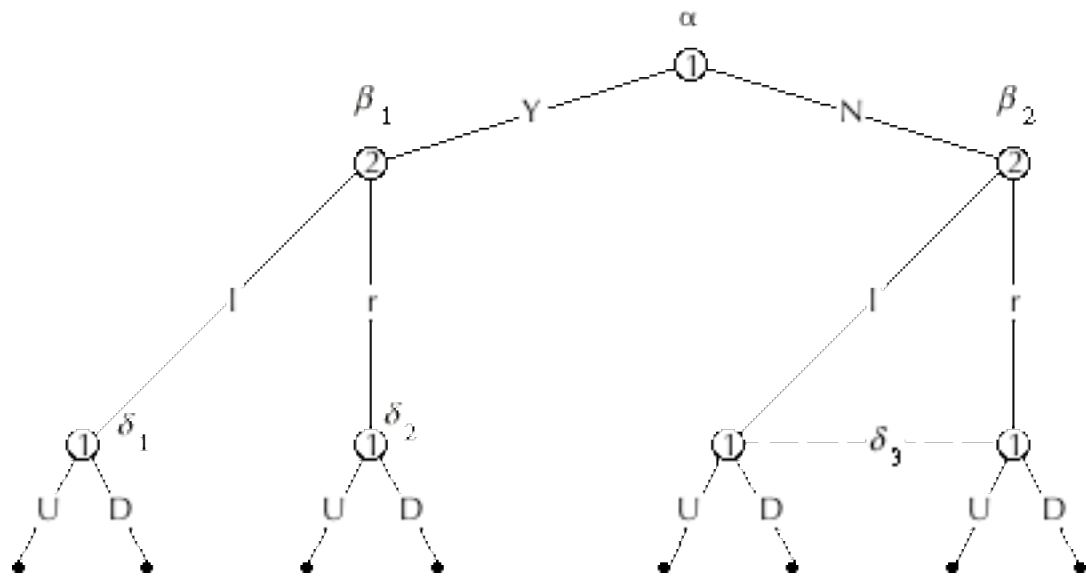


Figure 1

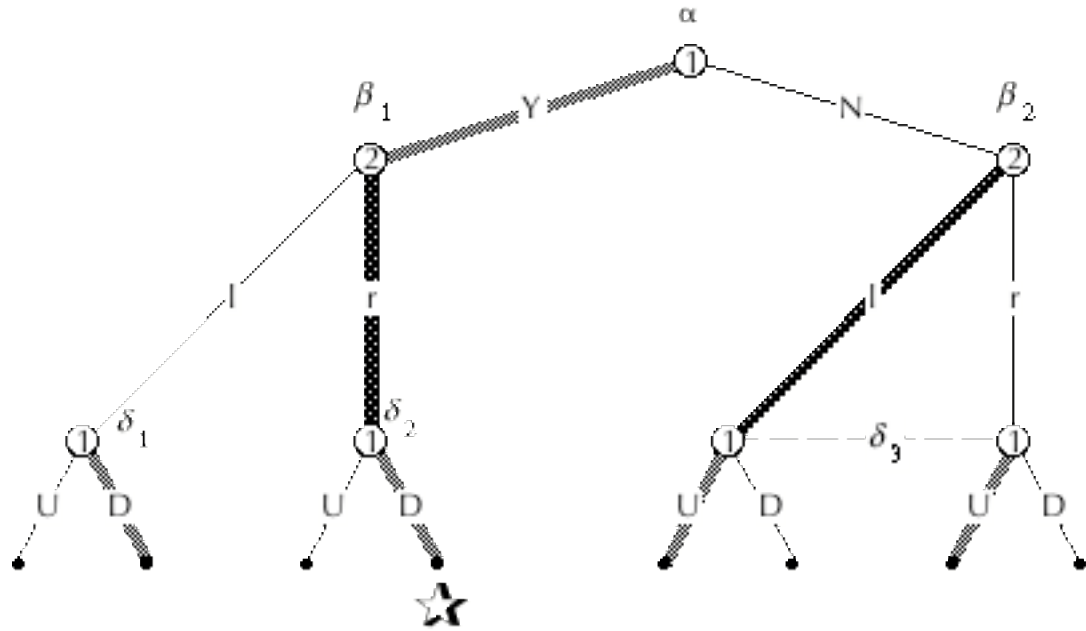
Similarly, player 2 has two information sets β_1 and β_2 (both singletons), i.e. $H_2 = \{\beta_1, \beta_2\}$. His strategies are of the form

$$s_2 = (s_2(\beta_1), s_2(\beta_2)). \quad (3)$$

For example the strategy

$$s_2 = (r, l), \quad (4)$$

is graphically depicted by the darker thick line segments in Figure 2.

Figure 2: The strategy profile $(Y, D, D, U; r, l)$.

A *strategy profile* is an n -tuple of individual strategies, one strategy for each player. Specifying a strategy profile completely determines the outcome of the game. We can write a strategy profile as a concatenation of the strategies for each player; i.e. it is of the form:

$$s = (s_1(\alpha), s_1(\delta_1), s_1(\delta_2), s_1(\delta_3); s_2(\beta_1), s_2(\beta_2)).^2 \quad (5)$$

The strategy profile corresponding to our exemplar strategies (2) and (4) would be

$$s = (Y, D, D, U; r, l). \quad (6)$$

This strategy profile is represented in Figure 2 by the union of the gray segments and the darker thick line segments. Notice that the graphical representation of this strategy profile is not connected; i.e. there are disjoint pieces. There is only one component of this union which reaches from the initial node α to a terminal node.³ This is the *path* which corresponds to the strategy profile s . Its outcome is the starred terminal node.

We can be more formal in defining a player's extensive-form strategies. Recall that H_i is the set of player i 's information sets and, for any player- i information set $h \in H_i$, $A(h)$ is the set of actions available to her at that information set.⁴ A strategy s_i for player i specifies an action for each information set; therefore for each $h \in H_i$ we represent player i 's choice at h by $s_i(h) \in A(h)$. A strategy s_i is an $\#H_i$ -tuple of actions. The set of actions which player i has available *somewhere* is the union of the actions she has

² The semicolon serves to separate the two strategies in the profile.

³ You might find it challenging to prove this in general using graph theory.

⁴ See "Extensive-Form Games."

available at each information set, viz. $\bar{A}_i \equiv \bigcup_{h \in H_i} A(h)$; so we can describe her strategy as a function $s_i: H_i \rightarrow \bar{A}_i$. Her strategy space is the Cartesian product

$$S_i = \prod_{h \in H_i} A(h). \quad (7)$$

As we did in the case of strategic-form games, we denote the space of strategy profiles by $S \equiv \prod_{i \in I} S_i$.

Each strategy profile $s \in S$ results in a particular terminal node $z \in Z$ being reached with certainty. We denote the identity of this terminal node by $\hat{z}(s)$; i.e. $\hat{z}: S \rightarrow Z$ is a function from strategy profiles into terminal nodes. Player i 's utility associated with this strategy profile is her utility for the terminal node reached when that strategy profile is played. Therefore we define for each player i the function $u_i: S \rightarrow \mathbb{R}$, which specifies player i 's utility for any pure-strategy profile by: $\forall s \in S$,

$$u_i(s) \equiv \mu_i(\hat{z}(s)). \quad (8)$$

As usual we let $u: S \rightarrow \mathbb{R}^n$ be the n -tuple of individual utility functions u_i .

The strategic form of an extensive-form game

Recall that when we studied strategic-form games we specified a game by a triple: (I, S, u) , where I was the player set, S was the space of strategy profiles, and u was an n -tuple of player utility functions $u_i: S \rightarrow \mathbb{R}$.⁵ In this framework, strategy was a primitive concept: A pure-strategy s_i had no internal structure and a player's strategy space S_i was part of the specification of the game. In the extensive form, on the other hand, the game tree, the player partition $\mathcal{X}$ (which assigns decision nodes to players), the information partition H (which assigns decision nodes to information sets), and the sets of actions $A(x)$ available at the decision nodes $x \in X$ are primitive specifications of the game. A player's strategy space S_i is *derived* from these according to (7).

How then can we construct a strategic-form representation of an extensive-form game? We need to provide the constituents of the (I, S, u) triple. The player set I can obviously be supplied directly. Likewise, once we have constructed each player i 's strategy space S_i from the extensive-form primitives according to (7), we can simply supply their Cartesian product S as the space of strategy profiles for the strategic-form representation. To complete the strategic-form specification we pass along the utility function $u: S \rightarrow \mathbb{R}^n$ constructed from the utility functions u_i from (8).

One consequence of this one-to-one correspondence between strategic-form and extensive-form strategies is that we can find the pure-strategy Nash equilibria of an extensive-form game in extensive-form strategies by finding the pure-strategy Nash equilibria of the game's strategic form. (Consider a conjectured Nash-equilibrium pure-strategy profile s^* . When viewed as extensive-form strategies, player i will have a better strategy than s_i^* against s_{-i}^* if and only if she does when they are viewed as

⁵ See "Strategic-Form Games."

strategic-form strategies.)

Example: Constructing the strategic form of an extensive-form game.

Consider the extensive-form game in Figure 3a. Player 1's strategy space is $S_1 = \{U, D\}$; player 2's is $S_2 = \{(l, l), (l, r), (r, l), (r, r)\}$, where we write a typical strategy for player 2 as a pair: (action if player 1 chooses U , action if player 1 chooses D).

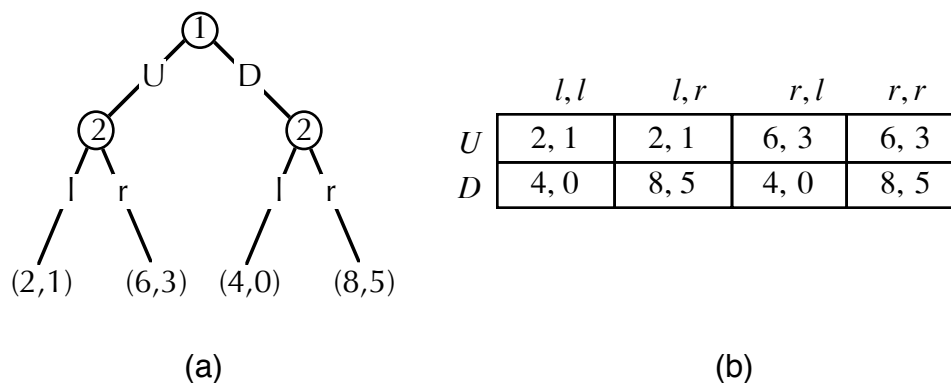


Figure 3: A game in (a) extensive form and (b) strategic form.

In order to cast an extensive-form game into strategic form we simply identify each extensive-form pure strategy s_i with a strategic-form pure strategy. To construct the bimatrix representation of this game we assign player 1 to rows and player 2 to columns, listing each player's strategies along her side of the bimatrix. To find the payoff vector corresponding to any given cell (strategy profile) we identify the path determined by that strategy profile; the payoff vector located at that terminal node is the one entered into the bimatrix. For example, consider the strategy profile $(D; r, l)$. This strategy pair is displayed in Figure 4; the payoff vector $(4, 0)$ located at the end of its path is entered into the corresponding cell of the bimatrix of Figure 3b. The payoff vectors for the other strategy profiles are calculated similarly.

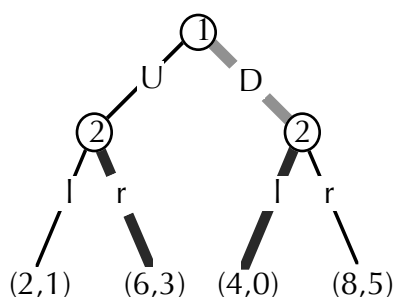


Figure 4: The strategy profile $(D; l, r)$.

Note that there are four payoff vectors in the extensive-form representation in the game in Figure 3a, and there are eight payoff vectors in its strategic form. For any strategy choice by player 1, one of player 2's information sets will be "off the path." For example, when player 1 chooses Up, player 2's right-hand information set is off the path. Player 2's choice at an off-the-path information set is irrelevant to

the payoff. This explains why more than one strategy profile can result in the same outcome—for example why the payoff vectors corresponding to $(U; l, l)$ and $(U; l, r)$ are the same.

Several extensive forms can have the same strategic form. For example, you can easily show that both extensive forms in Figure 5, which represent a simultaneous-move game, have the same strategic form.

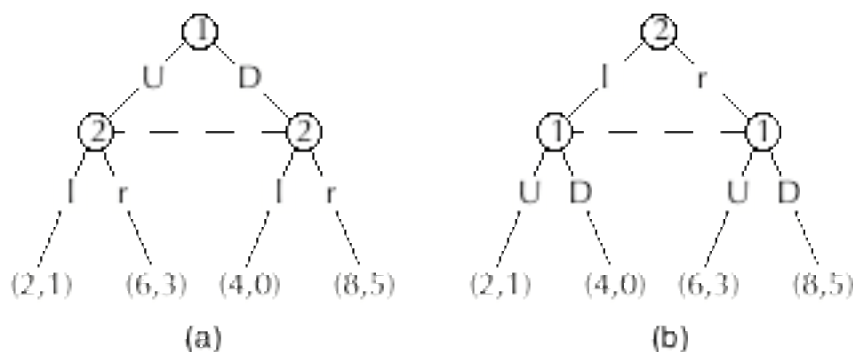


Figure 5: Two extensive forms of the same simultaneous-move game.

When Nature takes her turn at bat

Our game-playing agents do not always completely control their destinies; significant events can occur which are exogenously determined. If these events occur and are revealed to the players in such a way as to make them common knowledge prior to the commencement of the game, the occurrences can be built into the structure and payoffs of the extensive-form game. In this class of cases our present framework is adequate without enhancement.

However, it is possible that the players know that some event has, will, or might occur, but that one or more of the players are uncertain, when called upon to move, of exactly which event occurred. For example, when two oligopolistic firms simultaneously choose their output levels for their new no-cholesterol French fries, they are uncertain whether the University of Arizona College of Medicine will soon issue a report showing that cholesterol is actually health promoting. (See, for example, Allen [1973].) In an alternative scenario one firm, thanks to a spy in the relevant research laboratory, would know whether or not this will happen, while the other firm would not be privy to this information.

We incorporate uncertain exogenous events into the extensive form by introducing Nature as a nonstrategic player who acts randomly. By saying that Nature is nonstrategic I mean that her action is not influenced by the actions of the “real,” strategic players. Nature is not an optimizer; we don’t need to specify payoffs for her. Figure 6 represents the extensive form for the two-firm case of the above-described game in which neither firm knows whether bad news will be published.⁶

⁶ Note that all four of player 2’s nodes are in one information set. He can see neither Nature’s move nor his opponent’s.

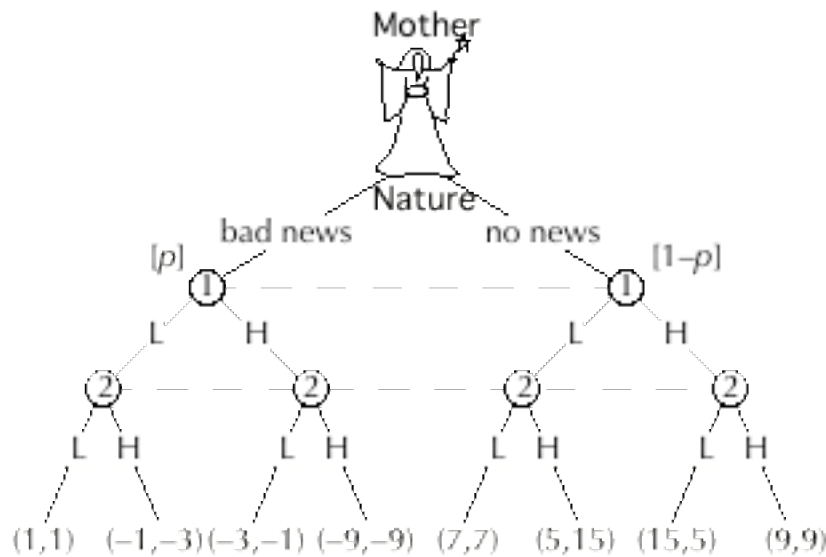


Figure 6: Cholesterol: friend or foe?

How do we construct the strategic form of an extensive-form game when Nature has a turn? In the game of Figure 6 Nature's random choice induces a probability distribution over the nodes in player 1's information set. Let p be the probability that Nature sends bad news. Each player has one information set and two actions at each information set; therefore each player's strategy space contains two strategies. Therefore the strategic form will be a 2×2 bimatrix. Let $p = \frac{1}{2}$ and let's compute the expected payoff vector to the pure-strategy profile $(L; H)$; i.e. when firm 1 chooses the Low quantity and firm 2 chooses the High quantity. (This strategy profile is depicted in Figure 7.)

There are two terminal nodes which can be reached when player 1 chooses L and player 2 chooses H , viz. the second and sixth from the left with payoff vectors $(-1, -3)$ and $(5, 15)$, respectively. The first of these is reached under the pure-strategy profile (L, H) when Nature sends bad news; the second is reached when "no news is good news." Therefore the expected payoff vector to (L, H) is

$$\frac{1}{2}(-1, -3) + \frac{1}{2}(5, 15) = (2, 6). \quad (9)$$

The expected payoff vectors for the other three pure-strategy profiles are calculated the same way. This strategic form is shown in Figure 8.

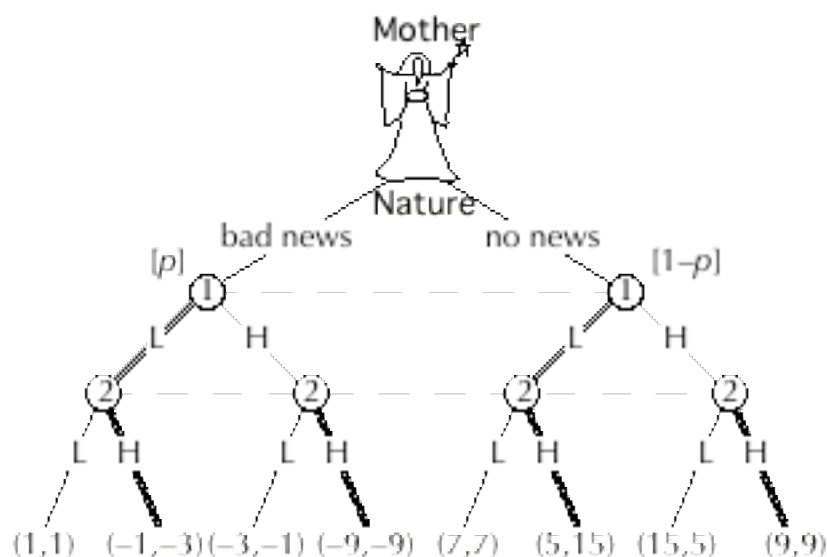


Figure 7: Firm 1 chooses Low quantity and firm 2 chooses High quantity.

	L	H
L	4,4	2,6
H	6,2	0,0

Figure 8: The strategic form of the extensive form of Figure 6, when Nature sends bad news half of the time.

Randomized strategies in extensive-form games

Behavior strategies

When we studied strategic-form games we saw that randomized strategies, viz. mixed strategies, were useful because randomized strategies could be Nash equilibria and because without randomized strategies Nash equilibria could fail to exist. We will find randomized strategies useful in the extensive form as well. We define a natural conception of strategic randomization in the extensive form called *behavior strategies*, in which players randomize independently at each information set.

It is easy to accept that there is a one-to-one correspondence between pure strategies in the extensive form and pure strategies in the strategic form. It is less obvious what correspondence exists between behavior strategies in the extensive form and mixed strategies in the strategic form. We will see that there is a useful correspondence, though not one-to-one, provided our game is one of perfect recall: any probability distribution over outcomes which can be achieved by a mixed strategy can be achieved by a behavior strategy and vice versa. This result gives us freedom to analyze an extensive-form game in whichever framework—strategic or extensive form—is more convenient. In particular we can find the set of Nash equilibria in an extensive-form game by computing the mixed-strategy Nash equilibria in the

strategic form.⁷

How should we define a randomized strategy in the extensive form? A pure strategy specifies a particular action at each of a player's information sets. One natural extension of this concept is the *behavior strategy*, which specifies at each information set a conditional probability distribution over the actions available at that information set; i.e. it specifies the probability with which each action would be taken conditional upon that information set being reached. (A pure strategy would be a degenerate case in which at each information set the probability distribution at that information set placed full weight upon a single action available at that information set.) We stipulate that the randomizations at the player's information sets are performed independently of one another.⁸

For example, in the game from Figure 1, a possible behavior strategy for player 1 would specify that she play Y two-thirds of the time at information set α , play D with certainty at information set δ_1 , mix with equal probability at information δ_2 , and play U four-fifths of the time at each of the two nodes in information set δ_3 . Referring to her information sets in the same order as in the pure-strategy specification (1), we could write player 1's behavior strategy as

$$b_1 = (\frac{2}{3} \circ Y \oplus \frac{1}{3} \circ N, D, \frac{1}{2} \circ U \oplus \frac{1}{2} \circ D, \frac{4}{5} \circ U \oplus \frac{1}{5} \circ D). \quad (10)$$

Similarly, referring to (3), we could write a behavior strategy for player 2 as

$$b_2 = (\frac{1}{2} \circ l \oplus \frac{1}{2} \circ r, \frac{3}{4} \circ l \oplus \frac{1}{4} \circ r). \quad (11)$$

Rather than indicate a deterministic path as we did in Figure 2 for the pure-strategy profile in (2), we depict the behavior-strategy profile $b = (b_1, b_2)$ by labeling in brackets each branch of the tree with the probability that the branch is chosen conditional on its source node being reached. See Figure 9.

What are these behavior strategies formally? Player i 's behavior strategy b_i specifies at each information set $h \in H_i$ a probability for each action $a \in A(h)$ available at h . We denote this probability by $b_i(a | h)$. The set of probability distributions over the set of actions $A(h)$ is $\Delta(A(h))$.⁹ Therefore for each $h \in H_i$, $b_i(\cdot | h) \in \Delta(A(h))$. Therefore for every player $i \in I$ and for every information set $h \in H_i$,

$$\sum_{a \in A(h)} b_i(a | h) = 1. \quad (12)$$

Player i 's behavior strategy b_i specifies such a distribution for each information set in the finite set H_i . Therefore we can write B_i , the space of player- i behavior strategies, as the Cartesian product:

⁷ Kuhn [1953] is the seminal paper concerning these issues. See also Kreps [1990: 380–384], Fudenberg and Tirole [1991: 85–90], and Myerson [1991: 154–163] for discussions of behavior strategies and their relation to mixed strategies.

⁸ Note that this is a stronger sense of independence than we encountered with mixed strategies of strategic-form games. There we assumed that each player randomized independently from every other player. Here we are saying in addition that each player randomizes at each of her information sets independently from her randomizations at other information sets.

⁹ For any finite set T we denote by $\Delta(T)$ the set of probability distributions over T .

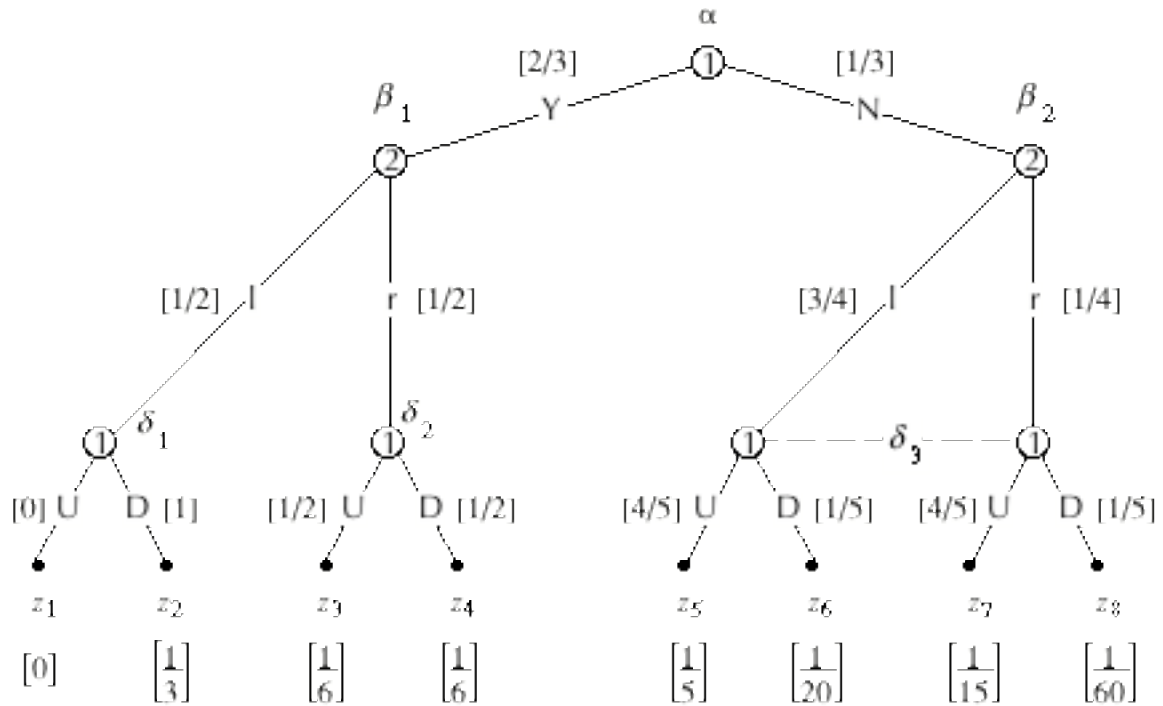


Figure 9: The conditional probabilities specified by, and the probability distribution over outcomes determined by, the example behavior-strategy profile.

$$B_i = \sum_{h \in H_i} \Delta(A(h)). \quad (13)$$

[Compare this to (7), which describes player i 's pure-strategy space.]

When a behavior-strategy profile is played, there need not be a single terminal node reached with probability one; instead there will be a probability associated with each terminal node. The resulting probability distribution over outcomes is of interest because it determines the expected payoffs to the players. To compute the probability distribution over terminal nodes associated with the behavior-strategy profile we calculate the probability with which each path will be traversed. For a particular terminal node z we identify the unique path from the initial node to z . At each node along this path a particular action is required to keep the play on this path. We refer to the relevant player's behavior strategy to obtain the probability with which this required action is chosen conditional on reaching that node's information set. Because the randomizations at the various information sets are independent of one another, the probability that this path is traversed from start to finish is the product of the probabilities associated with the required actions at all the nodes along the path.

For example in Figure 9 to arrive at terminal node z_3 player 1 must choose Y at information set α , player 2 must choose r at β_1 , and player 1 must finally choose U at δ_2 . The probabilities associated with these actions, according to the behavior-strategy profile $b = (b_1, b_2)$ from (10) and (11), are $\frac{2}{3}$, $\frac{1}{2}$, and $\frac{1}{2}$, respectively. Their product, $1/6$, is the probability attached to terminal node z_3 . The entire probability distribution over terminal nodes is calculated in the same way. The resulting probabilities are displayed

in brackets next to each terminal node in Figure 9. Note that, as required, the sum of these terminal-node probabilities is one.

Let's formalize this computation a little. Consider any terminal node z . The path from the initial node to z encounters a set of nonterminal nodes, which we denote by $\hat{X}(z)$. Each nonterminal node $x \in \hat{X}$ along this path belongs to some player $\bar{I}(x) \in I$, and to some information set $h_x \equiv \hat{h}(x) \in H_{\bar{I}(x)}$. At each node $x \in \hat{X}(z)$, a particular action $a \in A(h_x)$ is required to keep the play on the path to z ; call this required action $\hat{a}(x, z) \in A(h_x)$. The probability that player $\bar{I}(x)$ chooses the action $\hat{a}(x, z)$, conditional on reaching x 's information set h_x , is $b_{\bar{I}(x)}(\hat{a}(x, z) | h_x)$. The probability associated with the terminal node z when the behavior-strategy profile $b = (b_1, \dots, b_n)$ is played, then, is

$$\mathbb{P}(z | b) = \prod_{x \in \hat{X}(z)} b_{\bar{I}(x)}(\hat{a}(x, z) | \hat{h}(x)). \quad (14)$$

Mixed strategies

Implicit within our discussion of extensive-form games has been another notion of randomized strategies. When we transformed an extensive-form game into strategic form, we identified each extensive-form pure strategy with a strategic-form pure strategy. Randomized strategies in the strategic form are *mixed* strategies: probability distributions over pure strategies. Therefore mixed strategies—in the strategic-form sense—are an alternative to behavior strategies as a vehicle to introduce randomization into extensive-form play.

At first glance, behavior and mixed strategies seem distinctly different. When a player implements a mixed strategy, she spins the roulette wheel a single time; the outcome of this spin determines which pure strategy (set of deterministic choices at each information set) she will play. When she implements a behavior strategy, she independently spins the roulette wheel every time she reaches a new information set. The space of player- i behavior strategies B_i is given in (13). The space of player- i mixed strategies is given by $\Sigma_i = \Delta(S_i)$, which we can rewrite in terms of the actions available at each of her information sets using (7) and then contrast to B_i :

$$B_i = \prod_{h \in H_i} \Delta(A(h)), \quad \Sigma_i = \Delta \left(\prod_{h \in H_i} A(h) \right). \quad (15)$$

We will see that mixed strategies permit correlations across information sets which behavior strategies cannot accommodate (because they involve only independent randomizations). So it's not at all obvious that a restriction of attention to behavior strategies would not involve a loss of generality. However, we will also see that—in games of perfect recall—the two types of randomized strategies *can* be used interchangeably in the sense that any probability distribution over outcomes which can be achieved by a mixed strategy can be achieved by a behavior strategy and vice versa. We will find as this course progresses that mixed strategies in the strategic form will provide a more useful perspective in some contexts and behavior strategies in the extensive form will be more useful in others.

First we define what it means for two strategies to be equivalent. We will see how, in games of perfect recall, a mixed strategy generates an “essentially unique” equivalent behavior strategy. Then we will see how, given a behavior strategy, we can find an equivalent mixed strategy. In fact we will see that the same behavior strategy can be generated by many different mixed strategies. We will summarize with a theorem which guarantees that, for games of perfect recall, any probability distribution over outcomes which can be achieved by a behavior strategy can be achieved by a mixed strategy and vice versa.

Consider a strategy profile $\gamma = (\gamma_1, \dots, \gamma_n)$, where, for each player $i \in I$, γ_i may be either a pure, mixed, or behavior strategy. The strategy profile γ implies a probability distribution over terminal nodes. Consider two strategies for player i , e.g. γ_i' and γ_i'' , $\gamma_i', \gamma_i'' \in (B_i \cup \Sigma_i)$ —perhaps one mixed and one behavior. We say that γ_i' and γ_i'' are *equivalent* if, no matter what strategies $\gamma_{-i} \in \prod_{j \in I \setminus \{i\}} (B_j \cup \Sigma_j)$ the opponents choose, the strategy profiles (γ_i', γ_{-i}) and $(\gamma_i'', \gamma_{-i})$ both induce the same probability distribution over terminal nodes.

Mixed strategy $\rightarrow$ behavior strategy: the role of perfect recall

A behavior strategy $b_i \in B_i$ for player i is a specification, for every player- i information set $h \in H_i$ and every action $a \in A(h)$ feasible at h , of a conditional probability for that action a at the information set h . To say that the behavior strategy $b_i \in B_i$ and the mixed strategy $\sigma_i \in \Sigma_i$ are equivalent would require that, if player i has chosen the mixed strategy σ_i , then for every $h \in H_i$ and every $a \in A(h)$, $b_i(a|h)$ is the probability that player i would choose action a whenever she reaches information set h .

Note that nothing in the above paragraph refers to what the other players are doing. Equivalence between a mixed and behavior strategy, then, requires that the conditional probability distribution over actions at any information set implied by a particular mixed strategy be independent of the strategy choices of the opponents. Now I'll show you an extensive-form game in which such an equivalence cannot possibly hold in general because the conditional probability distribution associated with a particular mixed strategy depends crucially on the strategy of the opponent.

Consider the two-player game in Figure 10. Player 1's strategy space is $S_1 = \{(l, a), (l, b), (r, a), (r, b)\}$. Let h be player 1's two-node information set. Consider the player-1 mixed strategy $\sigma_1 = \frac{1}{2} \circ (l, a) \oplus \frac{1}{2} \circ (r, b)$. If player 2 chooses D , one-half the time player 1 will be playing the pure strategy (l, a) and the left-hand node of information set h will be reached, at which player 1 will then choose a . The other half of the time player 1 is playing the pure strategy (r, b) , in which case the right-hand node of h is reached, at which player 1 chooses b . Therefore if player 2 chooses D , the probability of player 1 choosing a conditional on reaching h is one-half; i.e. $\mathbb{P}(a|h) = \mathbb{P}(b|h) = \frac{1}{2}$.

However, if player 2 chooses U , information set h is reached only half of the time, viz. when player 1's randomization results in the pure strategy (r, b) . So, if player 2 chooses U then, *conditional on information set h being reached*, player 1 will always choose b at h ; i.e. $\mathbb{P}(a|h) = 0$ and $\mathbb{P}(b|h) = 1$. Therefore the conditional probabilities at information set h are *not* independent of player 2's strategy.

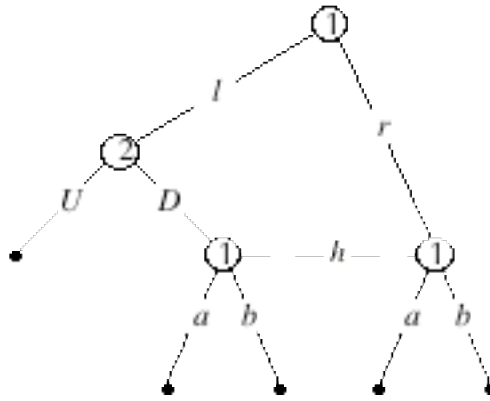


Figure 10: The conditional probability distribution over actions at an information set implied by a mixed strategy is not necessarily independent of the opponent's strategy.

Player 1's mixed strategy allowed her to direct probability from information set h in different directions depending on whether player 2's strategy was relevant. (When player 1 chose r , player 2's strategy was irrelevant, and player 1 chose b at information set h . When player 1 chose l , player 2's strategy was relevant, and player 1 chose a at information set h .) As a result, the direction in which probability flows out of information set h , and thus the conditional probability distribution over actions at h , depends on the action of player 2.

Even when the probability distribution over actions implied by a mixed strategy is independent of the opponents' strategies, it is possible that the behavior strategy which implements those conditional probabilities at each information set will generate a terminal-node distribution which differs from the distribution produced by the mixed strategy; i.e. the behavior and mixed strategies would not be equivalent.

To see this consider the one-player game in Figure 11.¹⁰ Consider the mixed strategy $\sigma_1 = \frac{1}{2} \circ (U, l) \oplus \frac{1}{2} \circ (D, r)$. This induces the probability distribution over terminal nodes $\frac{1}{2} \circ z_1 \oplus \frac{1}{2} \circ z_4$. The information set α is always reached; in one-half the cases player 1 chooses U ; in the others she chooses D . Therefore the conditional probability distribution over actions at information set α is given by $\mathbb{P}(U|\alpha) = \mathbb{P}(D|\alpha) = \frac{1}{2}$. The information set β is always reached; in one-half the cases player 1 chooses l ; in the others she chooses r . Therefore the conditional probability distribution over actions at information set β is given by $\mathbb{P}(l|\beta) = \mathbb{P}(r|\beta) = \frac{1}{2}$. The behavior strategy b_1 which implements these conditional probabilities is defined by $b_1(U|\alpha) = b_1(D|\alpha) = b_1(l|\beta) = b_1(r|\beta) = \frac{1}{2}$. However, this behavior strategy results in the probability distribution over terminal nodes in which each node is reached $\frac{1}{4}$ of the time. In fact, no behavior strategy can duplicate the distribution achieved by the mixed strategy. Any such behavior strategy would have to play each of U and D and each of l and r with nonzero probability at α and β , respectively. However, since the behavior strategy cannot distinguish between the two nodes in information set β , such a behavior strategy must put nonzero probability upon reaching each of z_2 and z_3 .

¹⁰ Of course it seems unreasonable to call this a game, since there is only one player. One player suffices to make the point, and I want to keep the situation as simple as possible in order not to cloud the issue with needless complexities.

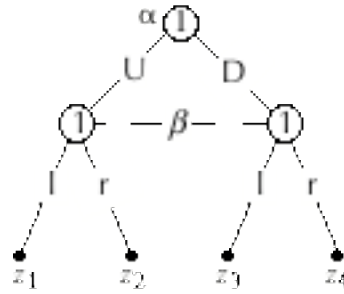


Figure 11: No behavior strategy can replicate the terminal-node distribution of the mixed strategy $\sigma_1 = \frac{1}{2} \circ (U, l) \oplus \frac{1}{2} \circ (D, r)$.

In the game of Figure 11 player 1's mixed strategy allows her to create a different conditional probability distribution over actions at each of the nodes in information set β : At the left-hand node she chooses l with probability one; at the right-hand node she chooses l with probability zero. A behavior strategy on the other hand imposes the same conditional probability distribution over all nodes within the same information set. It is not surprising then that the added flexibility of mixed strategies can generate probability distributions over terminal nodes which cannot be duplicated by behavior strategies.

The games in Figure 10 and Figure 11 were troublesome for finding a behavior strategy which was equivalent to some mixed strategy. A common feature of both games is that they do not satisfy perfect recall. (In both cases, at player 1's second information set, she has forgotten the action she took at the initial node.) A mixed strategy specifies a probability distribution over pure strategies, and each pure strategy can stipulate a particular pair of actions at any pair of information sets. When two player- i nodes within the same information set are distinguished by an earlier action of player i 's, a mixed strategy can effectively dictate a different action at each node, even though those nodes are in the same information set. In games of perfect recall this is not possible: If one player- i information set follows another, then every node of the later information set must be reached by the same action at the earlier information set.

Requiring that our games satisfy perfect recall certainly squashes the counterexamples of Figure 10 and Figure 11. It turns out that requiring perfect recall is sufficient to squash all other counterexamples as well; it is sufficient to guarantee that mixed and behavior strategies are equivalent in the sense we have defined.

Given that mixed strategies can correlate behavior across information sets, why should behavior strategies be able to duplicate every terminal-node distribution which can be generated by mixed strategies? Consider two information sets h and h' for a player. There are only two classes of cases of interest: either 1 h and h' cannot be reached along the same path [i.e. every pair of nodes $(x, x') \in h \times h'$ is unordered by precedence] or 2 there is a path which encounters h and later encounters h' (or vice versa). If there is no path which reaches both h and h' , a behavior strategy's inability to correlate its randomizations at these two information sets does not limit the distributions over terminal nodes it can achieve, because at most one of these information sets will be reached in any single play of the game,

and the randomization at the unreached information set has no influence on the outcome. If there is a path which encounters first h , say, and then h' , then a behavior strategy could effectively correlate its actions at the two information sets by conditioning its h' action on its knowledge that h was reached first. But this correlation would require that the player remember she had reached h , and this recollection is guaranteed when the game is one of perfect recall.

Mixed strategy $\rightarrow$ behavior strategy: computing the conditional probabilities

Assume now, and for the rest of the semester (unless directed otherwise on an exam or problem set) that our game satisfies perfect recall. I will show heuristically how to take a given mixed strategy for a player and derive an equivalent behavior strategy; then I will formalize, generalize, and qualify the presentation. (For a more rigorous, nonheuristic demonstration, see the Appendix.) Consider the game in Figure 12. The players' strategy spaces are $S_1 = \{(L, U), (L, D), (R, U), (R, D)\}$ and $S_2 = \{g, s\}$. Consider any mixed-strategy profile $\sigma = (\sigma_1, \sigma_2)$, where, for any player $i \in \{1, 2\}$ and any player- i pure strategy $s_i \in S_i$, $\sigma_i(s_i)$ is the probability with which player i chooses the pure strategy s_i . What are the corresponding behavior strategies b_1 and b_2 ?

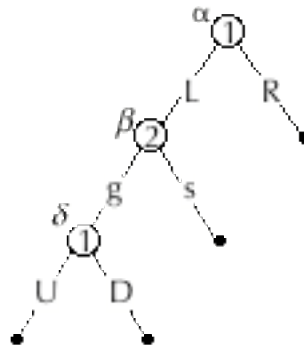


Figure 12: Generating a behavior strategy from a mixed strategy.

We can think of probability as a flow that goes into a node and then is parceled out among its outgoing branches. Computing the conditional probability of a particular action at a node is the same as asking what fraction of the entering probability exits along that action's branch. We first seek $b_1(L|\alpha)$, the probability with which player 1 chooses L at the initial node α conditional upon her reaching node α . (Obviously the conditioning is ineffectual at this node, since player 1 always reaches it.) The only probability which exits node α along the L branch is that associated with pure strategies which specify that player 1 choose L at α , viz. (L, U) and (L, D) . Therefore

$$b_1(L|\alpha) = \frac{\sigma_1((L, U)) + \sigma_1((L, D))}{\sigma_1((L, U)) + \sigma_1((L, D)) + \sigma_1((R, U)) + \sigma_1((R, D))} = \sigma_1((L, U)) + \sigma_1((L, D)). \quad (16)$$

The denominator is the sum of the probabilities associated with the pure strategies entering α . Because α has no predecessors belonging to player 1, this includes all of player 1's pure strategies; hence the denominator sums to unity.

Similarly, player 2's conditional probability of choosing g at node β is

$$b_2(g|\beta) = \frac{\sigma_2(g)}{\sigma_2(g) + \sigma_2(s)} = \sigma_2(g). \quad (17)$$

Note that we ignored the extent to which player 1 might be choosing L or R at α when calculating the denominator of (17), which I said was the probability entering β . So I'll now be more precise: the denominator is the probability which enters β over which player 2 has control.¹¹

Calculating player 1's local randomization toward U at δ , viz. $b_1(U|\delta)$, is more interesting. In this case we must take more care when identifying the probability which enters δ . We use the notion of "a strategy which does not preclude a particular information set." For example, the pure strategies for player 1 (R, U) and (R, D) preclude play ever reaching δ . (L, U) and (L, D) do not preclude information set δ .¹² In this case the denominator is the sum of the mixing probabilities associated with the pure strategies which do not preclude δ being reached. The numerator is the sum of the probabilities of the subset of these strategies which require player 1 to choose U , viz. (L, U) . Therefore

$$b_1(U|\delta) = \frac{\sigma_1((L, U))}{\sigma_1((L, U)) + \sigma_1((L, D))}. \quad (18)$$

Note a problem with the formula in (18): What if the mixed strategy σ_1 we are considering puts no weight upon either (L, U) or (L, D) ? Then the denominator would vanish, leaving the conditional probability undefined. In this case our specification of $b_1(U|\delta)$ and $b_1(D|\delta)$ is arbitrary, subject to the constraint that $b_1(\cdot|\delta)$ constitutes a probability distribution over the actions $A(\delta)$ available at δ . This requires only that $b_1(U|\delta) + b_1(D|\delta) = 1$. When the mixed strategy we are considering precludes an information set being reached, it is conventional that we determine the local behavior there by summing the probabilities associated with all the pure strategies which specify the desired action at the information set in question (whether any one of them precludes the information set or not).¹³ In this example, the pure strategies (L, U) and (R, U) both specify U at information set δ ; therefore, if $\sigma_1((L, U)) = \sigma_1((L, D)) = 0$, we would define

$$b_1(U|\delta) = \sigma_1((L, U)) + \sigma_1((R, U)) = \sigma_1((R, U)). \quad (19)$$

Now that we have seen in a simple example how a mixed strategy σ_i generates a behavior strategy b_i , we can formalize the process. For every information set $h \in H_i$ which belongs to player i , let $\hat{S}_i(h) \in S_i$ be the subset of player i 's pure-strategy space S_i whose strategies do not preclude h from being reached. In

¹¹ We can ignore the extent to which player 1 played L at α because that multiplicative effect on the incoming probability in the numerator is exactly the effect on the outgoing probability, and therefore the two effects divide out. This is shown explicitly in the Appendix.

¹² Choosing one of these strategies does not guarantee that play will reach δ , because player 2 could choose s . We are only interested in partitioning player 1's pure-strategy space into 1 strategies which by themselves (i.e. regardless of her opponent's play) rule out a particular information set being reached and 2 the remainder, for each of which there exists a strategy by her opponent such that the resulting strategy profile reaches the information set in question.

¹³ This is a simple way to guarantee that $b_i(\cdot|h)$ is a probability distribution over the actions at h .

other words, if $s_i \in \hat{S}_i(h)$, then there exists a deleted pure-strategy profile $s_{-i} \in S_{-i}$ such that the strategy profile (s_i, s_{-i}) reaches information set h . (I.e. $\hat{X}(\hat{z}(s_i, s_{-i})) \cap h \neq \emptyset$.)

Consider now some player $i \in I$ and one of her mixed strategies $\sigma_i \in \Sigma_i$. For any information set $h \in H_i$ we want to compute a conditional probability distribution $b_i(\cdot | h)$ over the actions $A(h)$ available at h . There are two cases: either 1 the mixed strategy σ_i is *compatible* with reaching h —it puts positive weight on at least one of the pure strategies which do not preclude h ; i.e. $\exists s_i \in \hat{S}_i(h)$ such that $\sigma_i(s_i) > 0$ or 2 σ_i puts no weight on these strategies; i.e. $\text{supp } \sigma_i \cap \hat{S}_i(h) = \emptyset$. When the mixed strategy is compatible with reaching h , the conditional probability of choosing some action $a \in A(h)$ is the ratio of the probability leaving along the branch a to the incoming probability to h over which i has control; otherwise we make the aforementioned conventional, arbitrary assignment when h is never reached by σ_i . In other words,

$$b_i(a | h) = \begin{cases} \frac{\sum_{\substack{s_i \in \hat{S}_i(h) \\ \text{s.t. } s_i(h)=a}} \sigma_i(s_i)}{\sum_{s_i \in \hat{S}_i(h)} \sigma_i(s_i)}, & \text{supp } \sigma_i \cap \hat{S}_i(h) \neq \emptyset, \\ \sum_{\substack{s_i \in S_i \\ \text{s.t. } s_i(h)=a}} \sigma_i(s_i), & \text{supp } \sigma_i \cap \hat{S}_i(h) = \emptyset. \end{cases} \quad (20)$$

You can verify that $\sum_{a \in A(h)} b_i(a | h) \equiv 1$ for every $i \in I$, $h \in H$, and $\sigma_i \in \Sigma_i$.¹⁴

We say that the mixed strategy $\sigma_i \in \Sigma_i$ determines an “essentially unique” behavior strategy $b_i \in B_i$ because the determination is unique at information sets which are compatible with σ_i . Although there is a high degree of arbitrariness in the definition at other information sets, the assignments at those are inconsequential because they do not affect the probability distribution over terminal nodes.

Behavior strategy $\rightarrow$ mixed strategy

Now we take a given behavior strategy b_i for player i and construct an equivalent mixed strategy σ_i . The mixed strategy we construct will be but one of many equivalent mixed strategies. We’ll look for a mixed strategy σ_i such that 1 at any information set h , the player’s choice of action from $A(h)$ has the marginal probability distribution specified by the behavior strategy, viz. $b_i(\cdot | h)$ and 2 the player’s choice at any information set is made independently of her choice at any other information set.

Remember that a pure strategy $s_i \in S_i$ in the extensive form is a specification at every information set

¹⁴ Implicit in the lower branch of the definition of $b_i(a | h)$, viz. the $\text{supp } \sigma_i \cap \hat{S}_i(h) = \emptyset$ case, is a denominator which sums $\sigma_i(s_i)$ over all pure strategies $s_i \in S_i$. This sum is equal to one, so it is omitted.

$h \in H_i$ of a feasible action $s_i(h) \in A(h)$. What is the probability that player i 's randomizations according to the behavior strategy b_i result in a realization of s_i ? Such a realization would require that at every player- i information set $h \in H_i$ the realized action was that specified by the pure strategy s_i , viz. $s_i(h)$. The probability that the action $s_i(h)$ is the realized action at h is $b_i(s_i(h) | h)$. Because the randomizations are independent across information sets, the probability that at every information set h the realized action is $s_i(h)$ is the product of the probabilities $b_i(s_i(h) | h)$. In other words symbols, the probability that s_i is realized by the behavior strategy b_i is

$$\mathbb{P}(s_i | b_i) = \prod_{h \in H_i} b_i(s_i(h) | h).$$

A mixed strategy $\sigma_i \in \Sigma_i$ would be equivalent to the behavior strategy b_i if it specified that each player- i pure strategy $s_i \in S_i$ was chosen with the same probability with which the pure strategy s_i would be realized by the behavior strategy b_i , i.e. if, for all $s_i \in S_i$, $\sigma_i(s_i) = \mathbb{P}(s_i | b_i)$ and therefore if

$$\sigma_i(s_i) = \prod_{h \in H_i} b_i(s_i(h) | h). \quad (21)$$

To verify that σ_i as defined in (21) really is equivalent to the behavior strategy b_i you can substitute (21) into (20) to obtain an identity in $b_i(a | h)$.¹⁵

Many mixed strategies give rise to the same distribution over outcomes

Consider the game in Figure 13. Consider two mixed strategies for player 2:

$$\sigma_2 = \frac{1}{4} \circ (L, U) \oplus \frac{1}{4} \circ (L, D) \oplus \frac{1}{4} \circ (R, U) \oplus \frac{1}{4} \circ (R, D), \quad (22)$$

$$\sigma_2' = \frac{1}{2} \circ (L, U) \oplus \frac{1}{2} \circ (R, D). \quad (23)$$

You can easily verify that both mixed strategies give exactly the same behavior strategy— $b_1(L | \alpha) = b_1(U | \delta) = \frac{1}{2}$, as computed by (16) and (18).

What is the origin of this degeneracy? Consider two information sets h and h' for player i on distinct paths (i.e. so that no node of h precedes or succeeds a node of h'). A mixed strategy allows player i to correlate his randomizations at the two information sets. For example, one-half the time she may choose the pure strategy that plays a at h and a' at h' , and the other half of the time she plays the pure strategy which chooses b and b' at h and h' , respectively. Never will she play, for example, a pure strategy which prescribes a at h and b' at h' . However, since h and h' are on distinct paths, player i will never reach both information sets in a single play of the game. How she correlates her randomizations across these two information sets is irrelevant to the distribution over terminal nodes. What's important for behavior strategies is only the marginal distribution over actions at each information set. Many different mixed strategies can result in identical marginal distributions.

¹⁵ You might also find it challenging to show directly that the sum of $\sigma_i(s_i)$, as defined in (17), over all $s_i \in S_i$ is unity.

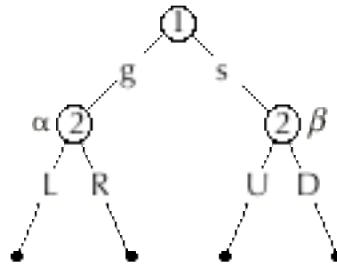


Figure 13

For example, the matrix in Figure 14a represents mixed strategies by player 2 in the game of Figure 13. The probability in each cell is the value of $\sigma_2(s_i)$ for one of the four pure strategies (L, U) , (L, D) , (R, U) , and (R, D) . The sum of each row is the marginal probability of that action at node α . The sum of each column is the marginal probability of that action being chosen at β . The marginal probabilities p_L , p_R , p_U , and p_D define a behavior strategy. In Figure 14b the probability of (L, U) is set to t and the entries in the remaining cells are constructed so as to satisfy the given marginal probabilities. We see that the parameter t is free to vary somewhat while still satisfying the marginal probabilities: there are many ways to construct mixed strategies which yield these marginal probabilities. For example, let all four marginal probabilities equal one-half. Then any mixed strategy of the form

$$\sigma_i = t \circ (L, U) \oplus (\tfrac{1}{2} - t) \circ (L, D) \oplus (\tfrac{1}{2} - t) \circ (R, U) \oplus t \circ (R, D), \quad (24)$$

where $t \in [0, \frac{1}{2}]$, will generate these marginal probabilities. The choices for σ_2 and σ_2' in (22) and (23) were but two examples from this family.

		U	D			U	D		
L		$\sigma_2((L, U))$	$\sigma_2((L, D))$	p_L	L	t	$p_L - t$	p_L	
R		$\sigma_2((R, U))$	$\sigma_2((R, D))$	$1 - p_L$	R	$p_U - t$	$1 - p_L - p_U + t$	$1 - p_L$	
		p_U	$1 - p_U$	1		p_U	$1 - p_U$	1	
		(a)				(b)			

Figure 14: (a) Representing player 2's mixed strategy with one information set's action on each dimension. (b) The one-parameter family of mixed strategies corresponding to particular marginal probabilities.

Kuhn's theorem: The equivalence of behavior and mixed strategies

We defined two strategies to be equivalent if, for any set of strategies by the other players, they result in the same distribution over terminal nodes. We just saw that, given a behavior strategy, we could construct many mixed strategies which are equivalent. Before that, for a given mixed strategy, we constructed an equivalent behavior strategy; however, we were unable to do so when the game was not one of perfect recall.

This exposition was merely suggestive and does not constitute a proof of anything. Kuhn [1953] proves the following theorem: Every behavior strategy is generated by some mixed strategy and every mixed strategy generates an essentially unique behavior strategy. If the game has perfect recall, then

every behavior strategy is equivalent to any mixed strategy which generates it.^{16,17}

A consequence of this theorem is that we can find the Nash equilibria of a game in whichever form—extensive or strategic—is more convenient. Consider a conjectured Nash-equilibrium mixed-strategy profile σ^* . If σ_i^* is not a best response to σ_{-i}^* , then there exists a $\bar{\sigma}_i$ such that $(\bar{\sigma}_i, \sigma_{-i}^*)$ yields player i a better distribution over terminal nodes (and hence a higher expected payoff). Now consider the behavior-strategy profile b^* such that, for all $j \in I$, σ_j^* and b_j^* are equivalent strategies. There exists a behavior strategy $\bar{b}_i$ which is equivalent to $\bar{\sigma}_i$. Therefore $(\bar{b}_i, b_{-i}^*)$ yields player i a higher expected payoff than b^* . So we see that a mixed-strategy profile is a Nash equilibrium if and only if an equivalent behavior-strategy profile is.

The restriction of a strategy to a subgame

Often when we analyze subgames we want to know what a strategy profile in the original game implies about play in the subgame. Let's discuss the subgame Γ' , of the game Γ from Figure 1, which is indicated by the shaded box in Figure 15. The strategy profile s of Γ from (6) is *not* a strategy profile for this subgame Γ' , because it specifies actions at information sets which don't even exist in the subgame. To make sense of what it would mean to discuss the strategy profile s with regard to this subgame we first *restrict* s , creating a new strategy profile s' , by throwing away all those actions which correspond to information sets that don't belong to the subgame. (We also say that s' is the *restriction of s to the subgame*.) In other words, s' is of the form

$$s' = (s_1(\delta_1), s_1(\delta_2); s_2(\beta_1)),^{18} \quad (25)$$

and corresponding to our particular example in (6) we write

$$s' = (D, D; r). \quad (26)$$

More generally.... Recall that, when we decompose a game in order to form a subgame, we create a new information partition $\tilde{H} \subset H$ by restriction of the original partition H to the residual set of nodes $\tilde{V}$; in other words $\tilde{H}$ contains those information sets of the original game which are also in the subgame.¹⁹ $\tilde{H}_i \subset \tilde{H}$ is the set of information sets in the subgame which belong to player $i \in I$.

¹⁶ See also Myerson [1991: 202–204] for another proof.

¹⁷ Kuhn uses a slightly weaker definition of perfect recall than we (and others) have used—and which is actually a misnomer because it tolerates some amnesia—and proves that his flavor of perfect recall is necessary and sufficient, rather than just sufficient as we state here, for the equivalence of behavior and mixed strategies. See Kreps [1990: 374–375] for a discussion.

¹⁸ This may seem an odd order of actions, because β_1 comes before δ_1 and δ_2 in the game tree. However, this is consistent with specifying player 1's strategy followed by 2's strategy.

¹⁹ See “Extensive-Form Games.”

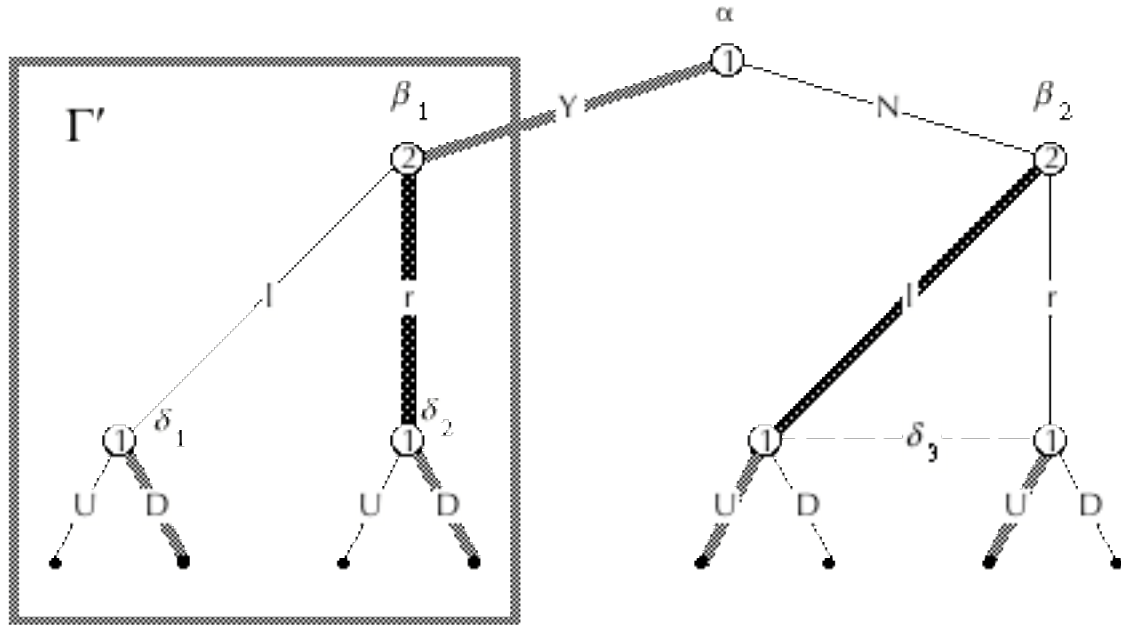


Figure 15. Strategy profile restricted to a subgame.

A pure strategy s_i for player i is a map from player- i information sets to feasible actions. Given any player- i pure strategy $s_i: H_i \rightarrow A_i$ in the original game, we form $\tilde{s}_i$, the restriction of s_i to the subgame, by restricting the domain of s_i to the new set of player- i information sets $\tilde{H}_i$; i.e. $\tilde{s}_i: \tilde{H}_i \rightarrow \bar{A}_i$ and $\forall h \in \tilde{H}_i$, $\tilde{s}_i(h) = s_i(h)$. Player i 's pure-strategy space in the subgame then is

$$\tilde{S}_i = \prod_{h \in \tilde{H}_i} A(h). \quad (27)$$

[Compare this to (7).]

A behavior strategy for player i is a map from player- i information sets to probability distributions over feasible actions; i.e. $\forall h \in H_i$, $b_i(\cdot | h) \in \Delta(A(h))$. Given any player- i behavior strategy b_i , we form its restriction to the subgame $\tilde{b}_i$ by restricting the domain of player- i information sets over which it is defined to the set $\tilde{H}_i$ of player- i information sets in the subgame: $\forall h \in \tilde{H}_i$, $\tilde{b}_i(\cdot | h) = b_i(\cdot | h) \in \Delta(A(h))$. Player i 's space of subgame behavior strategies then is

$$\tilde{B}_i = \prod_{h \in \tilde{H}_i} \Delta(A(h)). \quad (28)$$

[Compare this to (13).]

Return to the example behavior profile $b = (b_1, b_2)$ defined in (10) and (11) for the game of Figure 15 (as depicted in Figure 9). Similar to what we did in the pure-strategy case, to restrict the behavior strategy b to the subgame Γ' we simply throw out those probability distributions over actions which are

defined at information sets of the original game which are not present in the subgame. The restricted strategies are:

$$\tilde{b}_1 = (D, \tfrac{1}{2} \circ U \oplus \tfrac{1}{2} \circ D), \quad (29)$$

$$\tilde{b}_2 = (\tfrac{1}{2} \circ l \oplus \tfrac{1}{2} \circ r). \quad (30)$$

Note, when comparing (27) to (7) and comparing (28) to (13), that the restricted strategy spaces $\tilde{S}_i$ and $\tilde{B}_i$ are simply projections of the original spaces S_i and B_i , respectively, onto the restricted set of player- i information sets $\tilde{H}_i$.

Now I'll explain how to restrict a mixed strategy to a subgame—NOT! This is why we use behavior strategies in the first place. It's hard to make sense of what it would mean to restrict a mixed strategy to a subgame.

Appendix: Mixed strategy → Behavior strategy

Consider player i 's mixed strategy $\sigma_i \in \Sigma_i$ in an extensive-form game with perfect recall. Our task is to compute the conditional probability $b_i(a|h)$ with which player i will choose action a at information set h for every player- i information set $h \in H_i$ at which this is defined and action $a \in A(h)$ which is feasible at h . (Implicit in the statement of the problem is the belief that these conditional probabilities will be independent of the deleted strategy profile $\sigma_{-i} \in \Sigma_{-i}$ played by player i 's opponents.) If any behavioral strategy is equivalent to the mixed strategy σ_i , it must be consistent with the $b_i(a|h)$ so defined.²⁰ In fact, Kuhn's [1953] theorem guarantees that this behavior strategy is indeed equivalent to the mixed strategy σ_i .

For any node $v \in V$, let $\hat{X}(v) \subset X$ be the set of decision nodes on the unique path from the initial node to v .²¹ Let $\hat{H}(v) \subset H$ be the set of information sets encountered on the path $\hat{X}(v)$; i.e. $\hat{H}(v) \equiv \hat{h}(\hat{X}(v)) = \bigcup_{x \in \hat{X}(v)} \hat{h}(x)$. For any pure-strategy profile $s \in S$, let $\tilde{X}(s) \subset X$ be the set of decision nodes on the path of s ; i.e. $\tilde{X}(s) \equiv \hat{X}(\hat{z}(s))$.²² Let $\tilde{H}(s) \subset H$ be the set of information sets encountered on the path $\tilde{X}(s)$; i.e. $\tilde{H}(s) \equiv \hat{h}(\tilde{X}(s))$.

Our event space is the space S of pure-strategy profiles. Let $\hat{S}(x) \subset S$ be the event (i.e. set of strategy profiles such) that node $x \in X$ is reached; i.e.

$$\hat{S}(x) = \{s \in S: x \in \tilde{X}(s)\}.^{23} \quad (\text{A.1})$$

Let $\hat{S}(h) \subset S$ be the event that information set h is reached, i.e.

$$\hat{S}(h) = \{s \in S: h \in \tilde{H}(s)\} = \{s \in S: \tilde{X}(s) \cap h \neq \emptyset\} = \bigsqcup_{x \in h} \hat{S}(x).^{24} \quad (\text{A.2})$$

Because of perfect recall, all the nodes in any information set are unordered by precedence and therefore no strategy profile can generate a path which contains two distinct nodes of the same information set. I.e. $\forall h \in H, \forall x, x' \in h$ such that $x \neq x'$,

$$\hat{S}(x) \cap \hat{S}(x') = \emptyset.^{25} \quad (\text{A.3})$$

For any decision node $x \in X$, let $\hat{S}_i(x) \subset S_i$ be the projection of $\hat{S}(x) \subset S$ onto player i 's strategy space

²⁰ A behavioral strategy b_i' which is equivalent to σ_i only needs to agree with b_i at information sets at which the conditional probability is defined.

²¹ Recall that V is the set of all nodes and is partitioned into decision nodes X and terminal nodes Z .

²² Recall that $\hat{z}(s) \in Z$ is the terminal node reached by strategy profile $s \in S$.

²³ Therefore $s \in \hat{S}(x) \Leftrightarrow x \in \tilde{X}(s)$; i.e. strategy profile s encounters node x if and only if node x is encountered by strategy profile s .

²⁴ Therefore $\hat{S}(h)$ is just the image of h under $\hat{S}$, so there is no abuse of notation here.

²⁵ If to the contrary $\exists s \in (\hat{S}(x) \cap \hat{S}(x'))$, then $x, x' \in \tilde{X}(s)$; i.e. x and x' would be on the same path and therefore—since they are distinct nodes—ordered by precedence.

S_i ; i.e.

$$\hat{S}_i(x) = \{s_i \in S_i: \exists s_{-i} \in S_{-i}, (s_i, s_{-i}) \in \hat{S}(x)\} = \{s_i \in S_i: \exists s_{-i} \in S_{-i}, x \in \tilde{X}((s_i, s_{-i}))\}. \quad (\text{A.4})$$

These are the player- i pure strategies which do not preclude node $x \in X$. For any decision node $x \in X$ and any player- i pure strategy $s_i \in \hat{S}_i(x)$ which does not preclude x , let

$$\tilde{S}_{-i}(x, s_i) = \{s_{-i} \in S_{-i}: (s_i, s_{-i}) \in \hat{S}(x)\} = \{s_{-i} \in S_{-i}: x \in \tilde{X}((s_i, s_{-i}))\}. \quad (\text{A.5})$$

This is the set of deleted strategy profiles which can be combined with $s_i \in \hat{S}_i(x)$ in order to reach x .

Lemma 1

Consider any decision node $x \in X$ which is encountered by a strategy profile $s \in S$. The strategy profile $s' \in S$ encounters node x if and only if the two strategy profiles agree at every previously encountered information set. I.e. $\forall x \in X, \forall s, s' \in S$, such that $x \in \tilde{X}(s)$, we have $x \in \tilde{X}(s')$ if and only if $\forall h \in \hat{H}(x) \setminus \{\hat{h}(x)\}, s_{t(h)}(h) = s'_{t(h)}(h)$.

Proof

Omitted.

Lemma 2

For all player- i information sets $h \in H_i$ and for all pairs $x, x' \in h$ of nodes in this information set, the set of player- i strategies which do not preclude the first node is exactly the set of strategies which do not preclude the second. I.e. $\forall i \in I, \forall h \in H_i, \forall x, x' \in h, \hat{S}_i(x) = \hat{S}_i(x')$.

Proof

Assume to the contrary that, for some $i \in I, h \in H_i$, and some node pair $x, x' \in h$, there exists a player- i pure strategy $s_i \in \hat{S}_i(x)$ such that $s_i \notin \hat{S}_i(x')$. Therefore there exists a $s_{-i} \in S_{-i}$ such that $x \in \tilde{X}((s_i, s_{-i}))$. Because $\hat{S}_i(x') \neq \emptyset, \exists s'_i \in \hat{S}_i(x')$ such that $s'_i \neq s_i$. Consider the player- i information sets encountered on the path to x , viz. $(\hat{H}(x) \cap H_i) \setminus \{h\}$. The two strategies s_i and s'_i cannot agree at all of these information sets because, if they did, the two strategy profiles (s_i, s_{-i}) and (s'_i, s_{-i}) would agree at all the on-the-path information sets preceding x , viz. $\hat{H}(x) \setminus \{h\}$, and therefore (s'_i, s_{-i}) would also encounter x by Lemma 1, and this would violate perfect recall.

So s_i and s'_i must differ at some previous information set $h' \in (\hat{H}(x) \cap H_i) \setminus \{h\}$. But this would also violate perfect recall because x and x' share an information set, x ultimately succeeds h' via $s_i(h')$, and x' ultimately succeeds h' via a *different* action $s'_i(h')$ at h' . ☺

We use the result of Lemma 2 to justify writing

$$\forall h \in H_i, \forall x \in h, \hat{S}_i(h) = \hat{S}_i(x). \quad (\text{A.6})$$

I.e. we can meaningfully talk about the player- i pure strategies which do not preclude an information set $h \in H_i$.

Lemma 3

If neither of two player- i strategies preclude a given player- i node $x \in X_i$, then the two strategies agree at all on-the-path information sets prior to that node. I.e. if $s_i, s_i' \in \hat{S}_i(x)$, then $\forall h \in [(H_i \cap \hat{H}(x)) \setminus \{\hat{h}(x)\}]$, $s_i(h) = s_i'(h)$.

Proof

From (A.4) we know that there exist $s_{-i}, s_{-i}' \in S_{-i}$ such that $x \in (\tilde{X}((s_i, s_{-i})) \cap \tilde{X}((s_i', s_{-i}')))$. Therefore (s_i, s_{-i}) and (s_i', s_{-i}') agree for all on-the-path information sets preceding x by Lemma 1, and therefore s_i and s_i' agree for all information sets $h \in [(H_i \cap \hat{H}(x)) \setminus \{\hat{h}(x)\}]$. ☺

Lemma 4

Let $x \in X_i$ be a player- i node. If neither of two player- i strategies preclude node x , then the deleted strategy profiles which encounter x are the same for the two player- i strategies. I.e.

$$\forall s_i, s_i' \in \hat{S}_i(x), \tilde{S}_{-i}(x, s_i) = \tilde{S}_{-i}(x, s_i').$$

Proof

I will show that $s_{-i} \in \tilde{S}_{-i}(x, s_i) \Rightarrow s_{-i} \in \tilde{S}_{-i}(x, s_i')$. We have $x \in \tilde{X}((s_i, s_{-i}))$. From Lemma 3, s_i and s_i' must agree at all player- i on-the-path information sets encountered before x . Therefore (s_i, s_{-i}) and (s_i', s_{-i}) agree at all on-the-path information sets encountered before x . Therefore from Lemma 1 (s_i', s_{-i}) must encounter x and therefore $s_{-i} \in \tilde{S}_{-i}(x, s_i')$. ☺

Lemma 4 allows us to define for all $x \in X$

$$\check{S}_{-i}(x) = \tilde{S}_{-i}(x, s_i), \tag{A.7}$$

where $s_i \in \hat{S}_i(x)$ is any player- i strategy which does not preclude node x .

For all information sets $h \in H$ and actions $a \in A(h)$, let $\dot{S}(a, h) \subset S$ be the event that $s_{I(h)}(h) = a$; i.e.

$$\dot{S}(a, h) = \{s \in S: s_{I(h)}(h) = a\}. \tag{A.8}$$

This is the set of strategy profiles in which the owner of information set h chooses action a if play reaches h .

The conditional probability we seek is:

$$b_i(a|h) = \frac{\mathbb{P}(\dot{S}(a, h) \cap \hat{S}(h))}{\mathbb{P}(\hat{S}(h))}, \tag{A.9}$$

where we restrict attention to information sets h which are reached with positive probability by the mixed strategy σ_i . (I.e. we restrict attention to strategy/information-set pairs such that the mixed strategy is *compatible* with the information set.)

First we calculate the denominator. From (A.2) and (A.3), the probability that play reaches h is

$$\begin{aligned}
\mathbb{P}(\hat{S}(h)) &= \prod_{x \in h} \mathbb{P}(\hat{S}(x)) = \prod_{x \in h} \mathbb{P}\left(\prod_{s \in \hat{S}(x)} s\right) = \prod_{x \in h} \prod_{s \in \hat{S}(x)} \sigma(s) \\
&= \prod_{x \in h} \prod_{s_i \in \hat{S}_i(x)} \sigma_i(s_i) \prod_{s_{-i} \in \hat{S}_{-i}(x, s_i)} \prod_{j \in I \setminus \{i\}} \sigma_j(s_j).
\end{aligned} \tag{A.10}$$

From (A.6) and (A.7) we change the ranges of two summations according to

$$s_i \in \hat{S}_i(x) \rightarrow s_i \in \hat{S}_i(h), \tag{A.11}$$

$$s_{-i} \in \tilde{S}_{-i}(x, s_i) \rightarrow s_{-i} \in \check{S}_{-i}(x). \tag{A.12}$$

These changes allow a rearrangement and separation of the terms in (A.8), becoming

$$\mathbb{P}(\hat{S}(h)) = \left(\prod_{s_i \in \hat{S}_i(h)} \sigma_i(s_i) \right) \left(\prod_{x \in h} \prod_{s_{-i} \in \check{S}_{-i}(x)} \prod_{j \in I \setminus \{i\}} \sigma_j(s_j) \right). \tag{A.13}$$

Now to calculate the numerator.... We transform the event in the numerator to become

$$\dot{S}(a|h) \cap \hat{S}(h) = \dot{S}(a|h) \cap \left(\prod_{x \in h} \hat{S}(x) \right) = \prod_{x \in h} (\dot{S}(a|h) \cap \hat{S}(x)). \tag{A.14}$$

Then

$$\mathbb{P}(\dot{S}(a|h) \cap \hat{S}(h)) = \prod_{x \in h} \prod_{s \in (\dot{S}(a|h) \cap \hat{S}(h))} \sigma(s) = \prod_{x \in h} \prod_{\substack{s \in \hat{S}(h) \\ \text{s.t. } s_i(h)=a}} \sigma(s). \tag{A.15}$$

A very similar chain of reasoning (with an added finesse or two) to what we used in the calculation of the denominator yields

$$\mathbb{P}(\dot{S}(a|h) \cap \hat{S}(h)) = \left(\prod_{\substack{s_i \in \hat{S}_i(h) \\ \text{s.t. } s_i(h)=a}} \sigma_i(s_i) \right) \left(\prod_{x \in h} \prod_{s_{-i} \in \check{S}_{-i}(x)} \prod_{j \in I \setminus \{i\}} \sigma_j(s_j) \right). \tag{A.16}$$

Combining (A.14) and (A.11) yields

$$b_i(a|h) = \frac{\prod_{\substack{s_i \in \hat{S}_i(h) \\ \text{s.t. } s_i(h)=a}} \sigma_i(s_i)}{\prod_{s_i \in \hat{S}_i(h)} \sigma_i(s_i)}. \tag{A.17}$$

This is exactly the branch of (20) which corresponds to the compatible strategy case.

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Jim Ratliff's Game Theory Course:

4.3: Solution Concepts in Extensive-Form Games

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Abstract



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In real-world games we look ahead and plan what we would do in various contingencies. However, our plan is not a commitment. As the situation progresses, we constantly reassess our play in response to new information. Even if there is no new information, and the situation is exactly what we forecast it to be, we can still reassess the wisdom of the plan we made earlier. If at some point what we had planned to do no longer seems optimal, we must deviate from our original plan. Because there are situations in which we cannot commit to a particular decision in the future, a rational player should also be *sequentially rational*: Her planned action at any situation and point in time must actually be optimal at that time and in that situation given her beliefs. We will see that this requirement of *dynamic consistency* is a stronger form of rationality assumption than we have made thus far.

In order to impose dynamic consistency upon our solution concepts, we learn how to decompose an extensive-form game into a subgame and its complement, viz., its difference game. Then we learn how to restrict extensive-form game strategies to the subgame and to the difference game, as well as the reverse: how to compose a new extensive-form game strategy from a subgame strategy and a difference-game strategy.

We define the solution concept of *subgame-perfect equilibrium* as a refinement of Nash equilibrium that imposes the desired dynamic consistency. A subgame-perfect equilibrium of an extensive-form game is a behavior-strategy profile whose restriction to each subgame is a Nash equilibrium of that subgame.

An extensive-form game need not have a pure-strategy Nash equilibrium. However, we then consider the special case of extensive-form games of *perfect information*, i.e., where every information set contains exactly one decision node. We use Zermelo's backward-induction algorithm to prove that all such games of perfect information have a pure-strategy subgame-perfection equilibrium. This algorithm also provides a useful technique for finding the equilibria of actual games.



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Games](#)

[Course Outline](#)

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§5.1: Introduction to Repeated
Games](#)

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Extensive-Form Solution Concepts

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Extensive-form strategies

When we began our study of game theory with the strategic form, strategy was a primitive of the theory—it was a basic entity with no internal structure. The players each chose a strategy simultaneously, and those choices determined the players' payoffs. In the extensive form of a game, strategies can be very complex. A strategy for a player is a plan for perhaps many actions. Those actions may be spread out in time; between one action for a player and her next, other players' actions may intervene; she might observe those intervening actions perfectly, imperfectly, or not at all.

Game theorists are somewhat schizoid in their interpretation of extensive-form strategies. One popular interpretation is that an extensive-game strategy is a book of instructions which a player could give to a disinterested referee. The player could then go home and have the referee play her desired action whenever called upon to move. (E.g. in the pure-strategy case, there is a page for each of her information sets, on which is written the action to be taken at that information set.) Under this interpretation, the entire plan is not only conceived of before the game is played but, more significantly, a player effectively commits herself—when she files this book of instructions with the referee before the

game begins—to particular later actions in response to particular future contingencies.

But at the same time we are supposed to take decision nodes seriously: they mark the occasion at which the *decision* actually takes place (or perhaps better: the occasion at which the player commits to the decision). (In the “play book” interpretation the decision node marks the occasion at which the referee *implements* the previously made decision.) Under this interpretation, a player cannot commit now to a particular decision at a later node. (Any opportunities to commit at an earlier time to a future action can be modeled, but the resulting extensive form would have a decision node at the time of commitment rather than at the time of implementation.)

The utility of the “play book” interpretation is not that it represents how we should think of the game as actually being played. Its role is to specify the level of detail in which we must express a contingent plan in order for it to serve as a well-defined strategy.

Dynamic consistency

In real-world games we look ahead and plan what we would do in various contingencies. However, our plan is not a commitment. As the situation progresses, we constantly reassess our play in response to new information. Even if there is no new information, and the situation is exactly what we forecast it to be, we can still reassess the wisdom of the plan we made earlier. If at some point what we had planned to do no longer seems optimal, we must deviate from our original plan.

Because there are situations in which we cannot commit to a particular decision in the future, a rational player should also be *sequentially rational*: Her planned action at any situation and point in time must actually be optimal at that time and in that situation given her beliefs. We will see that this is a stronger form of rationality assumption than we have made thus far.

Consider the two-player extensive-form game in Figure 1a. Player 1 can choose U , ending the game immediately, or D , passing the move to player 2, who then chooses l or r , which ends the game. Figure 1b displays the strategic form of the game.

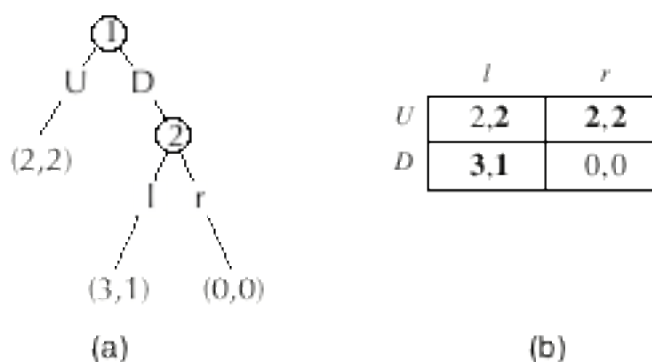


Figure 1: An extensive-form game and its strategic-form representation.

Since we have already developed techniques for analyzing games in strategic form, let's see what we can say about the extensive-form game in Figure 1a by studying its strategic form in Figure 1b. Let's

first ask what outcomes are consistent with common knowledge of rationality—what are the rationalizable outcomes. We note that neither player has a strongly dominated strategy, so the iterated elimination of strongly dominated strategies ends as soon as it begins. Because this is a two-player game and all players' strategies survive this process, every possible outcome is rationalizable.

Let's focus on the outcome in which player 1 chooses U , thus ending the game. Why is U rationalizable for player 1? It is a best response to player 2 choosing r . But that isn't enough. If r is not a rationalizable strategy for player 2, then U being a best response to r wouldn't make U rationalizable for player 1. However, r is a best response to player 1's U . We have therefore created a cycle of beliefs which rationalizes both U for player 1 and r for player 2.

So player 1 chooses U because she believes that player 2 would play r if player 2's node were reached. Note that if player 1 does choose U , then player 2's node is *not* reached. Therefore player 2's planned action at that node is never taken and is completely inconsequential to the payoffs. As you can see from Figure 1b, *any* choice by player 2 is a best response to U because player 2's choice is entirely irrelevant to the payoffs in that case. If play were to reach player 2's node, obviously the only rational thing for player 2 to do would be to choose l and receive 1 rather than the 0 he would receive by playing r . The only reason that player 2's plan to play r at his node is rational is because his beliefs tell him that he will never be called upon to execute this plan.¹

Player 2's plan to play r would violate his sequential rationality. If player 1 knew that player 2 was sequentially rational, then player 1 could not hold the belief that player 2 would choose r . This belief about 2's choice was the only possible underpinning for player 1's plan to play U . With it removed we see that U is no longer justified for player 1.

To provide ourselves with the analytical power to study arbitrarily complicated extensive-form games we now assume that the sequential rationality of the players is common knowledge.

We have previously seen how games can be decomposed into subgames and how a strategy can be restricted to a subgame. We say that a strategy is rationalizable in a subgame if its restriction to the subgame is rationalizable in the subgame. A strategy is sequentially rationalizable if it is consistent with common knowledge of sequential rationality. In order for a strategy to be sequentially rationalizable, it is clear that it must be rationalizable in every subgame.

In the game of Figure 1a, player 1 must believe that player 2 would play l at his information set if it were reached, because l is the only rationalizable action in that subgame. Player 1 must choose a best response to that belief, therefore the unique sequentially rational outcome of the game is (D, l) .

¹ You could feel secure promising to give up all your money on the day it snows in Tucson in July.

Decomposing an extensive form

The subgame

We have previously seen how to isolate a *subgame* from an extensive-form game: it's a subtree which inherits the player set, actions, payoffs, and the player and information partitions from the original game tree and which respects information sets.² In order to formalize the procedure of *backwards induction* we used above in the game in Figure 1 we need to also be able to talk about the part of the original game which is excluded from a given subgame; we call this the *difference game*.

Let Γ be an extensive-form game. Because it contains a tree, it contains a set V of vertices and a set E of edges. One node $\mathbb{O} \in V$ is designated to be the initial node and this choice determines the set $Z \subset V$ of terminal nodes and set $X \subset V$ of decision nodes. (We have $V = X \cup Z$, $X \cap Z = \emptyset$, and $\mathbb{O} \in X$.)

Consider any noninitial decision node $x \in X \setminus \{\mathbb{O}\}$ such that Γ can be decomposed into a subgame $\tilde{\Gamma}^x$ whose initial node is x . (As long as the identity of the node x at which the original game is being decomposed is clear from context, I'll suppress the “ x ” superscript on the $\tilde{\Gamma}$.) The nodes $\tilde{V}$ of the subgame $\tilde{\Gamma}$ are the node x and all of its successors in Γ . The subgame's set $\tilde{V}$ of nodes is partitioned into its decision nodes $\tilde{X}$ and its terminal nodes $\tilde{Z}$, which are defined by restriction of X and Z to $\tilde{V}$. The player set, actions available at each node and the action assignments to each edge, and payoffs to the terminal nodes $\tilde{Z}$ are inherited from the original game by restriction. The set $\tilde{H}$ of subgame information sets is also inherited by restriction from the original game. Because $\tilde{\Gamma}$ is a subgame (rather than just an arbitrary subtree), we know that information sets are respected: $\tilde{H} = \{h \in H : h \subset \tilde{X}\}$.

For example, in Figure 2, the extensive-form game Γ is decomposed at the player-2 information set α . This node α and all of its successors as well as the edges which represent actions at these nodes form the subtree for the subgame $\tilde{\Gamma}$.

² See “Extensive-Form Games,” October 5, 1993, pages 13–16.

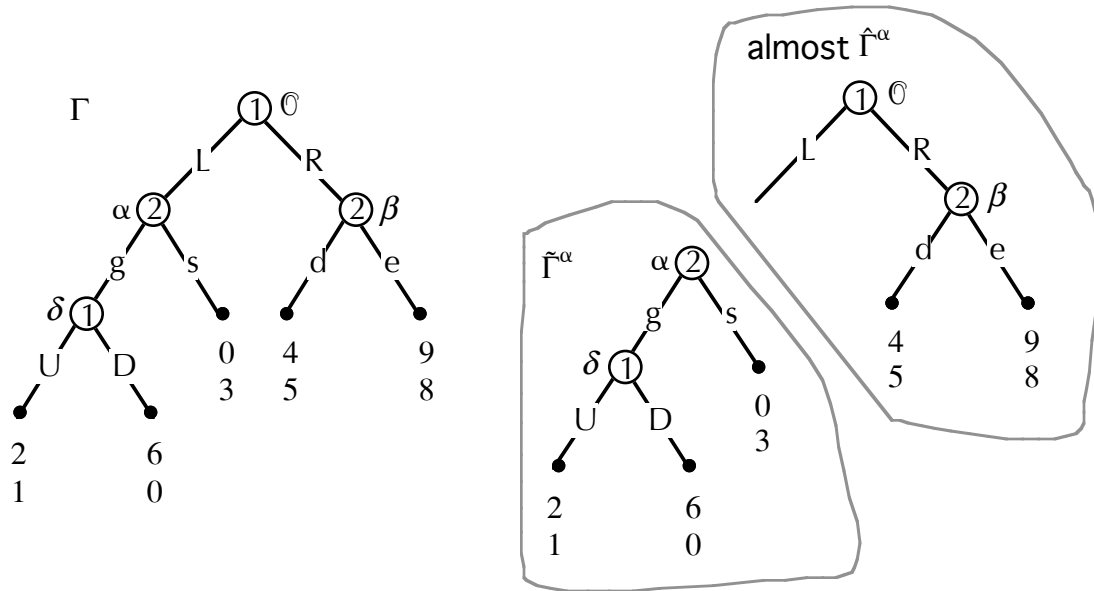


Figure 2: An extensive-form game Γ decomposed at node α into the corresponding subgame $\tilde{\Gamma}^\alpha$ and difference game $\hat{\Gamma}^\alpha$.

The difference game

The difference game $\hat{\Gamma}^x$ is partially defined by a set of vertices $\hat{V} \subset V$ and a set of edges $\hat{E} \subset E$. (Again I'll suppress the “ x ” superscript on the $\tilde{\Gamma}$ when no confusion will result.) The set of vertices $\hat{V}$ contains all of those and only those nodes of the original game which are not in the subgame; i.e. $\hat{V} \supset V \setminus \tilde{V}$ and $\hat{V} \cap \tilde{V} = \emptyset$.³ (It is because this set is defined in terms of a set-theoretic difference that the residual game is called the difference game.) The decision nodes $\hat{X}$ of the difference game are inherited by restriction from the original game: $\hat{X} = X \cap \hat{V}$. Note in particular that the node x belongs to the subgame $\tilde{\Gamma}$, i.e. $x \in \tilde{V}$, and therefore does not belong to $\hat{V}$. We also note that the original game Γ 's set X of decision nodes is partitioned by the decision nodes of the subgame $\tilde{\Gamma}$ and of the difference game $\hat{\Gamma}$; i.e. $X = \tilde{X} \cup \hat{X}$. The difference game's collection $\tilde{H}$ of information sets is defined by restriction: $\hat{H} = \{h \in H : h \subset \hat{X}\}$. The original game's information sets are partitioned by the information sets for the subgame and difference game: $H = \tilde{H} \cup \hat{H}$ and $\tilde{H} \cap \hat{H} = \emptyset$.

Consider the edge $e^x \in E$ of the original game Γ which joined node x to its immediate predecessor, say x' , in Γ ; i.e. $e^x = (x, x')$.⁴ Because node x is not a member of the difference game $\hat{\Gamma}$, in the difference game $\hat{\Gamma}$ the edge e^x is now incomplete: it joins x 's immediate predecessor x' on one end but has no node on its other end.⁵ (In Figure 2b note that the edge labeled L extending from the initial node $\textcircled{0}$ of the difference game tree has no node on its other end.) Therefore to make the difference game a legitimate extensive form we must augment its set of vertices with an additional terminal node $\hat{z}$ (which we'll soon place where the decision node x had been located). So the difference game's set of nodes is

³ You'll soon see why I don't more simply just say $\hat{V} = V \setminus \tilde{V}$.

⁴ Recall that an edge is an unordered pair of vertices. (See “Extensive-Form Games,” October 5, 1993, page 1.)

⁵ More properly, then, e^x is not an edge in $\hat{\Gamma}$ because it is not an unordered pair of vertices in $\hat{V}$.

$\hat{V} = (V \setminus \tilde{V}) \cup \{\hat{z}\}$ and the set of terminal nodes is $\hat{Z} = (Z \cap \hat{V}) \cup \{\hat{z}\}$.⁶ (See Figure 3.)

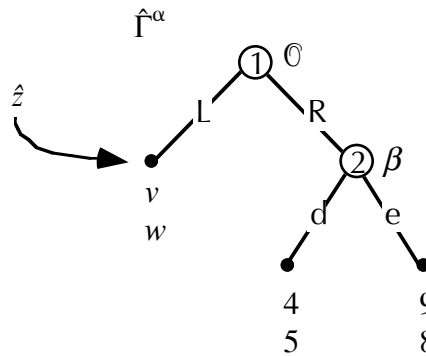


Figure 3: The difference game with artificially added terminal node $\hat{z}$ and redefined edge L connecting $\hat{z}$ to α 's immediate predecessor in Γ .

The set of edges $\hat{E}$ of the difference game are defined by first restricting the edges in E to the new set of vertices $\hat{V}$. This however excludes $e^x = (x, x')$, because $x \notin \hat{V}$. So we augment this set of edges with a new one $e' = (x', \hat{z})$ which connects x' 's immediate predecessor in Γ (viz. x') with the newly added terminal node $\hat{z}$. I.e. $\hat{E} = \{e = (y, z) \in E: y, z \in \hat{V}\} \cup \{e'\}$. The new edge inherits the action assignment of its now-incomplete forbear: $f(e') = f(e^x)$. [In the original game Γ of Figure 2, the edge L was defined by the nodes it directly connected, viz. $L = (\emptyset, \alpha)$. In Figure 3, L has been redefined to be $L = (\emptyset, \hat{z})$.]

Adding the terminal node $\hat{z}$ to the difference tree still doesn't quite succeed in making $\hat{\Gamma}$ a legitimate extensive-form game, because we still do not have a payoff vector assigned to $\hat{z}$. We want to assign $\hat{z}$ the payoff vector which the players would receive if indeed play did proceed to $\hat{z}$. If play did reach $\hat{z}$, that would imply that the subgame $\tilde{\Gamma}$ were reached, so we want to assign to $\hat{z}$ the payoff vector which would be received by the players if they played the subgame. However, there are typically many different payoff vectors which could be received in the subgame. Before we continue to resolve this indeterminacy, we need to discuss the notion of strategy in the subgame and difference game.

Subgame and difference-game strategies

A strategy profile in an extensive-form game is a map from information sets to actions; i.e. $s: H \rightarrow \bar{A}$, where $\bar{A}$ is the set of all actions available at any information set.⁷ Given a strategy profile s in the original game Γ , we can break it up into two components: a strategy profile $\tilde{s}$ in the subgame $\tilde{\Gamma}$ and a strategy profile $\hat{s}$ in the difference game $\hat{\Gamma}$. We do this by simply restricting the original strategy profile s to the subsets of information sets $\tilde{H}$ and $\hat{H}$, respectively; $\tilde{s}$ is the restriction of s to $\tilde{H} \subset H$ and $\hat{s}$ is the restriction of s to $\hat{H} \subset H$.⁸ Every strategy profile s is equivalent to the pair $(\hat{s}, \tilde{s})$ of difference-game and

⁶ This is why I earlier couldn't just say that $\hat{V} = V \setminus \tilde{V}$.

⁷ To be more explicit... For each player $i \in I$, a player- i strategy $s_i: H_i \rightarrow \bar{A}_i$ is a map from player- i information sets to player- i actions. Therefore a strategy profile s takes an arbitrary information set $h \in H$, finds the player $i(h)$ who owns that information set, and returns the action dictated by $s_{i(h)}(h) \in \bar{A}_i \subset \bar{A}$. Therefore the strategy profile is a map from $H \rightarrow \bar{A}$.

⁸ Let $f: X \rightarrow Z$ be a function and let $Y \subset X$ be a subset of X . Then we can define a function $\tilde{f}$, the *restriction* of f to Y , as a function whose domain is Y and which agrees with f for all points Y . I.e. $\tilde{f}: Y \rightarrow Z$ and $\forall x \in Y, \tilde{f}(x) = f(x)$.

subgame strategy profiles because no information is lost in the decomposition (because $H = \tilde{H} \cap \hat{H}$). In other words we can reconstruct s from the pair $(\hat{s}, \tilde{s})$ according to, for all $h \in H$,

$$s(h) = \begin{cases} \hat{s}(h), & h \in \hat{H}, \\ \tilde{s}(h), & h \in \tilde{H}. \end{cases}$$

We let $\tilde{S}$ and $\hat{S}$ be the space of all subgame and difference-game strategy profiles, respectively. When we restrict a subgame or difference-game strategy profile to player- i 's information sets, we define in the natural way player- i subgame and difference-game strategies $\tilde{s}_i \in \tilde{S}_i$ and $\hat{s}_i \in \hat{S}_i$ and the corresponding player- i strategy spaces $\tilde{S}_i$ and $\hat{S}_i$. We get the same equivalence between player- i strategies in the original game and pairs of strategies in the difference-game and subgame as we did between strategy profiles; i.e. every player- i strategy profile $s_i \in S_i$ in the original game determines a pair $(\hat{s}_i; \tilde{s}_i)$ of player- i difference-game and subgame strategies, and vice versa.

Consider a strategy profile $(t_i, s_{-i}) \in S$ of the original game in which player i chooses $t_i \in S_i$ while her opponents choose the deleted strategy profile $s_{-i} \in S_{-i}$. We can decompose each player's strategy into an individual difference-game and subgame strategy; i.e. $t_i = (\hat{t}_i; \tilde{t}_i)$ and $s_{-i} = (\hat{s}_{-i}; \tilde{s}_{-i})$.⁹ You can show that the following equivalence holds:

$$(t_i, s_{-i}) = ((\hat{t}_i; \tilde{t}_i), (\hat{s}_{-i}, \tilde{s}_{-i})) = ((\hat{t}_i, \hat{s}_{-i}); (\tilde{t}_i, \tilde{s}_{-i})). \quad (1)$$

For an example of the decomposition of strategy profiles, reconsider Figure 2. We can write a strategy profile in the original game Γ in the form

$$s = (s_1(\mathbb{O}), s_1(\mathcal{D}); s_2(\alpha), s_2(\beta)).$$

When we restrict $s \in S$ to the subgame and to the difference game, respectively, we obtain the strategy profiles $\tilde{s} \in \tilde{S}$ and $\hat{s} \in \hat{S}$, which we can write in the form

$$\tilde{s} = (s_1(\mathcal{D}); s_2(\alpha)),$$

$$\hat{s} = (s_1(\mathbb{O}); s_2(\beta)).$$

(Note that the pair $(\hat{s}, \tilde{s})$ contains all the information present in s .) For example, see Figure 4. Here I have shown the added terminal node $\hat{z}$ and its associated payoff vector $(v, w) \in \mathbb{R}^2$, whose values have yet to be determined. The strategy profile indicated in the original game Γ is $s = (L, U; s, e)$. When restricted to the subgame it becomes $\tilde{s} = (U; s)$. When restricted to the difference game, it becomes $\hat{s} = (L; e)$.

⁹ Note the distinction between the use of a comma and a semicolon in (t_i, s_{-i}) and $(\hat{t}_i; \tilde{t}_i)$, respectively.

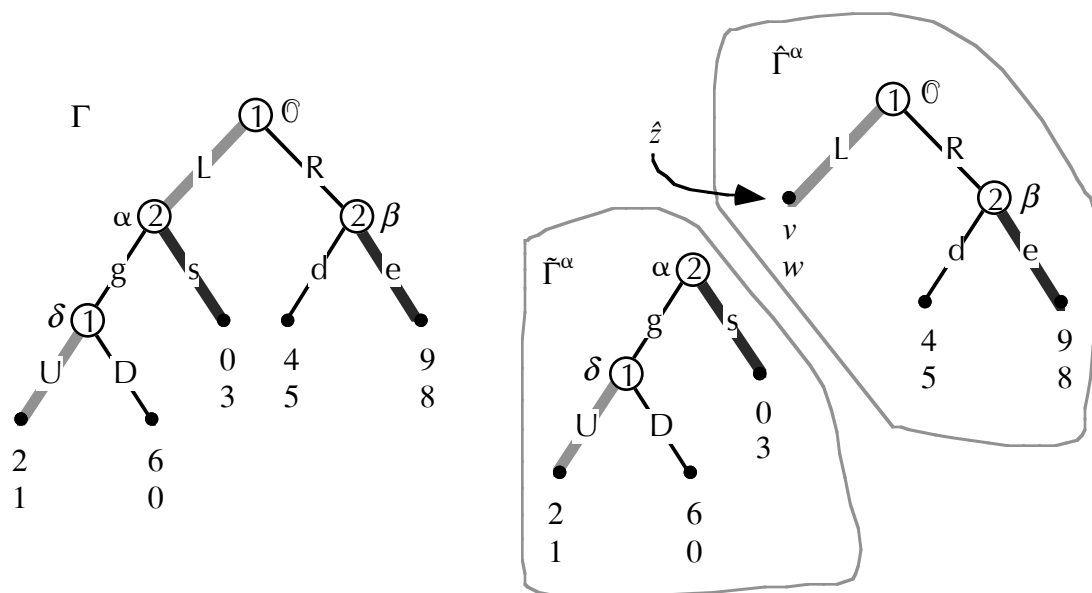


Figure 4: A strategy profile in the game Γ decomposed into strategy profiles for the subgame and difference games.

A subgame strategy $\tilde{s}$ determines a particular difference game

Now we can address the question of what payoff vector to assign to the artificially added terminal node $\hat{z}$. Every strategy profile $\tilde{s} \in \tilde{S}$ of the subgame determines a terminal node and therefore a payoff vector in the subgame. If we believe that the subgame would be played according to some subgame strategy profile $\tilde{s}$ if that subgame were reached, then we would want to award the newly added terminal node $\hat{z}$ in the difference game the payoff vector of the subgame which would be received if the subgame were reached and $\tilde{s}$ were played there. Each choice of subgame strategy profile $\tilde{s} \in \tilde{S}$ determines such a payoff and therefore completes the determination of a difference game $\hat{\Gamma}(\tilde{s})$. Formally we say that $\hat{\mu}(\hat{z}) = \tilde{u}(\tilde{s})$, where $\hat{\mu}: \hat{Z} \rightarrow \mathbb{R}^n$ is the function assigning utilities to outcomes in the difference game. For example, in Figure 4 the specification of the subgame strategy profile $\tilde{s} = (U; s)$ determines the payoff vector $(0, 3)$ in the subgame and therefore determines the payoff vector for $\hat{z}$, viz. $(v, w) = (0, 3)$, in the difference game $\hat{\Gamma}((U; s))$. See Figure 5.

Payoffs in the original game, the subgame, and the difference game

We let, as usual, $u: S \rightarrow \mathbb{R}^n$ be the payoff-vector function in the game Γ . We let $\tilde{u}: \tilde{S} \rightarrow \mathbb{R}^n$ be the payoff-vector function in the subgame $\tilde{\Gamma}$; i.e. $\tilde{u}(\tilde{s}) \in \mathbb{R}^n$ is the payoff vector in the subgame corresponding to the subgame strategy profile $\tilde{s} \in \tilde{S}$. We let $\hat{u}(\cdot; \tilde{s}): \hat{S} \rightarrow \mathbb{R}^n$ be the payoff-vector function in the particular difference game $\hat{\Gamma}(\tilde{s})$ determined by a particular subgame strategy profile $\tilde{s} \in \tilde{S}$; i.e. $\hat{u}(\hat{s}; \tilde{s}) \in \mathbb{R}^n$ is the payoff vector in the difference game $\hat{\Gamma}(\tilde{s})$ —defined by the subgame strategy profile $\tilde{s} \in \tilde{S}$ —when the difference-game strategy profile $\hat{s} \in \hat{S}$ is played in the difference game.

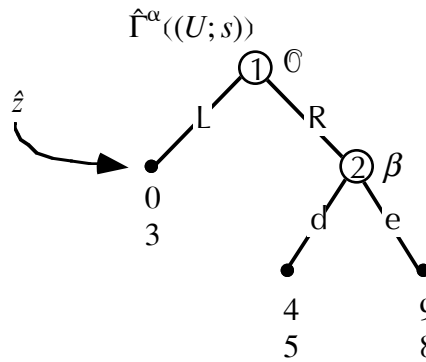


Figure 5: The difference game determined by the particular choice of subgame strategy profile $\tilde{s} = (U; s)$

Consider any subgame strategy profile $\tilde{s} \in \tilde{S}$ and the difference game $\hat{\Gamma}(\tilde{s})$ it defines. Now consider a difference-game strategy profile $\hat{s} \in \hat{S}$. This difference-game strategy profile results in a path in the difference game which either does or does not reach the node-of-decomposition x , and therefore either does or does not reach the subgame $\tilde{\Gamma}$. This dichotomy is useful in analyzing the difference-game payoff: If the subgame is reached by $\hat{s}$, the difference game's payoff is that of the subgame;¹⁰ if the subgame is not reached by $\hat{s}$, the difference game's payoff is independent of the subgame strategy $\tilde{s}$. I.e.

$$\hat{u}(\hat{s}; \tilde{s}) \begin{cases} = \tilde{u}(\tilde{s}), & \hat{s} \text{ reaches } x, \\ \text{is independent of } \tilde{s}, & \hat{s} \text{ does not reach } x. \end{cases} \quad (2)$$

The payoff in the original game to the strategy profile $s = (\hat{s}, \tilde{s})$ is exactly the payoff to the difference-game strategy profile $\hat{s}$ in the difference game defined by the subgame strategy profile $\tilde{s}$: $\forall \tilde{s} \in \tilde{S}, \forall \hat{s} \in \hat{S}$, where $s = (\hat{s}, \tilde{s}) \in S$,

$$u(s) \equiv \hat{u}(\hat{s}; \tilde{s}). \quad (3)$$

To see this consider any strategy profile $s \in S$ of the original game Γ and consider the path of s . There are two cases: either 1 the path of s encounters the node-of-decomposition x or 2 it does not. 1 If the path of s encounters node x , then the path continues (as determined by the subgame strategy profile $\tilde{s}$) until it reaches a terminal node of the original game which is also a terminal node of the subgame $\tilde{\Gamma}$. In this case the payoff to s in the original game Γ is exactly the subgame payoff corresponding to the subgame strategy profile $\tilde{s}$; $u(s) = \tilde{u}(\tilde{s}) = \hat{u}(\hat{s}; \tilde{s})$, by (2). (The subgame payoffs corresponding to subgame terminal nodes are inherited from the original game's terminal-node payoffs.) 2 If the path of s does not reach the subgame, then the path of s is exactly the path of the difference-game strategy profile $\hat{s}$. (The path of s only encounters information sets in $\hat{H}$. Because $\hat{s}$ is the restriction of s to $\hat{H}$, s and $\hat{s}$ agree on $\hat{H}$.) Therefore, again thanks to the inheritance of payoffs from the original game, the payoff to s in the original game is exactly the payoff to $\hat{s}$ in the difference game; i.e. $u(s) = \hat{u}(\hat{s}; \tilde{s})$. Therefore we have

¹⁰ If the subgame is reached, then the artificially added terminal node $\hat{z}$ is reached, and therefore the difference-game payoff is the payoff vector associated with $\hat{z}$. But this payoff is exactly the payoff of the subgame when the subgame strategy profile $\tilde{s}$ is played in the subgame.

established (3) in all cases.

Using (1) and (3) we see that

$$u((t_i, s_{-i})) = \hat{u}((\hat{t}_i, \hat{s}_{-i}); (\tilde{t}_i, \tilde{s}_{-i})). \quad (4)$$

Composing equilibria

The restriction of a Nash equilibrium to any on-the-equilibrium-path subgame is a Nash equilibrium of that subgame

Theorem 1

Consider an extensive-form game Γ , a subgame $\tilde{\Gamma}$ of Γ , and its associated difference extensive form $\hat{\Gamma}$. Let $s \in S$ be a Nash equilibrium of Γ and denote by $s = (\hat{s}; \tilde{s})$ the pair of difference-game and subgame strategy profiles which correspond to the equilibrium strategy profile s . If the difference-game strategy profile $\hat{s}$ reaches the subgame $\tilde{\Gamma}$, then the subgame strategy profile $\tilde{s}$ is a Nash equilibrium of the subgame $\tilde{\Gamma}$.

Proof

Because $\hat{s}$ reaches the subgame $\tilde{\Gamma}$, we know from (2) that the payoff to the entire strategy profile s is just the payoff to the subgame strategy profile $\tilde{s}$ in $\tilde{\Gamma}$; i.e. $\hat{u}(\hat{s}; \tilde{s}) = \tilde{u}(\tilde{s})$. Assume that $\tilde{s}$ is not a Nash equilibrium of $\tilde{\Gamma}$. Then there exists a player $i \in I$ and a subgame strategy $\tilde{t}_i \in \tilde{S}_i$ such that

$$\tilde{u}_i(\tilde{t}_i, \tilde{s}_{-i}) > \tilde{u}_i(\tilde{s}). \quad (5)$$

Let $\hat{t}_i = \hat{s}_i$ and let $t_i = (\hat{t}_i, \tilde{t}_i) \in S_i$. Using (4), $\hat{t}_i = \hat{s}_i$, (2), (5), (2), and (3), we write

$$u_i((t_i, s_{-i})) = \hat{u}_i((\hat{t}_i, \hat{s}_{-i}); (\tilde{t}_i, \tilde{s}_{-i})) = \hat{u}_i(\hat{s}; (\tilde{t}_i, \tilde{s}_{-i})) = \tilde{u}_i((\tilde{t}_i, \tilde{s}_{-i})) > \tilde{u}_i(\tilde{s}) = \hat{u}_i(\hat{s}; \tilde{s}) = u_i(s).$$

Therefore s , contrary to assertion, must not have been a Nash equilibrium of the original game Γ . ☺

A subgame equilibrium and a difference-game equilibrium form an equilibrium of the entire game

Theorem 2

Consider an extensive-form game Γ , a subgame $\tilde{\Gamma}$ of Γ , and its associated difference extensive form $\hat{\Gamma}$. Let $\tilde{s}$ be a Nash equilibrium of the subgame $\tilde{\Gamma}$ and let $\hat{s}$ be a Nash equilibrium of the particular difference game $\hat{\Gamma}(\tilde{s})$. Then $s = (\hat{s}, \tilde{s})$ is a Nash equilibrium of Γ . (Kuhn [1953: 208].)

Proof

We need to show that for all players $i \in I$, and for all player- i strategies $t_i = (\hat{t}_i; \tilde{t}_i) \in S_i$,

$$u_i(s) \geq u_i(t_i, s_{-i}). \quad (6)$$

Consider the difference-game strategy profile $(\hat{t}_i, \hat{s}_{-i})$ implied by player i 's proposed deviation. Because $\hat{s}$ is an equilibrium of $\hat{\Gamma}(\tilde{s})$, we have, using (3),

$$u_i(s) = \hat{u}_i(\hat{s}; \tilde{s}) \geq \hat{u}_i((\hat{t}_i, \hat{s}_{-i}); \tilde{s}). \quad (7)$$

There are two cases. 1 $(\hat{t}_i, \hat{s}_{-i})$ does not lead to the subgame $\tilde{\Gamma}$ and 2 $(\hat{t}_i, \hat{s}_{-i})$ does lead to the subgame $\tilde{\Gamma}$. In case 1, the difference-game payoff to $(\hat{t}_i, \hat{s}_{-i})$ is independent, using (2), of the subgame strategy profile played in $\tilde{\Gamma}$. Therefore, using (4), we have

$$\hat{u}_i((\hat{t}_i, \hat{s}_{-i}); \tilde{s}) = \hat{u}_i((\hat{t}_i, \hat{s}_{-i}); (\tilde{t}_i, \tilde{s}_{-i})) = u_i((t_i, s_{-i})). \quad (8)$$

Combining (7) and (8) yields (6), proving the proposition for case 1.

In case 2, the difference-game payoff is just the subgame payoff, using (2), so we have

$$\hat{u}_i((\hat{t}_i, \hat{s}_{-i}); \tilde{s}) = \tilde{u}_i(\tilde{s}). \quad (9)$$

Because $\tilde{s}$ is an equilibrium of $\tilde{\Gamma}$, using (2) and (4),

$$\tilde{u}_i(\tilde{s}) \geq \tilde{u}_i((\tilde{t}_i, \tilde{s}_{-i})) = \hat{u}_i((\hat{t}_i, \hat{s}_{-i}); (\tilde{t}_i, \tilde{s}_{-i})) = u_i((t_i, s_{-i})). \quad (10)$$

Combining (7), (9), and (10) yields the desired (6) for case 2. ☺

Subgame perfection

We saw in the game of Figure 1 that Nash equilibria of an extensive-form game can fail to exhibit dynamic consistency (or sequential rationality): they can rely on specification of actions at off-the-equilibrium-path information sets which would not rationally be implemented if, contrary to specification, the off-the-path information set were reached.

Subgame perfection has been offered as a refinement of Nash equilibrium which in some cases successfully weeds out such undesirable equilibria. (Selten [1965], Selten [1975])

Definition A behavior strategy profile σ of an extensive-form game Γ is a *subgame-perfect* equilibrium of Γ if, for every subgame $\tilde{\Gamma}$ of Γ , the restriction of σ to the subgame $\tilde{\Gamma}$ is a Nash equilibrium of the subgame $\tilde{\Gamma}$.

We saw in Theorem 1 that the restriction of any Nash equilibrium to any on-the-equilibrium-path subgame is a Nash equilibrium of the subgame. Therefore the only additional restrictions being imposed by subgame perfection are at off-the-equilibrium-path subgames.

We see that subgame perfection is a refinement of Nash equilibrium because every subgame-perfect equilibrium is a Nash equilibrium. (Let σ be a subgame-perfect equilibrium of Γ . Every game Γ is a subgame of itself, so the restriction of σ to the subgame Γ , which is simply just σ , must be a Nash equilibrium of Γ .)

To see how subgame perfection is sometimes successful in eliminating dynamically inconsistent Nash

equilibria, let's revisit the game of Figure 1. That game had two pure-strategy Nash equilibria, viz. (D, l) and (U, r) .¹¹ In Figure 6 I have indicated the only strictly smaller subgame of the whole game. We object to the (U, r) equilibrium because player 2 would not rationally choose r at his information set if he actually reached it. In order for (U, r) to be subgame perfect, its restriction to the subgame containing only player 2's information set—viz. the player-2 strategy r —must be a Nash equilibrium of that subgame. However, it fails that test.

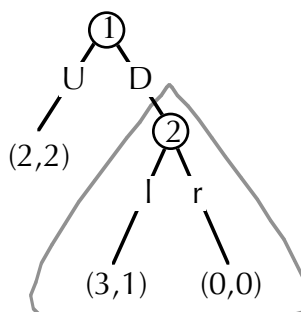


Figure 6: A subgame of the game from Figure 1.

Backward induction in games of perfect information

A finite game of perfect information has a pure-strategy equilibrium

We know that every finite game has a Nash equilibrium, however there need not exist a pure-strategy equilibrium. When the game has perfect information, however, the existence of a pure-strategy equilibrium is assured. In a game of perfect information every decision node $x \in X$ is a node at which the game can be decomposed into a subgame.

Theorem 3

Every finite game of perfect information has a pure-strategy equilibrium. (Kuhn [1953: 209].)

Proof

We will prove this by induction on the number m of decision nodes in Γ ; i.e. let $m = \#X$. If there is only one decision node, i.e. $m = 1$, then there exists a pure-strategy equilibrium in which the player who owns that node chooses an action which maximizes her payoff. (The other players, if any, have empty strategy spaces.) Now we establish that if every perfect-information game with less than m decision nodes has a pure-strategy equilibrium, then every game with exactly m decision nodes has a pure-strategy equilibrium. Then by induction, starting with the existence of a pure-strategy equilibrium for $m = 1$, all finite games of perfect information have pure-strategy equilibria.

Let $m > 1$. Because Γ has perfect information, we can decompose it at some noninitial decision node

¹¹ There is also a continuum of equilibria in which player 1 chooses U and player 2 chooses l with any probability weakly less than $\frac{2}{3}$. The intuition behind this continuum is the following: As long as player 1 chooses U , player 2 is indifferent to whether he chooses l or r . As long as player 2 does not choose l with too high a probability, U is a best response by player 1. Only when player 2 chooses l more than $\frac{2}{3}$ of the time will player 1's expected payoff to D exceed the 2 she receives for certain by playing U .

$x \in X \setminus \{\emptyset\}$ into a subgame $\tilde{\Gamma}$ with $m' < m$ decision nodes and a difference extensive form $\hat{\Gamma}$ with $m - m' < m$ decision nodes. (Both the subgame and the difference extensive form have perfect information.) Because every perfect-information game with less than m decision nodes has a pure-strategy equilibrium, $\tilde{\Gamma}$ does. Let $\tilde{s}$ be a pure-strategy equilibrium of $\tilde{\Gamma}$. Now consider the difference game $\hat{\Gamma}(\tilde{s})$ determined by this equilibrium subgame strategy profile. It is a perfect-information game with fewer than m decision nodes. Therefore it has a pure-strategy equilibrium. Let $\hat{s}$ be a pure-strategy equilibrium of $\hat{\Gamma}(\tilde{s})$. Then, by Theorem 2, $s = (\hat{s}, \tilde{s})$ is an equilibrium of the original game Γ . Therefore every perfect-information game with m decision nodes has a pure-strategy equilibrium. ☺

Zermelo's backward-induction algorithm

In order to solve a perfect-information game by backward induction we first need to identify the set of *penultimate* decision nodes. A decision node is penultimate if all of its immediate successors are terminal nodes. A subgame which has a penultimate decision node x as its initial node is just a single-player game; the decision problem belongs to the owner $\iota(x)$ of that node. That player will choose some action available at that node which maximizes her payoff over the terminal nodes which can be reached from x . After we specify the player's choice at x , we can delete the successors of x from the game. Then we change x from a decision node to a terminal node, assigning the new terminal node x the payoff vector associated with the former terminal node reached by the stipulated action.¹²

This process shrinks the game. We identify the new set of penultimate decision nodes in this smaller game and repeat the above process, keeping track of each player's chosen action at each of her penultimate decision nodes. Eventually the game is reduced until there is only one decision node left, viz. the initial node $\emptyset$. The owner of the initial node then makes her payoff-maximizing decision and this completes the determination of a backward-induction solution of the game. By this time every decision node in the original game has become a penultimate decision node for its owner, and so an action has been recorded for it. The union of all these decisions constitutes the strategy profile corresponding to the backward-induction solution. (This algorithm is described as a flowchart in Figure 7.)

For any decision node $x \in X$, let $N(x) \subset V$ be the set of immediate successors of x . Let $\check{v}(a|x) \in N(x)$ be the node reached directly when action $a \in A(x)$ is taken at node x .

Let $\bar{X} \subset X$ and $\bar{Z} \subset \bar{V}$ be sets of decision nodes and terminal nodes, respectively. Then the set $P(\bar{X}, \bar{Z})$ of penultimate decision nodes are those decision nodes whose immediate successors are all terminal nodes:

$$P(\bar{X}, \bar{Z}) = \{x \in \bar{X} : N(x) \subset \bar{Z}\}.$$

¹² This algorithm has been attributed to Zermelo [1912].

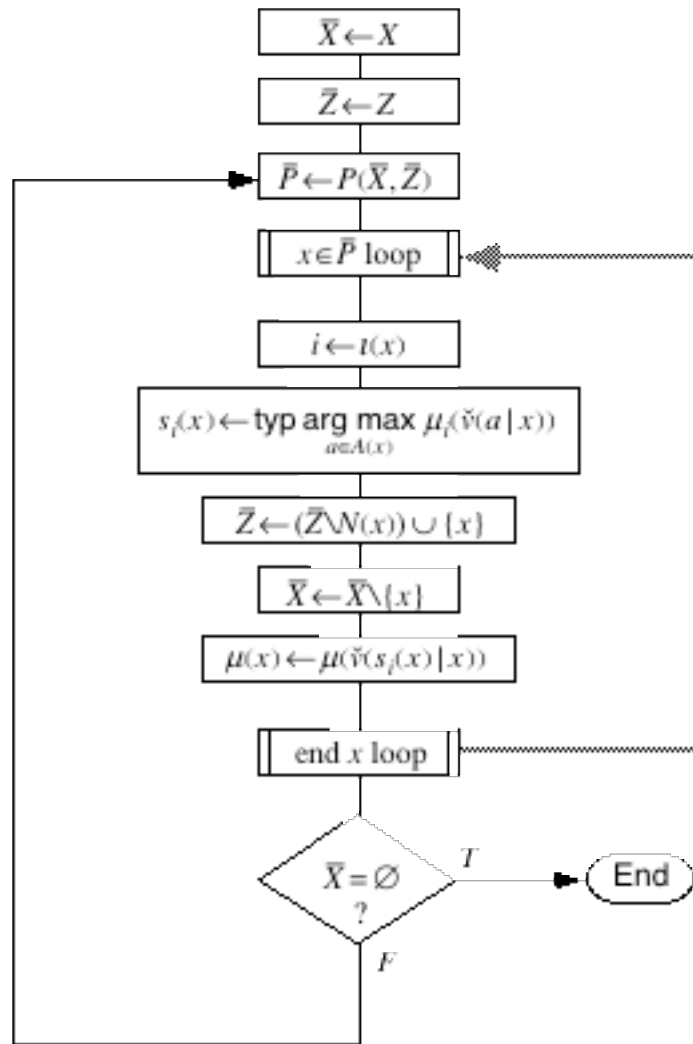


Figure 7: Zermelo's backward-induction algorithm.

Recall that $\mu: Z \rightarrow \mathbb{R}^n$ specifies the assignment of payoff vectors to terminal nodes in the original game. The backward-induction process will extend the domain of definition of μ so that it will also assign payoff vectors to each decision node at the time it makes the transition from penultimate decision node to new terminal node.

We let the sets of decision and terminal nodes at any point in the process be $\bar{X}$ and $\bar{Z}$, respectively. We initialize the process by assigning these to be the corresponding sets in the original game, i.e. $\bar{X} = X$ and $\bar{Z} = Z$. Let $\bar{P}$ be the set of penultimate decision nodes at any stage of the backward-induction process. A penultimate decision node $x \in \bar{P}$ belongs to player $i = I(x)$. Player i will choose an action $a \in A(x)$ available at this node such that

$$a \in \arg \max_{a' \in A(x)} \mu_i(\tilde{v}(a' | x)).$$

We record this choice made at node x by setting $s_i(x) = a$.¹³

Now we remove all of x 's immediate successors $N(x)$ from the game's set of terminal nodes and replace them in the set of terminal nodes with x ; i.e. $\bar{Z} \leftarrow (Z \setminus N(x)) \cup \{x\}$. Now that x is no longer a decision node, we remove x from the set of decision nodes; i.e. $\bar{X} \leftarrow \bar{X} \setminus \{x\}$. The payoff vector we assign to the new terminal node x is the payoff vector associated with the terminal node $\check{v}(s_i(x) | x)$ reached when player i takes action $s_i(x)$ at node x ; i.e. $\mu(x) = \mu(\check{v}(s_i(x) | x))$.

Every decision node eventually becomes a penultimate decision for its owner, an action is recorded for it, and it becomes a terminal node. The last decision node to become an penultimate decision node is the initial node $\mathbb{O}$. When an action is determined for the initial node it is removed from the set $\bar{X}$ of current decision nodes, making this set empty. This is the stopping test for the algorithm.

Theorem 2 can be used to show that any strategy profile derived by Zermelo's algorithm in a game of perfect information is a subgame-perfect equilibrium of that game.

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¹³ The function name **typ** in the flowchart might be interpreted as "typical" or "t[ake] y[our] p[ick]." It is, of course, crucial that **typ** truly be a function, i.e. uniquely defined. The only other important property of **typ** S , for $S \neq \emptyset$, is that **typ** $S \in S$. In this context it is just a way of expressing that we need a definite member of the **arg max**, but we don't care which member it is.

Jim Ratliff's Game Theory Course:

5.1: Introduction to Repeated Games

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Abstract



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One striking feature of many one-shot games we study (e.g., the Prisoners' Dilemma) is that the Nash equilibria are so noncooperative: each player would prefer to fink than to cooperate. Repeated games can incorporate phenomena which we believe are important but which aren't captured when we restrict our attention to static, one-shot games. In particular we can strive to explain how cooperative behavior can be established as a result of rational behavior.

We will develop a useful formalism, the semiextensive form, for analyzing repeated games, i.e. those which are repetitions of the same one-shot game (called the stage game). We will describe strategies for such repeated games as sequences of history-dependent stage-game strategies. The payoffs to the players in this repeated game will be functions of the stage-game payoffs. We will define the concept of Nash equilibrium and after identifying the subgames in this formalism the concept of a subgame-perfect equilibrium for a repeated game.

We will prove one result which applies to both finitely and infinitely repeated games, which makes it easy to identify at least some of the Nash equilibria of a repeated game: any sequence of stage-game Nash equilibria is a subgame-perfect repeated-game equilibrium.

We will then focus our attention on finitely-repeated games. We will exploit the existence of a final period to establish a necessary condition for a repeated-game strategy profile to be a Nash equilibrium: the last period's play must be Nash on the equilibrium path.

We then turn to subgame perfection within the finitely-repeated game context and prove two theorems about subgame-perfect equilibria. The first is a necessity theorem which is much stronger than the one we obtained for Nash equilibria: the last period's play must be Nash even off the equilibrium path. The second pertains to the special case in which the stage game has a unique Nash equilibrium payoff. In this case the subgame-perfect equilibria of the repeated game are any period-by-period repetitions of the stage-game Nash equilibria.



[Previous section:
§4.3: Solution Concepts in
Extensive-Form Games](#)

[Course Outline](#)

[Next section:
§5.2: Infinitely Repeated Games
with Discounting](#)

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Repeated Games

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Introduction

One striking feature of many one-shot games we study (e.g., the Prisoners' Dilemma) is that the Nash equilibria are so noncooperative: each player would prefer to fink than to cooperate. Real life is not a one-shot game; in fact, it's not even a disconnected series of one-shot games. Real life is a bigger game in which what a player does early on can affect what others choose to do later on. Repeated games can incorporate phenomena which we believe are important but which aren't captured when we restrict our attention to static, one-shot games. In particular we can strive to explain how cooperative behavior can be established as a result of rational behavior.

We will develop a useful formalism, the semiextensive form, for analyzing repeated games, i.e. those which are repetitions of the same one-shot game (called the stage game). We will describe strategies for such repeated games as sequences of history-dependent stage-game strategies. The payoffs to the players in this repeated game will be functions of the stage-game payoffs. We will define the concept of Nash equilibrium and—after identifying the subgames in this formalism—the concept of a subgame-perfect equilibrium for a repeated game.

We will prove one result which applies to both finitely and infinitely repeated games, which makes it easy to identify at least some of the Nash equilibria of a repeated game: any sequence of stage-game Nash equilibria is a subgame-perfect repeated-game equilibrium.

We will then focus our attention on finitely-repeated games. We will exploit the existence of a final period—which wouldn't exist in an infinitely-repeated game—to establish a necessary condition for a repeated-game strategy profile to be a Nash equilibrium: the last period's play must be Nash on the equilibrium path.

We then turn to subgame perfection within the finitely-repeated game context and prove two theorems about subgame-perfect equilibria. The first is a necessity theorem which is much stronger than the one we obtained for Nash equilibria: the last period's play must be Nash even off the equilibrium path. The second pertains to the special case in which the stage game has a unique Nash equilibrium payoff.¹ In this case the subgame-perfect equilibria of the repeated game are any period-by-period repetitions of the stage-game Nash equilibria.

Repeating the Stage Game

Consider a game G (which we'll call the *stage game* or the *constituent game*). As usual we let the player set be $I = \{1, \dots, n\}$. In our present repeated-game context it will be clarifying to refer to a player's stage-game choices as *actions* rather than strategies. (We'll reserve "strategy" for choices in the repeated game.) So each player has a pure-action space A_i . The space of action profiles is $A = \prod_{i \in I} A_i$. Each player has a von Neumann-Morgenstern utility function defined over the outcomes of G , $g_i: A \rightarrow \mathbb{R}$. (I want to reserve " u " for the payoff to the entire repeated game.)

Let G be played several times (perhaps an infinite number of times) and award each player a payoff which is the sum (perhaps discounted) of the payoffs she got in each period from playing G .² Then this sequence of stage games is itself a game: a *repeated game* or a *supergame*.³

¹ For example, the Prisoners' Dilemma or a Cournot duopoly with linear demand and constant marginal costs.

² This is the typical assumption, although a weaker assumption is sufficient for most results: What is really necessary is that every player's ultimate payoff be an additively separable function of her per-period payoffs.

³ I have conflicting reports on whether supergame is a more general concept than repeated game or vice versa. Friedman [1990: 108] says that "A repeated game is a supergame in which the same (ordinary) game is played at each iteration." On the other side, Mertens [1987: 551] says "'Supergame' is the original name for situations where the same game is played repetitively.... Repeated game is used for more general models...."

Two statements are implicit when we say that in each period we're playing the same stage game:

- a For each player the set of actions available to her in any period in the game G is the same regardless of which period it is and regardless of what actions have taken place in the past.
- b The payoffs to the players from the stage game in any period depend only on the action profile for G which was played in that period, and this stage-game payoff to a player for a given action profile for G is independent of which period it is played.⁴

Statements a and b are saying that the environment for our repeated game is *stationary* (or, alternatively, independent of time and history).⁵ This does *not* mean the actions themselves must be chosen independently of time or history.

We'll limit our attention here to cases in which the stage game is a one-shot, simultaneous-move game. Then we interpret a and b above as saying that the payoff matrix is the same in every period. Our stationarity assumption rules out some economically important situations, such as those which involve investment or learning. Modeling these situations requires a more general dynamic-game framework.

We make the typical “observable action” or “standard signalling” assumption that the play which occurred in each repetition of the stage game is revealed to all the players before the next repetition. Therefore even if the stage game is one of imperfect information (as it is in simultaneous-move games)—so that during the stage game one of the players doesn't know what the others are doing/have done that period—each player *does* learn what the others did before another round is played. This allows subsequent choices to be conditioned on the past actions of other players.

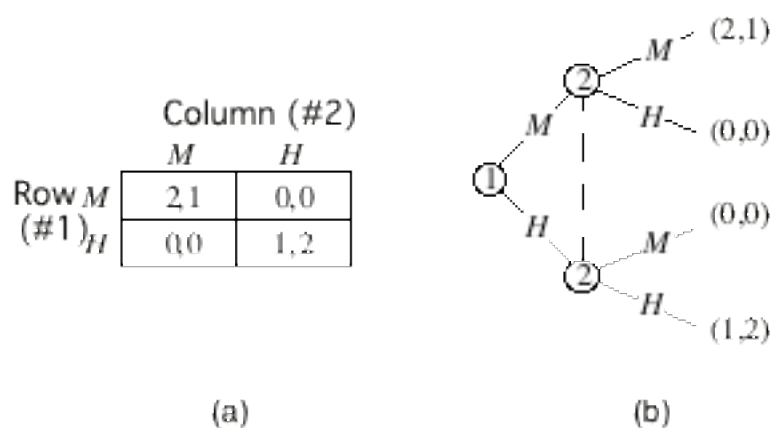
Example: A two-period repetition of the tubular coordination game

As an example, consider the tubular coordination game in normal form in Figure 1a.⁶ Figure 1b depicts an extensive-form game which corresponds to that normal form.

⁴ This doesn't rule out the discounting of payoffs from the stage game. Discounting is performed after the stage game's payoffs are calculated.

⁵ A dependence on time can be different from a dependence on history. The payoffs of the stage game could change with time in a way that didn't depend on what actions had been taken in the past. This would be time dependence. (For example, the payoff to a player to a particular action profile could be proportional to e^{-rt} .) Alternatively, the payoffs in a period could become double that of the previous period's payoffs if a particular action (applying for a patent, for instance) were taken the previous period *regardless of which particular period the action was taken*. This would be a dependence on history but not on time.

⁶ To make a long story short.... Tubular refers to the boob tube. M and H refer to MacNeil-Lehrer and Homer Simpson, respectively. Both of these programs are broadcast in Tucson at 7 p.m. Thursday nights.



Figures 1: A tubular coordination game in normal and extensive forms

We consider this game to be our stage game G . If we repeated G twice we would get the extensive-form game shown in Figure 2.⁷ Here we have assumed there is no discounting, so the payoffs at each terminal node are the sum of the payoffs from each period.⁸ You can see that even just a two-period repetition of the simplest normal-form game imaginable results in a rather complicated extensive-form representation.

The Semiextensive Form

Before we can talk about equilibrium strategies in repeated games, we need to get precise about what a strategy in a repeated game is. As we saw in the above example, even a few repetitions of a simple stage game would yield a very cumbersome game tree. We'll find it useful when studying repeated games to consider the *semiextensive form*.⁹ This is a representation in which we accept the normal-form description of the stage game but still want to retain the temporal structure of the repeated game.

⁷ Note that each player has a total of five information sets in this two-period game. One of the information sets is from the first period and four are from the second period. There are four information sets from the second period because there are four different ways the first round could have been played [viz. (U, L) , (U, R) , (D, L) , and (D, R)].

⁸ Note the italicized stage game payoff vectors with the "+" signs. These indicate the contribution to the players' final payoffs from each stage. Along each path two of these are encountered. Their sum is the payoff vector for the repeated game.

⁹ This terminology is from Friedman [1990: 109].

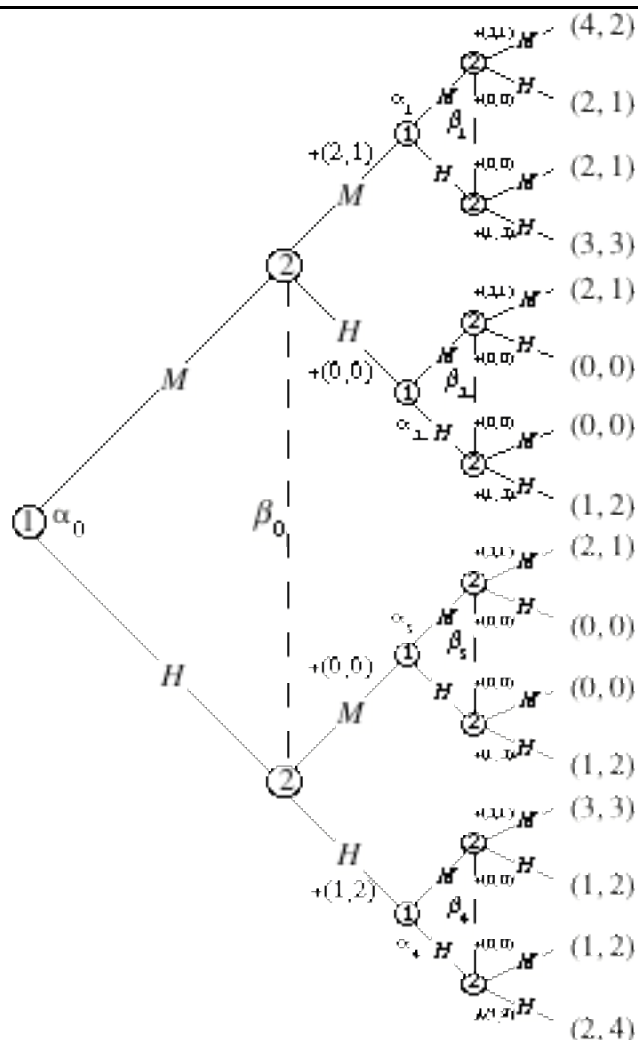


Figure 2: The extensive form for a twice-repeated tubular coordination game.

History and period- t stage-game strategies

I'll try to minimize the notation here, but a certain amount is unavoidable in the pursuit of clarity. We let the first period be labeled $t=0$.¹⁰ The last period, if one exists, is period T , so we have a total of $T+1$ periods in our game. We allow the case where $T=\infty$, i.e. we can have an infinitely repeated game.

We'll refer to the action of the stage game G which player i executes in period t as a_i^t . The *action profile* played in period t is just the n -tuple of individuals' stage-game actions

$$a^t = (a_1^t, \dots, a_n^t). \quad (1)$$

We want to be able to condition the players' stage-game action choices in later periods upon actions

¹⁰ This simplifies some of the discounting expressions we'll encounter later. However, this choice varies from paper to paper, with some adopting $t=1$ as the first period.

taken earlier by other players. To do this we need the concept of a *history*: a description of all the actions taken up through the previous period. We define the history at time t to be

$$h^t = (a^0, a^1, \dots, a^{t-1}).^{11} \quad (2)$$

In other words, the history at time t specifies which stage-game action profile (i.e., combination of individual stage-game actions) was played in each previous period. Note that the specification of h^t includes within it a specification of all previous histories $h^0, h^1, \dots, h^{t-1}$. For example, the history h^t is just the concatenation of h^{t-1} with the action profile a^{t-1} ; i.e. $h^t = (h^{t-1}; a^{t-1})$. The history of the entire game is $h^{T+1} = (a^0, a^1, \dots, a^T)$. Note also that the set of all possible histories h^t at time t is just

$$A^t = \prod_{j=0}^{t-1} A_j, \quad (3)$$

the t -fold Cartesian product of the space of stage-game action profiles A .

To condition our strategies on past events, then, is to make them functions of history.¹² So we write player i 's period- t stage-game strategy as the function s_i^t , where $a_i^t = s_i^t(h^t)$ is the stage-game action she would play in period t if the previous play had followed the history h^t . A player's stage-game action in any period and after any history must be drawn from her action space for that period, but because the game is stationary her stage-game action space A_i does not change with time. Therefore we write: $(\forall i \in I) (\forall t) (\forall h^t \in A^t) s_i^t(h^t) \in A_i$. Alternatively, we can write $(\forall i \in I) (\forall t) s_i^t: A^t \rightarrow A_i$. The period- t stage-game strategy profile s^t is

$$s^t = (s_1^t, \dots, s_n^t). \quad (4)$$

This profile can be described by $s^t: A^t \rightarrow A$. I.e. $\forall h^t \in A^t, s^t(h^t) = (s_1^t(h^t), \dots, s_n^t(h^t))$.

Strategies in the Repeated Game

So far we have been referring to stage-game strategies for a particular period. Now we can write, using these stage-game entities as building blocks, a specification for a player's strategy for the repeated game. We write player i 's strategy for the repeated game as

$$s_i = (s_i^0, s_i^1, \dots, s_i^T), \quad (5)$$

i.e. a $(T+1)$ -tuple of history-contingent player- i stage-game strategies. Each s_i^t takes a history $h^t \in A^t$ as its argument. The space S_i of player- i repeated-game strategies is the set of all such $(T+1)$ -tuples of player- i stage game strategies $s_i^t: A^t \rightarrow A_i$.

We can write a strategy profile s for the repeated game in two ways. We can write it as the n -tuple

¹¹ Again, this can vary from paper to paper. Some would call this history h^{t-1} .

¹² In the first period, when $t=0$, there is no history on which to condition, so the functional dependence is degenerate.

profile of players' repeated-game strategies

$$s = (s_1, \dots, s_n), \quad (6)$$

as defined in (5). Alternatively, we can write the repeated-game strategy profile s as

$$s = (s^0, s^1, \dots, s^T), \quad (7)$$

i.e., as a collection of stage-game strategy profiles, one for each period, as defined in (4). These two alternative perspectives of repeated-game strategy profiles are shown schematically in Figure 3.¹³

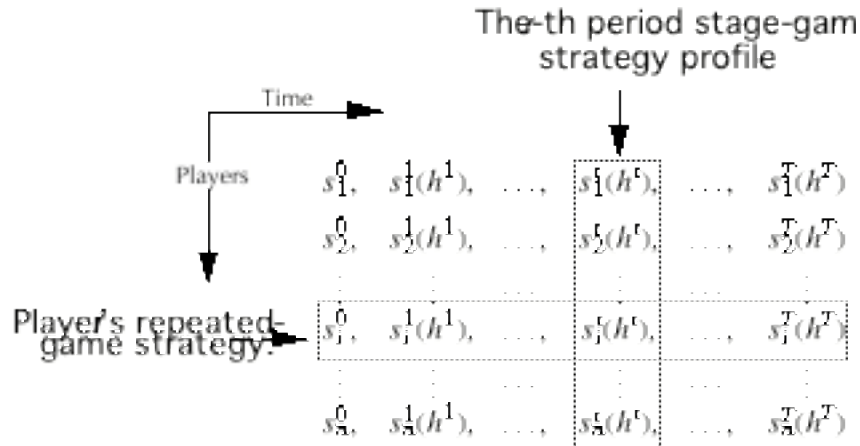


Figure 3: A repeated-game strategy profile viewed alternatively a as players' repeated-game strategies or b as a sequence of stage-game strategy profiles.

Playing the repeated game

In (6) and (7) we have written a repeated-game strategy profile as a collection of functions of histories. It is interesting to note that a repeated-game strategy profile is not itself a function of histories. The reason is simple: Once every one of the individual players' repeated-game strategies s_i is specified (or alternatively once each time period's history-contingent stage-game strategy profile s^t is specified), the sequence of actions, and therefore the history of the entire game, is determined. Perhaps this will become clearer when we see below explicitly how the game is played out.

Let's see how this repeated game is played out once every player has specified her repeated-game strategy s_i . It is more convenient at this point to view this repeated-game strategy profile as expressed in (7), i.e. as a sequence of $T + 1$ history-dependent stage-game strategy profiles. When the game starts, there is no past play, so the history h^0 is degenerate: Every player executes her $a_i^0 = s_i^0$ stage-game strategy from (5). This zero-th period play generates the history $h^1 = (a^0)$, where $a^0 = (a_1^0, \dots, a_n^0)$. This history is then revealed to the players so that they can condition their period-1 play upon the period-0 play. Each player then chooses her $t = 1$ stage-game strategy $s_i^1(h^1)$. Consequently, in the $t = 1$ stage

¹³ We will see that the second form, viz. a sequence of stage-game profiles, is more useful when defining the players' payoffs in the repeated game and in proving some theorems characterizing equilibrium in finitely-repeated games. The first form, viz. as a collection of individual repeated-game strategies, is more useful for defining equilibrium in the repeated game.

game the stage-game strategy profile $a^1 = s^1(h^1) = (s_1^1(h^1), \dots, s_n^1(h^1))$ is played. In order to form the updated history this stage-game strategy profile is then concatenated onto the previous history: $h^2 = (a^0, a^1)$. This new history is revealed to all the players and they each then choose their period-2 stage-game strategy $s_i^2(h^2)$. And so it goes.... This process is shown schematically in Figure 4. We say that h^{T+1} is the *path* generated by the repeated-game strategy profile s .

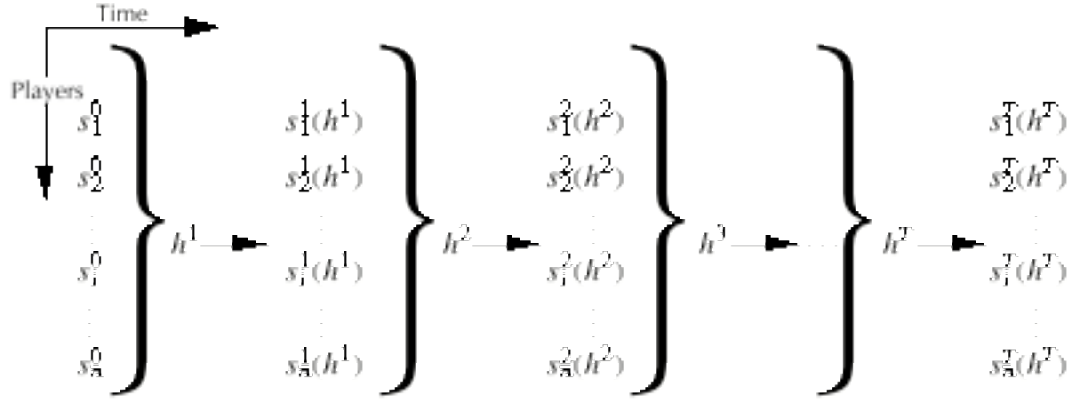


Figure 4: Playing out the repeated game according to a repeated-game strategy.

Payoffs in the Repeated Game

To complete our description of the semiextensive form we need to define the payoffs for this repeated game. Consider some repeated-game strategy profile s and some time period t . As the game is played out according to this repeated-game strategy profile a history h^t of actions up to this period is compiled. The history-contingent stage-game strategy profile to be played this period is $s^t(h^t)$. The payoff to player i from the period- t stage game when each player executes her component of the stage-game strategy profile $s^t(h^t)$ is $g_i(s^t(h^t))$.¹⁴ We define the payoff to player i for the repeated game, when the repeated-game strategy profile s [as expressed in (7)] is played, to be

$$u_i(s) = \sum_{t=0}^T \delta^t g_i(s^t(h^t)), \quad (8)$$

where δ is the common discount factor.¹⁵ Note the appearance of the histories h^t in the summand of (8). These are the histories which are generated sequentially as the game is played, as described in Figure 4. I.e. $h^t = (h^{t-1}; s^t(h^{t-1}))$.

¹⁴ Note that this g_i doesn't require a superscript " t " because the game is stationary and hence the stage-game payoff functions don't change with time.

¹⁵ Assuming a common discount factor is done for simplicity only. The analysis can be carried out allowing the discount factor to vary between players and also across time for each player.

Equilibrium in Repeated Games

Nash Equilibrium in the Repeated Game

Now that we know what strategies and payoffs are in the repeated game, we are in position to make clear what a Nash equilibrium is in this semiextensive form of a repeated game. Now we will use the expression of a repeated-game strategy profile as a collection of players' repeated-game strategies, as in (6). As usual, we say that a strategy profile $\bar{s}$ for the repeated game is a Nash equilibrium if each player's part of $\bar{s}$ —i.e., the strategy $\bar{s}_i$, as given by (5)—is a best response given that the other players are playing their parts of $\bar{s}$. More formally, we say that the repeated-game strategy profile $\bar{s}$ is a Nash equilibrium if for all players i

$$\bar{s}_i \in \arg \max_{s_i \in S_i} u_i(s_i, \bar{s}_{-i}). \quad (9)$$

Let $\bar{h}^{T+1}$ be the history generated by the repeated-game strategy profile $\bar{s}$; i.e. let $\bar{h}^{T+1}$ be the path associated with $\bar{s}$. If $\bar{s}$ is a Nash-equilibrium strategy profile, then $\bar{h}^{T+1}$ is its associated *equilibrium path*.

Note from (8) that $u_i(\bar{s})$ depends only upon stage-game strategy profiles played along the equilibrium path. Therefore in a Nash equilibrium each player's repeated-game strategy need only be optimal along the equilibrium path.

Subgame-Perfect Equilibrium in the Repeated Game

We know that a subgame-perfect equilibrium strategy profile $\bar{s}$ is one such that the restriction of $\bar{s}$ to any subgame is a Nash-equilibrium strategy profile in that subgame. We know what a subgame is in the familiar extensive form of a game, but what is a subgame in this semiextensive form?

A subgame, as we know, is a piece of the bigger game which—if we detach it—still makes sense as a game by itself and, if reached, it would be common knowledge that this was the game remaining to be played. Clearly, we could begin our repeated game at some other period, say the k -th period, and the shorter repeated game would still make sense as a game by itself. It is tempting to say, then, that we can find subgames by simply jumping in at any period and playing the game from there. Although tempting, this solution to the “what are the subgames?” problem isn't quite right. We'll see that a subgame of a repeated game is identified by a beginning period *plus* a specification of the history of actions up to that period.

To see why our definition of a subgame of a repeated game must include a specification of the history we can look back at Figure 2, which depicts the extensive form for our twice-repeated tubular coordination game. Clearly each of the second-period repetitions of the stage game (the parts of the tree beginning at the nodes labeled $\alpha_1 \rightarrow \alpha_4$) is a subgame. Even though all four instances of this second-period repetition of the stage game look very similar—because they *are* exactly the same stage game—the four do not constitute four clones of a single subgame but rather they are four distinguishable subgames. (For example, they lead to different payoffs.) Note that the history of actions leading to each

one is different. For example, the history leading to the repetition corresponding to the information set α_1 is $h^1 = (a^0) = (M, M)$, but the history leading to the α_4 repetition is $h^1 = (a^0) = (H, H)$. Therefore we see that we need to specify the history of actions in order to identify a particular subgame within the set of those subgames which begin at a particular period. Or another way to see this: When a subgame is reached in the game, the players know the history of play. Therefore when we analyze the subgame, we must allow the players all the information they would have in the original game if they encountered the subgame.

Consider some strategy profile $\bar{s}$ of the original full-length repeated game in extensive form. It specifies an action at each information set and can be written in the form:

$$\bar{s} = (\alpha_0, \alpha_1, \alpha_2, \alpha_3, \alpha_4; \beta_0, \beta_1, \beta_2, \beta_3, \beta_4). \quad (10)$$

In general $\bar{s}$ can specify a different action by player #1 at each of the information sets $\alpha_1 \rightarrow \alpha_4$. Similarly, it can specify a different action by player #2 at each of the information sets $\beta_1 \rightarrow \beta_4$.

If we want to determine whether $\bar{s}$ is a subgame-perfect equilibrium strategy profile of the full-length repeated game, we need to examine its restrictions to every subgame. The restriction of $\bar{s}$ to the subgame beginning with α_k is of the form (α_k, β_k) , $k \in \{1, \dots, 4\}$. Because these (α_k, β_k) pairs are not necessarily the same (because they could be specified independently in $\bar{s}$), the restriction of $\bar{s}$ to each subgame is a possibly different pair of actions. Asking whether $\bar{s}$ is a subgame-perfect strategy profile is the same as asking whether the restriction of $\bar{s}$ to each subgame is a Nash equilibrium of that subgame. But this is the same as asking whether the pair of actions (α_k, β_k) is a Nash-equilibrium strategy profile for the stage game G . Because all four of these (α_k, β_k) pairs can be different, we need to ask this question four times, once for each $k \in \{1, \dots, 4\}$. Therefore it is possible that the restriction of $\bar{s}$ to one subgame will be a Nash equilibrium in the stage game, but that the restriction to a different subgame will not be a Nash equilibrium in that stage game.

We see that it's not enough to say that we're interested in the subgame which begins with period k . There are many subgames corresponding to the repeated game beginning with period k ; there are as many such subgames as there are histories up until that point.

How then do we decide whether some repeated-game strategy profile $\bar{s}$ is a subgame-perfect equilibrium? First we must proceed period by period to begin the process of decomposing the repeated game into its subgames. For each period t , then, we must consider every possible history h^t , because there exists a subgame which corresponds to that history. Now that we have identified a subgame beginning at period t with history h^t , we must ask whether the restriction to this subgame of the original repeated-game strategy profile $\bar{s}$ is a Nash equilibrium of this subgame.

A sequence of stage-game Nash equilibria is a repeated-game subgame-perfect equilibrium

It will be useful to distinguish a particularly simple class of repeated-game strategies: open-loop strategies. When a player's strategy depends on the history h^t , we say that it is a *closed-loop* strategy. If for every t her stage game strategy at time t depends only on t but not on previously taken actions, then

her strategy is an *open-loop* strategy. To be more explicit... a repeated-game strategy $s_i^t: A^t \rightarrow A_i$ for player i is an open-loop strategy if

$$(\forall t) (\exists a_i^t \in A_i) (\forall h^t \in A^t) \square s_i^t(h^t) = a_i^t.^{16} \quad (11)$$

In this case we can simply write player i 's repeated-game strategy as a $(T + 1)$ -tuple of stage-game actions $s_i = (a_i^0, a_i^1, \dots, a_i^T)$. If player i is playing an open-loop strategy, then no action of player j 's will ever influence a later action by player i . You should be able to show that if all other players are implementing open-loop strategies, then the remaining player might as well play an open-loop strategy as well. (Because all other players' actions are known, and these actions won't be changed as a result of your choices, then you are free to optimize against others' actions on a myopic, period-by-period basis.)

Our first theorem states a sufficient condition for a strategy profile of a repeated game to be a subgame-perfect equilibrium:

Theorem 1

A sequence of history-independent (i.e. open-loop) stage-game Nash-equilibrium strategy profiles is a subgame-perfect equilibrium in the (possibly infinitely) repeated game. More formally: Let $\bar{s} = (\bar{a}^0, \dots, \bar{a}^T)$ be a history-independent strategy profile for the repeated game.¹⁷ If each of the stage-game strategy profiles $\bar{a}^t$ of $\bar{s}$ is a Nash-equilibrium strategy profile for the stage game, then $\bar{s}$ is a subgame-perfect equilibrium strategy profile for the repeated game.

Proof

(By contradiction): First we show that $\bar{s}$ is a Nash equilibrium of the repeated game. Assume not. Then for some player i there is an alternative open-loop repeated-game strategy $\hat{s}_i = (\hat{a}_i^0, \dots, \hat{a}_i^T)$ which is different (in at least one period) from her part $\bar{s}_i = (\bar{a}_i^0, \dots, \bar{a}_i^T)$ of $\bar{s}$ such that her payoff in the repeated game is higher, given that everyone else plays their parts of $\bar{s}$.¹⁸ In order that her repeated-game payoff be higher, there must be at least one period t in which her stage-game payoff using $\hat{s}_i^t$ is higher than the stage-game payoff she would get from playing $\bar{s}_i^t$. But if that is true, then $\bar{s}_i^t$ could not have been a part of a Nash-equilibrium strategy profile of the stage game, because some other strategy for player i , viz. $\hat{s}_i^t$, would have been better for i . This contradicts the hypothesis that every component of $\bar{s}$ is a Nash equilibrium of the stage game.

Now consider any subgame of this repeated game. For any t and any history h^t , the restriction of $\bar{s}$ to the subgame associated with history h^t still prescribes a history-independent sequence of stage-game Nash-equilibrium profiles. We have already proved that such a sequence is a Nash equilibrium of a repeated game. Therefore the restriction of $\bar{s}$ to any subgame is a Nash equilibrium in that subgame. Therefore $\bar{s}$ is subgame perfect. ☺

¹⁶ Alternatively... $\forall t, \forall h^t, \tilde{h}^t \in A^t, s_i^t(h^t) = s_i^t(\tilde{h}^t)$.

¹⁷ In other words this strategy profile calls upon each player to take the specified action in each period without regard to what any other player actually did in previous periods.

¹⁸ Note that I wrote the alternative strategy $\hat{s}_i$ as an open-loop strategy (i.e. history independent), which is less general than a closed-loop strategy. I gave a sketch of the justification above.

Finitely-repeated games

We now turn to finitely-repeated games. Here, when we say that a game is repeated a finite number of times we not only mean that the game eventually ends but usually also that the stage game is played a given, fixed number of times T and that this number of repetitions is common knowledge. We can exploit the existence of a commonly known last period in order to obtain some results dramatically different from infinitely repeated games.

Nash Equilibria

We now establish a necessary condition for a repeated-game strategy profile $\bar{s}$ to be a Nash equilibrium of the repeated game.

On the equilibrium path the last period's play must be a stage-game Nash equilibrium

Theorem 2

Let $\bar{s} = (\bar{s}^0, \bar{s}^1, \dots, \bar{s}^T)$ be a Nash-equilibrium strategy profile for the repeated game and let $\bar{h}^T$ be its associated equilibrium path (i.e. the history which is generated when all players play their parts of $\bar{s}$). Then on the equilibrium path the last period's play must be a Nash equilibrium of the stage game G ; i.e. the stage-game strategy profile $\bar{s}^T(\bar{h}^T)$ must be a Nash equilibrium of the stage game G .

Proof

(By contradiction): Assume that $\bar{s}^T(\bar{h}^T)$ is not a Nash equilibrium of the stage game. Then for some player i , there is a stage-game action $\hat{a}_i \in A_i$ which is different in the last period from her part of $\bar{s}$ along the equilibrium path, i.e., $\hat{a}_i \neq \bar{s}_i^T(\bar{h}^T)$, such that she would increase her payoff in period T by playing this other strategy $\hat{a}_i$, rather than playing $\bar{s}_i^T(\bar{h}^T)$, given that the other players play $\bar{s}_{-i}^T(\bar{h}^T)$. But we see from the definition of the payoff for the repeated game in (8) that if player i increases her payoff in the T -th period (without changing any of her other periods' payoffs), she increases her repeated-game payoff.¹⁹ Therefore, she would do better in the repeated game to play the strategy

$$\hat{s}_i = (\bar{s}_i^0, \bar{s}_i^1, \dots, \bar{s}_i^{T-1}, \hat{a}_i), \quad (12)$$

rather than the entirety of her part of the alleged equilibrium strategy profile. Therefore, contrary to assumption, $\bar{s}$ must not have been a Nash equilibrium of the repeated game. 😊

Subgame-Perfect Equilibria

We learned above that in a Nash equilibrium of a repeated game the play in the last period along the equilibrium path must be a Nash equilibrium in the stage game. If our strategy profile $\bar{s}$ of the repeated

¹⁹ How do we know that her defection from the equilibrium doesn't affect any of her other payoffs? There are no later payoffs to consider, and previous payoffs depend on previous actions which are conditioned at most on *previous* histories. Therefore no action in period T can affect an earlier stage-game payoff.

game is not only a Nash equilibrium but is also a subgame-perfect equilibrium of the repeated game, then we can prove a theorem which states a stronger necessary condition for $\bar{s}$.

Subgame perfection requires that the last period's play be a stage-game Nash equilibrium for every possible history

Theorem 3

In a subgame-perfect equilibrium of the repeated game the last-period play for *any* history (i.e., not just along the equilibrium path) must be a Nash equilibrium of the stage game. More formally: If $\bar{s} = (\bar{s}^0, \bar{s}^1, \dots, \bar{s}^T)$ is a subgame-perfect equilibrium strategy profile of the repeated game, then, for any conceivable history $h^T \in A^T$, the last period's stage-game strategy profile, $\bar{s}^T(h^T)$, must be a Nash equilibrium of the stage game.

Note the difference between Theorem 3 and Theorem 2. Theorem 2 only required—in order that a repeated-game strategy profile was a Nash equilibrium—that the last period's stage-game strategy profile *along the equilibrium path*—i.e. given that everyone played what she was supposed to play according to $\bar{s}$ —was a Nash-equilibrium stage-game strategy profile. Theorem 3 requires—in order that $\bar{s}$ be a subgame-perfect equilibrium strategy profile for the repeated game—that no matter what actually occurs in periods $0 \rightarrow (T - 1)$ every player is compelled by $\bar{s}$ to play their part of a stage-game Nash equilibrium in the last period. This theorem is one consequence of subgame perfection's requirement that behavior be optimal even off the equilibrium path.

Proof

(By definition): As we have seen, if we specify the history by which we reach the last period, then we have completely specified a subgame which consists solely of one play of the stage game. A subgame-perfect equilibrium of the repeated game must induce Nash-equilibrium behavior in every subgame, therefore in particular its restriction to the last period for any possible history must be a Nash equilibrium of the stage game. ☺

A unique stage-game Nash equilibrium payoff vector is repeated in every repeated-game subgame-perfect equilibrium

A special case arises when the stage game has a unique Nash-equilibrium payoff vector.²⁰ (Of course, this includes the case where the stage game has a unique Nash equilibrium.) Then we can use a backward-induction argument to show that every subgame-perfect equilibrium strategy profile of the repeated game involves a repetition every period of a Nash-equilibrium stage-game strategy profile and hence a repetition of the unique Nash-equilibrium stage-game payoff vector.

²⁰ I.e. there may be multiple Nash equilibria, but each one yields the same payoff vector.

Theorem 4

Let $\bar{A}$ be the set of stage-game Nash equilibria. Further, let $\bar{A}$ have the property that there exists a payoff vector $\bar{u} \in \mathbb{R}^n$ such that, for all stage-game Nash equilibria $\bar{a} \in \bar{A}$ and for all players $i \in I$, $u_i(\bar{a}) = \bar{u}_i$.

Let $\bar{s} = (\bar{s}^0, \bar{s}^1, \dots, \bar{s}^T)$ be any subgame-perfect equilibrium strategy profile of the repeated game. Then for all t and all $h^t \in A^t$,

$$\bar{s}^t(h^t) \in \bar{A}. \quad (13)$$

Proof

Since we know that $\bar{s}$ is subgame perfect, we know that regardless of how we reach the last period, i.e. for whatever $h^T \in A^T$, when we get there a stage-game Nash-equilibrium strategy profile $\bar{s}^T(h^T) \in \bar{A}$ will be played. This yields the payoff vector $\bar{u}$. Therefore the payoffs to each player from the final stage game will be independent of the history, and in particular the payoffs will be independent of the play in the penultimate, $T - 1$, period.

For any history h^{T-1} , the restriction of $\bar{s}$ must be a Nash equilibrium in the subgame h^{T-1} defines. Therefore a stage-game Nash-equilibrium strategy profile will be played in this subgame in period $T - 1$. (Assume not. Then some player would be able to increase her stage-game payoff in that period. But that would increase her repeated-game payoff as well, because her final-period payoff is independent of anything she chooses now. So if something other than a stage-game Nash equilibrium were played in any subgame beginning in period $T - 1$, then the restriction of $\bar{s}$ would not be a Nash equilibrium of the subgame.) So for any history h^{T-1} , the players receive the unique stage-game Nash-equilibrium payoff vector $\bar{u}$ in period $T - 1$.

Now their payoffs in the last two periods are determined independently of their previous actions. So the same argument goes through for the $T - 2$ period, and so on, establishing that every period's play after any history is a stage-game Nash-equilibrium profile. 😊

Example: A two-period repeated game

Consider the two-player 3×3 stage game in Figure 5. The stage game has two pure-strategy Nash equilibria, viz. (U, l) and (M, m) , and importantly these Nash equilibria result in different payoff vectors.²¹ Let's determine the set of repeated-game subgame-perfect equilibrium payoffs of the repeated game in which this stage game is repeated twice. (We will simply use the sum of the two per-period payoffs, without discounting, as the repeated-game payoff.)

²¹ If there were only one stage-game Nash equilibrium payoff vector, Theorem 4 would tell us that, in any subgame-perfect repeated-game equilibrium, a stage-game Nash equilibrium would be played in every period for every history.

	<i>l</i>	<i>m</i>	<i>r</i>
<i>U</i>	5,5	-1,-1	-2,-2
<i>M</i>	-1,-1	0,0	-2,-2
<i>D</i>	-2,-2	-2,-2	-6,-6

Figure 5: A two-player game with two pure-strategy Nash equilibria.

A simple punishment structure

We will only consider repeated-game strategy profiles which can be expressed in a particularly simple form: as four pairs of stage-game action profiles of the form

$$s = \langle (A, b), (C, d); (E, f), (G, h) \rangle, \quad (14)$$

where $A, C, E, G \in S_R = \{U, M, D\}$ and $b, d, f, g \in S_C = \{l, m, r\}$. We interpret this repeated-game strategy profile in the following way: The first two stage-game action profiles, viz. $\langle (A, b), (C, d) \rangle$ represent the path of the strategy profile. (In other words, in the first period Row should play A and Column should play b . If both players conformed in the first period, then in the second period Row should play C and Column should play d .)

The third and fourth stage-game action profiles are *punishment* profiles to be played in the second period if one player unilaterally deviates in the first period from the prescribed action profile (A, b) ; the punishment (E, f) is directed against Row if she deviates unilaterally from the prescribed first-period action profile (A, b) ; (G, h) is directed against Column if he deviates unilaterally from the prescribed first-period action profile (A, b) .

To be perfectly clear... Let the observed first-period action profile be $(X, y) \in S_R \times S_C$. We say that Row unilaterally deviated from (A, b) if $(X \neq A \text{ and } y = b)$; Column unilaterally deviated from (A, b) if $(X = A \text{ and } y \neq b)$. If $(X \neq A \text{ and } y \neq b)$, then neither player unilaterally deviated. [In this case we interpret the strategy profile s as instructing the players to play their on-the-path action profile (C, d) in the second period. In other words, play continues as if they had both conformed in the first period.²²] If Row unilaterally deviates in the first period (by choosing some $X \neq A$ when Column plays b), then s dictates that in the second period Row plays E and Column plays f . If Column unilaterally deviates in the first period, then in the second period Row should play G and Column should play h .

Equilibrium implications

What would an open-loop repeated-game strategy profile look like in this game? Being open loop implies that the second-period play would be independent of history (i.e. independent of first-period play and therefore independent of whether the players conformed to the first-period prescription). Therefore s is an open-loop repeated-game strategy profile if and only if

²² It is unimportant what specification we make for the second period following a bilateral deviation in the first period. When we check whether a strategy profile is a Nash equilibrium, we only compare a player's conformity payoff with her payoffs to possible unilateral deviations. Nowhere does her payoff to a multilateral deviation appear in the equilibrium-verification computations.

$$(C, d) = (E, f) = (G, h). \quad (15)$$

Requiring s to be a repeated-game Nash-equilibrium strategy profile puts an immediate restriction on one of the action profiles, viz. the second-period on-the-path action profile (C, d) . From Theorem 2, we know that this action profile must be a stage-game Nash-equilibrium action profile. In our example, then, if s is a repeated-game Nash-equilibrium,

$$(C, d) \in \{(U, l), (M, m)\}. \quad (16)$$

Requiring s to be a repeated-game subgame-perfect equilibrium imposes restrictions on the two punishment action profiles, viz. (E, f) and (G, h) , as well. [Of course, (C, d) is still constrained by (16) because any subgame-perfect equilibrium is also a Nash equilibrium. Alternatively, (C, d) is still constrained because it is a last-period action profile.] Both of these are action profiles for the last period, albeit corresponding to off-the-path information sets. Theorem 3 tells us that in order for s to be subgame perfect both of these action profiles must be stage-game Nash-equilibrium action profiles. So for s to be subgame perfect it is necessary that

$$(C, d), (E, f), (G, h) \in \{(U, l), (M, m)\}. \quad (17)$$

Subgame-perfect equilibrium payoffs via open-loop strategy profiles

Recall that Theorem 1 gave us a way to identify some of the repeated game's subgame-perfect equilibria: Construct open-loop strategy profiles which always choose stage-game Nash-equilibrium action profiles. For example, we can achieve the repeated-game payoff vector $(10, 10)$ as a subgame-perfect equilibrium via the open-loop strategy profile

$$\langle (U, l), (U, l); (U, l), (U, l) \rangle. \quad (18)$$

This is an open-loop sequence of stage-game Nash-equilibrium action profiles according to (15).

Similarly we can support the repeated-game payoff vector $(0, 0)$ as a subgame-perfect equilibrium via the open-loop strategy profile

$$\langle (M, m), (M, m); (M, m), (M, m) \rangle. \quad (19)$$

We can achieve the repeated-game payoff vector $(5, 5)$ as a subgame-perfect equilibrium via either of the following two open-loop strategy profiles:

$$\langle (M, m), (U, l); (U, l), (U, l) \rangle \text{ or } \langle (U, l), (M, m); (M, m), (M, m) \rangle. \quad (20)$$

These three repeated-game payoff vectors, viz. $(10, 10)$, $(5, 5)$, and $(0, 0)$, exhaust the repeated-game payoff vectors which can be supported as subgame-perfect equilibrium payoffs using open-loop strategy profiles.

Non-open-loop subgame-perfect equilibria: general principles

Consider a repeated-game strategy profile which is not an open-loop sequence of stage-game Nash-

equilibrium action profiles. By choosing each second-period action profile to be a stage-game Nash equilibrium [i.e. $(C, d), (E, f), (G, h) \in \{(U, l), (M, m)\}$ in accordance with (17)] we ensure that no player will choose to deviate in the second period no matter what transpires in the first period. (When the second period is reached, the remaining game is simply a one-shot play of the stage game. No player has an incentive to unilaterally deviate from the stage-game Nash equilibrium in the second period.)

The challenge, however, is to construct the four action profiles describing a strategy profile so that no player will want to deviate from the prescribed *first*-period action profile. Let's see under what conditions a player would choose to unilaterally deviate from her first-period prescription. A player will deviate in the first period if she prefers a the combination of first-period deviance followed by second-period punishment to b conformity with the path in both periods.

Under our four-action-profile framework in (14), a player is punished for a unilateral deviation independently of precisely which deviation she commits. Therefore we can without loss of generality assume that she picks her myopically most advantageous defection. For example let's consider some proposed strategy profile $s = \langle (A, b), (C, d); (E, f), (G, h) \rangle$. Consider the Row player for definiteness. (The analysis for the Column player is parallel.) Let $\bar{g}_R(b; A)$ be the first-period value of Row's most advantageous stage-game defection from some first-period prescribed action profile (A, b) . I.e.

$$\bar{g}_R(b; A) = \max_{X \in \{U, M, D\} \setminus \{A\}} g_R(X, b). \quad (21)$$

If Row unilaterally deviates in the first period, she can secure herself a payoff of $\bar{g}_R(b; A)$ in the first period. Then she will be subjected to the punishment action profile (E, f) in the second period. Row will be willing to conform to the strategy profile s , then, if

$$\bar{g}_R(b; A) + g_R(E, f) \leq g_R(A, b) + g_R(C, d). \quad (22)$$

For a given prescribed path $\langle (A, b), (C, d) \rangle$, we see from (17) and (22) that we can without loss of generality choose Row's punishment to be the stage-game Nash-equilibrium action profile which is worst for Row. I.e. for Row's punishment we are free to choose (E, f) to solve

$$(E, f) \in \arg \min_{(X, y) \in \{(U, l), (M, m)\}} g_R(X, y). \quad (23)$$

“Without loss of generality” here means the following: Let (E', f') be some Row-punishing action profile which gives Row a sufficiently small payoff $g_R(E', f')$ so that (22) is satisfied and therefore that Row is deterred from a first-period deviation. Then (E, f) as defined in (23) is also such that (22) is satisfied and therefore also deters Row from deviating.

In this game, the same stage-game Nash-equilibrium action profile is the worst for both players, viz. (M, m) . Therefore we set

$$(E, f) = (G, h) = (M, m). \quad (24)$$

We want to determine the set of repeated-game payoffs which can be supported as subgame-perfect

equilibrium payoffs. To construct candidate subgame-perfect repeated-game strategy profiles we use (24) to define the punishments and (16) to constrain our choice for the second-period on-the-path action profile. Our choice for the first-period on-the-path action profile (A, b) is unconstrained.

The subgame-perfect equilibrium payoffs

We have already seen how to support $(10, 10)$, $(5, 5)$, and $(0, 0)$ as subgame-perfect two-period repeated-game payoffs using open-loop strategy profiles. We can also support as a subgame-perfect equilibrium the two-period payoff vector $(-1, -1)$. The path we choose to support is $\langle (D, r), (U, l) \rangle$ which yields $(-6, -6) + (5, 5) = (-1, -1)$. Using the punishments from (24) gives us the strategy profile $s = \langle (D, r), (U, l); (M, m), (M, m) \rangle$. To verify that this is indeed a subgame-perfect equilibrium we first verify that (17) is satisfied. To check that Row would not choose to deviate in the first period we calculate her most advantageous deviation against Column's action r to be either U or M which yields Row a first-period deviation payoff of $\bar{g}_R(r; D) = -2$. We verify that (22) holds, then, by noting that indeed $-2 + 0 \leq -6 + 5$. An exactly parallel computation confirms that Column would not choose to deviate either.

You can show that we can also support a two-period payoff of $(3, 3)$ as a subgame-perfect equilibrium via the path $\langle (M, r), (U, l) \rangle$.

However, we cannot support $(4, 4)$ because such a two-period payoff can only be achieved by a first-period payoff of $(-1, -1)$ followed by $(5, 5)$.²³ However, a payoff of $(-1, -1)$ requires either (M, l) or (U, m) . From either of these action profiles one player would be able to deviate in order to receive 5 in the first period. Because this already exceeds the total conformity payoff, conformity would never be preferred.

We also cannot support a payoff of $(-2, -2)$. This would require a sequence of payoffs on the equilibrium path of $(-2, -2)$ followed by $(0, 0)$. Therefore (M, m) is the implied action profile in the second period on the path; but this is already each player's worst stage-game Nash equilibrium. Therefore there is no effective punishment to wield against a player who defects in the first period, because she's already getting her worst punishment even if she conforms. More formally... from (22), if the second-period on-the-path prescription is also a worst-for-Row Nash equilibrium, i.e. $g_R(C, d) = g_R(E, f)$, then subgame perfection requires that $g_R(A, b) \geq \bar{g}_R(b; A)$. I.e. A must be a best response by Row to Column's choice b . When this holds for both players, as it does here, this implies that if the second-period prescription is (M, m) then the first-period prescription must be a stage-game Nash equilibrium.

We can also show that any subgame-perfect equilibrium of the repeated game must give each player at least -2 for a two-period repeated-game payoff. Calculate the minimum value of Row's most advantageous defections for each of Column's three stage-game actions l , m , and r . For example, if Column is playing l , Row can earn at least -1 by defecting; $\min \{ \bar{g}_R(l; X) : X \in \{U, M, D\} \} = -1$. (You

²³ We can't have $(5, 5)$ followed by $(-1, -1)$ because the second-period payoff must correspond to a stage-game Nash equilibrium.

might say “Row can earn at least 5 by defecting against l .” But if Row is supposed to be playing U already, then U would not be a defection.) Similarly $\min \{ \bar{g}_R(m; X) : X \in \{U, M, D\} \} = -1$. And against r , $\min \{ \bar{g}_R(r; X) : X \in \{U, M, D\} \} = -2$. So, for all $(X, y) \in \{U, M, D\} \times \{l, m, r\}$,

$$\bar{g}_R(y; X) \geq -2. \quad (25)$$

Reconsider the condition (22) for Row’s conformity. Note that the right-hand side is Row’s two-period payoff to the strategy profile s , and that Row’s payoff to her worst stage-game Nash-equilibrium action profile (M, m) is 0. Condition (22) then implies that Row’s two-period payoff in any subgame-perfect equilibrium must exceed $\bar{g}_R(b; A) \geq -2$. An exactly parallel analysis holds for Column.

So we have found all of the subgame-perfect equilibrium payoffs.

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Jim Ratliff's Game Theory Course:

5.2: Infinitely Repeated Games with Discounting

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Abstract



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We'll now discuss repeated games which are "infinitely repeated." This need not mean that the game never ends, however. We will see that this framework is appropriate for modeling situations in which the game eventually ends (with probability one) but the players are uncertain about exactly when the last period is (and they always believe there's some chance the game will continue to the next period).

Infinite repetitions of the stage game potentially pose a problem: a player's repeated-game payoff may be infinite. We ensure the finiteness of the repeated-game payoffs by introducing discounting of future payoffs relative to earlier payoffs. Such discounting can be an expression of time preference and/or uncertainty about the length of the game. We introduce the average discounted payoff as a convenience which normalizes the repeated-game payoffs to be "on the same scale" as the stage-game payoffs.

Infinite repetition can be key to obtaining behavior in the stage games which could not be equilibrium behavior if the game were played once or a known finite number of times. For example, finking every period by both players is the unique equilibrium in any finite repetition of the prisoners' dilemma. When repeated an infinite number of times, however, cooperation in every period is an equilibrium if the players are "sufficiently patient."

We first show that cooperation is a Nash equilibrium of the infinitely repeated game for sufficiently patient players. Then we discuss subgames of infinitely repeated games and show that, in our uniformly discounted payoff scenario, a subgame is the same infinitely repeated game as the original game. We derive how to restrict a repeated-game strategy to a subgame. This allows us to complete the analysis of the example by showing that cooperation in every period by both players is a subgame-perfect equilibrium of the infinitely repeated prisoners' dilemma for sufficiently patient players.



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§5.1: Introduction to Extensive-
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Infinitely Repeated Games with Discounting

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Introduction

We'll now discuss repeated games which are "infinitely repeated." This need not mean that the game never ends, however. We will see that this framework is appropriate for modeling situations in which the game eventually ends (with probability one) but the players are uncertain about exactly when the last period is (and they always believe there's some chance the game will continue to the next period).

We'll call the stage game G and interpret it to be a simultaneous-move matrix game which remains exactly the same through time. As usual we let the player set be $I = \{1, \dots, n\}$. Each player has a pure action space A_i .¹ The space of action profiles is $A = \prod_{i \in I} A_i$. Each player has a von Neumann-Morgenstern utility function defined over the outcomes of G , $g_i: A \rightarrow \mathbb{R}$.

The stage game repeats each period, starting at $t = 0$. Although each stage game is a simultaneous-move game, so that each player acts in ignorance of what her opponent is doing that period, we make the "observable action" or "standard signaling" assumption that the play which occurs in each repetition of the stage game is revealed to all the players before the next repetition. Combined with perfect recall, this allows a player to condition her current action on all earlier actions of her opponents.

We can think of the players as receiving their stage-game payoffs period-by-period. Their repeated-game payoffs will be an additively separable function of these stage-game payoffs. Right away we see a potential problem: There is an infinite number of periods and, hence, of stage-game payoffs to be added

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¹ I want to reserve "s" and "S" to represent typical strategies and strategy spaces, respectively, in the repeated game. So I'm using "a" and "A" for stage-game actions and action spaces.

up. In order that the players' repeated-game payoffs be well defined we must ensure that this infinite sum does not blow up to infinity. We ensure the finiteness of the repeated-game payoffs by introducing discounting of future payoffs relative to earlier payoffs. Such discounting can be an expression of time preference and/or uncertainty about the length of the game. We introduce the average discounted payoff as a convenience which normalizes the repeated-game payoffs to be “on the same scale” as the stage-game payoffs.

Infinite repetition can be key to obtaining behavior in the stage games which could not be equilibrium behavior if the game were played once or a known finite number of times. For example, finking every period by both players is the unique equilibrium in any finite repetition of the prisoners' dilemma.² When repeated an infinite number of times, however, cooperation in every period is an equilibrium if the players are “sufficiently patient.”

We first show that cooperation is a Nash equilibrium of the infinitely repeated game for sufficiently patient players. Then we discuss subgames of infinitely repeated games and show that, in our uniformly discounted payoff scenario, a subgame is the same infinitely repeated game as the original game. We derive how to restrict a repeated-game strategy to a subgame. This allows us to complete the analysis of the example by showing that cooperation in every period by both players is a subgame-perfect equilibrium of the infinitely repeated prisoners' dilemma for sufficiently patient players.

Discounting the future

When we study infinitely repeated games we are concerned about a player who receives a payoff in each of infinitely many periods. In order to represent her preferences over various infinite payoff streams we want to meaningfully summarize the desirability of such a sequence of payoffs by a single number. A common assumption is that the player wants to maximize a weighted sum of her per-period payoffs, where she weights later periods less than earlier periods.

For simplicity this assumption often takes the particular form that the sequence of weights forms a geometric progression: for some fixed $\delta \in (0, 1)$, each weighting factor is δ times the previous weight. (δ is called her *discount factor*.) If in each period t player i receives the payoff u_i^t , we could summarize the desirability of the payoff stream $u_i^0, u_i^1, \dots$ by the number^{3,4}

$$\sum_{t=0}^{\infty} \delta^t u_i^t. \quad (1)$$

Such an intertemporal preference structure has the desirable property that the infinite sum of the

² See Theorem 4 of “Repeated Games.”

³ It's imperative that you not confuse superscripts with exponents! You should be able to tell from context. For example, in the summation (1), the t in “ δ^t ” is an exponent; in x_i^t , it is a superscript identifying the time period.

⁴ Note that this assigns a weight of unity to period zero. It's the relative weights which are important (why?), so we are free to normalize the weights any way we want.

weighted payoffs will be finite (since the stage-game payoffs are bounded). You would be indifferent between a payoff of x^t at time t and a payoff of $x^{t+\tau}$ received τ periods later if

$$x^t = \delta^\tau x^{t+\tau}. \quad (2)$$

A useful formula for computing the finite and infinite discounted sums we will encounter is

$$\sum_{t=T_1}^{T_2} \delta^t = \frac{\delta^{T_1} - \delta^{T_2+1}}{1 - \delta}, \quad (3)$$

which, in particular, is valid for $T_2 = \infty$.⁵

Interpreting the discount factor

One way we can interpret the discount factor δ is as an expression of traditional time preference. For example, if you received a dollar today you could deposit it in the bank and it would be worth $\$(1+r)$ tomorrow, where r is the per-period rate of interest. You would be indifferent between a payment of $\$x^t$ today and $\$x^{t+\tau}$ received τ periods later only if the future values were equal: $(1+r)^\tau x^t = x^{t+\tau}$. Comparing this indifference condition with (2), we see that the two representations of intertemporal preference are equivalent when

$$\delta = \frac{1}{1+r}. \quad (4)$$

We can relate a player's discount factor δ to her *patience*. How much more does she value a dollar today, at time t , than a dollar received $\tau > 0$ periods later? The relative value of the later dollar to the earlier is $\delta^{t+\tau}/\delta^t = \delta^\tau$. As $\delta \rightarrow 1$, $\delta^\tau \rightarrow 1$, so as her discount factor increases she values the later amount more and more nearly as much as the earlier payment. A person is more patient the less she minds waiting for something valuable rather than receiving it immediately. So we interpret higher discount factors as higher levels of patience.

There's another reason you might discount the future. You may be unsure about how long the game will continue. Even in the absence of time preference *per se*, you would prefer a dollar today rather than a promise of one tomorrow because you're not sure tomorrow will really come. (A payoff at a future time is really a conditional payoff—conditional on the game lasting that long.)

For example, assume that your beliefs about the length of the game can be represented by a constant *conditional continuation probability* ρ ; i.e. conditional on time t being reached, the probability that the game will continue at least until the next period is ρ .⁶ Without loss of generality we assign a probability

⁵ See the "Appendix: Discounting Payoffs" for a derivation of this formula.

⁶ Note that there is no upper bound which limits the length of the game; for any value of $T \in \mathbb{Z}_{++}$, there is a positive probability of the game lasting more than T periods. If, on the other hand, players' beliefs about the duration are such that there is a common knowledge upper bound $\bar{T}$ on the length of the game, the game is similar to a finitely repeated game with a common knowledge and definite

of one to the event that we reach period zero.⁷ Denote by $\mathbb{P}(t)$ the probability that period t is reached and by $\mathbb{P}(t|t')$ the probability that period t is reached conditional on period t' being reached; so $\mathbb{P}(t+1|t) = \rho$ for all t . The probability of reaching period 1 is $\mathbb{P}(1) = \mathbb{P}(1|0)\mathbb{P}(0) = \rho \cdot 1 = \rho$. The probability of reaching period 2 is $\mathbb{P}(2) = \mathbb{P}(2|1)\mathbb{P}(1) = \rho^2$. In general, the probability of reaching period t is $\mathbb{P}(t) = \rho^t$. The probability that the game ends is $1 - \mathbb{P}(\infty) = 1$, i.e. one minus the probability that it continues forever; therefore the game is finitely repeated with probability one.

Consider the prospect of receiving the conditional payoff of one dollar at time t , given your time preference δ and the conditional continuation probability ρ . The probability that the payoff will be received is the probability that period t is reached, viz. ρ^t . Its discounted value if received is δ^t and zero if not received. Therefore its expected value is $\rho^t \delta^t = \delta^t$, where $\delta \equiv \rho\delta$ is the discount factor which is equivalent to the combination of the time preference with continuation uncertainty. So, even though the game is finitely repeated, due to the uncertainty about its duration we can model it as an infinitely repeated game.

The average discounted payoff

If we adopted the summation (1) as our players' repeated-game utility function, and if a player received the same stage-game payoff v_i in every period, her discounted repeated-game payoff, using (3), would be $v_i/(1 - \delta)$. This would be tolerable, but we'll find it more convenient to transform the repeated-game payoffs to be "on the same scale" as the stage-game payoffs, by multiplying the discounted payoff sum from (1) by $(1 - \delta)$. So we define the *average discounted value* of the payoff stream $u_i^0, u_i^1, \dots$ by

$$(1 - \delta) \sum_{t=0}^{\infty} \delta^t u_i^t. \quad (5)$$

Now if a player receives the stage-game payoff v_i in every period, her repeated-game payoff is v_i as well. What could be simpler?⁸

Let's make some sense of why (5) is interpreted as an average payoff. Consider some time varying sequence of payoffs u_t , for $t = 0, 1, 2, \dots$, and let V be the resulting value of the payoff function (5). Now replace every per-period payoff u_t with the value V . The value of the payoff function (5) associated with this new constant-payoff stream of $V, V, V, \dots$ is

$$(1 - \delta) \sum_{t=0}^{\infty} \delta^t V = V(1 - \delta) \sum_{t=0}^{\infty} \delta^t = V.$$

duration in the following sense: If the stage game has a unique Nash equilibrium payoff vector, every subgame-perfect equilibrium of the repeated game involves Nash equilibrium stage-game behavior in every period.

⁷ If we don't get to period zero, we never implement our strategies. Therefore we can assume that we have reached period zero if we do indeed implement our strategies.

⁸ We will see at a later date that an additional advantage to this normalization is that the convex hull of the feasible repeated-game payoffs using the average discounted value formulation is invariant to the discount factor.

In other words, the average discounted payoff of a payoff stream $u_0, u_1, u_2, \dots$ is that value V which, if you received it every period, would give you the same average discounted payoff as the original stream.

We'll often find it convenient to compute the average discounted value of an infinite payoff stream in terms of a leading finite sum and the sum of a trailing infinite substream. For example, say that the payoffs v_i^t a player receives are some constant payoff v_i' for the first t periods, viz. $0, 1, 2, \dots, t-1$, and thereafter she receives a different constant payoff v_i'' in each period $t, t+1, t+2, \dots$. The average discounted value of this payoff stream is

$$(1-\delta) \sum_{\tau=0}^{\infty} \delta^{\tau} v_i^{\tau} = (1-\delta) \left(\sum_{\tau=0}^{t-1} \delta^{\tau} v_i^{\tau} + \sum_{\tau=t}^{\infty} \delta^{\tau} v_i^{\tau} \right) = (1-\delta) \left(\frac{v_i'(1-\delta^t)}{1-\delta} + \frac{v_i''\delta^t}{1-\delta} \right) = (1-\delta^t)v_i' + \delta^t v_i''. \quad (6)$$

We see that the average discounted value of this stream of bivalued stage-game payoffs is a convex combination of the two stage-game payoffs.

We can iterate this procedure in order to evaluate the average discounted value of more complicated payoff streams. For example—and this one will be particularly useful—say you receive v_i' for the t periods $0, 1, \dots, t-1$ as before, and then you receive v_i'' only in period t and receive v_i''' every period thereafter. The average discounted value of the stream beginning in period t (discounted to period t) is $(1-\delta)v_i'' + \delta v_i'''$. Substituting this for v_i'' in (6), we find that the average discounted value of this three-valued payoff stream is

$$(1-\delta^t)v_i' + \delta^t[(1-\delta)v_i'' + \delta v_i''']. \quad (7)$$

Example: Cooperation in the Prisoners' Dilemma (Nash Equilibrium)

In the one-shot prisoners' dilemma the players cannot avoid choosing their dominant strategy Fink. (See Figure 2.) Even when this game is finitely repeated—because the stage game has a unique Nash equilibrium—the unique subgame-perfect equilibrium has both players Finking every period.⁹ However, when the players are sufficiently patient we can sustain cooperation (i.e. keeping Mum) in every period as a subgame-perfect equilibrium of the infinitely repeated game. First we will see that such cooperation is a Nash equilibrium of the repeated game. Later we will show that this cooperation is a subgame-perfect equilibrium.

	<i>M</i>	<i>F</i>
<i>M</i>	1, 1	-1, 2
<i>F</i>	2, -1	0, 0

Figure 2: A Prisoners' Dilemma

When an infinitely repeated game is played, each player i has a repeated-game strategy s_i , which is a sequence of history-dependent stage-game strategies s_i^t ; i.e. $s_i = (s_i^0, s_i^1, \dots)$, where each $s_i^t: A^t \rightarrow A_i$. The

⁹ See Theorem 4 in the "Repeated Games."

n -tuple of individual repeated-game strategies is the repeated-game strategy profile s ; i.e. $s = (s_1, \dots, s_n)$.

The repeated-game strategies I exhibit, which are sufficient to sustain cooperation, have the following form: Cooperate (i.e. play Mum) in the first period. In later periods, cooperate if both players have always cooperated. However, if either player has ever Finked, Fink for the remainder of the game. More precisely and formally, we write player i 's repeated-game strategy $\bar{s}_i = (\bar{s}_i^0, \bar{s}_i^1, \dots)$ as the sequence of history-dependent stage-game strategies such that in period t and after history h^t ,

$$\bar{s}_i^t(h^t) = \begin{cases} M, & t=0 \text{ or } h^t = ((M, M)^t), \\ F, & \text{otherwise.} \end{cases} \quad (8)$$

Recall that the history h^t is the sequence of stage-game action profiles which were played in the t periods $0, 1, 2, \dots, t-1$. “ $((M, M)^t)$ ” is just a way of writing “ $((M, M), (M, M), \dots, (M, M))$ ”, where “ (M, M) ” is repeated t times.

We can simplify (8) further in a way that will be useful in our later analysis: Because h^0 is the null history, we adopt the convention that, for any action profile $a \in A$, the statement $h^0 = (a)^0$ is true.¹⁰ With this understanding, $t=0$ implies that $h^t = ((M, M)^t)$. Therefore

$$\bar{s}_i^t(h^t) = \begin{cases} M, & h^t = ((M, M)^t), \\ F, & \text{otherwise.} \end{cases} \quad (9)$$

First, I'll show that for sufficiently patient players the strategy profile $\bar{s} = (\bar{s}_1, \bar{s}_2)$ is a Nash equilibrium of the repeated game. Later I'll show that for the same required level of patience these strategies are also a subgame-perfect equilibrium.

If both players conform to the alleged equilibrium prescription, they both play Mum at $t=0$. Therefore at $t=1$, the history is $h^1 = (M, M)$; so they both play Mum again. Therefore at $t=2$, the history is $h^2 = ((M, M), (M, M))$, so they both play Mum again. And so on.... So the path of s is the infinite sequence of cooperative action profiles $((M, M), (M, M), \dots)$. The repeated-game payoff to each player corresponding to this path is trivial to calculate: They each receive a payoff of 1 in each period, therefore the average discounted value of each player's payoff stream is 1.

Can player i gain from deviating from the repeated-game strategy $\bar{s}_i$ given that player j is faithfully following $\bar{s}_j$? Let t be the period in which player i first deviates. She receives a payoff of 1 in the first t periods $0, 1, \dots, t-1$. In period t , she plays Fink while her conforming opponent played Mum, yielding player i a payoff of 2 in that period. This defection by player i now triggers an open-loop Fink-always response from player j . Player i 's best response to this open-loop strategy is to Fink in every period

¹⁰ Here is a case of potential ambiguity between time superscripts and exponents. “ $(a)^0$ ” means “ a ” raised to the zero power; i.e. the profile a is played zero times.

herself. Thus she receives zero in every period $t+1, t+2, \dots$. To calculate the average discounted value of this payoff stream to player i we can refer to (7), and substitute $v_i' = 1$, $v_i'' = 2$, and $v_i''' = 0$. This yields player i 's repeated-game payoff when she defects in period t in the most advantageous way to be $1 - \delta^t(2\delta - 1)$. This is weakly less than the equilibrium payoff of 1, for any choice of defection period t , as long as $\delta \geq \frac{1}{2}$. Thus we have defined what I meant by “sufficiently patient:” cooperation in this prisoners’ dilemma is a Nash equilibrium of the repeated game as long as $\delta \geq \frac{1}{2}$.

Restricting strategies to subgames

Consider the subgame which begins in period τ after a history $h^\tau = (a^0, a^1, \dots, a^{\tau-1})$. This subgame is itself an infinitely repeated game; in fact, it is the same infinitely repeated game as the original game in the sense that 1 the same stage game G is played relentlessly and 2 a unit payoff tomorrow is worth the same as a payoff of δ received today. Another way to appreciate 2 is to consider an infinite stream of action profiles $a^0, a^1, \dots$. If a player began in period τ to receive the payoffs from this stream of action profiles, the contribution to her total repeated-game payoff would be δ^τ times its contribution if it were received beginning at period zero. If player i were picking a sequence of history-dependent stage-game strategies to play beginning when some subgame h^τ were reached, against fixed choices by her opponents, she would face exactly the same maximization problem as she would if she thought she was just starting the game.¹¹

What behavior in the subgame is implied by the repeated-game strategy profile s ? The restriction of the repeated-game strategy profile s to the subgame h^τ tells us how the subgame will be played. Let's denote this restriction by $\hat{s}$. If we played the infinitely-repeated game from the beginning by having each player play her part of $\hat{s}$, we would see the same infinite sequence of action profiles as we would see if instead s were being played and we began our observation at period τ .

For example, in the initial period of the subgame (viz. period τ on the original game's timeline and period zero on the subgame's timeline), player i chooses the action dictated by her subgame strategy $\hat{s}_i$, viz. $\hat{s}_i^0$. This action must be what her original repeated-game strategy s_i would tell her to do after the history h^τ by which this subgame was reached. I.e.

$$\hat{s}_i^0 = s_i^\tau(h^\tau). \quad (10)$$

After all the players choose their actions, a stage-game action profile results. We can denote this action profile in two ways: as $\hat{a}^0$, which refers to the initial period of the subgame, or as a^τ , which refers to the timeline of the original game. (See Figure 3.) This action profile $\hat{a}^0 = a^\tau$ then becomes part of the history which conditions the next period's choices. Period 1 of the subgame is period $\tau+1$ of the original game.

¹¹ This paragraph may be unclear, but it's not a trivial statement! Its validity relies crucially on the structure of the weighting coefficients in the infinite summation. Let α_t be the weight for period t 's per-period payoff. In the discounted payoff case we consider here, we have $\alpha_t = \delta^t$. The fact that the “continuation game” is the same game as the whole game is a consequence of the fact that, for all t and τ , the ratio $\alpha_{t+\tau}/\alpha_t$ depends only upon τ but not upon t . In our case, this ratio is just δ^τ . On the other hand, for example, if the weighting coefficients were $\alpha_t = \delta^t/t$, we could not solve each continuation game just as another instance of the original game.

The history in the subgame is $\hat{h}^1 = (\hat{a}^0)$; the history in the original game is

$$h^{\tau+1} = (a^0, a^1, \dots, a^{\tau-1}, a^\tau) = (h^\tau; a^\tau) = (h^\tau; \hat{a}^0) = (h^\tau; \hat{h}^1), \quad (11)$$

where $(h^\tau; a^\tau)$ is the *concatenation* of h^τ with a^τ .¹² The action chosen next by player i can be denoted equivalently as $\hat{s}_i^1(\hat{h}^1)$ or $s_i^{\tau+1}((h^\tau; \hat{h}^1))$. Therefore we must have

$$\hat{s}_i^1(\hat{h}^1) = s_i^{\tau+1}((h^\tau; \hat{h}^1)). \quad (12)$$

Continuing the analysis period-by-period, we see that we can define the restriction of s_i to the subgame determined by h^τ by

$$\hat{s}_i^t(\hat{h}^t) = s_i^{\tau+t}((h^\tau; \hat{h}^t)), \quad (13)$$

for every $t \in \{0, 1, \dots\}$.



Figure 3: Histories and history-dependent stage-game strategies in two time frames: the original game's and a subgame's.

Example: Cooperation in the Prisoners' Dilemma (Subgame-perfect Equilibrium)

To verify that $\bar{s}$ is a subgame-perfect equilibrium of the repeated prisoners' dilemma we need to check that this strategy profile's restriction to each subgame is a Nash equilibrium of that subgame. Consider a subgame, beginning in period τ with some history h^τ . What is the restriction of $\bar{s}_i$ to this subgame? Denoting the restriction by $\hat{s}_i$ and using definitions (13) and (9), we find

$$\hat{s}_i^t(\hat{h}^t) = \bar{s}_i^{\tau+t}((h^\tau; \hat{h}^t)) = \begin{cases} M, & h^\tau = ((M, M)^\tau) \text{ and } \hat{h}^t = ((M, M)^t), \\ F, & \text{otherwise.} \end{cases} \quad (14)$$

We can partition the subgames of this game, each identified by a beginning period τ and a history h^τ , into two classes: A those in which both players chose Mum in all previous periods, i.e. $h^\tau = ((M, M)^\tau)$, and B those in which a defection by either player has previously occurred.¹³

For those subgames in class A, the sequence of restrictions $\hat{s}_i^t(\hat{h}^t)$ from (14) reduces to the sequence of

¹² To *concatenate* two sequences a with b is simply to append b to the end of a .

¹³ Note that class A includes the "subgame of the whole," which begins in period zero. (Because there are no previous periods, it is trivially true that (M, M) was played in all previous periods.)

original stage-game strategies $\bar{s}_i^t(h^t)$ from (9). I.e. for all τ and $h^\tau = ((M, M)^\tau)$,¹⁴¹⁵

$$\hat{s}_i^t(\hat{h}^t) = \begin{cases} M, & h^\tau = ((M, M)^\tau) \text{ and } \hat{h}^t = ((M, M)^t), \\ F, & \text{otherwise.} \end{cases} = \begin{cases} M, & \hat{h}^t = ((M, M)^t), \\ F, & \text{otherwise.} \end{cases} = \bar{s}_i^t(\hat{h}^t) \quad (15)$$

Because $\bar{s}$ is a Nash-equilibrium strategy profile of the repeated game, for each subgame h^τ in class A, the restriction $\hat{s}$ is a Nash-equilibrium strategy profile of the subgame when $\delta \geq \frac{1}{2}$.

For any subgame h^τ in class B, $h^\tau \neq ((M, M)^\tau)$. Therefore the restriction $\hat{s}_i$ of $\bar{s}_i$ specifies $\hat{s}_i^t = F$ for all $t \in \{0, 1, \dots\}$. In other words, in any subgame reached by some player having Finked in the past, each player chooses the open-loop strategy “Fink always.” Therefore the repeated-game strategy profile $\hat{s}$ played in such a subgame is an open-loop sequence of stage-game Nash equilibria. From Theorem 1 of the Repeated Games handout we know that this is a Nash equilibrium of the repeated game and hence of this subgame.

So we have shown that for every subgame the restriction of $\bar{s}$ to that subgame is a Nash equilibrium of that subgame for $\delta \geq \frac{1}{2}$. Therefore $\bar{s}$ is a subgame-perfect equilibrium of the infinitely repeated prisoners’ dilemma when $\delta \geq \frac{1}{2}$.

¹⁴ When $h^\tau = ((M, M)^\tau)$, $[h^\tau = ((M, M)^\tau) \text{ and } h^t = ((M, M)^t)] \Leftrightarrow h^t = ((M, M)^t)$.

¹⁵ There is no distinction between the roles played between h^t and $\hat{h}^t$ in these two expressions. They are both dummy variables.

Appendix: Discounting Payoffs

Often in economics—for example when we study repeated games—we are concerned about a player who receives a payoff in each of many (perhaps infinitely many) periods and we want to meaningfully summarize her entire sequence of payoffs by a single number. A common assumption is that the player wants to maximize a weighted sum of her per-period payoffs, where she weights later periods less than earlier periods. For simplicity this assumption often takes the particular form that the sequence of weights forms a geometric progression: for some fixed $\delta \in (0, 1)$, each weighting factor is δ times the previous weight. (δ is called her *discount factor*.) In this handout we’re going to derive some useful results relevant to this kind of payoff structure. We’ll see that you really don’t need to memorize a bunch of different formulas. You can easily work them all out from scratch from a simple starting point.

Let’s set this problem up more concretely. It will be convenient to call the first period $t = 0$. Let the player’s payoff in period t be denoted u_t . By convention we attach a weight of 1 to the period 0 payoff.¹⁶ The weight for the next period’s payoff, u_1 , then, is $\delta \cdot 1 = \delta$; the weight for the next period’s payoff, u_2 , then, is $\delta \cdot \delta = \delta^2$; the weight for the next period’s payoff, u_3 , then, is $\delta \cdot \delta^2 = \delta^3$. You see the pattern: in general the weight associated with payoff u_t is δ^t . The weighted sum of all the payoffs from $t = 0 \rightarrow t = T$ (where T may be ∞) is

$$u_0 + \delta u_1 + \delta^2 u_2 + \cdots + \delta^T u_T = \sum_{t=0}^T \delta^t u_t. \quad (1)$$

The infinite summation of the discount factors

We sometimes will be concerned with the case in which the player’s period payoffs are constant over some stretch of time periods. In this case the common u_t value can be brought outside and in front of the summation over those periods because those u_t values don’t depend on the summation index t . Then we will be essentially concerned with summations of the form

In a hurry?

Here’s a formula we’ll eventually derive and from which you can derive a host of other useful discounting expressions. In particular, it’s valid when $T_2 = \infty$.

$$\sum_{t=T_1}^{T_2} \delta^t = \frac{\delta^{T_1} - \delta^{T_2+1}}{1 - \delta}.$$

¹⁶ It’s the relative weights which are important (why?), so we are free to normalize the sequence of weights any way we want.

period T , but after that she receives a constant payoff per period. To handle this latter constant-payoff stage we'll want to know the value of the summation

$$\sum_{t=T+1}^{\infty} \delta^t. \quad (7)$$

Now that we know the value from (4) for the infinite sum that starts at $t=0$, we can use that result and the technique of *change of variable* in order to find the value of (7).

If only the lower limit on the summation (7) were $t=0$ instead of $t=T+1$, we'd know the answer immediately. So we define a new index τ , which depends on t , and transform the summation (7) into another summation whose index τ runs from $0 \rightarrow \infty$ and whose summand is δ^τ . So, let's define

$$\tau = t - (T + 1). \quad (8)$$

We chose this definition because when $t = T + 1$ —like it does for the first term of (7)—then $\tau = 0$, which is exactly what we want τ to be for the first term of the new summation.¹⁹ For purposes of later substitution we write (8) in the form

$$t = \tau + (T + 1). \quad (9)$$

Substituting for t using (9), summation (7) can now be rewritten as

$$\sum_{t=T+1}^{\infty} \delta^t = \sum_{\tau+(T+1)=T+1}^{\infty} \delta^{\tau+(T+1)} = \delta^{T+1} \sum_{\tau=0}^{\infty} \delta^\tau = \frac{\delta^{T+1}}{1-\delta}. \quad (10)$$

where we used (4) to transform the final infinite sum into its simple reciprocal equivalent.²⁰

The finite summation

Our game might only last a finite number of periods. In this case we want to know the value of

$$\sum_{t=0}^T \delta^t. \quad (11)$$

We can immediately get this answer by combining the two results—(4) and (10)—which we've already found. We need only note that the infinite sum in (3) is the sum of two pieces: 1 the piece we want in (11) and 2 the infinite piece in (7). So we immediately obtain

$$\sum_{t=0}^T \delta^t = \sum_{t=0}^{\infty} \delta^t - \sum_{t=T+1}^{\infty} \delta^t = \frac{1}{1-\delta} - \frac{\delta^{T+1}}{1-\delta} = \frac{1-\delta^{T+1}}{1-\delta}. \quad (12)$$

¹⁹ Of course, we could have achieved this goal by defining $\tau = (T + 1) - t$. But we want τ to *increase* with t , so that when $t \rightarrow \infty$, we'll also have $\tau \rightarrow \infty$.

²⁰ The “ t ” in (4) is just a dummy variable; (4) is still true if you replace every occurrence of “ t ” with “ τ ”.

Endless possibilities

The techniques we have used here of employing a change of variable and combining previous results is very general and can be used to obtain the value of summations of the general form in (2). The particular results we derived, viz. (10) and (12), are merely illustrative cases.

As an exercise, you should be able to easily combine results we have already found here to show that the value of the general summation in (2) is

$$\sum_{t=T_1}^{T_2} \delta^t = \frac{\delta^{T_1} - \delta^{T_2+1}}{1 - \delta}. \quad (13)$$

Note that (13) is valid even when $T_2 = \infty$, because

$$\sum_{t=T_1}^{\infty} \delta^t = \lim_{T_2 \rightarrow \infty} \sum_{t=T_1}^{T_2} \delta^t = \lim_{T_2 \rightarrow \infty} \frac{\delta^{T_1} - \delta^{T_2+1}}{1 - \delta} = \frac{\delta^{T_1}}{1 - \delta}, \quad (14)$$

where we used the fact that

$$\lim_{t \rightarrow \infty} \delta^t = 0, \quad (15)$$

for $\delta \in [0, 1)$.

You should verify that, by assigning the appropriate values to T_1 and T_2 in (13), we can recover all of our previous results, viz. (4), (10), and (12).

Jim Ratliff's Game Theory Course:

5.3: A Folk Theorem Sampler

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Abstract



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We have successfully shown how cooperation can arise in a repeated relationship where it could not in a "single-shot" context. Can other outcomes besides "everyone always cooperate" or "everyone always fink" be sustained in equilibrium in the infinitely repeated prisoners' dilemma? What about other games? What outcomes can be sustained in equilibrium when they are infinitely repeated? The original motivation for developing a theory of repeated games was to show that cooperative behavior was an equilibrium. The theoreticians were all too clever, for, as we will see, they showed that in many cases a huge multiplicity of even very "noncooperative" stage-game payoffs could be sustained on average as an equilibrium of the repeated game.

These findings are made precise in numerous *folk theorems*. Each folk theorem considers a class of games and identifies a set of payoff vectors each of which can be supported by some equilibrium strategy profile. There are many folk theorems because there are many classes of games and different choices of equilibrium concept. Some folk theorems identify sets of payoff vectors which can be supported by Nash equilibria; of course, of more interest are those folk theorems which identify payoffs supported by subgame-perfect equilibria.

The strongest folk theorems are of the following loosely stated form: "Any *strictly individually rational* and *feasible* payoff vector of the stage game can be supported as a subgame-perfect equilibrium average payoff of the repeated game." These statements often come with qualifications such as "for discount factors sufficiently close to 1" or, for finitely repeated games, "if repeated sufficiently many times."

First we will precisely define the terms feasible and individually rational. A payoff vector's individual rationality relies on the concept of a player's minmax value—a number useful for characterizing the worst punishment to which a deviating myopic player can be subjected in a single player—so we will define it as well. Then we will state and prove two folk theorems, one Nash and one perfect, which have the virtue of being relatively easy to prove because their proofs rely only on simple "grim trigger" strategies. Then we will prove a perfect folk theorem stronger than the first two using more complicated strategies.



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A Folk Theorem Sampler

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Recapitulation

Consider the prisoners' dilemma game of Figure 1. The unique Nash equilibrium (in fact, the dominance-solvable outcome) requires both players to Fink, which is Pareto dominated by both players cooperating (i.e. playing Mum). We often observe cooperation in the real world; what must we add to our model in order that cooperation become rational? Perhaps cooperation would be rational when players acknowledge that they are in a repeated relationship: that there is actually a sequence of stage games, where a player's behavior in a stage can be conditioned upon the treatment she has received from other players in the past.

	<i>M</i>	<i>F</i>
<i>M</i>	1, 1	-1, 2
<i>F</i>	2, -1	0, 0

Figure 1: A Prisoners' Dilemma

Disconcertingly, when the prisoners' dilemma is repeated a commonly known finite number of times, a backwards induction argument implies that in the unique subgame-perfect equilibrium of this repeated game the players will still Fink in every period.^{1,2} The key to sustaining cooperative behavior in this game is to let the stage game be infinitely repeated.³ In particular, when we assumed that players' preferences over stage-game payoff streams could be represented by the discounted sum of the stage-game payoffs, we saw that cooperation by both players in every period was a subgame-perfect equilibrium of the infinitely repeated prisoners' dilemma, as long as the players were "sufficiently patient"—their discount factors were sufficiently close to one.⁴

Cooperation was achieved in the infinitely repeated prisoners' dilemma by having both players adopt "grim trigger" strategies: Each starts out playing the desired Mum and continues to choose this cooperative action as long as the other player has always played Mum as well. If her opponent ever plays Fink, though, she switches to the open-loop strategy of Finking in every period.

To see that these trigger strategies constitute a Nash equilibrium of the repeated game we analyze the tradeoff that a player faces when deciding whether to deviate from her appointed actions along the alleged equilibrium path: Playing Fink rather than Mum benefits the deviant in that period because she receives two instead of one.⁵ However, because upon a defection her opponent would switch to the open-loop punishment phase of Fink every period, the deviant can do no better than accept zero in every remaining period. So the prospective deviant asks herself which is more desirable: receiving two today and zero forever afterward? or conforming and receiving one today and forever? The answer depends on her discount factor δ . If she is very impatient, valuing the future very little, i.e. δ is close to zero, the single-period gain in the defecting period will outweigh the loss in later periods. If, on the other hand, she is more patient, receiving one rather than zero in every subsequent period outweighs the temptation of today grabbing two rather than one. We saw that in the infinitely repeated version of the game in Figure 1 cooperation in every period was a Nash equilibrium as long as each player's discount factor exceeded one-half.

-
- ¹ This is Theorem 4 of the "Repeated Games." The result follows because there is a unique Nash-equilibrium payoff vector in the stage game. As we saw in the two-period example in "Repeated Games," it does not hold when the stage game has more than one Nash-equilibrium payoff vector. See also Benoît and Krishna [1985].
 - ² In fact, (Fink, Fink) in every period is also the unique Nash equilibrium of the finitely repeated prisoners' dilemma. This requires more reasoning and relies upon the fact that the stage-game Nash equilibrium yields the minmax payoff vector. See Sorin [1986: 156, Proposition 13] or Nalebuff [1988: 150, 153–154, Puzzle 3].
 - ³ Or the game could be finitely repeated with probability one, but players are uncertain about how long the game will continue and in every period they assess a positive probability to the game continuing at least one more period. See the "Infinitely Repeated Games with Discounting" handout.
 - ⁴ See the "Infinitely Repeated Games with Discounting" handout.
 - ⁵ We analyze a deviation assuming that all other players choose the actions prescribed by the conjectured equilibrium strategies. In this case, then, a player contemplating deviation in some period assumes that her opponent will play Mum that period.

These trigger strategies constitute a subgame-perfect equilibrium of the repeated game, for the same range of discount factors, because their restrictions to any subgame form a Nash equilibrium of the subgame. In particular, in any subgame in which either player had ever deviated, the equilibrium strategies dictate that the players choose (Fink, Fink) every period. Because this is just an open-loop repetition of a stage-game Nash equilibrium, the restrictions of the repeated-game strategies to such a subgame constitute a Nash equilibrium of the subgame.^{6,7}

Introduction

So we have successfully shown how cooperation can arise in a repeated relationship where it could not in a “single-shot” context. This raises additional questions. Can other outcomes besides “everyone always cooperate” or “everyone always fink” (which remains equilibrium behavior for all discount factors) be sustained in equilibrium in the infinitely repeated prisoners’ dilemma? What about other games? What outcomes can be sustained in equilibrium when they are infinitely repeated? The original motivation for developing a theory of repeated games was to show that cooperative behavior was an equilibrium. The theoreticians were all too clever, for, as we will see, they showed that in many cases a huge multiplicity of even very “noncooperative” stage-game payoffs could be sustained on average as an equilibrium of the repeated game.

These findings are made precise in numerous *folk theorems*.^{8,9} Each folk theorem considers a class of games and identifies a set of payoff vectors each of which can be supported by some equilibrium strategy profile. There are many folk theorems because there are many classes of games and different choices of equilibrium concept. For example, games may be repeated infinitely or only finitely many times. There are many different specifications of the repeated game payoffs. For example, there is the Cesaro limit of the means, the Abel limit (Aumann [1985: 210]), the overtaking criterion (Rubinstein [1979]) as well as the average discounted payoff, which we have adopted. They may be games of complete information or they might be characterized by one of many different specifications of incomplete information. Some folk theorems identify sets of payoff vectors which can be supported by Nash equilibria; of course, of more interest are those folk theorems which identify payoffs supported by subgame-perfect equilibria.

The strongest folk theorems are of the following loosely stated form: “Any *strictly individually rational* and *feasible* payoff vector of the stage game can be supported as a subgame-perfect equilibrium average payoff of the repeated game.” These statements often come with qualifications such as “for discount factors sufficiently close to 1” or, for finitely repeated games, “if repeated sufficiently many times.”

⁶ See Theorem 1 of the “Repeated Games” handout.

⁷ Their restrictions to these deviation subgames are Nash equilibria of these subgames for all discount factors. The qualification on the discount factor is inherited by the requirement that the strategy profile be a Nash equilibrium of the entire game.

⁸ They are so named because results of their type were widely believed by game theorists prior to published formal statements or proofs. Myerson [1991: 332] suggests that these be called *general feasibility* theorems. He correctly points out that “naming a theorem for the fact that it was once unpublished conveys no useful information to the uninitiated reader.” However, I wouldn’t hold my breath waiting for Myerson’s suggestion to become universally accepted. Game theorists are a terminologically stubborn lot.

⁹ There is a large folk-theorem literature. For an introduction to it see Fudenberg and Tirole [1991: 150–160].

First we will precisely define the terms *feasible* and *individually rational*. A payoff vector's individual rationality relies on the concept of a player's *minmax value*—a number useful for characterizing the worst punishment to which a deviating myopic player can be subjected in a single player—so we will define it as well. Then we will state and prove two folk theorems, one Nash and one perfect, which have the virtue of being relatively easy to prove because their proofs rely only on simple “grim trigger” strategies. Then we will prove a perfect folk theorem stronger than the first two using more complicated strategies.

Infinitely repeated games with discounting

We'll call the stage game G and interpret it to be a simultaneous-move matrix game which remains exactly the same through time. As usual we let the player set be $I = \{1, \dots, n\}$. Each player has a pure action space A_i . The space of action profiles is $A = \prod_{i \in I} A_i$. Each player has a von Neumann-Morgenstern utility function defined over the outcomes of G , $g_i: A \rightarrow \mathbb{R}$. We let g be the n -tuple of players' stage-game payoff functions, i.e. $g = \prod_{i \in I} g_i$ so that $g(a) = (g_1(a), \dots, g_n(a))$.¹⁰

The stage game repeats each period, starting at $t = 0$. At the conclusion of each period's play, the action profile which occurred is revealed to all the players. Combined with perfect recall, this allows a player to condition her current action on all earlier actions of her opponents. A repeated-game strategy $s_i \in S_i$ for player i is a sequence $s_i = (s_i^0, s_i^1, \dots)$, where each s_i^t is a history-dependent stage-game strategy, $s_i^t: A^t \rightarrow A_i$. The history at time t , viz. $h^t \in A^t$, is the sequence of action profiles $h^t = (a^0, a^1, \dots, a^{t-1})$.

We can think of the players as receiving their stage-game payoffs period-by-period, where the players discount future payoffs according to a common discount factor $\delta \in (0, 1)$. We adopt the *average discounted payoff* representation of players' preferences over streams of stage-game payoffs. A player's average discounted payoff for an infinite stream of stage-game payoffs $v_i^\infty \equiv v_i^0, v_i^1, \dots$ is

$$\Lambda(v_i^\infty, \delta) = (1 - \delta) \sum_{t=0}^{\infty} \delta^t v_i^t. \quad (1)$$

(The δ may be suppressed as an argument if it is not a parameter of interesting variation.) This formulation normalizes the repeated-game payoffs to be “on the same scale” as the stage-game payoffs. We refer to the infinitely repeated game as G^∞ . Because the discount factor δ enters the players' utility functions, it is part of the definition of the game. When we need to emphasize the role of the discount factor we more explicitly refer to the game as $G^\infty(\delta)$.

Any repeated-game strategy profile s generates a path $h^\infty = (a^0, a^1, \dots)$, where each $a^t = s^t(h^t)$ and the histories are generated recursively by concatenation as $h^t = (h^{t-1}; s^{t-1}(h^{t-1}))$. Player i 's repeated-game payoff to player i along this path is, from (1),

¹⁰ See the “Infinitely Repeated Games with Discounting” handout. I want to reserve “ s ”, “ S ”, and “ u ” to refer to typical strategies, strategy spaces, and utility functions, respectively, in the repeated game. So I'm using “ a ”, “ A ”, and “ g ” for the corresponding stage-game entities.

$$u_i(s) = (1 - \delta) \sum_{t=0}^{\infty} \delta^t g_i(a^t). \quad (2)$$

We denote the n -tuple of repeated-game payoff functions by $u = (u_1, \dots, u_n)$. If we need to emphasize the role of the discount factor we will write $u_i(s; \delta)$.

A useful formula for computing the infinite sums we will encounter is

$$\sum_{t=T_1}^{T_2} \delta^t = \frac{\delta^{T_1} - \delta^{T_2+1}}{1 - \delta}, \quad (3)$$

which, in particular, is valid for $T_2 = \infty$.¹¹ If the payoffs v_i^t a player receives are some constant payoff v_i' for the first t periods, viz. $0, 1, 2, \dots, t-1$, and thereafter she receives a different constant payoff v_i'' in each period $t, t+1, t+2, \dots$, the average discounted value of this payoff stream is¹²

$$(1 - \delta^t) v_i' + \delta^t v_i'', \quad (4)$$

i.e. a convex combination of the two stage-game payoffs. If she receives v_i' for the t periods $0, 1, \dots, t-1$ as before, then v_i'' only in period t , and v_i''' every period thereafter, the average discounted value of this three-valued payoff stream is

$$(1 - \delta^t) v_i' + \delta^t [(1 - \delta) v_i'' + \delta v_i''']. \quad (5)$$

The no one-stage improvement principle

In order to check whether a repeated-game strategy profile $s \in S$ is a subgame-perfect equilibrium of a repeated game, we in principle need to examine the restriction $s|_{h^t}$ of that strategy profile s to every possible subgame defined by an arbitrary history h^t . In order to check whether $s|_{h^t}$ is a Nash equilibrium of the subgame, we need to check whether any player $i \in I$ has a profitable deviation from her part of $s|_{h^t}$. A deviation can be very complicated; it can involve a counter-to-specification action at any or all of an infinite number of histories. The set of possible such deviations can be huge.

Fortunately there is a result from dynamic programming which greatly simplifies our deviation-checking task. We need only check every possible deviation of an extremely simple class. If none of those deviations are profitable for a deviating player, then the strategy profile is subgame perfect. The simple deviations we consider are called “one-stage” deviations. In a one-stage deviation, the deviating player disobeys her component of the specified strategy profile at only a single history h^t . At all other information sets (i.e. histories), she obeys the prescription.

The intuition behind the result is loosely the following: The change in the repeated-game payoff due

¹¹ See the “Appendix: Discounting Payoffs,” in the “Infinitely Repeated Games with Discounting” handout for a derivation of this formula.

¹² See the “Infinitely Repeated Games with Discounting” handout for the details.

to a complicated deviation from the prescribed strategy is a sum of the changes due to one-stage deviations. If a complicated deviation is to result in an increased repeated-game payoff, then some one-stage deviation must also result in an increased repeated-game payoff. If no one-stage deviation does, then no complicated deviation will be profitable either.

We say that the repeated-game strategy for player i , $\hat{s}_i \in S_i$, is a *one-stage deviant* of $s_i \in S_i$ if there exist a time $\hat{t}$ and a history $\hat{h}^{\hat{t}} \in A^{\hat{t}}$ such that 1 $\forall t \neq \hat{t}, \hat{s}_i^t = s_i^t$, 2 $\forall h^{\hat{t}} \in A^{\hat{t}} \setminus \{\hat{h}^{\hat{t}}\}, \hat{s}_i^{\hat{t}}(h^{\hat{t}}) = s_i^{\hat{t}}(h^{\hat{t}})$, and 3 $\hat{s}_i^{\hat{t}}(\hat{h}^{\hat{t}}) \neq s_i^{\hat{t}}(\hat{h}^{\hat{t}})$. Let $\hat{S}_i(s_i)$ be the space of all one-stage deviants of s_i .

We say that s satisfies the “no one-stage improvement (NOSI)” property if for all $i \in I$, for all one-stage deviants $\hat{s}_i \in \hat{S}_i(s_i)$ of s_i , for all t , and for all $h^t \in A^t$, $\hat{s}_i$ is no better than s_i against s_{-i} , conditional on reaching the history h^t .

Theorem

Let s be a strategy profile for a finitely repeated game or an infinitely repeated game with discounting. Then s is a subgame-perfect equilibrium if and only if s satisfies the no one-stage improvement property.

Proof

See Appendix.¹³

The sufficient-patience limit

In proofs of the various folk theorems we’ll be considering we will often be taking limits of average discounted payoffs as the discount rate tends toward unity. If there is a point in time beyond which a player receives some constant payoff for every period thereafter, such a limit is very simple: it is simply that constant payoff. For example, we calculated in (4) the average discounted payoff to a player if she received v_i' for a finite number of periods and v_i'' for all remaining periods. It is clear in (4) that, in the limit as $\delta \rightarrow 1$, the first term vanishes leaving the limit to be the infinitely repeated payoff v_i'' . A similar analysis shows that the limit of (5) is v_i''' .

We say that a payoff sequence $v_i^\infty \equiv v_i^0, v_i^1, \dots$ has a *terminal subsequence* $\bar{v}_i$ if there exists a time period τ such that, for all $t > \tau$, $v_i^t = \bar{v}_i$.

Lemma 1

Let $v_i^\infty \equiv v_i^0, v_i^1, \dots$ be an infinite sequence of payoffs with terminal subsequence $\bar{v}_i$. Then

$$\lim_{\delta \rightarrow 1} \Lambda(v_i^\infty, \delta) = \bar{v}_i. \quad (6)$$

Proof

The proof is left as an exercise for you. Hint: use (3). ☺

Sometimes we will consider two infinite sequences of payoffs, e.g. one for conforming and one for deviating, and we will wish to compare them in the infinite-patience limit. The following result can be

¹³ Warning! The proof I provide needs a lot of work to motivate it and make it more accessible.

very useful for performing this comparison.

Theorem 1

Let $\hat{v}_i^\infty$ and $\check{v}_i^\infty$ be two infinite payoff sequences with terminal subsequences $\hat{v}_i$ and $\check{v}_i$, respectively. If $\hat{v}_i > \check{v}_i$, then, for a sufficiently patient player, the payoff stream $\hat{v}_i^\infty$ is strictly preferred to the payoff stream $\check{v}_i^\infty$. In other words, there exists a $\underline{\delta} \in (0, 1)$ such that, for all $\delta \in (\underline{\delta}, 1)$,

$$\Lambda(\hat{v}_i^\infty, \delta) > \Lambda(\check{v}_i^\infty, \delta).^{14}$$

Proof

(To simplify the notation, let's suppress the “ i ” subscripts.) Let $\varepsilon = \frac{1}{2}(\hat{v} - \check{v})$. From Lemma 1, we know that we can find $\delta', \delta'' \in (0, 1)$ such that, for all $\delta \in (\delta', 1)$,

$$|\hat{v} - \Lambda(\hat{v}^\infty, \delta)| < \varepsilon, \quad (\heartsuit.1)$$

and, for $\delta \in (\delta'', 1)$,

$$|\check{v} - \Lambda(\check{v}^\infty, \delta)| < \varepsilon. \quad (\heartsuit.2)$$

Let $\underline{\delta} = \max\{\delta', \delta''\}$. Then, for all $\delta \in (\underline{\delta}, 1)$,

$$\Lambda(\hat{v}^\infty, \delta) > \hat{v} - \varepsilon, \quad (\heartsuit.3)$$

$$\Lambda(\check{v}^\infty, \delta) < \check{v} + \varepsilon, \quad (\heartsuit.4)$$

from which we obtain

$$\Lambda(\check{v}^\infty, \delta) < \frac{1}{2}(\hat{v} + \check{v}) < \Lambda(\hat{v}^\infty, \delta), \quad (\heartsuit.5)$$

as desired. (See Figure 18.) ☺

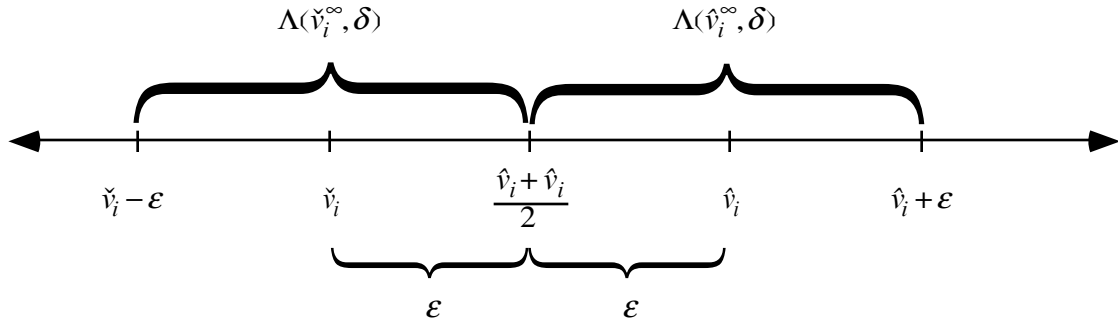


Figure 18: Taking δ sufficiently close to one ensures that the payoff with the higher terminal subsequence is more preferred.

¹⁴ Note that the weak-inequality form of this theorem is *not* true. In other words, if two payoff sequences have identical terminal subsequences, they need not have identical average discounted values, no matter how patient the player is. (However, the limit, as $\delta \rightarrow 1$, of the difference between the two average discounted values will go to zero.)

Feasible stage-game payoffs

Say that we wish to support some payoff vector $v \in \mathbb{R}^n$ as an equilibrium of an infinitely repeated game with discounting. We must find an equilibrium strategy profile of the repeated game which generates an equilibrium-path history such that each player $i \in I$ receives a repeated-game payoff, from (2), of v_i .

We have already seen that the average discounted value of a constant stage-game payoff stream is independent of the discount factor δ . An analytical inconvenience arises if the equilibrium path generates payoffs to player i which change from period to period, i.e. when $g_i(a^t)$ varies with t . In this case, unlike the constant-stream case, the value of the average discounted value in (2) depends on the discount factor δ . For example, say that the equilibrium path generates stage-game payoffs v' in even numbered periods and v'' in odd numbered periods. The average discounted value of this stream is

$$(1 - \delta)(v' + \delta v'' + \delta^2 v' + \delta^3 v'' + \dots) = \frac{v' + \delta v''}{1 + \delta}, \quad (7)$$

which strictly increases (respectively, decreases) with δ when $v'' > v'$ (respectively, $v' > v''$).

What's so problematic about the dependence on δ of the average discounted value? In arguments we will employ later, we fix an average-discounted payoff vector $v \in \mathbb{R}^n$ and take limits as $\delta \rightarrow 1$. If the payoff vector for a given equilibrium path changed with δ , we would have to adjust the equilibrium strategies at every point in the limit so that, for each value of δ , v was achieved by those strategies. That would lead to a bona fide Excedrin™ intractability headache!

We can avoid this problem by limiting our attention to repeated-game strategy profiles whose equilibrium paths prescribe the same stage-game action profile in every period; i.e. $a^t = \bar{a}$ for all t . Then $g_i(a^t)$ is constant in time, and we know that the average-discounted value of this path for player i is simply $g_i(\bar{a})$, which is conveniently independent of the discount factor δ .

But at what cost is this proposed restriction of attention to strategy profiles which prescribe exactly the same stage-game profile in every period along the equilibrium path? Is this without loss of generality? In other words might there not be payoff vectors which could be sustained by time-varying equilibrium paths but which cannot be sustained by time-independent paths? The answer is: yes, this could be the case. We will next see the origin of the problem, and then we will augment our model with a public randomizing device in order to restore generality to our results. Of course, this augmentation has its own cost—what if the economic situation being studied does not admit such a device? (See Fudenberg and Tirole [1991: 152] for references to research which attacks this issue.)

Stage-game payoffs with uncorrelated randomization

Consider the game of Figure 2. The game's three distinct pure-strategy payoff vectors are plotted in “payoff space” in Figure 3. The union of the two shaded regions is the *convex hull* of these three vectors.¹⁵ Note that the $(0, 1) \rightarrow (1, 0)$ edge can be achieved by player 1 choosing Down and player 2

¹⁵ Let X be a finite set of m vectors in a real topological linear space Y (e.g., $Y = \mathbb{R}^n$), $X = \{x^1, \dots, x^m\} \subset Y$, and let Δ^k be the k -dimensional unit simplex. The *convex hull* of X , viz. $\text{co } X$, is the set of vectors in Y each of which is a convex combination of the vectors in X . In other words

mixing between left and right. It can also be achieved by Column choosing left and Row mixing between Up and Down. Similarly, the $(0, 0) \rightarrow (1, 0)$ edge can be achieved in either of two ways: 1 by player 1 choosing Up and player 2 mixing between right and left and 2 player 2 choosing right and player 1 mixing between Up and Down. In these cases each pair of adjacent extreme points of the convex hull corresponds to a pair of strategy profiles such that one player chooses the same pure strategy and one player chooses different pure strategies.¹⁶

However, this is not the case for the $(0, 0) \rightarrow (0, 1)$ edge. The $(0, 0)$ payoff vector is obtained only from the strategy profile (U, r) , and the $(0, 1)$ vector is obtained only from (D, l) . In this case each player chooses different pure strategies in the two strategy profiles.

$\text{co } X = \{y \in Y: \exists \beta \in \Delta^{m-1}, \text{ such that } \sum_{i=1}^m \beta_i x^i = y\}.$

¹⁶ Let X be a convex set. A point $x \in X$ is an *extreme point* of X if x cannot be expressed as the midpoint of two distinct points in X ; i.e. if $\nexists y, z \in X$ such that $y \neq z$ and $x = (y + z)/2$.

	$l: [q]$	$r: [1-q]$
$U: [p]$	1,0	0,0
$D: [1-p]$	0,1	1,0

Figure 2: A game whose pure-strategy payoffs' convex hull cannot be achieved with uncorrelated mixed strategies.

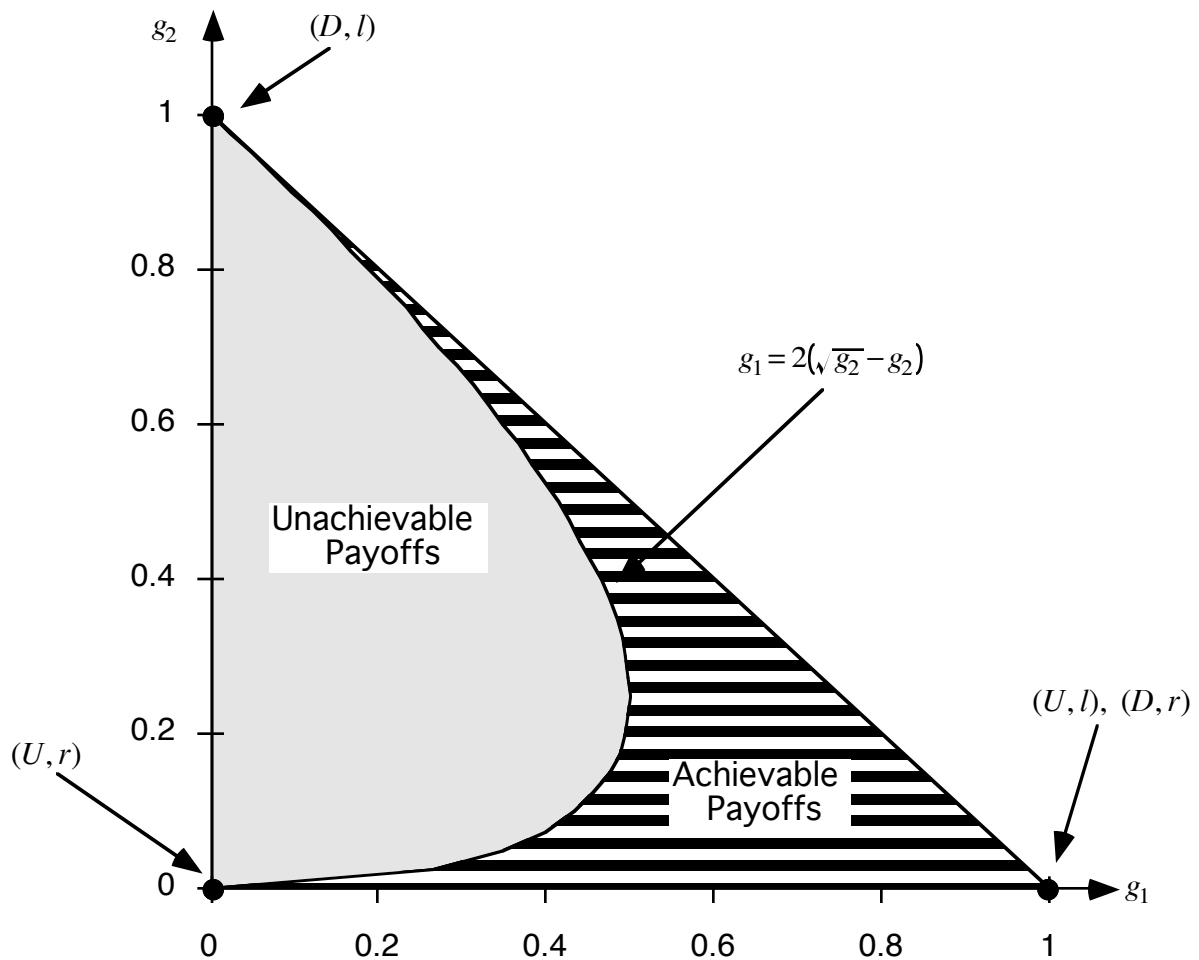


Figure 3: The payoff space for the game of Figure 1, indicating the unachievable and achievable subsets of the convex hull of the pure-strategy payoffs.

It is immediate to see why this edge cannot be achieved by uncorrelated randomizations. This edge is characterized by $g_1 = 0$. The only way such a payoff can be assured by the first player is if the only two strategy profiles receiving positive probability are (D, l) and (U, r) . If a point $(0, v_2)$, $v_2 \in (0, 1)$, is to be achieved, player 1 or player 2 must strictly mix. (The only values of g_2 obtained by pure-strategy profiles are 0 and 1.) Such a strategy profile always results in positive weight applied to a strategy profile which yields player 1 a payoff of 1. Therefore an expected payoff for player 1 of zero cannot be

obtained for $v_2 \in (0, 1)$.¹⁷

Public randomization device

We saw that in the game of Figure 2 we could not achieve any payoff vector of the form $(0, v_2)$ where $v_2 \in (0, 1)$ when both players were limited to independently randomized mixed strategies. So we now introduce a publicly observable randomization device, which will allow the players to correlate their randomized actions. Specifically, assume that there exists a unit roulette wheel which, when spun, lands on a number between zero and one. Its pointer can be considered a random variable ω which is uniformly distributed on $[0, 1]$. It is perched atop A-Mountain and therefore visible to all players. Before each period, the wheel is spun and all players observe the realization of ω before they choose their actions for that period. Because ω becomes part of all players' information, they can condition their actions upon ω .

For example, we could achieve the expected payoff vector $(0, v_2)$ if we could coax the players into playing the action profile (D, l) with probability v_2 and the action profile (U, r) with probability $(1 - v_2)$. The expected payoff vector from such correlated actions is, as desired,

$$g(v_2 \circ (D, l) \oplus (1 - v_2) \circ (U, r)) = v_2 g((D, l)) + (1 - v_2) g((U, r)) = v_2(0, 1) + (1 - v_2)(0, 0) = (0, v_2). \quad (8)$$

In order to implement this coordination we assign the players the following strategies:

$$a_R(\omega) = \begin{cases} D, & \omega \in [0, v_2), \\ U, & \omega \in [v_2, 1], \end{cases} \quad a_C(\omega) = \begin{cases} l, & \omega \in [0, v_2), \\ r, & \omega \in [v_2, 1]. \end{cases} \quad (9)$$

Because ω is uniformly distributed on the unit interval, the probability that the pointer takes on a value in some interval $\langle c, d \rangle \subset [0, 1]$ is simply $d - c$.¹⁸ Therefore the probability that $\omega \in [0, v_2)$ is v_2 . When this event occurs, Row will choose D and Column will choose l ; therefore the action profile (D, l) will occur with probability v_2 . Similarly, (U, r) will occur with probability $(1 - v_2)$. Therefore the pair of stage-game strategies in (9) results in the expected payoff vector $(0, v_2)$ as calculated in (8), even when $v_2 \in (0, 1)$.

More generally, consider the set of pure-action payoff vectors $g(A) \subset \mathbb{R}^n$.¹⁹ Let $g(A) = \{\tilde{v}^1, \tilde{v}^2, \dots, \tilde{v}^m\}$, where $m = \#g(A)$.²⁰ Each payoff vector $\tilde{v}^k$ can be achieved by some pure-action profile $\tilde{a}^k \in A$; i.e. $g(\tilde{a}^k) = \tilde{v}^k$. Let V be the convex hull of these pure-action payoff vectors; i.e.

$$V = \text{co } g(A) = \left\{ v \in \mathbb{R}^n : \exists \beta \in \Delta^{m-1}, v = \sum_{k=1}^m \beta_k \tilde{v}^k \right\}. \quad (10)$$

To achieve any vector $v \in V$ as an expected payoff vector of the players' stage-game correlated-action

¹⁷ You can show that the left-most frontier of the achievable payoffs in Figure 3 is defined by $\{(g_1, g_2) : g_1 = 2(\sqrt{g_2} - g_2), \forall g_2 \in [0, 1]\}$.

¹⁸ Interpret the “ $\rightarrow$ ” as being either “(” or “[” and similarly for “ $\leftarrow$ ”. In other words, because the probability distribution is continuous, it doesn't matter whether the interval includes none, one, or both of its endpoints.

¹⁹ Let $f: X \rightarrow Y$. The *image of X under f*, $f(X)$, is the set of values in the range of f which are achieved for some value in the domain. I.e. $f(X) = \{y \in Y : \exists x \in X, f(x) = y\}$.

²⁰ Note that $\#g(A) \leq \#A$.

profile, we find a corresponding vector β of convex coefficients and assign each player i the stage-game strategy

$$a_i(\omega) = \begin{cases} \tilde{a}_i^1, & \omega \in [0, \beta_1), \\ \tilde{a}_i^2, & \omega \in [\beta_1, \beta_1 + \beta_2), \\ \tilde{a}_i^3, & \omega \in [\beta_1 + \beta_2, \beta_1 + \beta_2 + \beta_3), \\ \vdots & \vdots \\ \tilde{a}_i^k, & \omega \in \left[\sum_{j=1}^{k-1} \beta_j, \sum_{j=1}^k \beta_j \right), \\ \vdots & \vdots \\ \tilde{a}_i^m, & \omega \in [\beta_1 + \cdots + \beta_{m-1}, 1]. \end{cases} \quad (11)$$

For example, if the roulette wheel is spun and the realization ω lies in the interval $(\beta_1 + \cdots + \beta_{k-1}, \beta_1 + \cdots + \beta_k)$, then each player $i \in I$ will choose $\tilde{a}_i^k$; i.e. the action profile $\tilde{a}^k$ will be chosen by the players. We note that the probability that any given realization of ω falls in the interval corresponding to the action profile $\tilde{a}^k$, and thus to the payoff vector $\tilde{v}^k$, is, as desired,

$$\sum_{j=1}^k \beta_j - \sum_{j=1}^{k-1} \beta_j = \beta_k. \quad (12)$$

Because every payoff in the convex hull of the pure-action payoffs can be achieved by some correlated action profile $a(\omega)$, we call V the set of *feasible* outcomes.

ω -augmented histories

Players can condition their current action upon everything they know. Before we introduced the public randomization device, we assumed the only knowledge players gained in the course of play was their observations of other players' past pure actions. Now they also learn in each period the realization ω^t of the random variable ω in that period. When a player is choosing her period- t action, then she knows all of the action profiles played in previous periods as well as all realizations of ω in previous periods *and* in the current period. (The current period's realization is revealed prior to taking that period's action.) Therefore we write the history in period t as the $(2t + 1)$ -tuple

$$h^t = (a^0, a^1, \dots, a^{t-1}; \omega^0, \omega^1, \dots, \omega^t). \quad (13)$$

Before we performed this augmentation, the space of period- t histories was simply A^t , the t -fold Cartesian product of action-profile spaces. Now it is more complicated. Each ω^t is a number on the unit interval. Therefore the space of augmented histories is

$$H^t = A^t \times [0, 1]^{t+1}. \quad (14)$$

A repeated-game strategy for player i is still a sequence $s_i = (s_i^0, s_i^1, \dots)$, but now each history-dependent stage-game strategy in the sequence is a function $s_i^t: H^t \rightarrow A_i$, where H^t is as defined in (14).

Minmax punishments

In order to implement some payoff vector $v \in V$ as an average discounted payoff of a repeated-game equilibrium, we will first find a correlated-action profile $a(\omega)$ whose expected payoff is v (where the expectation is taken with respect to ω); i.e.

$$\mathbb{E}_{\omega} g(a(\omega)) = v. \quad (15)$$

In equilibrium, then, in every period t we want each player i to choose the action $a_i(\omega^t)$, where ω^t is the realization of the random variable ω in period t .

However, we can't necessarily just prescribe that each player choose $a(\omega)$ every period regardless of what her opponents do. In other words, we can't necessarily prescribe open-loop strategies to the players. The reason is that it need not be the case that for all ω (or even for any ω) the action profile $a(\omega)$ is a Nash equilibrium of the stage game. Therefore there could be some ω for which some player i would receive a higher stage-game payoff by playing some $\hat{a}_i \in A_i$ rather than $a_i(\omega)$; i.e. $g_i(\hat{a}_i, a_{-i}(\omega)) > g_i(a(\omega))$. If her opponents were playing open-loop strategies, she could deviate to this $\hat{a}_i$ with impunity because it would not affect her opponents' future actions. Therefore the open-loop prescription of $a(\omega)$ every period would not be an equilibrium of the repeated game.

To encourage each player i to play her part $a_i(\omega)$ of the equilibrium-path prescription, the other players must plan to punish her if she deviates. It will be useful to identify the most severe punishment with which a player can be inflicted. Consider the case where all players other than i set out to punish player i as severely as they can.

We will allow the punishers to choose mixed stage-game actions, so we denote by $\mathcal{A}_j \equiv \Delta(A_j)$ the space of player- j mixed actions in the stage game.²¹ A typical mixed action for player j is $\alpha_j \in \mathcal{A}_j$. A typical deleted mixed-action profile by the punishers is $\alpha_{-i} \in \mathcal{A}_{-i} \equiv \prod_{j \in \mathcal{N} \setminus \{i\}} \mathcal{A}_j$.

The minmax vector

In any equilibrium all players know the strategies of the other players. Therefore whatever punishing actions α_{-i} the punishers choose, player i knows α_{-i} and will play a best response to it. Player i will therefore receive a payoff $\bar{g}_i(\alpha_{-i})$ defined by

$$\bar{g}_i(\alpha_{-i}) \equiv \max_{a_i \in A_i} g_i(a_i, \alpha_{-i}). \quad (16)$$

[Make sure you understand why it is without loss of generality that we can restrict attention to player- i 's pure-strategy space in solving (16).] Her punishers know that player i will choose a_i which solves (16), so they choose their deleted mixed-action punishment profile m_{-i}^i to minimize the payoff player i receives given that she optimizes against the punishment, i.e. to solve

²¹ “ $\mathcal{C}(X)$ ” denotes the set of all probability distributions over the finite set X .

$$m_{-i}^i \in \arg \min_{\alpha_{-i} \in \mathcal{A}_{-i}} \left(\max_{a_i \in A_i} g_i(a_i, \alpha_{-i}) \right) = \arg \min_{\alpha_{-i} \in \mathcal{A}_{-i}} \bar{g}_i(\alpha_{-i}).^{22} \quad (17)$$

Player i 's best response to m_{-i}^i , then, is some $m_i^i \in A_i$ satisfying

$$m_i^i \in \arg \max_{a_i \in A_i} g_i(a_i, m_{-i}^i), \quad (18)$$

where we have substituted m_{-i}^i for α_{-i} in (16). The mixed-action profile $m^i = (m_i^i, m_{-i}^i)$ is player i 's *minmax vector*. For any $j \in I$, m_j^i is player j 's component of the minmax mixed-action stage-game action profile which punishes player i as severely as possible (given that player i is myopically resisting the punishment). Note that the minmax mixed-action profile is a strategic specification for both the punished and the punishers. Player i 's *minmax payoff* v_i is the lowest expected stage-game payoff to which she can be held by players trying to punish her as severely as possible; this value is

$$v_i = g_i(m^i) = \bar{g}_i(m_{-i}^i) = \min_{\alpha_{-i} \in \mathcal{A}_{-i}} \left(\max_{a_i \in A_i} g_i(a_i, \alpha_{-i}) \right) = \min_{\alpha_{-i} \in \mathcal{A}_{-i}} \bar{g}_i(\alpha_{-i}). \quad (19)$$

We denote the n -tuple of the players' minmax values, the *minmax vector*, by $v = (v_1, \dots, v_n)$. (The minmax vector need not be feasible; i.e. it could be the case that $v \notin V$.)

Example: Calculating the minmax vector in a 3×2 game

Consider the two-player, 3×2 stage-game below.

	$l: [l]$	$r: [1-l]$
$U: [p]$	-2, 2	1, -2
$M: [q]$	1, -2	-2, 2
$D: [1-p-q]$	0, 1	0, 1

First let's look for each player's pure-action minmax value—the lowest payoff to which she can be held if her opponents can only play pure actions. To find Row's pure-action minmax value v_R^p , we consider each of Column's pure strategies, finding Row's best-response payoff for each. We have

$$\bar{g}_R(l) = 1 \text{ and } \bar{g}_R(r) = 1.$$

Therefore Row's pure-action minmax payoff is

$$v_R^p = \min \{ \bar{g}_R(l), \bar{g}_R(r) \} = 1.$$

Similarly, Column's pure-action minmax payoff is

$$v_C^p = \min \{ \bar{g}_C(U), \bar{g}_C(M), \bar{g}_C(D) \} = \min \{ 2, 1, 1 \} = 1.$$

Therefore the players' pure-action minmax payoff vector is $v^p = (1, 1)$. We will see that their mixed-action minmax payoff vector is strictly lower.

²² The occurrence of “min max” explains the origin of the term minmax.

Let the probability with which Column chooses l be t , and let the probabilities with which Row chooses U and M be p and q , respectively.

To find Row's minmax value v_R we compute Row's expected utility for each of her pure actions as a function of Column's mixed action t . The upper envelope of these three functions of t is itself a function of t and represents Row's expected utility when she maximizes against Column's action. The minimum value of this upper envelope will be Row's minmax value. Any minimizer (i.e. a value of t) of this upper envelope will be a punish-Row minmax deleted action profile m_C^R . We have

$$u_R(U; t) = -2t + (1 - t) = 1 - 3t,$$

$$u_R(M; t) = t - 2(1 - t) = -2 + 3t,$$

$$u_R(D; t) = 0.$$

See Figure Ex. 1.

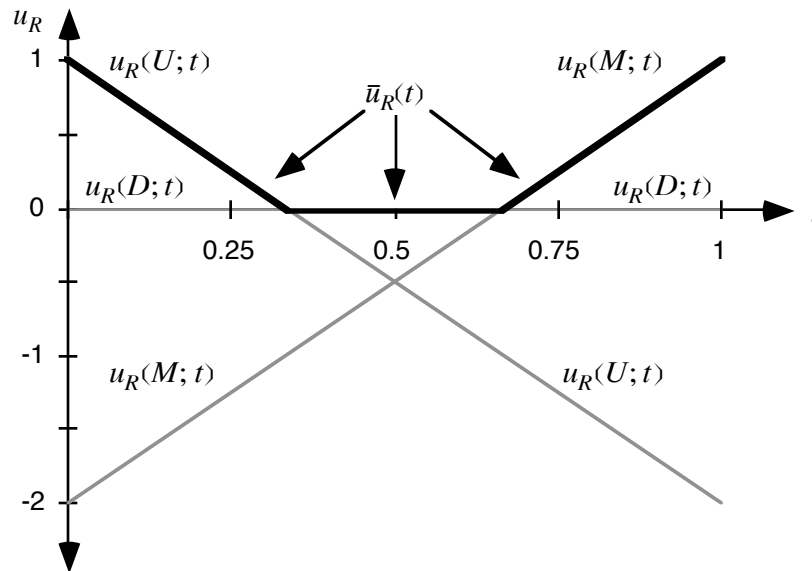


Figure Ex. 1: Row's expected payoffs to her pure actions as a function of Column's mixed action t .

The upper envelope of these functions is

$$\bar{u}_R(t) = \max \{u_R(U; t), u_R(M; t), u_R(D; t)\} = \begin{cases} 1 - 3t, & t \in [0, 1/3), \\ 0, & t \in [1/3, 2/3], \\ -2 + 3t, & t \in (2/3, 1]. \end{cases}$$

The minimum value of this upper envelope is zero, which is achieved for any $t \in [1/3, 2/3]$. Therefore Row's minmax payoff is $v_R = 0$. Therefore one minmax punish-Row deleted action profile would be $m_C^R = \frac{1}{2} \circ l \oplus \frac{1}{2} \circ r$. Row's best response to this mixed action of Columns is $m_C^C = D$.

To find Column's minmax value $\underline{v}_C$ we compute Column's expected utility for each of her two pure actions as a function of Row's mixing probabilities p and q . We have

$$u_C(l; p, q) = 2p - 2q + (1 - p - q) = 1 + p - 3q,$$

$$u_C(r; p, q) = -2p + 2q + (1 - p - q) = 1 - 3p + q.$$

To determine the upper envelope of this function we note that $u_C(l; p, q) \geq u_C(r; p, q)$ if and only if $p \geq q$. Let $\Delta \equiv \{(p, q) \in \mathbb{R}_+^2 : p + q \leq 1\}$, $\Delta^l = \Delta \cap \{(x, y) \in \mathbb{R}_+^2 : x \geq y\}$, and $\Delta^r = \Delta \cap \{(x, y) \in \mathbb{R}_+^2 : x < y\}$. (See Figure Ex. 2.) The two triangles Δ^l and Δ^r form a partition of Δ such that 1 when restricted to Δ^l , $u_C(l; p, q) \geq u_C(r; p, q)$ and 2 when restricted to Δ^r , $u_C(r; p, q) > u_C(l; p, q)$. Then

$$\bar{u}_C(p, q) = \max\{u_C(l; p, q), u_C(r; p, q)\} = \begin{cases} 1 + p - 3q, & (p, q) \in \Delta^l, \\ 1 - 3p + q, & (p, q) \in \Delta^r. \end{cases}$$

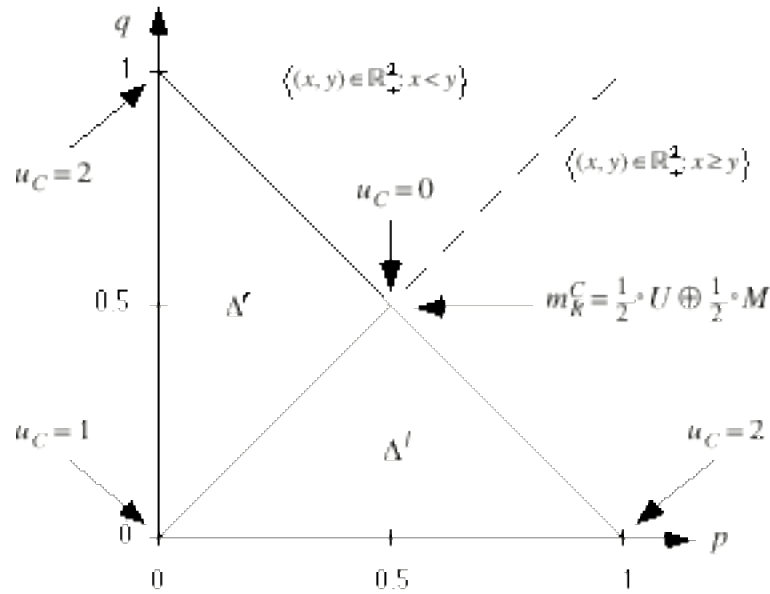


Figure Ex. 2: Row's mixed-actions as points in a two-dimensional simplex.

To find the minimum of $\bar{u}_C$ over Δ we can simply find the minimum of $\bar{u}_C$ over each cell of the partition and then take the minimum of these two minimum values. I.e.

$$\min_{(p,q) \in \Delta} \bar{u}_C(p, q) = \min \left\{ \min_{(p,q) \in \Delta^l} \bar{u}_C(p, q), \min_{(p,q) \in \Delta^r} \bar{u}_C(p, q) \right\}.$$

Each of the simple minimization problems, viz. $\min \{\bar{u}_C(p, q) : (p, q) \in \Delta^a\}$ for $a \in \{l, r\}$, is simply a minimization of a linear function over a convex polyhedron.²³ The minimum value must occur at a vertex of the convex polyhedron.²⁴ The vertices of Δ^l are $(0, 0)$, $(1, 0)$, and $(\frac{1}{2}, \frac{1}{2})$. The values of $\bar{u}_C$ at these vertices are 1, 2, and 0, respectively. Therefore the minimum of $\bar{u}_C$ over Δ^l is 0, which is achieved

²³ A set is a *convex polyhedron* if it is the convex hull of a finite set.

²⁴ See Intriligator [1971: 75].

at $(\frac{1}{2}, \frac{1}{2})$. The vertices of Δ^r are $(0, 0)$, $(0, 1)$, and $(\frac{1}{2}, \frac{1}{2})$. The values of Δ^r at these vertices are 1, 2, and 0, respectively. Therefore the minimum of $\bar{u}_C$ over Δ^r is 0 which is achieved at $(\frac{1}{2}, \frac{1}{2})$. Therefore the minimum of $\bar{u}_C$ over Δ is 0, and is achieved at $(\frac{1}{2}, \frac{1}{2})$.

Therefore Column's minmax payoff is $v_C = 0$. The minmax punish-Column deleted action profile then is $m_R^C = \frac{1}{2} \circ U \oplus \frac{1}{2} \circ M$. Column is indifferent between l and r against this m_R^C , so we can arbitrarily choose any Column mixed action for m_C^C .

Therefore the minmax payoff vector is $v = (0, 0)$, which is strictly less than the pure-action minmax payoff vector; i.e. $v \ll v^p$.

Individual rationality

A player's minmax value v_i establishes a lower bound for the payoff she can be forced to receive in any stage-game Nash equilibrium as well as for the average discounted payoff she can receive in any repeated-game equilibrium. More formally....

Lemma 2

Let $\bar{\alpha} \in \mathcal{A}$ be a stage-game mixed-action profile such that $\bar{\alpha}_i$ is a best response by player i to $\bar{\alpha}_{-i}$. Then, $g_i(\bar{\alpha}) \geq v_i$.

Proof

Because $\bar{\alpha}_i$ is a best response to $\bar{\alpha}_{-i}$,

$$g_i(\bar{\alpha}) = \bar{g}_i(\bar{\alpha}_{-i}) \geq \min_{\alpha_{-i} \in \mathcal{A}_{-i}} \bar{g}_i(\alpha_{-i}) = v_i,$$

where we have used (16), the definition of the minimum, and (19).

Theorem 2

Let $\bar{\alpha} \in \mathcal{A}$ be a stage-game Nash-equilibrium mixed-action profile. Then, for all $i \in I$, $g_i(\bar{\alpha}) \geq v_i$, or equivalently, $g(\bar{\alpha}) \geq v$.

Proof

For every $i \in I$, $\bar{\alpha}_i$ must be a best response to $\bar{\alpha}_{-i}$; therefore the conclusion follows immediately from Lemma 2. ☺

Theorem 3

Let $\bar{\sigma}$ be a Nash-equilibrium mixed-strategy profile of an infinitely repeated game with discounting. Then, for every $i \in I$, $u_i(\bar{\sigma}) \geq v_i$.

Proof

Because $\bar{\sigma}$ is a Nash equilibrium, for all $i \in I$, $\bar{\sigma}_i$ is a best response to $\bar{\sigma}_{-i}$. Therefore for all player- i repeated-game strategies $s_i \in S_i$,

$$u_i(\bar{\sigma}) \geq u_i(s_i, \bar{\sigma}_{-i}). \quad (9.1)$$

Let $BR_i: \mathcal{A}_{-i} \rightarrow A_i$, be player i 's stage-game pure-action best-response correspondence. Consider the player- i repeated-game strategy $s_i = (s_i^0, s_i^1, \dots)$, where in each period player- i chooses a myopic best response to the actions of the other players, $\bar{\sigma}_{-i}^t(h^t) \in \mathcal{A}_{-i}$. I.e. for all $t \in \{0, 1, \dots\}$ and all $h^t \in A^t$,

$$s_i^t(h^t) \in \text{BR}_i(\bar{\sigma}_{-i}^t(h^t)). \quad (9.2)$$

From Lemma 2 we know that, for all t and h^t ,

$$g_i(s_i^t(h^t), \bar{\sigma}_{-i}^t(h^t)) = \bar{g}_i(\bar{\sigma}_{-i}^t(h^t)) \geq v_i. \quad (9.3)$$

Player i 's repeated-game payoff is

$$u_i(s_i, \bar{\sigma}_{-i}) = (1 - \delta) \sum_{t=0}^{\infty} \delta^t g_i(s_i^t(h^t), \bar{\sigma}_{-i}^t(h^t)) \geq (1 - \delta) \sum_{t=0}^{\infty} \delta^t v_i = v_i, \quad (9.4)$$

where the h^t are computed recursively in the usual fashion. The conclusion follows from (9.1). ☺

Theorem 3 tells us that it would be hopeless to try to sustain a payoff vector v as an equilibrium of the repeated game if this would imply that some player were receiving less than her minmax value; i.e. that $\exists i \in I$ such that $v_i < v_i$. Further, such a player would not even be optimizing against the strategies of her opponents. So we say that a payoff vector v is *individually rational* if $v \geq v$; i.e. if for all $i \in I$, $v_i \geq v_i$. We say that v is *strictly individually rational* if $v \gg v$; i.e. for all $i \in I$, $v_i > v_i$.²⁵ We denote by V^* the intersection of the set V of feasible payoffs and the set of strictly individually rational payoffs, i.e.

$$V^* = \{v \in V: v \gg v\}. \quad (20)$$

Grim trigger strategy folk theorems

We now present two folk theorems, which have the virtue of being particularly easy to prove. The strategies they employ to sustain a given payoff vector as an equilibrium have the following form: Play begins in the “normal phase,” in which all players repeat every period their part of a stage-game action profile which achieves v . If a single player deviates from the normal phase, play switches to an open-loop, punish-the-defector phase. If players are sufficiently patient, the threat of the punishment phase will deter them from yielding to the temptation of normal-phase defection.

Adopting trigger strategies provides an important analytical simplification. The punishment phase kicks in when a deviation occurs; however, the details of the punishment phase depend only on the identity of the defector; they are independent of which particular deviation the deviator commits and independent of when she commits it. Therefore we can just consider the myopically best single-period deviation, because the punishment will be the same no matter what.

The first theorem chooses an arbitrary feasible and strictly individually rational payoff vector $v \in V^*$ and uses minmax threats to enforce a Nash-equilibrium path which yields the players an expected payoff

²⁵ My terminology is somewhat at variance with the published literature, I'm afraid. What I refer to as the strictly individually rational set is usually called simply the individually rational set. However, there is nothing necessarily irrational about a player receiving exactly her minmax payoff rather than something strictly greater than that. My terminology also highlights a theoretical lacuna: The folk theorems we will study at most show that the strictly individually rational payoffs are equilibria. However, the larger set of (weakly) individually rational payoffs cannot be ruled out as equilibria. Therefore the folk theorems do not necessarily fully characterize the set of equilibrium payoffs.

of v . However, as we have already observed, the minmax action profile need not be a Nash equilibrium of the stage game. Therefore if a player does defect from the normal phase, the equilibrium strategies can prescribe an open-loop repetition of a stage-game action profile which is not a Nash equilibrium of the stage game. This behavior will not be a Nash equilibrium of that subgame.²⁶ Therefore this theorem will establish that the chosen payoff vector v is a Nash-equilibrium payoff but will not show that it is a subgame-perfect equilibrium payoff. (We will later see a stronger folk theorem, which uses strategies more complicated than grim trigger strategies, which will show that these same feasible and strictly individually rational payoffs are subgame perfect.)

The second theorem focuses on a subset of the feasible and strictly individually rational payoffs V^* . We consider those payoffs which strictly Pareto dominate some stage-game Nash equilibrium payoff. (We actually look at a somewhat larger set, but I'll be more precise later.) Rather than punishing with the minmax strategy profile, if a player defects from the normal phase, play switches to the open-loop repetition of a stage-game Nash equilibrium which is worse for the deviator than her equilibrium payoff. The disadvantage of using stage-game Nash equilibria as the punishment profiles is that this method supports only a possibly proper subset of V^* as equilibrium payoffs. The advantage is that now the punishment phase is a repeated-game Nash equilibrium in any subgame in which a player previously defected; therefore this subset of V^* is shown to be composed not only of Nash-equilibrium payoffs but, more strongly, of subgame-perfect payoffs.

Grim trigger strategies defined

Consider a payoff vector $v \in V^* \subset V = \text{co } g(A)$. We know that there exists a correlated stage-game action profile $\bar{a}(\omega)$ which realizes the payoff vector v in expectation over the random variable ω . In every period t of the “normal phase” of the game we want each player i to choose her part of $\bar{a}(\omega^t)$. We say that player j was the *solo deviator* in period t if she alone did not play her part of $\bar{a}(\omega^t)$, i.e. if the actual stage-game action profile played in that period, $a^t \in A$, was such that $a^t_{-j} = \bar{a}_{-j}(\omega^t)$ but $a^t_j \neq \bar{a}_j(\omega^t)$. We say that player j was the *earliest solo deviator* if in some period t' player j was the solo deviator and, for all prior periods $t'' \in \{0, 1, \dots, t' - 1\}$, there was no player $i \in I$ who was a solo deviator in period t'' .

To be very formal... for any period t , define the earliest-solo-deviator function $\hat{j}^t: H^t \rightarrow (I \cup \{0\})$, so that $\hat{j}^t(h^t)$ is the earliest solo deviator, if any, according to the augmented history h^t . If no player ever deviated solo in periods $0, 1, \dots, t-1$, set $\hat{j}^t(h^t) = 0$. You can see that $\hat{j}^0(h^0) = 0$, i.e. in period zero no player has previously deviated. Further, once a player has earned the distinction of being the earliest solo deviator, she always has that distinction. In other words, if $\hat{j}^t(h^t) \in I$, then for every extension $h^{t+\tau} = (h^t; a^{t+1}, \dots, a^{t+\tau-1})$ of h^t , $\hat{j}^{t+\tau}(h^{t+\tau}) = \hat{j}^t(h^t)$.

Consider a set of n mixed-action profiles $b^i \in \mathcal{A}$, $i \in I$. The interpretation is that b^i is a stage-game profile intended to punish player i if she deviates from the normal phase. (“ b ” stands for “bad.”) We define a grim trigger-strategy profile $\bar{\sigma}$ by specifying for each player i and each period t a history-dependent stage-game action $\bar{\sigma}_i^t(h^t)$ according to

²⁶ If $\bar{s}$ is an open-loop strategy profile for the infinitely repeated game, then $\bar{s}$ is a Nash equilibrium of the infinitely repeated game if and only if every period's stage-game action profile $\bar{s}^t$ is a Nash equilibrium of the stage game.

$$\bar{\sigma}_i^t(h^t) = \begin{cases} \bar{a}_i(\omega^t), & \hat{f}^t(h^t) = 0, \\ b_i^{\hat{f}^t(h^t)}, & \hat{f}^t(h^t) \in I. \end{cases} \quad (21)$$

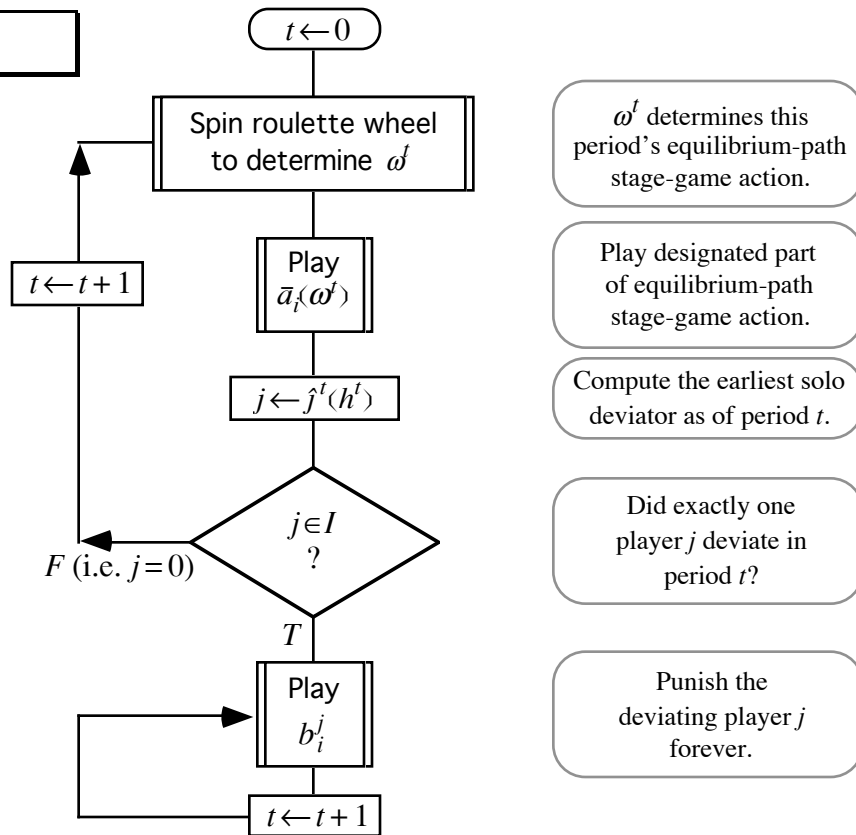
This says to begin in period zero by choosing your part of the equilibrium-path correlated action profile $\bar{a}(\omega^t)$. Continue to choose this ω^t -dependent stage-game action as long as no player has ever deviated alone. If some player j ever deviates alone, then switch to your part of the stage-game mixed-action profile b_i^j which punishes player j , and play that stage-game mixed-action forever, regardless of any future developments. Player i 's strategy is shown in flowchart form in Figure 4.

Nash grim trigger-strategy folk theorem with minmax threats

Let's discover under what conditions $\bar{\sigma}$ is a Nash equilibrium for sufficiently patient players. If all players conform, $\bar{a}(\omega^t)$ is chosen every period. By construction the expectation with respect to the random variable ω of $g(\bar{a}(\omega))$ is v . Therefore the expected repeated-game payoff to each player i if she conforms to $\bar{\sigma}$, given that the other player are conforming as well, is v_i . If instead she deviates in some period, play will switch to the open-loop punishment phase. Given that her opponents are playing open-loop strategies in the punishment phase, player i 's repeated-game best response is to choose myopic stage-game best responses in each period. She thus receives $\bar{g}_i(b_{-i}^i)$ in every period after her deviation. In other words her conformity and deviation payoff streams have terminal subsequences v_i and $\bar{g}_i(b_{-i}^i)$, respectively. From Theorem 1 we know that a sufficiently patient player will weakly prefer to conform to her part of $\bar{\sigma}$ as long as, for all $i \in I$,

$$v_i > \bar{g}_i(b_{-i}^i). \quad (22)$$

For a given set of n punishment vectors $b^1, \dots, b^n$, inequalities (22) fully characterize the set of payoffs which can be sustained as Nash equilibria by grim trigger strategies of the form (21). How can we choose the punishment vectors b^i so as to make this equilibrium payoff set as large as possible? Clearly we can do this by making the right-hand side of each inequality (22) as unrestrictive as possible, which we accomplish by minimizing it through our choice of punishment strategy profile. We observe from (17) that m_{-i}^i is exactly the deleted action-profile which minimizes the right-hand side of (22). Therefore by choosing $b^i = m^i$ for each $i \in I$, we have $\bar{g}_i(b_{-i}^i) = \bar{g}_i(m_{-i}^i) = v_i$ and therefore establish that every payoff vector $v \gg \underline{v}$ can be sustained as a Nash equilibrium of the repeated game for sufficiently patient players. This is expressed in the following theorem.

Theorem 4Figure 4: A grim trigger strategy for player i .

For any feasible and strictly individually rational payoff vector $v \in V^*$, there exists a repeated-game strategy profile which is a Nash equilibrium of the repeated game for sufficiently patient players and yields the expected payoff vector v . I.e., $\forall v \in V^*$, $\exists \bar{\sigma} \in \Sigma$ such that $\exists \delta \in (0, 1)$ such that $\forall \delta \in (\delta, 1)$, $\bar{\sigma}$ is a Nash equilibrium of $G^\infty(\delta)$, and $u(\bar{\sigma}; \delta) = v$.

A remark about the definition of the equilibrium strategies is in order: The equilibrium strategy profile $\bar{\sigma}$ specified punishments only when exactly one player deviated from the equilibrium prescription. Why isn't punishment called for if two or more players deviate? When assessing whether she would deviate from her part of an alleged equilibrium strategy profile, each player i asks whether deviation would be profitable given that all other players faithfully fulfill their part of the profile. (This is straight from the definition of Nash equilibrium.) In other words each player i 's calculations concern only unilateral deviations by player i herself. The prescriptions in cases of multilateral deviations are of no consequence; we could have assigned any actions in these cases. So why did we expend the effort to define the earliest solo deviator concept? The punishments we specified were tuned to individual players. Unless there was a solo deviator, the target of the punishment would have been ambiguous. In order to use open-loop punishments, we could not allow the target of the punishment to change in response to later player actions; therefore we employed the earliest solo deviator criterion, which has the property that the identity of the punished player never changes.

Perfect grim trigger-strategy folk theorem with Nash threats

The weakness of Theorem 4 is that it said nothing about whether any feasible and individually rational payoffs are subgame-perfect equilibrium payoffs. We now address the perfection question, while staying within the context of grim trigger strategies. Under what conditions will $\bar{\sigma}$ be a subgame-perfect equilibrium? There are two types of subgames: A normal-phase subgames in which no player has ever deviated solo from the equilibrium path and B punishment-phase subgames in which an open-loop punishment is being played. We already established that no sufficiently patient player would deviate from the normal phase as long as (22) were satisfied for all players. Consider the subgames in class B. These can be further broken down into n groups on the basis of which b^j profile is being played. When all players choose open-loop strategies, the resulting repeated-game strategy profile is a Nash equilibrium of the repeated game if and only if the stage-game action profile played in every period is a Nash equilibrium of the stage game. Therefore, in order that every punishment phase be a Nash equilibrium of that subgame, each b^i must be a Nash-equilibrium action profile of the stage game. So, in summary, $\bar{\sigma}$ is a subgame-perfect equilibrium if, for all $i \in I$,

$$v_i > \bar{g}_i(b_{-i}^i) \text{ and } b^i \text{ is a Nash equilibrium of the stage-game.} \quad (23)$$

Let $\mathcal{N} \subset \mathcal{A}$ be the set of stage-game mixed-action Nash equilibria. As we did above with respect to Nash equilibria of the repeated game, we now seek the largest set of payoffs which can be sustained as subgame-perfect equilibria for sufficiently patient players using grim trigger strategies of the form in (21). Again we want to minimize for each player i the right-hand side of inequality (22) through our choice of punishment profile, but now we are restrained to choose each b^i from the Nash equilibrium set $\mathcal{N}$. For each player i , we say that a stage-game Nash equilibrium $\eta^i \in \mathcal{N}$ is worst-for- i if

$$\eta^i \in \arg \min_{\eta \in \mathcal{N}} g_i(\eta). \quad (24)$$

For any Nash equilibrium η , $g_i(\eta) = \bar{g}_i(\eta_{-i})$; therefore η^i_{-i} is the minimizer of $\bar{g}_i(b_{-i}^i)$. So choosing, for each $i \in I$, $b^i = \eta^i$, we obtain the following folk theorem.

Theorem 5

For each player $i \in I$, let η^i be a stage-game mixed-action Nash equilibrium which is worst-for- i . Define the payoff vector $\tilde{v}$ by $\tilde{v} = (g_1(\eta^1), \dots, g_n(\eta^n))$. For any feasible and strictly individually rational payoff vector v such that $v \gg \tilde{v}$, there exists a repeated-game strategy profile which is a subgame-perfect equilibrium of the repeated game for sufficiently patient players and yields the expected payoff vector v . I.e., $\forall v \in \{v' \in V: v' \gg \tilde{v}\}$, $\exists \bar{\sigma} \in \Sigma$ such that $\exists \underline{\delta} \in (0, 1)$ such that $\forall \delta \in (\underline{\delta}, 1)$, $\bar{\sigma}$ is a subgame-perfect equilibrium of $G^\infty(\delta)$ and $u(\bar{\sigma}; \delta) = v$.

We know from Theorem 2 that any stage-game Nash equilibrium must give player i at least her minmax value; therefore, for all i , $\tilde{v}_i \geq \underline{v}_i$, and therefore $\tilde{v} \geq \underline{v}$.²⁷ Therefore the set of payoffs supported as subgame-perfect equilibria for sufficiently patient players by Theorem 5 is weakly smaller than the set V^* supported as Nash equilibria by Theorem 4. This leaves open the question of whether the remainder

²⁷ Recall that, for vectors $x, y \in \mathbb{R}^k$, $x \geq y$ means that, for all $i \in \{1, \dots, k\}$, $x_i \geq y_i$.

of V^* can be supported in subgame-perfect equilibrium. This will be answered in the affirmative in our last folk theorem.

A weaker version of Theorem 5 was proved by Friedman [1971]. He showed that any payoff vector $v \in V^*$ which strictly Pareto dominates a Nash-equilibrium payoff vector can be supported as a subgame-perfect equilibrium.²⁸ His punishment phase was the open-loop repetition of a stage-game Nash equilibrium which was dominated by v . In other words, he used the same Nash equilibrium to punish any deviator.

For any stage-game Nash-equilibrium action profile η , $\tilde{v} \leq g(\eta)$. There are games such that, for all stage-game Nash equilibria $\eta \in \mathcal{N}$, there is some player $i \in I$ such that $\tilde{v}_i < g_i(\eta)$; in other words, the set of payoffs supported in Theorem 5 is a strictly larger set than that supported by Friedman's theorem. Theorem 5 was able to support the larger set because its punishment profiles were chosen specifically according to the identity of the deviator rather than being "one punishment fits all."

The ultimate perfect folk theorem

We will now show, subject to a technical qualification concerning the dimensionality of the feasible payoff set V , that any feasible and strictly individually rational payoff $v \in V^*$ can be supported as a subgame-perfect equilibrium of an infinitely repeated game with discounting for sufficiently patient players.

Here is the challenge we face.... In order to enforce some payoff vector $v \in V^*$ as an equilibrium, we must be able to punish any deviating player i with a sequence of actions which yields her an average discounted payoff strictly worse than v_i . (Otherwise, she would deviate and gladly accept the punishment, since the punishment alone is as least as good as what she would receive in equilibrium.)

Consider a feasible and strictly individually rational payoff vector $v \in V^*$ which we *cannot* support as a subgame-perfect equilibrium for sufficiently patient players through the grim trigger strategies with Nash-equilibrium threats of Theorem 5. In other words, there is some player $i \in I$ for whom the payoff v_i she would receive in the specified payoff vector is weakly less than the payoff she would receive in any stage-game Nash-equilibrium. Therefore there is no way that we could punish player i with a stage-game Nash equilibrium so that she would receive a punishment payoff strictly less than v_i . Therefore we cannot support this $v \in V$ as a subgame-perfect equilibrium average payoff in the repeated game using open-loop repetitions of stage-game Nash equilibria for punishments. Therefore we cannot support v using grim trigger strategies.

The perfect folk theorem of Fudenberg and Maskin [1986], which we discuss here, solves the problem by punishing a deviator with her minmax profile for only a finite number of periods.

Theorem 6

Let $\dim V = n$. For any feasible and strictly individually rational payoff vector v , there exists a repeated-game strategy profile which is a subgame-perfect

²⁸ He actually claimed only that they were Nash-equilibrium payoffs. However, his strategies were subgame perfect, so the payoffs are too.

equilibrium of the repeated game for sufficiently patient players and yields the expected payoff vector v . I.e., $\forall v \in V^*, \exists \bar{\sigma} \in \Sigma$ such that $\exists \underline{\delta} \in (0, 1)$ such that $\forall \delta \in (\underline{\delta}, 1)$, $\bar{\sigma}$ is a subgame-perfect equilibrium of $G^\infty(\delta)$, and $u(\bar{\sigma}; \delta) = v$.

Proof Besides the payoff vector v to be supported and the minmax vector $\underline{v}$, which has already been defined, we need to identify $n + 1$ other payoff vectors in order to define the equilibrium strategies. We pick a $\check{v} \in V^*$ which is strictly “between” the minmax vector $\underline{v}$ and the equilibrium payoff v , i.e. $\underline{v} \ll \check{v} \ll v$, or equivalently $\forall i \in I$,

$$\underline{v}_i < \check{v}_i < v_i. \quad (\spadesuit.1)$$

Now fix some $\varepsilon > 0$, and for each player $i \in I$ define the vector $\check{v}^i \in \mathbb{R}^n$ by

$$\check{v}_j^i = \begin{cases} \check{v}_j, & j = i, \\ \check{v}_j + \varepsilon, & j \neq i, \end{cases} \quad (\spadesuit.2)$$

for all $j \in I$. In other words, $\check{v}^i$ is a payoff vector which is ε better than $\check{v}$ for every player except player i . See Figure 5. We further require that every such $\check{v}^i$ be a feasible vector; i.e. $\forall i \in I, \check{v}^i \in V$. (The existence of such a set of vectors for some $\check{v}$ satisfying $(\spadesuit.1)$ and for some $\varepsilon > 0$ is guaranteed by the full-dimensionality assumption.)

In order to keep this presentation of the proof as simple as possible, we’ll assume that 1 there is a pure action $\bar{a} \in A$ which yields v , i.e. $g(\bar{a}) = v$, 2 each player i ’s minmax profile m^i is a pure-action profile, and 3 each of the $\check{v}^i$ can be achieved via some pure action $\check{a}^i \in A$, i.e. $g(\check{a}^i) = \check{v}^i$.

The proposed strategies

The proposed equilibrium strategies $\bar{s}_i, i \in I$, can best be thought of as specifying a particular stage-game action for player i as a function of what *phase* the game is in. (The phase will be a function of the history; therefore the phase-dependent strategies are history-dependent as well.) The game begins, and in equilibrium remains, in the *normal* phase N . In the normal phase players play the action profile $\bar{a}$ each period, which results in the equilibrium payoff vector v . If a player j ever deviates solo from her equilibrium-strategy prescription at any point, play switches to the *punish- j* phase P^j . This consists of some number τ of periods of playing the minmax profile m^j in order to punish player j for deviating. (The value of τ will be specified later.) If this phase ends successfully (no player deviates solo from playing her part of m^j during the τ periods), the game switches to the *reprieve- j* phase R^j . In the R^j phase the action profile $\check{a}^j$ is played each period. Note that play would switch to phase P^j if player j were the sole deviator in any phase—whether she defected from her prescription in the normal phase, in the punishment of another during $P^i, i \neq j$, in her own punishment during P^j , or during a reprieve phase R^i or R^j .

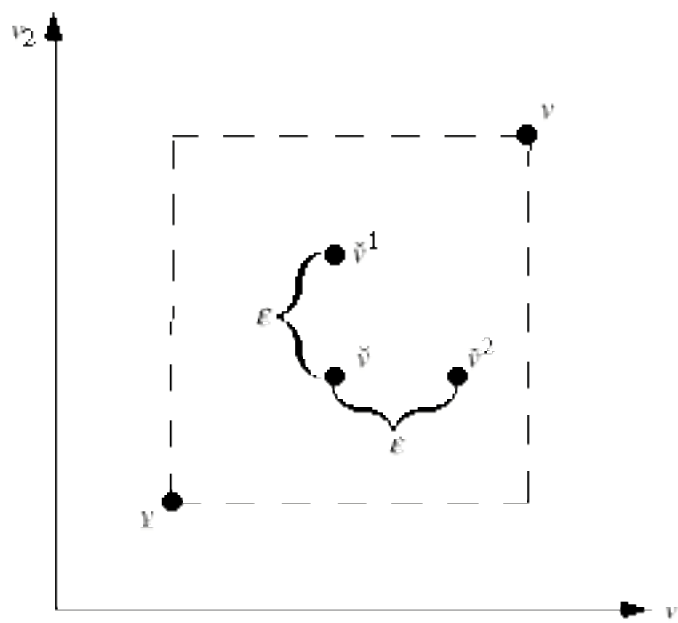


Figure 5: The minmax, equilibrium, and relieve-phase payoffs.

We let ϕ^t be the phase at the beginning of period t , i.e. before the players choose their period- t action. We propose the strategy profile $\bar{s} = (\bar{s}_1, \dots, \bar{s}_n)$, where each player i 's period- t phase-dependent stage-game action is defined by

$$\bar{s}_i^t(\varphi^t) = \begin{cases} \bar{a}_i, & \varphi^t = N, \\ m_i^j, & \varphi^t = P^j, \\ \check{a}_i^j, & \varphi^t = R^j. \end{cases} \quad (\spadesuit.3)$$

This rule tells us the action for each phase, but it does not specify how the phase changes during the course of the game.

So we need a phase transition rule. First we define $\tilde{j}(a, a')$ as the solo deviator, if any, between the pair of action profiles a and a' . I.e. $\tilde{j}(a, a') = j$ if $a_j \neq a'_j$ and $a_{-j} = a'_{-j}$. If no player is the solo deviator, then $\tilde{j}(a, a') = 0$. We also introduce a *counter*, λ , for the punishment phases. We set $\varphi^0 = N$ and define φ^t for $t \in \{1, 2, \dots\}$ by

$$(\varphi^t, \lambda) = \begin{cases} (N, 0), & \varphi^{t-1} = N \text{ and } \tilde{j}(a^{t-1}, \bar{a}) = 0, \\ (P^j, 1), & \tilde{j}(a^{t-1}, \tilde{s}^{t-1}(\varphi^{t-1})) = j, \\ (P^j, \lambda + 1), & \varphi^{t-1} = P^j, \lambda < \tau, \text{ and } \tilde{j}(a^{t-1}, m^j) = 0, \\ (R^j, 0), & \varphi^{t-1} = P^j, \lambda = \tau, \text{ and } \tilde{j}(a^{t-1}, m^j) = 0, \\ (R^j, 0), & \varphi^{t-1} = R^j \text{ and } \tilde{j}(a^{t-1}, \check{a}^j) = 0. \end{cases} \quad (\blacklozenge.4)$$

Let's interpret all this notation a little. The first line says that if the game was in normal phase N in the previous period and no player deviated alone from the normal-phase prescription $\bar{a}$, then play remains in the normal phase. The last line gives similar instructions if the previous period's play had been in a

reprieve phase.²⁹ The second line says that, if in the previous period some player j was the sole deviator from whatever the equilibrium prescription was, viz. $\bar{s}^{t-1}(\phi^{t-1})$, then switch to the punish- j phase and start the punishment counter at $\lambda = 1$.³⁰ The third line says that, if play was in a punish- j phase last period and if no player deviated alone from the appropriate punishment profile m^j , and as long as there is at least one scheduled punishment unadministered (i.e. $\lambda < \tau$), remain in that punishment phase but increment the punishment counter λ by one. The fourth line says that if the last period was the last scheduled punishment of a punishment phase and if that period's punishment was successfully carried out, then switch to the appropriate reprieve phase.

The easy arguments

Now we show that the strategy profile $\bar{s}$ defined by (◆.3) and the phase-transition rule (◆.4) is a subgame-perfect equilibrium. We invoke the one-stage deviation principle: we check whether there is any player i who can profit by deviating from $\bar{s}_i$ at a single period t and history h^t and then returning to her equilibrium strategy $\bar{s}_i$. If there are no such i , t , and h^t , $\bar{s}$ is a subgame-perfect equilibrium.

Consider what happens when some player i deviates from her equilibrium prescription in some subgame. Play immediately switches to the punishment phase P^i . Because all the other players are assumed to be choosing their equilibrium strategies and because we are looking only at one-stage deviations for player i , the remainder of the play will be according to $\bar{s}$. We can therefore predict that the punishment phase P^i will be completed successfully and the reprieve phase R^i will be reached and continue forever. This reprieve phase yields the players the payoff vector $\check{v}^j$ in every period. Therefore any player i contemplating a single-stage deviation will face an infinite payoff sequence with a terminal subsequence of $\check{v}_i^j = \check{v}_i$.

For many of this game's subgames we will be able to establish that a sufficiently patient player i would not wish to deviate from $\bar{s}_i$ by the following technique: We will show that conformity leads to an infinite payoff sequence with a terminal subsequence whose value is strictly higher than $\check{v}_i$. Then Theorem 1 tells us that a sufficiently patient player will prefer the conformity payoff stream.

For example, consider any subgame in the normal phase. Conformity to $\bar{s}$ implies that the game will remain in phase N forever, which yields player i a constant payoff stream of v_i . (This trivially has a terminal subsequence of v_i .) By construction, see (◆.1), $v_i > \check{v}_i$, so no player i will wish to deviate from the normal phase at any point. (I.e. we have shown that no player would deviate at any on-the-equilibrium-path subgame.)

Similarly, consider any subgame in the reprieve phase R^j , where $j \neq i$. Conformity would result in player i receiving a payoff of $\check{v}_i^j = \check{v}_i + \varepsilon$ forever. As before, deviation would result in an infinite sequence

²⁹ The counter $\neg$ is not used in either the Normal phase or any reprieve phase, so setting it to zero is an arbitrary decision. The point was just to have the ordered pair (ϕ, λ) completely defined.

³⁰ The second argument of every occurrence of the $\tilde{s}(\cdot, \cdot)$ function could be replaced by $\bar{s}^{t-1}(\phi^{t-1})$. I use this notation in the second line so that a deviation will be caught, whatever the correct action profile for the phase is. In the other occurrences, on the other hand, I more specifically state what $\bar{s}^{t-1}(\phi^{t-1})$ should be for the previous period's phase. This is intended to aid your comprehension, not to puzzle you further.

with a terminal subsequence of $\check{v}_i$. Clearly the conformity terminal subsequence has a strictly higher value than that offered by deviation. Therefore no player i would choose to deviate during any reprieve- $j, j \neq i$, phase.

The same argument can be made if the game is in the punishment phase $P^j, j \neq i$. Conformity by player i implies that the punishment phase will be successfully completed and then the game will switch to the reprieve phase R^j , where it will remain. Therefore conformity will also result in player i receiving an infinite payoff sequence with a terminal subsequence of $\check{v}_i + \varepsilon$, which strictly exceeds the $\check{v}_i$ offered by deviation.

So we have shown that a sufficiently patient player i would never deviate from a subgame in phase N, R^j , or P^j , when $j \neq i$. However, we still need to investigate whether player i would deviate from a subgame in phase R^i or P^i . In other words, would player i participate in her own reprieve and punishment? We cannot use the previous technique for the following reason. Assume the game is in phase R^i or P^i . If player i conforms she faces an infinite stream of payoffs with a terminal subsequence of $\check{v}_i$. As we saw above, if she deviates in any subgame she also faces a terminal subsequence of $\check{v}_i$. So whether she conforms or deviates, her terminal subsequence is the same. Therefore Theorem 1 is agnostic with respect to whether conformity or deviance is preferable.

It is particularly easy to see that player i would not deviate from her own punishment phase P^i , thanks to the way the punishment profiles are constructed. When player i 's opponents are minmaxing her with $m_{-i}^i, \bar{s}_i$ instructs player i to choose her best response m_i^i . So player i would gain nothing in the deviation period—because she is already playing a best response—and in fact she would merely prolong the duration of her punishment and postpone her reprieve. To see this more explicitly, assume the game is at a subgame in which there are $\tau' \leq \tau$ periods remaining in the punishment phase. If player i conforms she will earn a continuation payoff of

$$(1 - \delta^{\tau'})\underline{v}_i + \delta^{\tau'}\check{v}_i. \quad (\spadesuit.5)$$

[See (4).] If she deviates this period, she will earn at most $\underline{v}_i$ this period (since her opponents are minmaxing her), she will earn $\underline{v}_i$ for the next τ periods of her renewed punishment phase, and then earn $\check{v}_i$ forever. This continuation payoff is at most

$$(1 - \delta^{\tau+1})\underline{v}_i + \delta^{\tau+1}\check{v}_i. \quad (\spadesuit.6)$$

Since $\tau' < \tau + 1$, the deviation convex combination ($\spadesuit.6$) puts more weight on the strictly lower, minmax payoff than does the conformity payoff ($\spadesuit.5$), therefore it is strictly lower. Therefore no player i would deviate from her own punishment phase P^i .³¹ This argument is shown more graphically in Figure 6. There we see that the conformity and deviation payoff streams are identical except for $(\tau + 1) - \tau'$ periods in which the conformity stage-game payoff is strictly larger than the deviation stage-game payoffs. Therefore the conformity stream dominates the deviation stream for all discount factors.

³¹ Note that this conclusion required no sufficient-patience limit argument.

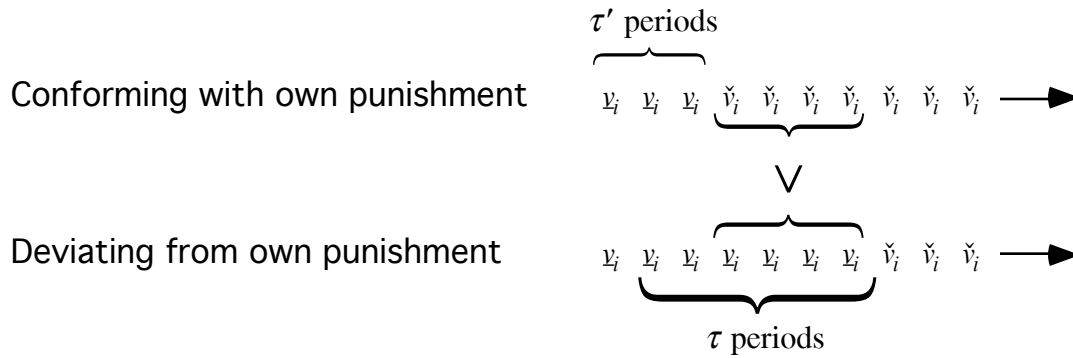


Figure 6: A player would never deviate from her own punishment because the initial deviation gains her nothing and it only prolongs her punishment.

The harder argument

The only subgames we still have not checked for vulnerability to defection by player i are those in player i 's reprieve phase R^i . Up to this point we have not needed to specify the duration τ of the punishment phases; now we will.

If player i conforms with R^i , she receives an infinite string of payoffs of v_i . If she deviates in a single stage and conforms thereafter, she receives some unspecified payoff in that period, followed by τ periods of her minmax payoff from the punishment phase, and then a terminal subsequence of v_i . After the first $\tau + 1$ periods (including the deviation period), the two streams are the same: v_i forever. To determine which stream is preferable, then, we compare the finite discounted sum over the first $\tau + 1$ periods.

During the reprieve- i phase the other players are choosing the deleted action profile $\check{a}_{-i}^i$. The unspecified deviation-period payoff is certainly bounded above, using (16), by

$$\bar{v}_i \equiv \bar{g}_i(\check{a}_{-i}^i). \quad (\spadesuit.7)$$

Using (3), we calculate the discounted sum of the payoffs during these $\tau + 1$ periods for both deviation and conformity and require that conformity be weakly preferable. This results in the inequality

$$v_i + \frac{\delta(1-\delta^\tau)}{1-\delta} v_i \geq \bar{v}_i + \frac{\delta(1-\delta^\tau)}{1-\delta} l_i, \quad (\spadesuit.8)$$

where the left-hand side is written unnecessarily expansively to make it more readily comparable to the right-hand side. We rewrite this condition as

$$\Psi(\delta, \tau) \equiv \frac{\delta(1-\delta^\tau)}{1-\delta} \geq \frac{\bar{v}_i - v_i}{v_i - l_i} \equiv \rho > 0. \quad (\spadesuit.9)$$

The question we must answer in the affirmative, if conformity is to be weakly preferable for a sufficiently patient player, is whether for arbitrary $\rho > 0$ there exist δ and τ such that the left-hand side of $(\spadesuit.9)$ exceeds ρ . Two limits are illuminating:

$$\lim_{\tau \rightarrow \infty} \Psi(\delta, \tau) = \frac{\delta}{1 - \delta}, \quad (\spadesuit.10)$$

$$\lim_{\delta \rightarrow 1} \Psi(\delta, \tau) = \lim_{\delta \rightarrow 1} \frac{1 - (\tau + 1)\delta^\tau}{-1} = \tau, \quad (\spadesuit.11)$$

where we used l'Hôpital's rule to evaluate the second limit.³²

Although the left-hand side function $\Psi(\delta, \tau)$ increases with τ for fixed δ , ($\spadesuit.10$) shows that it is still bounded above by $\delta/(1 - \delta)$ which need not exceed ρ . Therefore we cannot satisfy ($\spadesuit.9$) for arbitrary $\delta \in (0, 1)$ just by taking the punishment duration τ sufficiently large. Similarly, although, for fixed τ , $\Psi(\delta, \tau)$ increases with the discount rate δ , ($\spadesuit.11$) shows that $\Psi(\delta, \tau)$ is still bounded above by τ , which need not exceed ρ . Therefore we cannot satisfy ($\spadesuit.9$) for an arbitrary punishment duration τ merely by invoking a sufficient-patience argument.

So we need a combination of taking δ sufficiently close to 1 and τ sufficiently large. Limit ($\spadesuit.11$) tells us that $\Psi(\delta, \tau)$ can be made arbitrarily close to τ by taking δ sufficiently close to one. By choosing τ strictly greater than ρ , we guarantee that ($\spadesuit.9$) is satisfied by a sufficiently patient player i . Hence player i would not deviate from her own reprieve phase. ☺

³² Alternatively, note that $(1 - \delta^\tau)/(1 - \delta) = 1 + \delta + \delta^2 + \dots + \delta^{\tau-1}$.

Appendix: The No One-Stage Improvement Principle

Let $s_i = (s_i^0, s_i^1, \dots, s_i^T) \in S_i$ be a repeated-game strategy for player $i \in I$ in the repeated game “ending” in period T , where T may be ∞ , and where each $s_i^t: A^t \rightarrow A_i$. Consider a period $t \in \mathbb{T} \equiv \{0, 1, \dots, T\}$, a history $h^t = (a^0, \dots, a^{t-1}) \in A^t$, and a strategy profile $s \in S \equiv \prod_{i \in I} S_i$. The payoff to player i to the profile s conditional upon the history h^t being reached is

$$u_i(s | h^t) = \prod_{\tau=0}^{t-1} \delta^\tau g_i(a^\tau) + \prod_{\tau=t}^T \delta^\tau g_i(s^\tau(h^\tau)), \quad (1)$$

where the h^τ , for $\tau > t$, are defined recursively by concatenation as

$$h^\tau = (h^{t-1}, s^{t-1}(h^{t-1})). \quad (2)$$

The *continuation* payoff to player i to the profile s for the subgame determined by the history h^t is

$$\check{u}_i(s | h^t) = \prod_{\tau=t}^T \delta^{\tau-t} g_i(s^\tau(h^\tau)). \quad (3)$$

We can relate player i 's continuation payoff in adjacent periods by observing from (3) that

$$\check{u}_i(s | h^{t-1}) = \prod_{\tau=t-1}^T \delta^{\tau-(t-1)} g_i(s^\tau(h^\tau)) = g_i(s^{t-1}(h^{t-1})) + \delta \prod_{\tau=t}^T \delta^{\tau-t} g_i(s^\tau(h^\tau)), \quad (4)$$

which is equivalent to

$$\check{u}_i(s | h^{t-1}) = g_i(s^{t-1}(h^{t-1})) + \delta \check{u}_i(s | (h^{t-1}, s^{t-1}(h^{t-1}))), \quad (5)$$

where we have used (2).

A repeated-game strategy profile s is a subgame-perfect equilibrium iff for all players $i \in I$, for all periods $t \in \mathbb{T}$, for all histories $h^t \in A^t$, and for all player- i repeated-game strategies $\tilde{s}_i \in S_i$,

$$u_i(\tilde{s}_i, s_{-i} | h^t) \leq u_i(s | h^t). \quad (6)$$

Clearly, inequality (6) can be replaced by

$$\check{u}_i(\tilde{s}_i, s_{-i} | h^t) \leq \check{u}_i(s | h^t). \quad (7)$$

We say that the repeated-game strategy for player i , $\hat{s}_i: A^T \rightarrow A_i^{T+1}$,³³ is a *one-stage deviant* of s_i if there exist a $\hat{t}$ and a history $\hat{h}^{\hat{t}} \in A^{\hat{t}}$ such that 1 $\forall t \neq \hat{t}$, $\hat{s}_i^t = s_i^t$, 2 $\forall h^{\hat{t}} \neq \hat{h}^{\hat{t}}$, $\hat{s}_i^{\hat{t}}(h^{\hat{t}}) = s_i^{\hat{t}}(h^{\hat{t}})$, and 3 $\hat{s}_i^{\hat{t}}(\hat{h}^{\hat{t}}) \neq s_i^{\hat{t}}(\hat{h}^{\hat{t}})$. Let $\hat{S}_i(s_i)$ be the space of all one-stage deviants of s_i .

We say that s satisfies the “no one-stage improvement (NOSI)” property if for all $i \in I$, for all one-stage deviants $\hat{s}_i \in \hat{S}_i(s_i)$ of s_i , for all $t \in \mathbb{T}$, and for all $h^t \in A^t$, $\hat{s}_i$ is no better than s_i against s_{-i} , conditional

³³ This notation seems somewhat nonsensical when $T = \infty$!

on reaching the history h^t , i.e.

$$\check{u}_i((\hat{s}_i, s_{-i}) \mid h^t) \leq \check{u}_i(s \mid h^t). \quad (8)$$

It's clear that (8) is a weaker condition than (7); therefore if s is subgame perfect, it satisfies NOSI.

We say that the repeated-game strategy for player i , $\tilde{s}_i: A^T \rightarrow A_i^{T+1}$, is a *finite-stage deviant* of s_i if $\tilde{s}_i$ differs from s_i in at most a finite number of stages. We denote by $\tilde{S}_i(s_i)$ the space of finite-stage deviants of s_i . Formally, $\tilde{s}_i \in \tilde{S}_i(s_i)$ iff $\tilde{s}_i \neq s_i$ and there exists a $\bar{T} < \infty$ such that for all $t \in \{\bar{T} + 1, \bar{T} + 2, \dots, T\}$, $\tilde{s}_i^t = s_i^t$.³⁴

Lemma Let $s \in S$ satisfy the no one-stage improvement property and let $t \in \{1, \dots, T\}$ be such that for all $i \in I$, for all $h^t \in A^t$, and for all player- i repeated-game strategies $\tilde{s}_i \in S_i$,

$$\check{u}_i((\tilde{s}_i, s_{-i}) \mid h^t) \leq \check{u}_i(s \mid h^t). \quad (9)$$

Then for all $i \in I$, for all $h^{t-1} \in A^{t-1}$, and for all $\tilde{s}_i \in S_i$,

$$\check{u}_i((\tilde{s}_i, s_{-i}) \mid h^{t-1}) \leq \check{u}_i(s \mid h^{t-1}). \quad (10)$$

Proof Assume not. Then there exist a player $i \in I$, a history $\tilde{h}^{t-1} \in A^{t-1}$, and a player- i repeated game strategy $\tilde{s}_i \in S_i$ such that

$$\check{u}_i((\tilde{s}_i, s_{-i}) \mid \tilde{h}^{t-1}) > \check{u}_i(s \mid \tilde{h}^{t-1}). \quad (3.1)$$

Rewriting the left-hand side of (3.1), using (5), we obtain

$$\check{u}_i((\tilde{s}_i, s_{-i}) \mid \tilde{h}^{t-1}) = g_i((\tilde{s}_i^{t-1}, s_{-i}^{t-1})(\tilde{h}^{t-1})) + \delta \check{u}_i((\tilde{s}_i, s_{-i}) \mid \tilde{h}^t), \quad (3.2)$$

where we have defined

$$\tilde{h}^t = (\tilde{h}^{t-1}; (\tilde{s}_i^{t-1}, s_{-i}^{t-1})(\tilde{h}^{t-1})). \quad (3.3)$$

Focus now on the last term of (3.2). From (9) we have

$$\check{u}_i(s \mid \tilde{h}^t) \geq \check{u}_i((\tilde{s}_i, s_{-i}) \mid \tilde{h}^t). \quad (3.4)$$

Therefore, using (3.2) and (3.1),

$$g_i((\tilde{s}_i^{t-1}, s_{-i}^{t-1})(\tilde{h}^{t-1})) + \delta \check{u}_i(s \mid \tilde{h}^t) > \check{u}_i((\tilde{s}_i, s_{-i}) \mid \tilde{h}^{t-1}) > \check{u}_i(s \mid \tilde{h}^{t-1}). \quad (3.5)$$

Let $\hat{s}_i \in \hat{S}_i(s_i)$ be the one-stage deviant of s_i defined by

$$\hat{s}_i^{t-1}(\tilde{h}^{t-1}) = \tilde{s}_i^{t-1}(\tilde{h}^{t-1}). \quad (3.6)$$

³⁴ For a finitely repeated game, just take $\bar{T} = T$.

We observe from (5) that player i 's continuation payoff to $\hat{s}_i$ in the subgame determined by $\tilde{h}^{t-1}$, $\check{u}_i((\hat{s}_i, s_{-i}) \mid \tilde{h}^{t-1})$, is exactly the left-hand side of (3.5), because s_i and $\hat{s}_i$ agree for time periods from t onward. Therefore from (3.5)

$$\check{u}_i((\hat{s}_i, s_{-i}) \mid \tilde{h}^{t-1}) > \check{u}_i(s \mid \tilde{h}^{t-1}); \quad (3.7)$$

which contradicts that s satisfies NOSI because $\hat{s}_i$ is a one-stage deviant of s_i . ☺

Theorem 1

If the repeated-game strategy profile s satisfies the no one-stage improvement property then, for every player i and in every subgame h^t , s_i is as good for player i against s_{-i} as any finite-stage deviant $\tilde{s}_i \in \tilde{S}_i(s_i)$, conditional upon h^t being reached. In other words, if $s \in S$ satisfies NOSI, then $\forall i \in I, \forall t \in \mathbb{T}, \forall h^t \in A^t, \forall \tilde{s}_i \in \tilde{S}_i(s_i)$,

$$u_i((\tilde{s}_i, s_{-i}) \mid h^t) \leq u_i(s \mid h^t). \quad (11)$$

Proof

Let T' be such that for all $t \in \{T' + 1, T' + 2, \dots, T\}$, $\tilde{s}_i^t = s_i^t$. Let $\bar{T} \equiv \min \{T', T\}$.³⁵ We establish the premise (9) of the Lemma for the last possibly distinct period $\bar{T}$ and use induction to establish (7) for all $t < \bar{T}$. (Inequality (9) is trivially satisfied for $\bar{T} < t \leq T$.)

Let $t = \bar{T}$ and assume that the premise of the Lemma is not satisfied; i.e. for some $i \in I$, history $\tilde{h}^{\bar{T}}$, and finite-stage deviant $\tilde{s}_i \in \tilde{S}_i(s_i)$,

$$\check{u}_i((\tilde{s}_i, s_{-i}) \mid \tilde{h}^{\bar{T}}) > \check{u}_i(s \mid \tilde{h}^{\bar{T}}). \quad (9.1)$$

From (3) we see that (9.1) is equivalent to

$$g_i((\tilde{s}_i^{\bar{T}}, s_{-i}^{\bar{T}})(\tilde{h}^{\bar{T}})) > g_i(s^{\bar{T}}(\tilde{h}^{\bar{T}})), \quad (9.2)$$

because $\tilde{s}_i$ and s_i agree after period $\bar{T}$. Let $\hat{s}_i \in \hat{S}_i(s_i)$ be the one-stage deviant of s_i defined by $\hat{s}_i^{\bar{T}}(\tilde{h}^{\bar{T}}) = \tilde{s}_i^{\bar{T}}(\tilde{h}^{\bar{T}})$. We can now rewrite (9.2) as

$$g_i((\hat{s}_i^{\bar{T}}, s_{-i}^{\bar{T}})(\tilde{h}^{\bar{T}})) > g_i(s^{\bar{T}}(\tilde{h}^{\bar{T}})). \quad (9.3)$$

Again appealing to (3) we rewrite (9.3) as

$$\check{u}_i((\hat{s}_i, s_{-i}) \mid \tilde{h}^{\bar{T}}) > \check{u}_i(s \mid \tilde{h}^{\bar{T}}), \quad (9.4)$$

because $\hat{s}_i$ and s_i agree after period $\bar{T}$. However, (9.4) would violate NOSI. Therefore the premise of the Lemma must be satisfied for $t = \bar{T}$.

Now, inductive use of the Lemma establishes (9) for all $t \in \mathbb{T}$ such $t \leq \bar{T}$ and therefore for all $t \in \mathbb{T}$. This condition is exactly (11). ☺

³⁵ We did not require that T' was the earliest such period. In particular, in a finitely repeated game, it could be the case that $T' > T$.

Corollary 1

Let s be a strategy profile for a finitely repeated game. The profile s is a subgame-perfect equilibrium of the repeated game if and only if s satisfies the no one-stage improvement property.

Proof

As we observed above, satisfaction of NOSI is necessary for s to be a subgame-perfect equilibrium. In a finitely repeated game, every player- i strategy $\tilde{s}_i \neq s_i$ is a finite-stage deviant of s_i . Therefore condition (11) is exactly condition (7) for subgame perfection. ☺

Now consider an infinitely repeated game and let h^∞ and $\tilde{h}^\infty$ be any two infinite histories. Let h^t and $\tilde{h}^t$ be their respective restrictions to the first t periods. A game is *continuous at infinity* if

$$\lim_{t \rightarrow \infty} \sup_{\substack{h^\infty, \tilde{h}^\infty \in A^\infty \\ h^t = \tilde{h}^t}} |u_i(h^\infty) - u_i(\tilde{h}^\infty)|. \quad (12)$$

It is easily verified that an infinitely repeated game with discounting is continuous at infinity.

Theorem 2

Let s be a strategy profile for a finitely repeated game or an infinitely repeated game with discounting. Then s is a subgame-perfect equilibrium if and only if s satisfies the no one-stage improvement property.

Proof

As we observed above, satisfaction of NOSI is necessary for s to be a subgame-perfect equilibrium. Corollary 1 proved sufficiency of NOSI for subgame perfection in the case of finitely repeated games.

Consider an infinitely repeated game with discounting. Assume that s satisfies NOSI but is not a subgame perfect equilibrium. Then there exists a player i , a period $\hat{t}$, a history $h^{\hat{t}}$, and a player- i repeated-game strategy $\hat{s}_i$ such that, for some $\varepsilon > 0$,

$$u_i((\hat{s}_i, s_{-i}) | h^{\hat{t}}) - u_i(s | h^{\hat{t}}) = \varepsilon > 0. \quad (\heartsuit.1)$$

(We know that $\hat{s}_i$ is not a finite-stage deviant of s_i ; this is ruled out by Theorem 1.)

Define

$$\Sigma_{\hat{t}_1}^{\hat{t}_2} = \prod_{t=\hat{t}_1}^{\hat{t}_2} \delta^t (g_i((\hat{s}_i^t, s_{-i}^t) | h^{\hat{t}}) - g_i(s | h^{\hat{t}})). \quad (\heartsuit.2)$$

From $(\heartsuit.1)$, we have

$$\varepsilon = \Sigma_{\hat{t}}^\infty = \Sigma_{\hat{t}}^{\hat{t}-1} + \Sigma_{\hat{t}}^\infty, \quad (\heartsuit.3)$$

for any $\hat{t} \in \{\hat{t}, \hat{t} + 1, \dots\}$. Because the game is continuous at infinity, we can choose $\hat{t}$ sufficiently large

Forcing $\bar{T} \leq T$ ensures that the strategies are defined for $t = \bar{T}$.

that $\Sigma_{\bar{t}}^{\infty} < \frac{1}{2}\varepsilon$. Therefore $\Sigma_{\bar{t}}^{\bar{t}-1} > \frac{1}{2}\varepsilon$.

Now construct a finite-stage deviant of s_i by defining

$$\tilde{s}_i^t = \begin{cases} \hat{s}_i^t, & t < \bar{t}, \\ s_i^t, & t \geq \bar{t}. \end{cases} \quad (\heartsuit.4)$$

We observe that

$$u_i(\tilde{s}_i, s_{-i} | h^{\bar{t}}) - u_i(s | h^{\bar{t}}) = \Sigma_{\bar{t}}^{\bar{t}-1} > 0, \quad (\heartsuit.5)$$

because $\tilde{s}_i$ agrees with $\hat{s}_i$ prior to period $\bar{t}$ and agrees with s_i from $\bar{t}$ onward. Because $(\heartsuit.5)$ is positive, the conclusion of Theorem 1 is violated and therefore, contrary to assumption, s must not satisfy NOSI. ☺

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Jim Ratliff's Game Theory Course:

6.1: Static Games of Incomplete Information

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Abstract



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In many economically important situations the game may begin with some player having private information about something relevant to her decision making. These are called games of *incomplete information*, or *Bayesian* games. (Incomplete information is not to be confused with *imperfect* information in which players do not perfectly observe the actions of other players.) Although any given player does not know the private information of an opponent, she will have some beliefs about what the opponent knows, and we will assume that these beliefs are common knowledge.

In many cases of interest we will be able to model the informational asymmetry by specifying that each player knows her own payoff function, but that she is uncertain about what her opponents' payoff functions are.

We will introduce the notion of a player's *type* to describe her private information. A player's type fully describes any information she has which is not common knowledge. Each player knows her own type with complete certainty. Her beliefs about other players' types are captured by a common-knowledge joint probability distribution over the others' types.

We can think of the game as beginning with a move by Nature, who to each player assigns a type. Nature's move is imperfectly observed, however: each player observes the type which Nature has bestowed upon her, but no player directly observes the type bestowed upon any other player. We can think of the game which follows as a strategic-form game being played by a single type of each player, where at least one player doesn't know which type of some other player she is facing.

A strategy for a player in the incomplete-information game is a strategy for each of that player's types for the appropriate strategic-form game.

A Bayesian equilibrium of a static game of incomplete information is a strategy profile such that every type of every player is maximizing her expected utility given the type-contingent strategies of her opponents and the probability distribution over types of each player.

We consider an industrial-organization example with two firms, where one firm, the incumbent, knows privately whether its investment cost is low or high. We compute the Bayesian equilibrium to predict whether the challenger will enter the market and whether the incumbent firm will modernize its technology.



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[§5.3: A Folk Theorem Sampler](#)

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Static Games of Incomplete Information

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Introduction

In many economically important situations the game may begin with some player having private information about something relevant to her decision making. These are called games of *incomplete information*, or *Bayesian* games. (Incomplete information is not to be confused with *imperfect* information in which players do not perfectly observe the actions of other players.) For example in a currently monopolized market the incumbent firm knows its marginal cost, which would influence its production decision if a potential rival entered the market, but the rival does not know the incumbent's cost. Or in a sealed-bid auction each player knows how much she values the object for sale but does not know her opponents' valuations. Although any given player does not know the private information of an opponent, she will have some beliefs about what the opponent knows, and we will assume that these beliefs are common knowledge.

In many cases of interest we will be able to model the informational asymmetry by specifying that each player knows her own payoff function, but that she is uncertain about what her opponents' payoff functions are. In the monopolized market example above the incumbent's knowledge of its own costs translates into knowledge of its profit for any given combination of production decisions by the two firms. In the sealed-bid auction example, knowing her own valuation of the object is equivalent to knowing her utility if she is the successful bidder for any given price paid, and therefore to knowing her utility for any set of players' bids.

We will introduce the notion of a player's *type* to describe her private information. A player's type fully describes any information she has which is not common knowledge. A player may have several types—even an infinity of types, one for each possible state of her private information. Each player knows her own type with complete certainty. Her beliefs about other players' types are captured by a common-knowledge joint probability distribution over the others' types. In the currently monopolized-market example, if the incumbent firm's cost were restricted to being either “low” or “high,” it would have two types, e.g. $\{c, \bar{c}\}$. If its cost could be any value in some interval $[c, \bar{c}]$, it would have a continuum of types.

We can think of the game as beginning with a move by Nature, who to each player assigns a type. Nature's move is imperfectly observed, however: each player observes the type which Nature has bestowed upon her, but no player directly observes the type bestowed upon any other player. We can think of the game which follows as being played by a single type of each player, where at least one player doesn't know which type of some other player she is facing.

A Bayesian game

Let $I = \{1, \dots, n\}$ be the set of players. We refer to a type of player i by θ_i , where this type is a member of player i 's *type space* Θ_i ; i.e. $\theta_i \in \Theta_i$. We denote an n -tuple of types, one for each player—or *type profile*—by $\theta = (\theta_1, \dots, \theta_n) \in \Theta \equiv \prod_{i \in I} \Theta_i$, where Θ is the *type-profile space*. When we focus on the types of a player's opponents, we consider deleted type profiles of the form $\theta_{-i} = (\theta_1, \dots, \theta_{i-1}, \theta_{i+1}, \dots, \theta_n) \in \Theta_{-i}$.

For our current purposes we will consider the game played after Nature's type assignments as one in strategic form; we are considering static (i.e. simultaneous-move) games of incomplete information.¹ At a later time we will extend this framework to include dynamic games.

The “bigger” game begins when Nature chooses each player's type and reveals it to her alone. In the strategic-form game which follows, each player i ultimately chooses some pure action $a_i \in A_i$. The n -tuple of actions chosen is the action profile $a \in A \equiv \prod_{i \in I} A_i$.² The payoff player i receives in general depends on the actions a of all players as well as the types θ of all players; i.e. $u_i(a, \theta)$, where $u_i: A \times \Theta \rightarrow \mathbb{R}$.³ For most of our discussion we will assume that the action and type spaces are finite sets. We denote by $\mathcal{A}_i \equiv \Delta(A_i)$ the space of player- i mixed actions. A typical mixed action for player i is $\alpha_i \in \mathcal{A}_i$. A typical deleted mixed-action profile by player i 's opponents is $\alpha_{-i} \in \mathcal{A}_{-i} \equiv \prod_{j \in I \setminus \{i\}} \mathcal{A}_j$; also $\mathcal{A} \equiv \prod_{i \in I} \mathcal{A}_i$.

Beliefs

We assume that there is an objective probability distribution $p \in \Delta(\Theta)$ over the type space Θ , which Nature consults when assigning types.⁴ In other words, the probability with which Nature draws the type profile $\theta = (\theta_1, \dots, \theta_n)$ —and hence assigns type θ_1 to player 1, type θ_2 to player 2, etc.—is $p(\theta)$. The

¹ See Fudenberg and Tirole [1991: Chapter 6] and/or Gibbons [1992: Chapter 3] for more on static Bayesian games.

² When studying repeated games we reserved the symbol s to refer to strategies more complicated than stage-game actions, viz. for repeated-game strategies, which were sequences of history-dependent stage-game actions. We do this again, letting a denote actions in the strategic-form game succeeding Nature's revelation of types, and reserving s to refer to a more complicated strategic object, which depends on type, which we'll need in the larger game.

³ It's easy to see why other players' actions should enter into player i 's payoff function, but why should other players' *types* enter into player i 's payoffs? Sure, another player's type can influence his action; but this indirect influence of another's type on i 's payoff would be captured by the direct effect the other's action has on i 's payoff. A reason we should allow in general for a direct dependence of player i 's payoff upon others' types is given by the following example. Assume that one firm j has private information about demand, which therefore is captured by his type θ_j . When firm i competes with firm j , the market outcome and hence firm i 's profit will depend on the demand and therefore on firm j 's type.

⁴ For any finite set T we denote by $\Delta(T)$ the set of probability distributions over T .

marginal distribution of player i 's type is $p_i \in \Delta(\Theta_i)$, where

$$p_i(\theta_i) = \sum_{\theta_{-i} \in \Theta_{-i}} p(\theta_i, \theta_{-i}). \quad (1)$$

This number $p_i(\theta_i)$ is the probability that Nature draws a type profile θ whose i -th component is θ_i .⁵ For later technical convenience we remove from a player's type space any impossible types; i.e. we require, for all $i \in I$ and all $\theta_i \in \Theta_i$, that $p_i(\theta_i) > 0$.

We assume that the probability distribution p , which generates the type profile, is common knowledge and that each player derives from p her subjective beliefs about the types of her opponents. We represent these beliefs by a conditional probability $\hat{p}_i(\theta_{-i} | \theta_i)$ that the opponents' types are a particular deleted type profile $\theta_{-i} \in \Theta_{-i}$ given that player i 's known type is θ_i . By Bayes' Rule we have⁶

$$\hat{p}_i(\theta_{-i} | \theta_i) = \frac{\mathbb{P}(\theta_i \& \theta_{-i})}{\mathbb{P}(\theta_i)} = \frac{p(\theta)}{p_i(\theta_i)}. \quad (2)$$

Player i 's knowledge of her own type may or may not affect her beliefs about the types of her opponents. When players' types are independent, the probability of a particular type profile θ is just the product of the players' marginal distributions, each evaluated at the type θ_j specified by θ , i.e. $\forall \theta \in \Theta$,

$$p(\theta) = \prod_{j \in I} p_j(\theta_j). \quad (3)$$

Therefore, when players' types are independent, player i 's subjective beliefs about others' types are independent of her own type:

$$\hat{p}_i(\theta_{-i} | \theta_i) = \frac{p(\theta)}{p_i(\theta_i)} = \frac{\prod_{j \in I} p_j(\theta_j)}{p_i(\theta_i)} = \prod_{j \in I \setminus \{i\}} p_j(\theta_j), \quad (4)$$

which does not involve θ_i .

Example: Micro type spaces

Consider two players, Lucky and Dopey. They are in separate rooms and will compete against each other in a computer-intensive task. Clearly the brand of computer each employs will significantly affect her performance. Each player observes the brand of her own computer but not that of her opponent, so we describe each player's type by the brand of the computer with which she is endowed. The two available brands are Macintosh and IBM. Therefore both players have identical type spaces

⁵ The i subscript on the p_i does not mean that we are referring to player i 's beliefs. Indeed, player i is the only player guaranteed to know player i 's type. The subscript actually serves only a formal requirement: identifying the domain of this function, viz. Θ_i , and the deleted type profile over which the summation should extend.

⁶ Here the subscript i on the p_i happens to identify the holder of these beliefs; however, it actually refers to the component of θ on which this probability is conditioned. Note also that the denominator is positive, and therefore this quantity is well defined, because we have excluded impossible types.

$$\Theta_L = \Theta_D = \{\text{Mac}, \text{IBM}\}.$$

The type-profile space is

$$\Theta = \Theta_L \times \Theta_D = \{(\text{Mac}, \text{Mac}), (\text{Mac}, \text{IBM}), (\text{IBM}, \text{Mac}), (\text{IBM}, \text{IBM})\}.$$

First assume computers are being handed out by independent fair-coin tosses such that heads dispenses a Macintosh and tails an IBM. In this scenario the types are independent; e.g. $p((\text{Mac}, \text{IBM})) = p_L(\text{Mac})p_D(\text{IBM}) = \frac{1}{4}$. Lucky's prior subjective beliefs—prior, that is, to observing her own type—about Dopey's computer are given by $p_D(\text{Mac}) = p((\text{Mac}, \text{Mac})) + p((\text{IBM}, \text{Mac})) = \frac{1}{2}$.⁷ After observing that she was awarded a Mac, her posterior beliefs about Dopey's computer are $\hat{p}_L(\text{Mac}|\text{Mac}) = p((\text{Mac}, \text{Mac}))/p_L(\text{Mac}) = p_L(\text{Mac})p_D(\text{Mac})/p_L(\text{Mac}) = p_D(\text{Mac}) = \frac{1}{2}$; i.e. the prior and the posterior are the same.

Alternatively, assume that there is only one Mac, and it is awarded to Lucky or Dopey on the basis of a coin toss. The unfortunate remaining player receives an IBM. In this case the probability distribution over the type-profile space is

$$p((\text{Mac}, \text{IBM})) = p((\text{IBM}, \text{Mac})) = \frac{1}{2}, \quad p((\text{Mac}, \text{Mac})) = p((\text{IBM}, \text{IBM})) = 0.$$

Although Lucky's prior about Dopey's computer is still $p_D(\text{Mac}) = p_D(\text{IBM}) = \frac{1}{2}$, after Lucky observes that she was awarded the Mac, she updates her prior to reflect her certainty that Dopey received the IBM:

$$\hat{p}_L(\text{IBM}|\text{Mac}) = \frac{p((\text{Mac}, \text{IBM}))}{p_L(\text{Mac})} = \frac{1/2}{1/2} = 1.$$

This joint probability distribution and the associated marginal probabilities are shown in the table below.

		Dopey		
		Mac	IBM	
Lucky	Mac	0	$\frac{1}{2}$	$\frac{1}{2} = p_L(\text{Mac})$
	IBM	$\frac{1}{2}$	0	$\frac{1}{2} = p_L(\text{IBM})$
		$\frac{1}{2}$	$\frac{1}{2}$	
		$= p_D(\text{Mac})$	$= p_D(\text{IBM})$	

Strategies

We will now see that it is not sufficient for a strategy for player $i \in I$ in a Bayesian game to merely specify an action for that player; it must specify an action for every type $\theta_i \in \Theta_i$ of player i . Player i 's payoff function depends upon her type. Therefore for given actions by, and types of, her opponent, each type of player i will be solving a different maximization problem yielding different best responses; each

⁷ Note, again, that the D subscript on the p_D refers to the component of θ with respect to which the marginal distribution is computed, not to the player whose beliefs we are discussing.

type of player i is playing a different game from her sisters. Therefore a pure strategy for player i in a static Bayesian game is *type contingent*; it is a function $s_i: \Theta_i \rightarrow A_i$. The space of all such functions and hence player i 's pure-strategy space is $S_i = A_i^{\Theta_i}$.⁸ For a particular type θ_i of player i , her strategy s_i specifies some action $a_i = s_i(\theta_i) \in A_i$. A mixed strategy $\sigma_i: \Theta_i \rightarrow \mathcal{A}_i$ for player i specifies a mixed action $\alpha_i \in \mathcal{A}_i$ for each type of player i ; i.e. $\forall \theta_i \in \Theta_i, \sigma_i(\theta_i) \in \mathcal{A}_i$. As usual $\Sigma_i = \mathcal{A}_i^{\Theta_i}$ is the space of all player- i mixed strategies.

At this point you might say: "Fine. I agree that different types would play different actions; but, since only one type of each player is participating in the game at any time, it is only necessary for an equilibrium to specify *that* type's action." In order to rebut this line of reasoning let's consider some player j , who wants to play a best response to i 's strategy. The problem for player j is that player i 's action will depend upon her type θ_i , and player j doesn't know what type of player i he is facing. He considers perhaps several types of player i as possibilities for his opponent. He needs to consider the actions of all those types of player i , because he can't rule any of them out. Since every player must be able to compute a best response, every player must know the planned actions of all types of all other players. Therefore any well-defined strategy profile must define an action for every type of every player. Therefore a strategy profile s maps type profiles into action profiles; i.e. $s: \Theta \rightarrow A, s \in S = \prod_{i \in I} S_i$.

Bayesian equilibrium

Consider a particular player $i \in I$ and a particular one of her types $\theta_i \in \Theta_i$. Assume that her $n - 1$ opponents' types are described by some deleted type profile $\theta_{-i} \in \Theta_{-i}$ and that they play some deleted action profile $a_{-i} \in A_{-i}$. If player i then chooses an action $a_i \in A_i$, her utility will be $u_i((a_i, a_{-i}), (\theta_i, \theta_{-i}))$. More generally, if the players choose a mixed-action profile $\alpha \in \mathcal{A}$, player i 's expected utility is $u_i((\alpha_i, \alpha_{-i}), (\theta_i, \theta_{-i}))$.

Now assume player i knows the type-contingent mixed strategies $\sigma_{-i} \in \Sigma_{-i}$ her opponents are playing; i.e. she knows what mixed actions they would take for any given set of types. However, she doesn't know their *realized* types, so she doesn't know the actual deleted mixed-action profile α_{-i} which will occur as a result of their type-contingent strategies. What action $a_i \in A_i$ should player i choose? Although player i doesn't know θ_{-i} , she does know the probability distribution p by which Nature generates type profiles; and she also knows her own type θ_i , upon which she conditions her subjective probability about the types θ_{-i} of her opponents. For any particular combination θ_{-i} of other players' types, player i assesses this combination the probability $\hat{p}_i(\theta_{-i} | \theta_i)$. Therefore she also adds this probability to the event that her opponents will choose the particular deleted mixed-action profile $\sigma_{-i}(\theta_{-i}) \in \mathcal{A}_{-i}$. Player i 's expected utility, then, given her knowledge of her own type θ_i and of her opponents' type-contingent strategies σ_{-i} , if she chooses the action $a_i \in A_i$, is

$$\sum_{\theta_{-i} \in \Theta_{-i}} \hat{p}_i(\theta_{-i} | \theta_i) u_i((a_i, \sigma_{-i}(\theta_{-i})), (\theta_i, \theta_{-i})). \quad (5)$$

⁸ A^B is the set of all functions from $B \rightarrow A$.

For a_i to be a best response by type θ_i of player i , that choice must maximize (5) over her action space A_i . We define player i 's best-response correspondence $BR_i: \Sigma_{-i} \times \Theta_i \rightarrow A_i$, which maps opponents' strategy profiles and player- i types into player- i actions, by

$$BR_i(\sigma_{-i}, \theta_i) = \arg \max_{a_i \in A_i} \prod_{\theta_{-i} \in \Theta_{-i}} \hat{p}_i(\theta_{-i} | \theta_i) u_i((a_i, \sigma_{-i}(\theta_{-i})), (\theta_i, \theta_{-i})). \quad (6)$$

When a strategy profile $\sigma \in \Sigma$ is such that every type of every player is maximizing her expected utility given the type-contingent strategies of her opponents, then we say that σ is a Bayesian-Nash equilibrium of this game of incomplete information. In other words, σ is a *Bayesian-Nash equilibrium* if $\forall i \in I, \forall \theta_i \in \Theta_i, \text{supp } \sigma_i(\theta_i) \subset BR_i(\sigma_{-i}, \theta_i)$.

Example: To build or not to build

Consider two firms and one market.⁹ Firm 1 will definitely produce in this market; firm 2 may decide to Enter the market or Refrain. Firm 1 faces an investment decision: should it make a costly investment in order to modernize its plant? If firm 1 modernizes, firm 2 will find the competition overwhelming and would be better off not entering. However, if firm 1 eschews investment—keeping antique technology—firm 2 would find it worthwhile to enter and compete with firm 1. Firm 1's investment cost is either low or high; firm 1 knows this cost, but firm 2 only has beliefs such that firm 1's cost is high with probability ρ and low with probability $1 - \rho$. The payoffs are shown in Figure 1.

	High investment cost [ρ]			Low investment c [$1 - \rho$]	
	Enter [y]	Refrain [$1 - y$]		Enter [y]	Refrain [$1 - y$]
Modern	0, -2	4, 0	Modern [x]	3, -2	7, 0
Antique	4, 2	6, 0	Antique [$1 - x$]	4, 2	6, 0

Figure 1: A Bayesian investment/entry game.

Firm 2 has no private information, therefore its type space Θ_2 is degenerate—a singleton—so we will ignore it. Firm 1 has private information about its cost, so its type space has two elements, which we'll denote $\underline{c}$ and $\bar{c}$, for low and high cost, respectively. A strategy for firm 1 is an action for each of its two types, i.e. $s_1(c) \in A_1 = \{M, A\}$, for $c \in \Theta_1 = \{\underline{c}, \bar{c}\}$. A strategy for firm 2 is just a single action $a_2 \in A_2 = \{E, R\}$.

We note immediately that the high-cost firm 1 has a strictly dominant action, viz. A . Therefore in any Bayesian equilibrium s we must have $s_1(\bar{c}) = A$. The best-response for the low-cost firm 1, however, depends on Firm 2's strategy. If firm 2 enters, firm 1 prefers A ; if firm 2 refrains, firm 1 prefers M . We denote by y the probability that firm 2 chooses to Enter. The low-cost firm 1's payoffs to Modern and Antique, respectively, as a function of firm 2's mixed strategy y , are

⁹ See Fudenberg and Tirole [1991: →6.2, 211–213] for an example of Bayesian equilibrium with a continuum of types.

$$u_1(M; y, c) = 3y + 7(1 - y) = 7 - 4y,$$

$$u_1(A; y, c) = 4y + 6(1 - y) = 6 - 2y.$$

The low-cost firm 1 weakly prefers M , then, when $y \leq \frac{1}{2}$. Denoting by x the probability that the low-cost firm 1 chooses M , we can write the low-cost firm 1's mixed-strategy best-response correspondence $x^*(y)$ as

$$x^*(y) = \begin{cases} \{1\}, & y < \frac{1}{2}, \\ [0, 1], & y = \frac{1}{2}, \\ \{0\}, & y > \frac{1}{2}. \end{cases}$$

Now we seek Firm 2's best-response correspondence. Firm 2 faces the high-cost firm 1 with probability ρ , in which case Firm 1 definitely chooses A , resulting in a payoff of 2 to firm 2 if he enters. With probability $1 - \rho$, Firm 2 faces the low-cost firm 1, who chooses M with some probability x . If Firm 2 chooses to Enter, its expected payoff is

$$u_2(E; x) = 2\rho + (1 - \rho)[-2x + 2(1 - x)] = 2 - 4(1 - \rho)x.$$

If Firm 2 chooses Refrain, its expected payoff is zero. Firm 2 weakly prefers to Enter, then, if $x \leq 1/2(1 - \rho) \equiv \bar{x}$. We note that $\bar{x} \in [\frac{1}{2}, \infty)$ for $\rho \in [0, 1]$. We can write Firm 2's mixed-strategy best-response correspondence, then, as

$$y^*(x) = \begin{cases} \{1\}, & x < \bar{x}, \\ [0, 1], & x = \bar{x}, \\ \{0\}, & x > \bar{x}. \end{cases}$$

We note that, unlike the low-cost firm 1's best-response correspondence x^* , firm 2's best-response correspondence y^* depends on ρ through y^* 's dependence upon $\bar{x}$.

Now we can easily find the Bayesian-Nash equilibria of the game determined by the high-cost probability ρ by finding the intersection of the graphs of the players' mixed-strategy best-response correspondences. Figure 2 plots the graphs of x^* and y^* for three cases: a when $\rho \in [0, \frac{1}{2})$, b when $\rho = \frac{1}{2}$, and c when $\rho \in (\frac{1}{2}, 1]$. The Bayesian-Nash equilibria are indicated with $\star$ s. We can write any equilibrium of this Bayesian game in the form of a triple:

$$(A, x \circ M \oplus (1 - x) \circ A; y \circ E \oplus (1 - y) \circ W),$$

where the first element is the high-cost firm 1's dominant action A , the second element is the low-cost firm 1's mixed action, and the third element (separated by a semicolon to indicate a different player) is firm 2's mixed action.

When firm 1 is more likely to be low cost, viz. when $\rho < \frac{1}{2}$, the character of the equilibria is determined by the low-investment-cost game on the right-hand side of Figure 1. There are two pure-

strategy equilibria $(A, M; R)$ and $(A, A; E)$, corresponding respectively to 1 modernization by the low-cost firm 1 and refraining from entry by firm 2 and 2 sticking with antique technology by the low-cost firm 1 and entry by firm 2. There is also a mixed-strategy equilibrium in which the low-cost firm 1 modernizes one-half the time and firm 2 enters with probability $\bar{x}$, which increases the more likely firm 1 is high-cost.

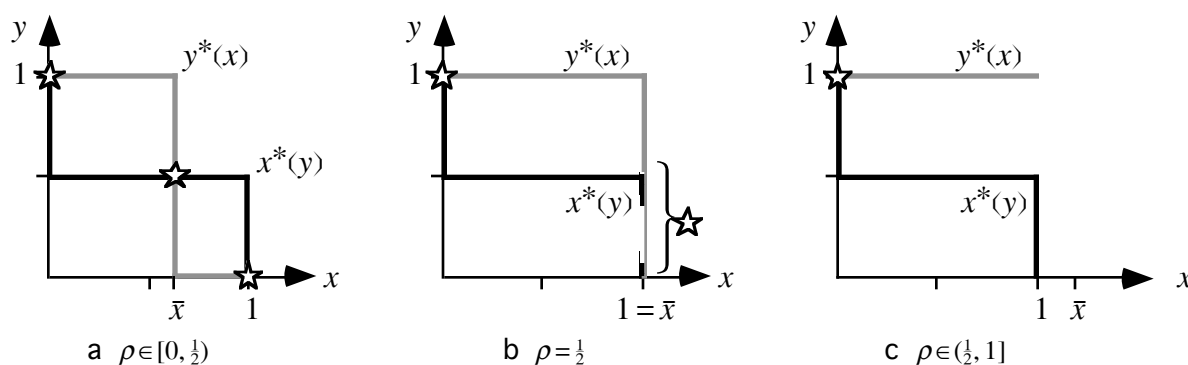


Figure 2: Best-response correspondences for the low-cost firm 1 and firm 2 for three regimes of the probability that firm 1 has high cost.

When it is equally likely that firm 1 is low or high cost, viz. when $\rho = \frac{1}{2}$, there is again a pure-strategy equilibrium in which both types of firm 1 stay with the antique technology and firm 2 enters. The remaining pure-strategy and mixed-strategy equilibria from case a have now merged together into a continuum of equilibria: The low-cost firm 1 modernizes, which makes firm 2 indifferent to entry. As long as firm 2 doesn't enter with too high a probability, the low-cost firm 1 still strictly prefers to modernize.

When it is more likely that firm 1 is high cost, the Bayesian game is dominated by the high-cost game on the left-hand side of Figure 1. It has a single pure-strategy equilibrium, determined by iterated strict dominance, in which both types of firm 1 stay with the antique technology and firm 2 enters.

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Jim Ratliff's Game Theory Course:

6.2: Perfect Bayesian Equilibria of Sender-Receiver (Signalling) Games

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Abstract



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We consider here the simplest dynamic games of incomplete information: *sender-receiver* games. There are only two players: a Sender (S) and a Receiver (R). The Sender's action will be to send a *message*, m , chosen from a message space M to the Receiver. The Receiver will observe this message m and respond to it by choosing an action a from his action space A .

To make this game a simple but nontrivial game of incomplete information we endow the Sender with some private information which we describe by her type. The Receiver has no private information, so he has but a single type, which we then have no need to mention further. The Receiver does have prior beliefs (i.e. prior to observing the Sender's message) about the Sender's type, which are described by the probability distribution over the Sender's type space. After the Receiver takes an action a , each player is awarded a payoff which can in general depend on the message m the Sender sent, the action a the Receiver took in response, and the type which Nature chose for the Sender. We can express this game of incomplete information as an extensive-form game of imperfect information by explicitly representing Nature, who chooses a type for the Sender.

A strategy for the Sender is a specification of a message (or distribution over messages) for each Sender type. A strategy for the Receiver is a pure or mixed action in response to each possible message in the message space. After observing the Sender's message, the Receiver will update his beliefs about the type of the Sender, using Bayes Rules when possible.

A Bayesian equilibrium of the Sender-Receiver game is (a) a strategy for each type of Sender, (b) a strategy for the Receiver, and (c) a conditional posterior belief system describing the Receiver's updated beliefs about the Sender's type as a function of the observed message, which satisfies two optimality conditions and a Bayes-consistency condition. Each Sender type's message must be optimal given the anticipated reaction by the Receiver. The Receiver's strategy must choose, for each "on the path" message, an action that is optimal given the Receiver's updated posterior beliefs about the Sender's type. And the Receiver's conditional posterior belief system must be consistent with Bayes' Rule whenever possible (i.e., on the equilibrium path). Perfect Bayesian equilibrium strengthens the Receiver-optimality condition to apply to off-the-path messages as well.

An example Sender-Receiver game motivates the claim that Perfect Bayesian equilibria can still be undesirable. Two additional refinements are considered and applied to example games: The Test of Dominated Messages, and the Intuitive Criterion.



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Information	Extensive-Form Games
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Perfect Bayesian Equilibrium in Sender-Receiver Games

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Introduction

We previously studied static games of imperfect information.¹ Each player $i \in I$ has private information which was summarized by her type $\theta_i \in \Theta_i$. Each player knows her own type but does not in general know the types of her opponents. Each player i 's beliefs about the types $\theta_{-i} \in \Theta_{-i}$ of her opponents are derived from her knowledge of her own type θ_i and a common prior belief $p \in \Delta(\Theta)$ over the space of type profiles.

Nature moves first, picking a type profile $\theta \in \Theta$ according to the probability distribution $p \in \Delta(\Theta)$. Nature then privately informed each player $i \in I$ of her type: player 1 is type θ_1 , player 2 is type $\theta_2, \dots$, player n is type θ_n . Then each player $i \in I$ of the n players simultaneously chooses an action $a_i \in A_i$ from her action space. A payoff $u_i(a, \theta)$ is then awarded to each player, which depends on the action profile $a \in A$ the players chose and the type profile θ Nature chose.

It was because the players simultaneously chose their actions that we called these games static. Now we want to generalize our analysis by considering dynamic games of incomplete information; i.e. we consider games in which some players take actions before others and these actions are observed to some extent by some other players.

Sender-receiver games

We consider here the simplest dynamic games of incomplete information: *sender-receiver* games. There

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¹ See the "Static Games of Incomplete Information" handout.

are only two players: a Sender (S) and a Receiver (R). The Sender's action will be to send a *message* $m \in M$ chosen from a *message space* M to the Receiver. The Receiver will observe this message m and respond to it by choosing an action $a \in A$ from his action space A .

To make this game a simple but nontrivial game of incomplete information we endow the Sender with some private information which we describe by her type $\theta \in \Theta$. The Receiver has no private information, so he has but a single type, which we then have no need to mention further. The Receiver does have prior beliefs (i.e. prior to observing the Sender's message) about the Sender's type, which are described by the probability distribution $p \in \Delta(\Theta)$ over the Sender's type space Θ . In other words, before observing the Sender's message, the Receiver believes that the probability that the Sender is some particular type $\theta \in \Theta$ is $p(\theta)$.

We will typically assume that the type space Θ , the message space M , and the Receiver's action space A are finite sets.

After the Receiver takes an action $a \in A$, each player is awarded a payoff which can in general depend on the message m the Sender sent, the action a the Receiver took in response, and the type θ which Nature chose for the Sender. The payoffs to the Sender and Receiver to a (message, action, type) triple $(m, a, \theta) \in M \times A \times \Theta$ are $u(m, a, \theta)$ and $v(m, a, \theta)$, respectively. I.e. $u, v: M \times A \times \Theta \rightarrow \mathbb{R}$.

We can express this game of incomplete information as an extensive-form game of imperfect information by explicitly representing Nature, who chooses a type $\theta \in \Theta$ for the Sender. Because the Sender observes this choice of Nature, every Sender information set is a singleton, and the number of Sender information sets is equal to the number of possible Sender types, viz. $\#\Theta$. The Receiver observes only the message sent by the Sender. Therefore the number of Receiver information sets is equal to the number of possible messages the Sender can transmit, viz. $\#M$. Within each of his information sets the Receiver cannot distinguish between the Sender's possible types; so each Receiver information set has a number of nodes equal to the number of possible Sender types, viz. $\#\Theta$. Therefore the total number of Receiver nodes is the product of the cardinalities of the message and type spaces, viz. $\#M \cdot \#\Theta$.

Example: Life's a Beach?

For example, the Sender might be a job applicant and the Receiver an employer. The Sender's decision could be a choice between going to College or going to the Beach. The particular choice the Sender makes is her message.² The employer will observe this decision and decide to Hire or Reject the applicant. In this example the Sender's message space is $M = \{\text{College, Beach}\}$ and the Receiver's action space is $A = \{\text{Hire, Reject}\}$.

The Sender's private information could concern her aptitude: she knows whether she is Bright or

² Of course in this case the message seems more than a mere message. The terms "message" for the Sender and "action" for the Receiver both refer to actions taken by a player. The distinction between the two is only interpretational. We use "message" for the Sender's action to acknowledge that the Sender realizes that the Receiver will respond to the Sender's action, and therefore the Sender can attempt to influence the Receiver's response through her choice of message.

Dull. Therefore her type space would be $\Theta = \{\text{Bright}, \text{Dull}\}$. The Receiver's prior beliefs concerning the probability that the Sender is Bright or Dull can be described by a single number $\gamma \in [0, 1]$. With probability γ , the Sender is Bright; with probability $1 - \gamma$, she is Dull. I.e. the Receiver's prior beliefs $p \in \Delta(\Theta)$ are defined by $p(\text{Bright}) = \gamma$ and $p(\text{Dull}) = 1 - \gamma$.

Consider the simple sender-receiver game shown in Figure 1. Note that the Sender has two information sets, corresponding to her two types (viz. Bright and Dull). The Receiver also has two information sets, but these correspond to the Sender's two possible messages (viz. Beach and College) rather than to the Sender's possible types. (The Receiver's left-hand information set is his Beach information set and his right-hand information set is his College information set.)

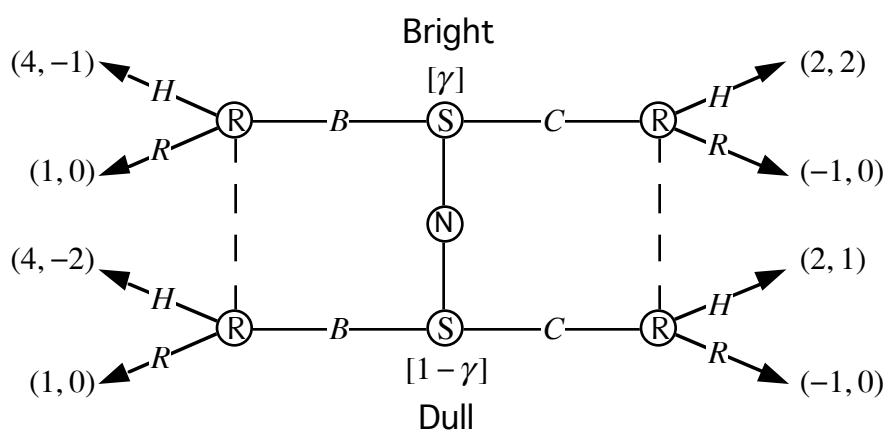


Figure 1: A simple sender-receiver game.

Let's interpret the payoffs shown in Figure 1. The first and second payoffs of each ordered pair are the Sender's and Receiver's payoffs, respectively, for a particular type/message/action triple. For a fixed type and Receiver action, the Sender's payoff to going to the Beach is always two greater than her payoff to going to College.³ For fixed educational and employment decisions, the Sender's payoff is independent of her type.⁴ For a fixed type and educational decision, the Sender receives a payoff from being Hired which is 3 greater than her payoff if she is Rejected.⁵ To summarize the Sender's payoffs (with the appropriate *ceteris paribus* qualifications): the Sender prefers the Beach over going to College, prefers being Hired over being Rejected, and is not discriminated against due to aptitude.

Whenever the Receiver Rejects an applicant, the Receiver gets a payoff of zero. Although the Sender's aptitude did not directly influence the Sender's payoffs, aptitude is payoff-relevant to the Receiver when he Hires: For a fixed educational decision, the Receiver's payoff to Hiring is 1 greater

³ For example, If the Bright applicant is Hired, she receives a payoff of 4 from the Beach but only 2 from College. If the Dull applicant is Rejected, she receives a payoff of 1 from the Beach but only -1 from College.

⁴ For example, if the applicant goes to the Beach and is Hired, she receives a payoff of 4 regardless of whether she is Bright or Dull.

⁵ For example, if the Bright applicant goes to College, she receives 2 if she is Hired and only -1 if she is Rejected. If the Dull applicant goes to the Beach, she receives 4 if she is Hired and only 1 if she is Rejected.

when the Sender is Bright than when she is Dull.⁶ Whereas the Sender disliked going to College, the Receiver appreciates a hired Sender's higher education: For a fixed type, the Receiver's payoff to Hiring is 3 greater when the Sender went to College rather than to the Beach.⁷ In fact, the Receiver's appreciation of a Sender's College education is so great that the Sender's educational decision is decisive in determining whether the Receiver should Hire or Reject: For both types of Sender, the Receiver's payoff to Hiring exceeds his payoff to Rejecting if and only if the Sender went to College.

Strategies in sender-receiver games

A pure strategy for a player in any extensive-form game is a mapping from her information sets to her available actions at the relevant information set. There is a one-to-one correspondence between the Sender's information sets and her type space Θ . Therefore a pure strategy for the Sender is a map $\bar{m}: \Theta \rightarrow M$ from her type space Θ to her message space M . There is a one-to-one correspondence between the Receiver's information sets and the Sender's message space. Therefore a pure strategy for the Receiver is a mapping $\bar{a}: M \rightarrow A$ from the Sender's message space M to the Receiver's action space A .

We can also define behavior strategies for the players. The Sender can send mixed messages. Let $\mathcal{M} \equiv \Delta(M)$ be the set of probability distributions over the Sender's message space M . A behavior strategy for the Sender is a map $\sigma: \Theta \rightarrow \mathcal{M}$ from her type space Θ to mixtures over her message space. Therefore, for all types $\theta \in \Theta$, $\sigma(\theta) \in \mathcal{M}$ is a mixture over messages. In particular, for any message $m \in M$, we denote by $\sigma(m|\theta)$ the probability, according to the Sender behavior strategy σ , that a type- θ Sender will send the message m .

For a given Sender strategy σ , a message m is *on the path* if, according to σ , there exists a type θ who sends m with positive probability. The set of on-the-path messages for Sender strategy σ is

$$M^+(\sigma) = \{m: \exists \theta \in \Theta, \sigma(m|\theta) > 0\} = \bigcup_{\theta \in \Theta} \text{supp } \sigma(\theta).^8 \quad (1)$$

The Receiver can also randomize his actions in response to his message observation. Let $\mathcal{A} \equiv \Delta(A)$ be the set of probability distributions over the Receiver's action space. Then a behavior strategy for the Receiver is a map $\rho: M \rightarrow \mathcal{A}$ from the Sender's message space M to the mixed-action space $\mathcal{A}$.⁹ Therefore, for all messages $m \in M$, $\rho(m) \in \mathcal{A}$ is a mixture over Receiver actions. In particular, for any action $a \in A$, we denote by $\rho(a|m)$ the probability, according to the Receiver behavior strategy ρ , that the Receiver will choose the action a conditional on having observed the message $m \in M$.

⁶ If the Sender goes to College, the Receiver's payoff to Hire is 2 when the Sender is Bright but only 1 when the Sender is Dull. If the Sender goes to the Beach, the Receiver's payoff to Hire is -1 when the Sender is Bright and -2 when she is Dull.

⁷ For example, if the Receiver hires the Dull Sender, the Receiver gains a payoff of 1 if the Sender went to College compared to a payoff of -2 if the Sender had gone to the Beach instead.

⁸ Note that the support of $\sigma(\theta)$ is the set of messages which a type- θ Sender sends with positive probability when she is playing according to the behavior strategy σ .

⁹ The symbols σ and ρ were selected for these behavior strategies to be mnemonically friendly—i.e. in the hope that “sigma” and “rho” would suggest “Sender” and “Receiver,” respectively.

Sender's best-response strategies

We first ask: When is a Sender strategy $\bar{m} \in M^\Theta$ a best response to some Receiver behavior strategy $\rho \in \mathcal{A}^M$?¹⁰ Consider the case where a type- θ Sender chooses to send a message $m \in M$ knowing that the Receiver will respond according to his behavior strategy $\rho \in \mathcal{A}^M$. This Sender's expected utility will be a convex combination of her payoffs to particular pure actions by the Receiver, viz. $\sum_{a \in A} \rho(a|m) u(m, a, \theta)$. A pure-strategy $\bar{m} \in M^\Theta$ will be a best response for the Sender to a Receiver behavior strategy $\rho \in \mathcal{A}^M$ if, for every type $\theta \in \Theta$ of Sender, the message specified by $\bar{m}(\theta)$ maximizes the expected utility of a type- θ Sender given that the Receiver will respond to the message $\bar{m}(\theta)$ according to the strategy ρ . For a given Receiver mixed strategy $\rho \in \mathcal{A}^M$, the set of optimal messages for a type- θ Sender is

$$\tilde{M}(\rho, \theta) \equiv \arg \max_{m \in M} \sum_{a \in A} \rho(a|m) u(m, a, \theta). \quad (2)$$

Therefore a Sender strategy $\bar{m} \in M^\Theta$ is a best response to the Receiver strategy $\rho \in \mathcal{A}^M$ if and only if, for all $\theta \in \Theta$, $\bar{m}(\theta) \in \tilde{M}(\rho, \theta)$. A sender behavior strategy $\sigma \in \mathcal{M}^\Theta$ is a best response to the Receiver behavior strategy $\rho \in \mathcal{A}^M$ if and only if, for all $\theta \in \Theta$, $\text{supp } \sigma(\theta) \subset \tilde{M}(\rho, \theta)$.

Receiver's best-response strategies

Now we ask: When is a Receiver strategy $\bar{a} \in A^M$ a best response to a Sender behavior strategy $\sigma \in \mathcal{M}^\Theta$?

Updating the Receiver's beliefs

The Receiver chooses an action after he observes the Sender's message. He wants to choose the action which is optimal given the best beliefs he can have concerning the Sender's type. The Receiver enters the game with prior beliefs $p \in \Delta(\Theta)$ concerning the Sender's type θ . Because the Receiver knows the Sender's type-contingent message-sending strategy $\sigma \in \mathcal{M}^\Theta$, the Receiver might be able to infer something more about the Sender's type and thereby update his beliefs.

As long as the observed message is not totally unexpected given that the Sender is playing the behavior strategy σ —i.e. there is some type which, according to σ , sends that message with positive probability—we can use Bayes' Rule to update the Receiver's prior beliefs $p \in \Delta(\Theta)$. Specifically, for any observed on-the-path message $m \in M^+(\sigma)$, we denote the Receiver's posterior belief that the Sender is type θ by $p^B(\theta|m)$, which is given from Bayes' Rule by

$$p^B(\theta|m) \equiv \frac{p(\theta) \sigma(m|\theta)}{\sum_{\theta' \in \Theta} p(\theta') \sigma(m|\theta')}.^{11} \quad (3)$$

We see the justification for the restriction to on-the-path (not totally unexpected) messages: if some

¹⁰ For any sets A and B , A^B is the set of all functions from $B \rightarrow A$.

¹¹ The numerator of the right-hand side is the probability of the event "the Sender is type θ and sends message m ." The denominator is the probability that message m is sent.

message m is never sent regardless of which type the Sender is, then the denominator of the right-hand side will vanish.

In general we can define the Receiver's posterior beliefs even after observing off-the-path (and therefore totally unexpected) messages. For every message $m \in M$ we let $\tilde{p}(m) \in \Delta(\Theta)$ be the Receiver's posterior beliefs—after observing the message m —about the Sender's type. In other words, the Receiver attaches the probability $\tilde{p}(\theta | m)$ to the event that the Sender has type $\theta \in \Theta$ conditional upon the Receiver observing the message $m \in M$. So $\tilde{p}: M \rightarrow \Delta(\Theta)$ is a conditional posterior belief system.

Where does the Receiver's conditional posterior belief system $\tilde{p}: M \rightarrow \Delta(\Theta)$ come from? It is derived from the Receiver's prior beliefs $p \in \Delta(\Theta)$ and updated in response to his observation of the Sender's message m . We require that this updating be done according to Bayes' Rule *whenever possible*. This means that, for all on-the-path messages $m \in M^+(\sigma)$ and for all types $\theta \in \Theta$, $\tilde{p}(m | \theta) = p^B(m | \theta)$. Alternatively but equivalently, we can say that the conditional posterior belief system $\tilde{p}: M \rightarrow \Delta(\Theta)$ is consistent with Bayes' Rule if the restriction of $\tilde{p}$ to the on-the-path messages $M^+(\sigma)$ is p^B .

Message-wise optimization

Consider a Receiver pure strategy $\hat{a} \in A^M$. If a type- θ Sender sends the message $m \in M$ and the Receiver responds according to his pure strategy $\hat{a}$, the Receiver's payoff will be $v(m, \hat{a}(m), \theta)$. The probability with which he receives this particular payoff is the probability of the event “the Sender is type θ and sends message m .” This probability is the probability that the Sender is type θ , viz. $p(\theta)$, multiplied by the probability that the Sender sends the message m conditional on the Sender being type θ , viz. $\sigma(m | \theta)$. Therefore the expected utility $V(\hat{a}, \sigma)$ to the Receiver who plays the strategy $\hat{a} \in A^M$ against the Sender behavior strategy $\sigma \in \mathcal{M}^\Theta$ is the sum of the probability-weighted payoffs $p(\theta)\sigma(m | \theta)v(m, \hat{a}(m), \theta)$ over all possible combinations of messages and types:

$$V(\hat{a}, \sigma) \equiv \prod_{m \in M} \prod_{\theta \in \Theta} p(\theta)\sigma(m | \theta)v(m, \hat{a}(m), \theta). \quad (4)$$

A Receiver strategy $\bar{a} \in A^M$ will be a best response to the Sender behavior strategy $\sigma \in \mathcal{M}^\Theta$ if and only if it maximizes the Receiver's expected utility over all possible Receiver pure strategies; i.e.

$$\bar{a} \in \arg \max_{\hat{a} \in A^M} V(\hat{a}, \sigma). \quad (5)$$

At first glance the optimization problem in (5) might appear problematic because it requires maximization over a function space. Fortunately the maximand, from (4), is additively separable in the various messages $m \in M$, so we'll be able to construct a best-response Receiver strategy $\bar{a} \in A^M$ via a message-by-message optimization to find individual best-response actions $\bar{a}(m)$ for each message $m \in M$. This simplification is justified by the following Lemma which you are invited to prove for yourself.

Lemma

Let A be a set and M be a finite set. Let f be a function $f: M \times A \rightarrow \mathbb{R}$. Then

$$\bar{a} \in \arg \max_{\hat{a} \in A^M} \prod_{m \in M} f(m, \hat{a}(m)) \quad (6)$$

if and only if, for all $m \in M$,

$$\bar{a}(m) \in \arg \max_{a \in A} f(m, a). \quad (7)$$

To apply this Lemma to the optimization problem (5) we define

$$f(m, a) \equiv \prod_{\theta \in \Theta} p(\theta) \sigma(m | \theta) v(m, a, \theta). \quad (8)$$

Now we have, from (4),

$$V(\hat{a}, \sigma) \equiv \prod_{m \in M} f(m, \hat{a}(m)). \quad (9)$$

Therefore, from (5), (9), and the Lemma, we see that the Receiver strategy $\bar{a} \in A^M$ is a best response to the Sender behavior strategy $\sigma \in \mathcal{M}^\Theta$ if and only if, for all $m \in M$,

$$\bar{a}(m) \in \arg \max_{a \in A} f(m, a). \quad (10)$$

If a message m is off the path, i.e. $m \in M \setminus M^+(\sigma)$, then it is sent by no type: for every type $\theta \in \Theta$, $\sigma(m | \theta) \equiv 0$. Therefore, $\forall m \in M \setminus M^+(\sigma)$, $\forall a \in A$, $f(m, a) \equiv 0$. Therefore all actions $a \in A$ are maximizers of $f(m, a)$ when m is an off-the-path message. I.e. $\forall m \in M \setminus M^+(\sigma)$,

$$A = \arg \max_{a \in A} f(m, a). \quad (11)$$

When $m \in M^+(\sigma)$ is an on-the-path message, it is useful to divide the maximand of (10) by the guaranteed-nonzero probability that m is sent, viz. $\sum_{\theta' \in \Theta} p(\theta') \sigma(m | \theta')$. This does not change the set of maximizers of (10). This division allows us, using (8) and (3), to express the condition (10) in terms of the Receiver's Bayes-updated beliefs: $\forall m \in M^+(\sigma)$,

$$\bar{a}(m) \in \arg \max_{a \in A} \prod_{\theta \in \Theta} p^B(\theta | m) v(m, a, \theta). \quad (12)$$

But this maximand is simply the Receiver's expected utility, given her Bayes-updated beliefs about the Sender's type, when she chooses the action $a \in A$ after observing the on-the-path message $m \in M^+(\sigma)$. Therefore condition (12) states that it is necessary and sufficient—in order that the Receiver strategy $\bar{a} \in A^M$ to be a best response to the Sender behavior strategy $\sigma \in \mathcal{M}^\Theta$ —that it specify for each on-the-path message an action which is a best response to that message given the Receiver's Bayes-updated beliefs.

For a given conditional posterior belief system $\tilde{p}: M \rightarrow \Delta(\Theta)$, the Receiver's expected utility to the action $a \in A$ conditional upon having observed the message $m \in M$ is $\sum_{\theta \in \Theta} \tilde{p}(\theta | m) v(m, a, \theta)$. Therefore for

a given conditional posterior-belief system $\tilde{p}$, the set of Receiver best-response actions to some message m is given by

$$\tilde{A}(\tilde{p}, m) = \arg \max_{a \in A} \prod_{\theta \in \Theta} \tilde{p}(\theta | m) v(m, a, \theta). \quad (13)$$

A Receiver pure strategy $\bar{a} \in A^M$ is a best response to the Sender behavior strategy $\sigma \in \mathcal{M}^\Theta$ if and only if, for all $m \in M$, $\bar{a}(m) \in \tilde{A}(\tilde{p}^B, m)$. A Receiver behavior strategy $\rho \in \mathcal{A}^M$ is a best response to the Sender behavior strategy $\sigma \in \mathcal{M}^\Theta$ if and only if, for all $m \in M$, $\text{supp } \rho(m) \subset \tilde{A}(\tilde{p}^B, m)$.

Bayesian equilibrium

Definition A Bayesian equilibrium of the sender-receiver game is a triple $(\sigma, \rho, \tilde{p}) \in \mathcal{M}^\Theta \times \mathcal{A}^M \times (\Delta(\Theta))^M$ satisfying the following three conditions:

1 For all types $\theta \in \Theta$,

$$\text{supp } \sigma(\theta) \subset \tilde{M}(\rho, \theta), \quad (14)$$

2 For all on-the-path messages $m \in M^+(\sigma)$,

$$\text{supp } \rho(m) \subset \tilde{A}(\tilde{p}, m), \quad (15)$$

3 The conditional posterior belief system $\tilde{p}$ is consistent with Bayes' Rule whenever possible in the sense that the restriction of $\tilde{p}$ to the on-the-path messages $M^+(\sigma)$ is $\tilde{p}^B$.

Note that optimality from the Receiver is required only at on-the-path information sets. Therefore the only Receiver information sets at which the specification of the conditional posterior belief system $\tilde{p}$ enters into the definition of Bayesian equilibrium is at on-the-path-message information sets, where these beliefs are just the ones derived from Bayes' Rule from (3).

Example: Bayesian equilibrium in Life is a Beach?

We consider further the game of Figure 1. In order to make the Receiver's posterior belief system explicit on the extensive form, we indicate by bracketed probabilities at each Receiver node the Receiver's belief at each of his information sets that he is located at that node conditional on having reached that information set. See Figure 2. For example, if the Receiver observes Beach, then $s \in [0, 1]$ is the probability the Receiver attaches to the event that the Sender is Bright. If the Receiver observes College, then $1 - t$ is the probability the Receiver attaches to the event that the Sender is Dull.

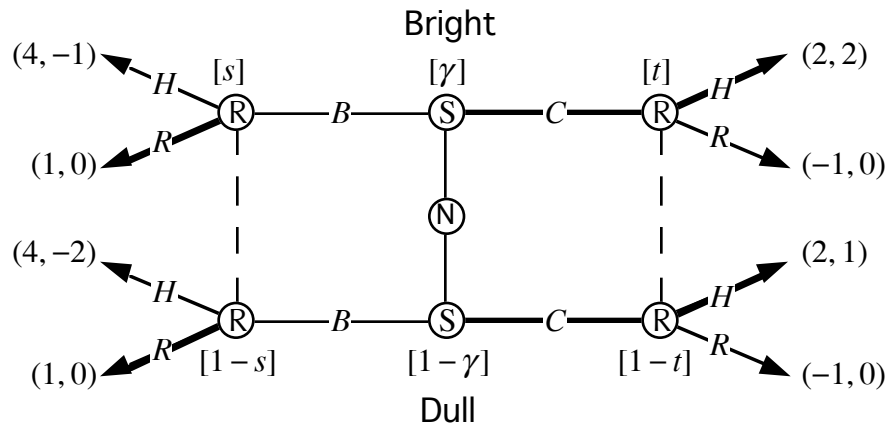


Figure 2: Life is a Beach? with the Receiver's conditional posterior beliefs indicated.

We can represent a strategy profile by the ordered sextuple $(X, Y; L, R; s, t)$, where

X = Sender's action if Bright,

Y = Sender's action if Dull,

L = Receiver's action if Beach is observed,

R = Receiver's action if College is observed,

s = Receiver's belief (probability), given that Beach is observed, that the Sender is Bright,

t = Receiver's belief (probability), given that College is observed, that the Sender is Bright.

Consider the following strategy profile: $(C, C; R, H; *, \gamma)$.¹² This strategy profile is depicted in Figure 2 by the thick line segments. (We note that according to this strategy profile the Beach message is never sent by any type of Sender and is therefore off-the-path. Therefore in order to evaluate whether this strategy profile is a Bayesian equilibrium we need not specify conditional posterior beliefs for the Receiver at this information set. Hence the "*" in the above specification.)

Let's verify that this strategy profile is a Bayesian equilibrium of this game. First we check whether any type of Sender wishes to deviate away from going to College in favor of going to the Beach instead, given the hiring policies of the Receiver. Each type of Sender receives a payoff of 2 from conforming to College. Each would receive a lower payoff of 1 instead if she went to the Beach. Therefore neither type of Sender would deviate.

To check whether the Receiver would prefer to change his hiring policy given the Sender's type-contingent strategy we need only check the only on-the-path information set, viz. the College information set. The easy way to see that Hiring is optimal at the College information set is to notice that

¹² I.e. both types of Sender go to College. The Receiver Rejects any Beachgoers and Hires any College graduates. If the Receiver observes College, he believes that the probability is γ that the Sender is Bright.

Hiring is better for the Receiver than Rejecting for each type of Sender separately. Therefore regardless of the Receiver's belief t , the corresponding convex combination of Hiring payoffs will exceed the zero he would get if he Rejects. More formally... For any Receiver beliefs $\gamma \in [0, 1]$ that the Sender is Bright conditional on observing College, the Receiver's expected payoff to Hiring at the College information set is $2\gamma + (1 - \gamma) > 0$. Therefore for any $\gamma \in [0, 1]$ the specified strategy profile is a Bayesian equilibrium.

The specification of posterior beliefs at the College information set, viz. $t = \gamma$, implies that, even after observing the Sender's message, the Receiver's beliefs about the Sender's type are unchanged from her prior beliefs. This no-updating result occurs because this is a *pooling* strategy profile—i.e. all types of the Sender send the same message. We can also use (5) to see formally that this specification $t = \gamma$ is consistent with Bayes' Rule. (This is the last step in verifying that the strategy profile is a Bayesian equilibrium.) Letting $m = \text{College}$ and $\theta = \text{Bright}$,

$$\begin{aligned} p^B(\text{Bright} | \text{College}) &= \frac{p(\text{Bright}) \sigma(\text{College} | \text{Bright})}{p(\text{Bright}) \sigma(\text{College} | \text{Bright}) + p(\text{Dull}) \sigma(\text{College} | \text{Dull})} \\ &= \frac{\gamma \cdot 1}{\gamma \cdot 1 + (1 - \gamma) \cdot 1} = \gamma. \end{aligned} \quad (16)$$

I.e. $\tilde{p}(\text{Bright} | \text{College}) = t = \gamma = p^B(\text{Bright} | \text{College})$, exactly as required by condition 3 for Bayesian equilibrium.

Now let's look at another strategy profile: $(B, B; R, R; \gamma, *)$. This strategy profile is indicated below in Figure 3. Each type of Sender is sending the optimal message, given the Receiver's hiring policy, by choosing Beach. (Because each type of Sender will be Rejected whatever message she sends, she'll choose the most pleasant message, viz. go to the Beach.) To check the optimality of the Receiver's hiring plans we need to check only the single on-the-path information set, viz. Beach. Uneducated Senders aren't worth hiring, so Rejection at this information set is optimal for the Receiver. You can also verify, similarly to the demonstration for the strategy profile of Figure 2, that the specification $s = \gamma$ is consistent with Bayes' Rule.

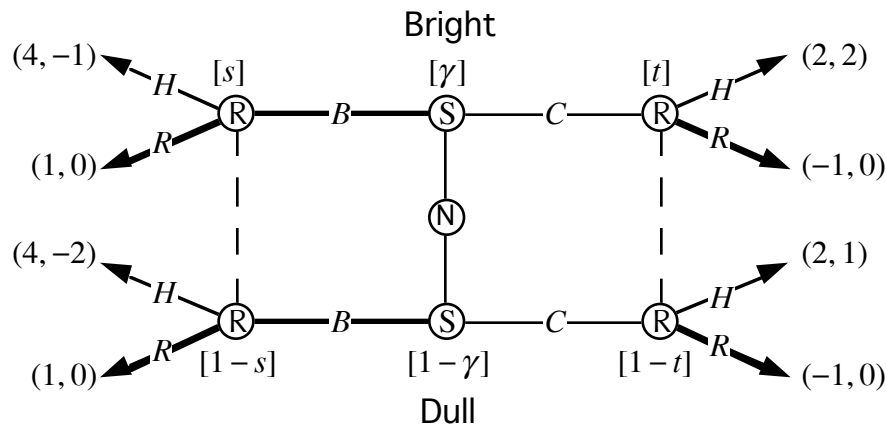


Figure 3: A less credible Bayesian equilibrium.

However, note why the above strategy profile's specification for the Sender is a best response to the

Receiver's hiring plans: Each type of Sender eschews College because the Receiver plans to Reject college-educated applicants. But regardless of the type of Sender the Receiver would be better off Hiring, rather than Rejecting, a college-educated applicant. No matter what off-the-path posterior belief $t \in [0, 1]$ we specified, Hiring would be the unique best response for the Receiver at his College information set. This equilibrium is undesirable because it relies on an incredible off-the-path action by the Receiver.

Perfect Bayesian equilibrium

We saw in the above example that the strategy profile depicted in Figure 3 was a Bayesian equilibrium of the game but was suspect because it relied on a nonoptimal action by the Receiver at an off-the-path Receiver information set. We can eliminate this strategy profile by a simple strengthening of our solution concept.

Definition A perfect Bayesian equilibrium of the sender-receiver game is a triple $(\sigma, \rho, \tilde{p}) \in \mathcal{M}^\Theta \times \mathcal{A}^M \times (\Delta(\Theta))^M$ satisfying the following three conditions:

1 For all types $\theta \in \Theta$,

$$\text{supp } \sigma(\theta) \subset \tilde{M}(\rho, \theta), \quad (17)$$

2 For all messages $m \in M$,

$$\text{supp } \rho(m) \subset \tilde{A}(\tilde{p}, m), \quad (18)$$

3 The conditional posterior belief system $\tilde{p}$ is consistent with Bayes' Rule whenever possible in the sense that the restriction of $\tilde{p}$ to the on-the-path messages $M^+(\sigma)$ is p^B .

Note that the only difference between this definition of perfect Bayesian and the earlier definition of Bayesian equilibrium is in the strengthening of the original Receiver-optimality condition (15)—which imposed optimality only at on-the-path-message information sets—resulting in (18), which requires optimality of the Receiver's strategy at all message information sets. Note from (13) that this also implies that now the Receiver's posterior beliefs are important even at off-the-path-message information sets. However, we aren't constrained by Bayes' Rule in the specification of these off-the-path beliefs.

The strategy profile from Figure 3 would fail to be a perfect Bayesian equilibrium regardless of how we specified $t \in [0, 1]$ because, as we saw in the analysis of the example of Figure 2, for any beliefs Hiring is better for the Receiver at the College information set is better than Rejecting there.

Also note that if all messages are on the path then if the strategy profile is a Bayesian equilibrium it is also a perfect Bayesian equilibrium.

Example: Perfect Bayesian equilibria can still be undesirable

Consider the same basic game we've been considering but with the different payoffs shown in Figure 4. Note that for a fixed hiring decision each type of Sender prefers going to Beach over going to College, but the Bright Sender finds College less onerous than the Dull Sender does. In fact this difference is extreme in the following sense: A Bright Sender is willing to incur the cost of College if it means that it makes the difference between being Hired and being Rejected.¹³ However, the Dull Sender finds College such a drag that she's unwilling to skip the Beach regardless of the effect her action has on the hiring decision of the Receiver.

For a fixed education decision the Receiver prefers to Hire the Bright Sender but prefers to Reject the Dull Sender. For a fixed type of Sender, the Receiver is indifferent between hiring a College-educated vs. a Beach-tanned Sender. Note that with this payoff structure education is unproductive. But because going to College has a higher cost for the lower-ability type of Sender, education might provide a costly signal of the Sender's type to the Receiver.

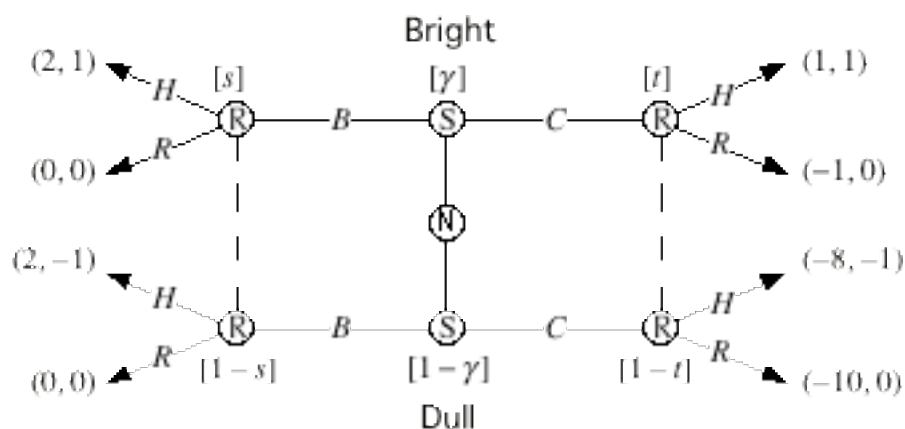


Figure 4: Education is unproductive but an effective signal of ability.

Consider the strategy profile $(C, B; R, H; 0, 1)$. This is not only a Bayesian equilibrium but also a perfect Bayesian equilibrium (because every message is on the path). This is a *separating* equilibrium because each type of Sender chooses a different action. (When each type of Sender sends a distinct message, the Receiver can deduce with certainty the identity of the Sender from her observed message.) You can use (3) to verify that the posterior-belief assignments $s = 0$ and $t = 1$ are those determined by Bayes' Rule.

Let $\gamma \in [0, 1/2)$. Consider the strategy profile $(B, B; R, R; \gamma, t)$, where $t \in [0, 1/2)$. This is a *pooling* strategy profile. This is a perfect Bayesian equilibrium. The off-the-path posterior beliefs imply that if a defection to College is observed, the defector is more likely to be Dull than Bright. However, such off-the-path beliefs are objectionable for the following reason: No matter what influence a deviation to

¹³ If going to the Beach implies that the Bright Sender will be Rejected, then going to the Beach implies a payoff of zero. If going to College is necessary to be Hired, then College implies a payoff of 1. Therefore the Bright Sender will go to College if that is necessary for being Hired.

College might have on the Receiver's hiring decision, the Dull Sender would never find going to College worthwhile. However, the Bright Sender would be willing to go to College if that convinced the Receiver that the Sender was indeed Bright and therefore should be hired.

The test of dominated messages

We saw in the above example that the pooling perfect Bayesian equilibrium profile was undesirable because it relied on the Receiver interpreting a deviation as coming from a type who would never find it optimal to deviate. The College message was *dominated* for the Dull type in the following sense: No matter how badly-for-the-Sender the Receiver might respond to the prescribed message Beach and no matter how favorably-for-the-Sender the Receiver might respond to the deviation message College, the Dull Sender would still prefer to send the prescribed message.

Denote the set of Receiver actions which are best responses, conditional on the message m , for *some* conditional posterior beliefs by

$$\hat{A}(m) = \bigcap_{\tilde{p} \in (\Delta(\Theta))^M} \tilde{A}(\tilde{p}, m).$$

(A Sender who sends the message $m \in M$ would never have to worry about a Receiver response which fell outside of the set $\hat{A}(m)$, because such an action would not be a best-response by the Receiver to any posterior belief she could possibly hold.)

Definition Message $m \in M$ is dominated for type $\theta \in \Theta$ if there exists a message $m' \in M$ such that

$$\min_{a \in \hat{A}(m')} u(m', a, \theta) > \max_{a \in \hat{A}(m)} u(m, a, \theta). \quad (19)$$

Definition Let $\psi \equiv (\sigma, \rho, \tilde{p})$ be a perfect Bayesian equilibrium. The equilibrium ψ fails the test of dominated messages if there exist types $\theta', \theta'' \in \Theta$ and an off-the-equilibrium-path message $m \in M \setminus M^+(\sigma)$ such that¹⁴

1 The receiver puts positive weight, conditional on m being observed, that the message was sent by type θ' , i.e. $\tilde{p}(\theta' | m) > 0$,

2 m is dominated for type θ' , and

3 m is not dominated for type θ'' .

Before we can reject an equilibrium because it puts positive weight on a deviant message originating

¹⁴ It is more common to allow m to be any message in M . The test stated here is equivalent because no on-the-equilibrium-path message m could possibly be dominated for a type θ' for whom $\tilde{p}(\theta' | m) > 0$, because, along the equilibrium path, $\tilde{p}$ is derived by Bayes' rule. (I.e. this would imply that $\sigma(m | \theta') > 0$, and thus that, in equilibrium, type θ' were sending a dominated message.) The statement given here simplifies the proof of the theorem to come.

from a type for whom the message is dominated, we must be able to identify a type of Sender for whom this message is *not* dominated. Otherwise this logic would force us to put zero weight on *all* types at this information set, and this would not be a legitimate conditional probability distribution.

We see that the pooling perfect Bayesian equilibrium fail the test of dominated messages.

Example: The separating equilibrium disappears and the pooling becomes reasonable.

Consider the example in Figure 5. Now College, though more costly for the Dull than for the Bright Sender, is not as costly for the Dull Sender as it was in the example of Figure 4. Going to College is no longer dominated for the Dull Sender; she would be willing to go to College if that made the difference between being Hired and being Rejected.

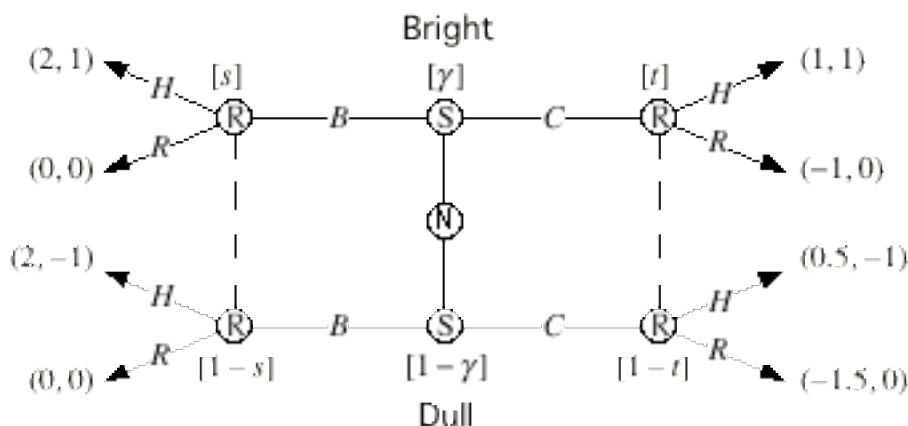


Figure 5: Pooling is now reasonable and separation is not.

The separating strategy profile $(C, B; R, H; 0, 1)$, which was a perfect Bayesian equilibrium in the game of Figure 4, is not an equilibrium of the present game, because the Dull Senders would now deviate to going to College.

The pooling equilibrium $(B, B; R, R; \gamma, t)$, where $\gamma, t \in [0, 1/2)$ of Figure 4 is not only still a perfect Bayesian equilibrium in this game, it is no longer rejected by the test of dominated messages.

Example: The test of dominated messages is not strong enough

Consider the game of Figure 6. The Receiver prefers that the Dull type be uneducated. The Bright Sender actually likes College, while the Dull Sender still finds it degrading. The Receiver strictly prefers to Hire, rather than Reject, a Bright Sender, and finds that College is unproductive when the Sender is Bright. The Receiver is indifferent between Hiring and Rejecting a Dull Beachbum Sender and strictly prefers to Reject a College-educated Dull Sender.

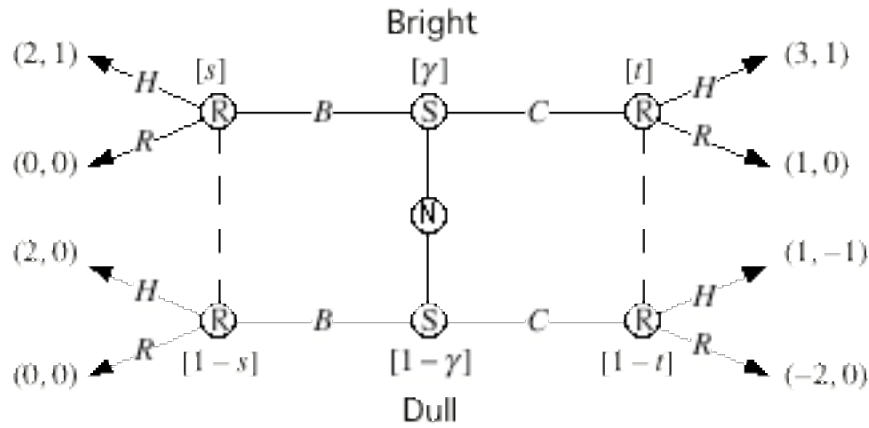


Figure 6: The test of dominated messages is not strong enough.

Consider the following equilibrium: $(B, B; H, R; \gamma, t)$, for $t \in [0, 1/2)$. I.e. if a deviation to College is observed, it is more likely that the deviator is a Dull Sender. This equilibrium passes the test of dominated messages because the Dull Sender could do worse by going to Beach (getting a zero) than by the most optimistic hopes for going to College, where she could get a 1. However, the Bright Sender could hope to gain by deviation relative to his equilibrium potential, but the Dull type cannot hope this. Therefore we shouldn't attribute positive probability to the Dull Sender deviating.

Definition Let $\psi \equiv (\sigma, \rho, \tilde{p})$ be a perfect Bayesian equilibrium. Let $\bar{u}(\theta)$ be the type- θ Sender's expected payoff in this equilibrium. Message $m \in M$ is equilibrium dominated, with respect to ψ , for type $\theta \in \Theta$ if

$$\bar{u}(\theta) > \max_{a \in \hat{A}(m)} u(m, a, \theta). \quad (20)$$

I quickly verify that domination implies equilibrium domination:

Fact If $m \in M$ is dominated for type $\theta \in \Theta$ then, for every perfect Bayesian equilibrium ψ , m is equilibrium dominated with respect to ψ for type θ .

Proof Let $m' \in M$ be a message which dominates m for type θ . For any equilibrium Receiver strategy ρ , the Sender's expected payoff to the message m' is

$$\int_{a \in \hat{A}(m')} \rho(a | m') u(m', a, \theta) \geq \min_{a \in \hat{A}(m')} u(m', a, \theta), \quad (21)$$

which is derived from (18) and $\tilde{A}(\tilde{p}, m') \subset \hat{A}(m')$. For any $m'' \in \text{supp } \sigma(\theta)$,

$$\bar{u} = \int_{a \in A} \rho(a | m'') u(m'', a, \theta). \quad (22)$$

Assume that m is not equilibrium dominated with respect to the equilibrium ψ . Then from (19), the converse of (20), (21), and (22),

$$\prod_{a \in A(m')} \rho(a | m') u(m', a, \theta) > \prod_{a \in A} \rho(a | m'') u(m'', a, \theta). \quad (23)$$

Therefore $m'' \notin \tilde{M}(\rho, \theta)$ —note in (2) that $\rho(a | m') = 0$ for $m' \in A \setminus \hat{A}(m')$ —which contradicts (17). ☺

Definition Let $\psi \equiv (\sigma, \rho, \tilde{p})$ be a perfect Bayesian equilibrium. The equilibrium ψ fails the refinement $\mathcal{J}$ —the Intuitive Criterion—if there exist types $\theta', \theta'' \in \Theta$ and an off-the-equilibrium-path message $m \in M \setminus M^+(\sigma)$ such that^{15,16}

- 1 The receiver puts positive weight, conditional on m being observed, that the message was sent by type θ' , i.e. $\tilde{p}(\theta' | m) > 0$,
- 2 m is equilibrium dominated with respect to ψ for type θ' , and
- 3 m is not equilibrium dominated with respect to ψ for type θ'' .

It is often asserted (or at least strongly suggested) that $\mathcal{J}$ is an equilibrium refinement of $\mathcal{D}$.¹⁷ However, a perfect Bayesian equilibrium strategy profile can pass the Intuitive Criterion yet fail the test of dominated messages. Yet, if a perfect Bayesian equilibrium survives the Intuitive Criterion, then there exists a perfect Bayesian equilibrium which yields the same outcome (i.e. probability distribution over terminal nodes) and which survives both the test of dominated messages and the Intuitive Criterion. (See Ratliff [1993].)

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¹⁵ As in the definition of refinement $\mathcal{D}$, the restriction of m to off-the-equilibrium-path messages is without loss of generality.

¹⁶ For more on these refinements see Cho and Kreps [1987] and Kreps [1990: 436].

¹⁷ For example, Fudenberg and Tirole [1989: 312] say “The idea is roughly to extend the elimination of weakly dominated strategies to strategies which are dominated relative to equilibrium payoffs. So doing eliminates more strategies and thus refines the equilibrium concept further.” Fudenberg and Tirole [1991: 446–447] suggest that replacing the equilibrium path by its payoff results in an equilibrium refinement whose rejection requirements are weaker and easier to apply. Kreps [1990: 436] says that the Intuitive Criterion is a “stronger test” than the test of dominated messages. Gibbons [1992: 213] says that, because equilibrium dominance is easier to satisfy than dominance, the Intuitive Criterion makes the test of dominated messages redundant.

Sender-Receiver Games,” mimeo.

Jim Ratliff's Game Theory Course:

6.3: Perfect Bayesian Equilibria of Extensive-Form Games

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Abstract



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We have seen that Nash equilibria of extensive-form games can be undesirable because they can rely on incredible threats at off-the-equilibrium-path subgames. We were sometimes able to refine away such undesirable equilibria by strengthening our solution concept--demanding subgame perfection, which requires that the restriction of a strategy profile to any subgame be a Nash equilibrium of that subgame.

I offer an example extensive-form game to demonstrate that subgame perfection will not eliminate all undesirable equilibria of extensive-form games.

The concept of Perfect Bayesian equilibrium for extensive-form games is defined by four Bayes Requirements. These requirements eliminate the bad subgame-perfect equilibria by requiring players to have beliefs, at each information set, about which node of the information set she has reached, conditional on being informed she is in that information set.



[Previous section:
§6.2: Perfect Bayesian Equilibria of Sender-Receiver
\(Signalling\) Games](#)

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Perfect Bayesian Equilibrium in Extensive-Form Games

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Introduction

We have seen that Nash equilibria of extensive-form games can be undesirable because they can rely on incredible threats at off-the-equilibrium-path subgames. We were sometimes able to refine away such undesirable equilibria by strengthening our solution concept—demanding subgame perfection, which requires that the restriction of a strategy profile to any subgame be a Nash equilibrium of that subgame.

Subgame perfection will not eliminate all undesirable equilibria of extensive-form games, however. Consider the extensive-form game of Figure 1. Analysis of its strategic form quickly shows that this game has two pure-strategy Nash equilibria: (U, l) and (A, r) . This game has only one subgame, viz. the entire game, so both of these Nash equilibria are also subgame perfect.

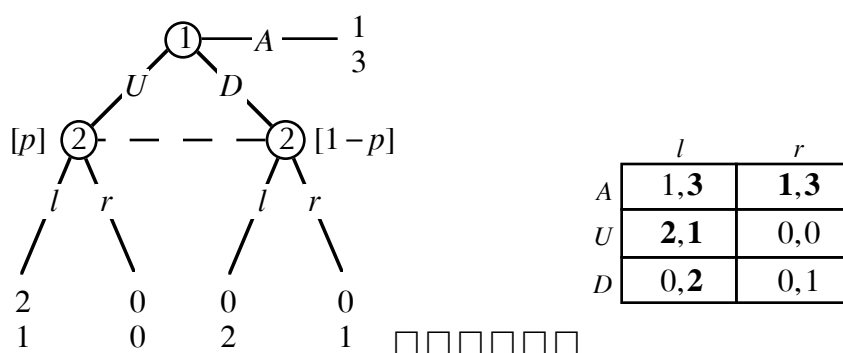


Figure 1: Subgame perfection admits undesirable equilibria.

The (A, r) equilibrium is objectionable for the following reason. [Note that player 2's information set is off-the-equilibrium path with respect to the (A, r) equilibrium—i.e. it is never reached when the players conform to the equilibrium specification.] If player 2's information set were ever reached, player 2 would be uncertain about whether it was reached via player 1 having chosen U or via player 1 having

chosen D . However, it doesn't matter to player 2's decision which move player 1 had chosen. No matter what player 2's beliefs about player 1's non- A choice, player 2 strictly prefers to choose l at his information set when it is reached. (If player 1 had chosen U , player 2 receives 1 from l and only 0 from r . If player 1 had chosen D , player 2 receives 2 from l and only 1 from r .) Because r is not a best response at player 2's information set for any possible beliefs which player 2 might have there, we say that r is dominated at player 2's information set. We can describe our dissatisfaction with the (A, r) equilibrium by objecting to its specification of an action at an information set which is dominated at that information set.

Let's formalize this reasoning. We begin by requiring that at every one of her information sets each player has *some* beliefs about the node at which she is located conditional on having reached that information set.

Bayes Requirement 1

For a particular strategy profile σ , we require that, for each player $i \in I$, and at each of her information sets $h_i \in H_i$, player i has beliefs $\rho_i(h_i) \in \Delta(h_i)$ about the node at which she is located conditional upon being informed that play has reached the information set h_i .

The beliefs $\rho_i(h_i) \in \Delta(h_i)$ are just a probability distribution over the nodes in the information set. Player i 's beliefs in this game, then, are a specification, for each player- i information set $h_i \in H_i$, of such conditional beliefs at that information set. The n -tuple $\rho = (\rho_1, \dots, \rho_n)$ of player beliefs is a *belief profile*.

In order to properly critique alleged equilibria we require that a candidate equilibrium be not just a strategy profile σ but be a strategy-belief profile (σ, ρ) . We want to state an equilibrium requirement that would loosely say something like: For every player $i \in I$ and every player- i information set $h_i \in H_i$, player i 's strategy is a best response given her beliefs $\rho_i(h_i) \in \Delta(h_i)$ at the information set h_i . However, this is too vague—at least to me!—so we must be more precise in our statement.

Recall that a subgame is formed by identifying a singleton information set and including all its successors from the original game. Information sets, actions, and payoffs for the subgame were derived from the original game by restriction. We now generalize the concept of a subgame and define a *continuation game*. A continuation game is an information set $h_i \in H_i$ for some player $i \equiv \iota(h_i)$ and all of its successor nodes from the original game. Again, information sets, actions, and payoffs in the continuation game are derived from the original game by restriction. If the designated initial information set is not a singleton, then this continuation is not a subgame.¹ And with good reason: This continuation game cannot be played as a game in its own right, because there is no initial node. So we include in the specification of the continuation game the probability distribution $\rho_i(h_i)$ over the nodes of the initial information set h_i given in the belief profile ρ . (Think of this continuation game as being preceded by a move of Nature's, where Nature chooses between the nodes of h_i according to the probability distribution $\rho_i(h_i)$.) We can restrict any strategy σ_j and any player beliefs ρ_j to this continuation game

¹ We're assuming perfect recall.

just as we restricted a strategy to a subgame: simply throw out its specifications at information sets which don't belong to the smaller game.

Bayes Requirement 2

Consider the continuation game defined by some player- i information set $h_i \in H_i$ and the conditional beliefs $\rho_i(h_i)$. The restriction of the strategy-belief profile (σ, ρ) to this continuation game must be a Nash equilibrium of the continuation game.

Definition

Let (σ, ρ) be a strategy-belief profile and let $h_i \in H_i$ be an information set for player $i = i(h_i)$. Let $(\tilde{\sigma}, \tilde{\rho})$ be the restriction of (σ, ρ) to the continuation game which begins at the information set h_i . We say that the player- i strategy σ_i is *strictly dominated beginning at the information set h_i* if there exists another player- i strategy σ_i' such that, for all other deleted strategy profiles σ_{-i}' for the opponents, player i 's expected payoff in the continuation game is strictly higher for $(\tilde{\sigma}_i', \tilde{\sigma}_{-i})$ than for $\tilde{\sigma}_i$.

Bayes Requirements 1 and 2 are sufficient to remove the undesirable equilibrium, viz. (A, r) , in Figure 1. To see this we construct the continuation game beginning at player 2's information set for some beliefs parameterized by $p \in [0, 1]$. See Figure 2. The strategic form of this continuation game is also shown in Figure 2, from which it is clear that l is the unique Nash equilibrium in the continuation game.

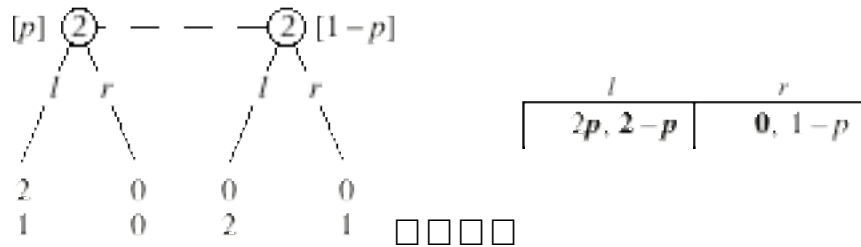


Figure 2: The continuation game beginning at player 2's information set.

More generally, Bayes Requirement 2 rejects all strategy profiles which specify at any information set an action which is dominated at that information set.

Example: Restricting a strategy-belief profile to a continuation game

Consider the strategy-belief profile $s = (U, a, d; l; p)$ for some $p \in [0, 1]$ in the extensive-form game in Figure 2a. Now consider the continuation game beginning at player 2's information set. Figure 2b depicts the restriction $\tilde{s}$ of this strategy-belief profile to the continuation game.

The expected payoff vector to the restriction $\tilde{s}$ is $p(2, 3) + (1-p)(3, 4) = (3-p, 4-p)$.

Let's evaluate whether the strategy profile s passes Bayes Requirement 2 with respect to the continuation game beginning at player 2's information set. We can construct a strategic form for the continuation game. For example, the expected payoff vector to the restricted strategy profile $(a, c; l)$ is $p(2, 3) + (1-p)(0, 1) = (2p, 2p+1)$. The expected payoff vector to $(*, *, r)$ is $p(1, 2) + (1-p)(1, 6) =$

$(1, 6 - 4p)$. Similar calculations for player 1's other strategies yield the payoff matrix in Figure 3.

In order that s passes Bayesian Requirement 2 with respect to the specified continuation game, $\tilde{s} = (a, d; l; p)$ must be a Bayesian Nash equilibrium of the continuation game. This requires that (a, d) be a best response by player 1 to l , i.e.

$$3 - p \geq \max \{2p, 0, 3 - 3p\},$$

which is satisfied for all $p \in [0, 1]$. In order that l be a best response by player 2 to (a, d) we must have

$$4 - p \geq 6 - 4p,$$

which is satisfied if and only if $p \in [\frac{2}{3}, 1]$. Therefore $\tilde{s}$ is a Bayesian Nash equilibrium of this continuation game if and only if $p \in [\frac{2}{3}, 1]$.

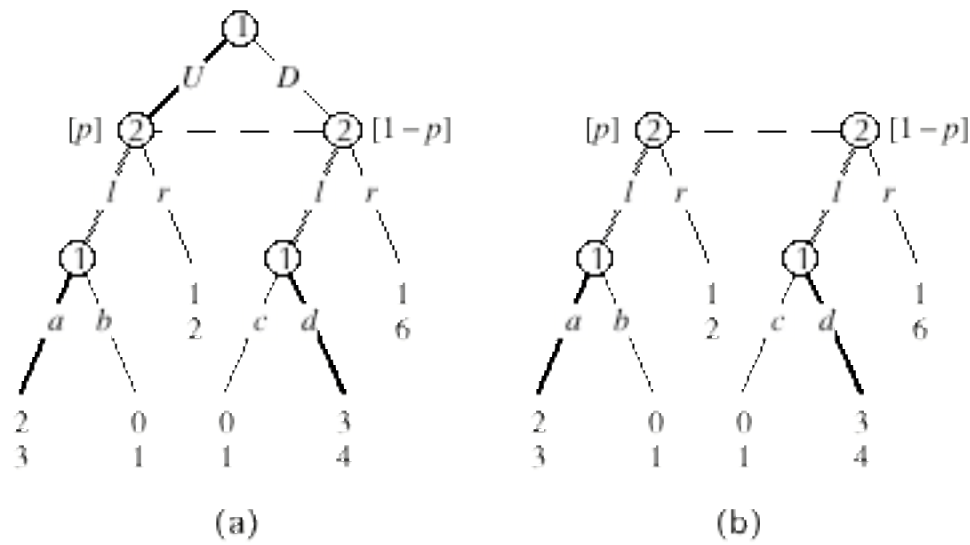


Figure 2: The restriction of the $(U, a, d; l; p)$ of (a) to the continuation game (b).

	l	r
(a, c)	$2p, 1 + 2p$	$1, 6 - 4p$
(a, d)	$3 - p, 4 - p$	$1, 6 - 4p$
(b, c)	$0, 1$	$1, 6 - 4p$
(b, d)	$3 - 3p, 4 - 3p$	$1, 6 - 4p$

Figure 3: The strategic-form matrix corresponding to the continuation game of Figure 2b.

We can also ask whether l is strictly dominated beginning at player 2's information set. This would require that

$$6 - 4p > \max \{1 + 2p, 4 - p, 1, 4 - 3p\} = 4 - p,$$

or $p < \frac{2}{3}$.

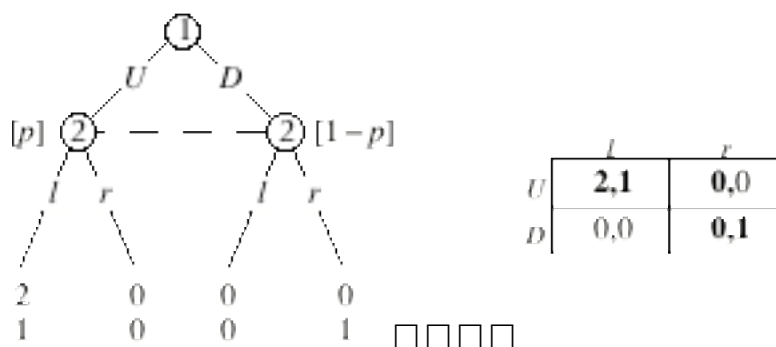


Figure 4: Requirements 1 and 2 don't even guarantee Nash equilibrium.

Bayes Requirements 1 and 2 are not strong enough however to generally capture even the concept of Nash equilibrium. Consider the game in Figure 4. The strategy-belief profile $(U, r; p=0)$ satisfies Bayes Requirements 1 and 2, yet (U, r) is not even a Nash equilibrium of the game.

So we add a third requirement. It ensures that in equilibrium each player's beliefs are correct.

Bayes Requirement 3

The beliefs at any on-the-path information set must be determined from the strategy profile according to Bayes' Rule. I.e. if $h_i \in H_i$ is a player- i information set reached with positive probability when the players conform to σ , then $\rho_i(h_i) \in \Delta(h_i)$ must be computed from σ using Bayes' Rule.

This requirement eliminates the non-Nash-equilibrium profile $(U, r; p=0)$ from the game of Figure 4 because, given that player 1 is choosing U , player 2's belief at his information set must put all weight on the node reached by U —i.e. we must have $p=1$ rather than $p=0$.

However, the addition of Bayes Requirement 3 does not guarantee that surviving strategy-belief profiles are even subgame-perfect equilibria of the game. Consider the game of Figure 5. Consider the subgame beginning with player 2's singleton information set. This subgame has a unique Nash equilibrium of (U, r) . Therefore the unique subgame-perfect equilibrium of the entire game is (B, U, r) . Every information set is on the path, and Bayes' Rule implies $p=1$. This strategy-belief profile $(B, U, r; p=1)$ satisfies Bayes Requirements 1, 2, and 3.

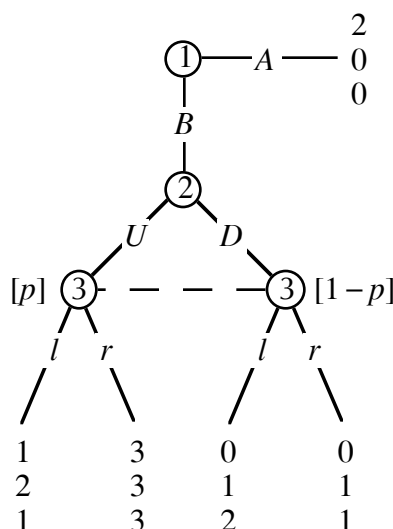


Figure 5: Requirements 1, 2, and 3 do not guarantee subgame perfection.

But now consider the strategy-belief profile $(A, U, l; p=0)$. This is a Nash equilibrium and satisfies Bayes Requirements 1, 2, and 3. (Note that player 3's information is off-the-path, and therefore Requirement 3 puts no restriction on p .) Yet this profile is not subgame perfect, because we have already seen that subgame perfection requires (U, r) by players 2 and 3. The problem with this profile can be traced to the beliefs at player 3's information set. The only way that player 3's information set could be reached is if player 1 chose B . In this case, according to the strategy profile, player 2 would choose U . Therefore player 3, conditional on reaching her information set, should infer that she is located at her left-hand node and therefore should believe that $p=1$ rather than $p=0$.

We now add an additional requirement, which eliminates the non-subgame-perfect equilibrium just considered.

Bayes Requirement 4

The beliefs at any off-the-path information set must be determined from the strategy profile according to Bayes' Rule whenever possible.

Bayes Requirements 1, 2, 3, and 4 taken together constitute the definition of perfect Bayesian equilibrium in extensive-form games. (For the sender-receiver games we studied earlier, Bayes Requirement 4 has no refining power. Therefore Bayes Requirements 1, 2, and 3 constitute the definition of perfect Bayesian equilibrium in sender-receiver games.)

As we saw with sender-receiver games, perfect Bayesian equilibrium can admit equilibria which are objectionable because they rely on off-the-path beliefs which are in some sense suspect. Consider the game in Figure 6. There are two classes of pure-strategy perfect Bayesian equilibria: 1 $(U, l; p=1)$ and 2 $\{(A, r; p): p \in [0, \frac{1}{2}]\}$.

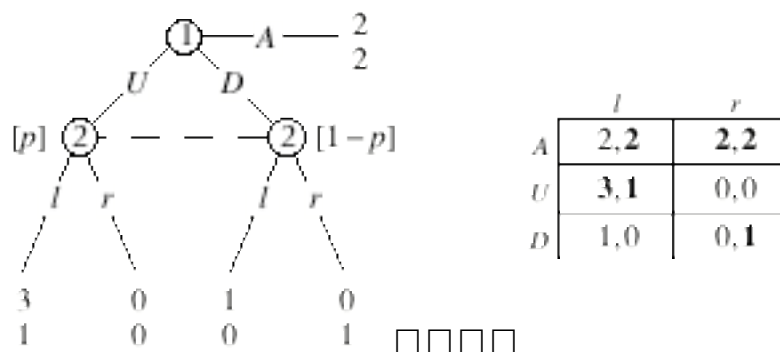


Figure 6: Perfect Bayesian equilibrium can be usefully refined.

In class 2 of these equilibria, player 2's information set is off-the-path. Player 2's beliefs at this information set are that, given that this information set was reached, there is a positive probability that the information set was reached via player 1 choosing D . But note from the strategic form that D is dominated for player 1 by A . On the other hand, U is not dominated. If player 2 observes that, contrary to his expectation, player 1 did not choose A , what should player 2 believe about player 1's actual choice? Did player 1 choose U or D ?

Player 1 could never profit by playing D instead of the dominating strategy A . However, player 1 could consider playing U in hopes that player 2 would play l , giving player 1 a payoff of 3 instead of 2. Therefore we should attach zero weight to the event that a deviation by player 1 was D rather than U . Therefore we should require, when player 2's information set is off-the-path, that $p = 1$.

A Refinement

If possible, each player's beliefs off the equilibrium path should put zero weight on nodes which can only be reached if another player plays a strategy that is strictly dominated beginning at some information set.